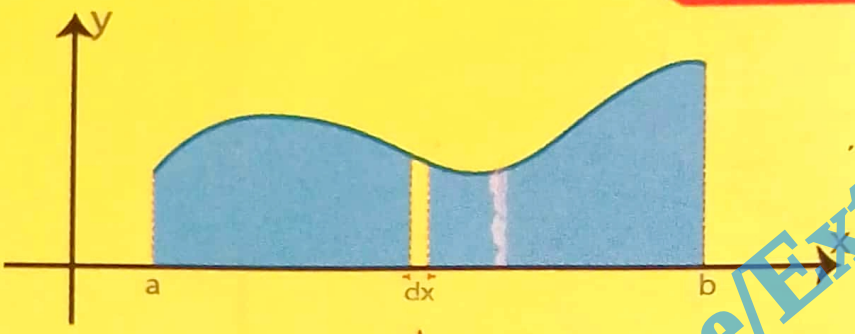
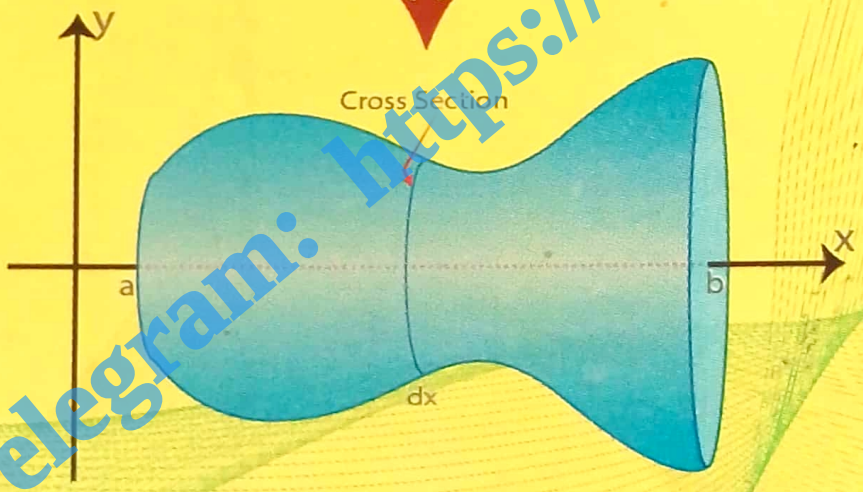
Extreme Series

MATHEMATICS

GRADE 11 - 12





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2017 E.C

Arebu Abdella

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UNIT 1

1. Relation and Function

1.1 What is a Relation

A relation describes the way how things are related. Usually the phrases like:

y is the brother of x

y is the father of x

y is the mother of x

y is less than x

y is greater than x

y is a multiple of x, ... etc.

This is day to day usual phrases to describe a relation.

Definition: A relation is a set whose elements are ordered pairs.

Note: The cartesian product of a set from A to B is denoted by $A \times B$ is a subset of a **relation.**

Example: Let A = {1,2,3}, B = {a,b,c}. Then the relation R from A to B is A × B = {(1,a), (1,b), (1,c), (2,a), (2,b), (2,c), (3 a), (3 b), (3 c)}

- The first component of the ordered pairs is called the **domain of relation.**

  ∴ Domain of R = {1,2,3}.

- The second component of the ordered pairs is called the **range of the relation.**

  ∴ Range of R = {a,b,c}

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Domain and Range of Relation

Definition: A set of ordered pairs (x,y) is called a **relation** from set A to set B, where $x \in A$ and $y \in B$.

- The set of x - values in the ordered pairs is called the **domain** of the relation.
- The set of y - values in the ordered pairs is called the **range** of the relation.

Symbol: Domain $= \{x : (x,y) \in R, \text{ for some } y\}$

Range $= \{y: (x,y) \in R \text{ for some } x\}$

Representing relations:

A relation can be represented by:

(i) as set of ordered pairs
(ii) as mapping (correspondence)
(iii) as a graph
(iv) as equation

Illustrative Example

1. Let $R = \{(x,y): y = x^3, x \in R, \text{ and } y \in R\}$

   a. Find the ordered pairs belonging to R, where $x \in \{-2, -1, 0, 1, 2\}$

   b. Fin the ordered pairs belonging to R, where $y \in \{-1, 0, 1, 2, 4, 8, 27\}$

   Solution

   a.

x	-2	-1	0	1	2
$y = x^3$	-8	-1	0	1	8

   $R = \{(x, x^3)\}$

   $\therefore R = \{(-2, -8), (-1, -1), (0,0), (1,1), (2,8)\}$.

   b. $y = x^3$ means $x = \sqrt[3]{x}$

   If $y = -1$, then $x = \sqrt[3]{-1} = -1$, $\therefore (-1, -1)$

   If $y = 0$, then $x = \sqrt[3]{0} = 0$, $\therefore (0, 0)$

   If $y = 1$, then $x = \sqrt[3]{1} = 1$, $\therefore (1, 1)$

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If $y = 2$, then $x = \sqrt[3]{2}$ $\therefore$ $(\sqrt[3]{2}, 2)$

If $y = 4$, then $x = \sqrt[3]{4}$ $\therefore$ $(\sqrt[3]{4}, 4)$

If $y = 8$, then $x = \sqrt[3]{8} = 2$ $\therefore$ $(2,8)$

If $y = 27$, then $x = \sqrt[3]{27} = 3$ $\therefore$ $(3, 27)$

$\therefore$ $R = \{(-1, -1),(0,0),(1,1),(\sqrt[3]{2}, 2),(\sqrt[3]{4},4),(2,8),(3,27)\}$

. Let $R = \{(x,y): x - y$ is an integral multiple of $3\}$.
which of the following ordered pairs belong to R?

(9,3), (3,9), (-1,2) (2,1), (1,5), (-7,-2), (-4-7), (-15)

Solution:

Note: a is an integral multiple of b if a = bk where k is an integer.

Multiple of 3 are {.... -6, -3, 0, 3, 6, 9, 12, ...}

$\therefore$ $9 - 3 = 6, 3 - 9 = -6, -1 - 2 = -3, -4 - (-7) = 3$

$\therefore$ $R = \{(9,3),(3,9), (-1,2), (-1,5), (-4,-7)\}$.

But $(2,1) \notin R, (1,5) \notin R, (-7,-2) \notin R$, because

$2 - 1 = 1, 1 - 5 = -4, -7 - (-2) = -5$ are not multiple of 3.

3. If the relation R is given by $R = \{(x,y): y = x^2 + 1\}$

where $x \in \{-2, -1, 1, 2, 3\}$, then list all elements of R and corresponding range.

Solution

$R = \{(x,y)\} = \{(x,x^2 + 1)\}$ and $R = \{(-2,5), (-1,2), (1,2), (2,5), (3,10))$

$\therefore$ Domain = {-2, -1, 1, 2, 3} and range = {2,5,10}

4. Let $A = \{1,2,3,4,5\}$ and $B = \{2,3,4\}$.

If R is the relation from A to B such that

$R = \{(x,y): x > y$, where $x \in A$ and $y \in B\}$. Then:

a. List all elements of relation R.

b. Find the domain and range of relation R.

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Solution

$A \times B = \{(1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,2), (3,3), (3,4), (4,2), (4,3), (4,4), (5,2), (5,3), (5,4)\}$.

A pair of (x,y) is in R, if $x > y$.

a. $R = \{(3,2), (4,2), (4,3), (5,2), (5,3), (5,4)\}$

b. Domain of $R = \{3, 4, 5\}$ and Range of $R = \{2, 3, 4\}$

5. Let $A = \{1,2,3,4,5,6,7,8\}$, $B = \{2,4,6,8\}$ and R is a relation from A to B such that $R = \{(x,y): y = 2x-4$, where $x \in A$ and $y \in B\}$ Then:

a. List all elements of R

b. Find the domain of and range of R

Solution

$(x,y) \in R$, if $y = 2x - 4$, $x \in A$ and $y \in B$.

Let $x = 1, \in A$, then $y = 2(1) - 4 = -2 \notin B$ $\therefore$ $(1,2) \notin R$

Let $x = 2 \in A$, then $y = 2(2) - 4 = 0 \notin B$ $\therefore$ $(2,0) \notin R$

Let $x = 3 \in A$, then $y = 2(3) - 4 = 2 \in B$ $\therefore$ $(3,2) \in R$

Let $x = 4 \in A$, then $y = 2(4) - 4 = 4 \in B$ $\therefore$ $(4,4) \in R$

Let $x = 5 \in A$, then $y = 2(5) - 4 = 6 \in B$ $\therefore$ $(5,6) \in R$

Let $x = 6 \in A$, then $y = 2(6) - 4 = 8 \in B$ $\therefore$ $(6,8) \in R$

Let $x = 7 \in A$, then $y = 2(7) - 4 = 10 \notin B$ $\therefore$ $(7,10) \notin R$

Let $x = 8 \in A$, then $y = 2(8) - 4 = 12 \notin B$ $\therefore$ $(8,12) \notin R$

So that

a. $R = \{(3,2), (4,4), (5,6) (6,8)\}$,

b. Domain of $R = \{3,4,5,6\}$ and range $= \{2, 4, 6, 8\}$

6. Let $A = \{1,2,3,4,5,6,7, ...\}$ and R is a relation from A to A defined by $R = \{(x,y): 2x + y = 9$, where $x \in A, y \in A\}$. Then:

a. Find all elements of relation R

b. Domain of relation R.

c. Range of relation R.

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Solution

$(x,y) \in R$, y is related to x, by $2x + y = 9$ or $y = 9 - 2x$, $x \in A$ and $y \in A$.

Let $x = 1 \in A$, then $y = 9 - 2(1) = 7 \in A$. $\therefore$ $(1,7) \in R$

Let $x = 2 \in A$, then $y = 9 - 2(2) = 5 \in A$ $\therefore$ $(2,5) \in R$

Let $x = 3 \in A$, then $y = 9 - 2(3) = 3 \in A$ $\therefore$ $(3,3) \in R$

Let $x = 4 \in A$, then $y = 9 - 2(4) = 1 \in A$ $\therefore$ $(4,1) \in R$

Let $x = 5 \in A$, then $y = 9 - 2(5) = -1 \notin A$. $\therefore$ $(5,-1) \notin R$

a. $R = \{(1,7), (2,5), (3,3), (4,1)\}$

b. Domain of $R = \{1,2,3,4\}$ c. Range of $R = \{1,3,5,7\}$

7. If $A = \{-2, -1, 0, 1, 2\}$ and R is a relation from A to A defined by $R = \{(x,y): x - y \in A$, where $x \in A$ and $y \in A\}$ then which of the following is **Not** true?

A. If $x \in A$, then $(x,x) \in R$. C. If $(x,y) \in R$, then $(y,x) \in R$

B. If $x \in A$, and $y \in A$, then $(x,y) \in R$ D. If $(x,y) \in R$, then $y - x \in R$

Solution:

A. The set of the ordered pair (x,x) are $\{(-2,-2),(-1,-1),(0,0),(1,1),(2,2)\}$.

$\therefore$ $-2 - (-2) = -1 - (-1) = 1 - 1 = 2 - 2 = 0 \in A$

$\therefore$ $(x,x) \in R$.

B. Let $x = 2 \in A$, and $y = -2 \in A$, then $x - y = 2 - (-2) = 2 + 2 = 4 \notin A$

$\therefore$ $(2,-2) \notin R$, because $x - y = 4 \notin A$

C. Since $(2,1) \in R$, $2 - 1 = 1 \in A$ and $(1,2) \in R$, $1 - 2 = -1 \in A$.

$\therefore$ if $(x,y) \in R$, then $(y,x) \in R$

Answer: B

8. If $A = \{0, 1, 2, 3, 4, 5\}$, $B = \{0, 1, 2, 3, 4\}$ are R is a relation from A to B such that $R = \{(x, y): y = \sqrt{x-1}$ where $x \in A$ and $y \in B\}$, then What is the range of R?

A. $\{1,2,5\}$ C. $\{0, 1, 2\}$

B. $\{0, 1, \sqrt{2}, \sqrt{3}, 2\}$ D. $\{0, 1, 2, 3\}$

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Solution

x is related to y by $y = \sqrt{x-1}$

Let $x = 0 \in A$, then $y = \sqrt{0-1} = \sqrt{-1} \notin B$

Let $x = 1 \in A$, then $y = \sqrt{1-1} = 0 \in B$ $\therefore$ $(1,0) \in R$

Let $x = 2 \in A$, then $y = \sqrt{2-1} = 1 \in B$ $\therefore$ $(2,1) \in R$

Let $x = 3 \in A$, then $y = \sqrt{3-1} = \sqrt{2} \notin B$

Let $x = 5 \in A$, then $y = \sqrt{5-1} = \sqrt{4} = 2 \in B$ $\therefore$ $(5,2) \in R$

$\therefore$ R = {(1,0), (2,1), (5,2)} and Range of R = {0, 1, 2}

Answer: C

Illustrative Example

9. Let R = $\{(x,y): 2x - 3y > 6\}$, then which of the following ordered pairs belongs to the relation R.

A. (-3, -2) B. (4, -1) C. (-1, 3) D. (-5, -3)

Solution

If (a,b) $\in$ R, then 2a - 3 b > 6.

A. R = {(-3, -2): 2(-3) -3(-2) = -6 + 6 = 0 > 6} is false

$\therefore$ (-3, -2) $\notin$ R.

B. R = {(4, -1): 2(4) -3(-1) = 8 + 3 = 11 > 6} is true

$\therefore$ (4, -1) $\in$ R

C. R = {(-1, 3): 2(-1) - 3 (3) = -2 - 9 = -11 > 6} is false

$\therefore$ (-1, 3) $\notin$ R

D. R = {(-5, -3): 2(-5) + 3 (3) = -10 + 9 = -1 > 6} is false

Answer: B

10. Which ordered pairs does not satisfy the relation R = $\{(x,y): x + 2y > 5$ and $3x - y < -2\}$

A. (-1,4) B. (0,2.5) C. (-1,4) D. (0,5)

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Solution

A. (-1,4): $-1 + 2(4) = 7 > 5$ is true and $3(-1) - 4 = -7 < -2$ is True

B. (0,2.5): $0 + 2(2.5) = 5 > 5$ is false and $3(0) - 2.5 < -2$ is true

Answer: B

1.1.1 The Graph of Relation

The graphs of a relation is the graph of its ordered pair. The graph gives a picture of the relation.

- When a relation R consist of a set of ordered pairs of a real numbers, we can represent R by a graph. Here, we going to see, how the region of the plane divided by the line.
- Every non vertical line divides the xy - plane in to two regions. One region is above the line, and the other is below the line.

The following table describes the graph of the relation:

$ax + by \geq c$ and $ax + by \leq c$

1. Solve the inequality for y, then the graph
   - $y \geq mx + b$ is satisfied above the line, $y = mx + b$
   - $y \leq mx + b$ is satisfied below the line, $y = mx + b$

11. Let $R = \{(x,y): y \geq x \text{ and } y + 2x \leq 4\}$, then which of the following is false

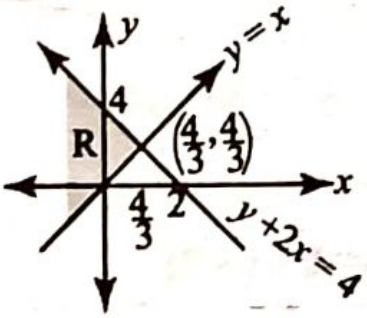
A. Domain of R is $\left(-\infty, \frac{4}{3}\right]$

B. Range of R is $\left(-\infty, \frac{2}{3}\right]$

C. $(-3,2) \in R$

D. For all real number y, $(-2, y) \in R$.

Solution



Graphing linear inequality:

(i) Solve for y, then graph the line $y = x$ and $y + 2x = 4$

- $y \leq -2x + 4$ is the region below

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the line $y = -2x + 4$.

• $y \geq x$ is the region above the line $y = x$.

(ii) solve the intersection point of the line $y = x$ and $y + 2x = 4$.

$\Rightarrow x + 2x = 4$, gives $x = \frac{4}{3}$ and $y = \frac{4}{3}$

• The set of all x - values in the shaded region is $\left(-\infty, \frac{4}{3}\right]$

$\therefore$ Domain of R = $\left(-\infty, \frac{4}{3}\right]$

The set of all y - values in the shaded region is $(-\infty, \infty)$.

$\therefore$ Range of R = $(-\infty, \infty)$.

Answer: B

12. Let R = $\{(x,y): y < x + 3$ and $y \geq x^2 - 2x - 1\}$ then which of the following is false.

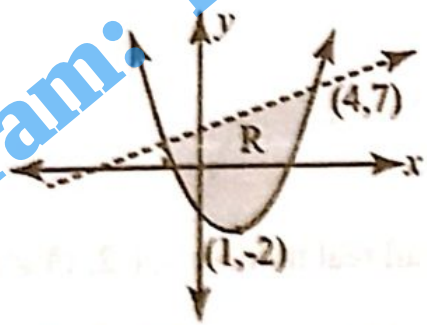
A. Domain of R is $\{x: -1 < x < 4\}$ C. $(2,5) \in R$

B. Range of R is $\{y: -2 \leq y < 7\}$ D. $(1,2) \in R$

Solution

To graph the inequality $y < x + 3$ and $y \geq x^2 - 2x - 1$,

• Solve the intersection point of the line $y = x + 3$ and $y = x^2 - 2x - 1$.



• Solving: $x + 3 > x^2 - 2x - 1$

$\Rightarrow x^2 - 3x - 4 < 0$

$\Rightarrow (x - 4)(x + 1) < 0$

$\Rightarrow \{x: -1 < x < 4\}$

To find the vertex of the parabola: $y = x^2 - 2x - 1$

$\Rightarrow y = x^2 - 2x + 1 - 1 - 1$

$\Rightarrow y = (x - 1)^2 - 2$

$\therefore$ Vertex is $(1, -2)$

• The range of the relation R is R = $\{y: -2 \leq y < 7\}$

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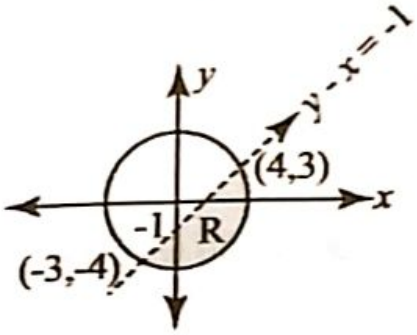
C. $(2,5) \notin R$, because $5 < 2 + 3$ is false **Answer: C**

13. The relation R is given by $R = \{(x,y): y - x < -1 \text{ and } x^2 + y^2 \leq 25\}$

Find the domain and range

Solution

To find the intersection point of the line $y - x = -1$ and the circle $x^2 + y^2 = 25$



$\Rightarrow y - x = -1$ is the same as $y = x - 1$

- Substitute $y = x - 1$ in the equation $x^2 + y^2 = 25$

$\Rightarrow x^2 + (x - 1)^2 = 25$

$\Rightarrow x^2 + x^2 - 2x + 1 = 25$

$\Rightarrow 2x^2 - 2x - 24 = 0$

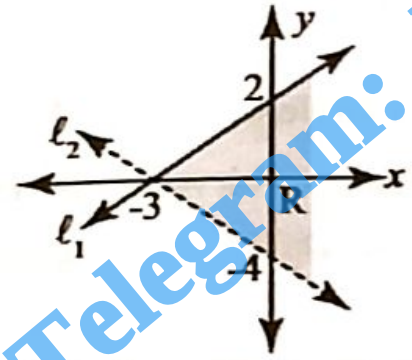
$\Rightarrow x^2 - x - 12 = 0$

$\Rightarrow (x - 4)(x + 3) = 0$

$\therefore \quad x = 4$ and $x = -3$

$\therefore$ Domain of R is $\{x: -3 < x < 4\}$ Range of R is $\{y: -5 \leq y < 3\}$

14. In figure below, the shaded region (R) is represented by the which of the following inequities.



A. $\ell_1 : 3y - 2x \geq 6$
$\ell_2 : 3y + 4x < -12$

B. $\ell_1 : 3y - 2x \leq 6$
$\ell_2 : 3y + 4x > -12$

C. $\ell_1 : 3y + 2x \leq 6$
$\ell_2 : 3y - 4x > -12$

D. $\ell_1 : 3y - 2x \geq 6$
$\ell_2 : 3y + 4x > -12$

Solution:- First write an equation of two line.

(i) The line ℓ_1 crosses the y - axis at (0,2) and the x - axis at (-3, 0).

- slope of $\ell_1 : m_1 = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{0 - 2}{-3 - 0} = \dfrac{2}{3}$

- The y - intercept is 2

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$\therefore$ The equation of the line ℓ_1 is $y = \frac{2}{3}x + 2$. The shaded region is below the solid line $y = \frac{2}{3}x + 2$ and write as $y \leq \frac{2}{3}x + 2$

• Multiplying both side by 3, it gives $3y \leq 2x + 6$, which is $3y - 2x \leq 6$

(ii) The line ℓ_2 crosses the y - axis at (0,- 4) and the x - axis at (-3,0).

Slope of ℓ_2 : $m_2 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - (-4)}{-3 - 0} = \frac{-4}{3}$

$\therefore$ The y - intercept is - 4. The equation of the line ℓ_2 is $y = \frac{-4}{3}x - 4$

The shaded region is above the broken line $y = \frac{-4}{3}x - 4$ and write as $y \geq \frac{-4}{3}x - 4$

$3y > -4x - 12$ ← Multiplying by 3

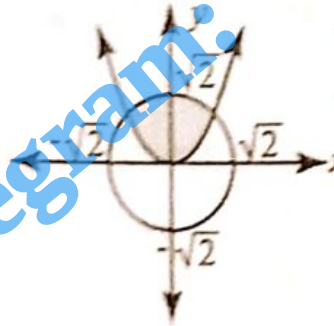
$3y + 4x > -12$

From (i) and (ii), the shaded region R is $3y - 2x \leq 6$ and $3y + 4x > 12$

Answer: B

15. Let $R = \{(x,y): y \geq x^2 \text{ and } x^2 + y^2 \leq 2\}$ then find the domain and range of R.

Solution



• The equation of $x^2 + y^2 = 2$ is circle with radius 4.

• The equation of $y = x^2$ is a parabola with vertex (0,0)

• To find the intersection of $x^2 + y^2 = 2$ and $y = x^2$, equating:

$y + y^2 = 2$ this implies $y^2 + y - 2 = 0$

$(y - 1)(y + 2) = 0$, $y = 1$, $y = -2$

Now, we have $y = x^2$ and $x^2 = 1, x = \pm 1$

$\therefore$ Domain = $\{x : -1 \leq x \leq 1\}$ and Range of R = $\{y : 0 \leq y \leq \sqrt{2}\}$

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1.1.2 Inverse Relations

The inverse of any relation is obtained by interchanging the coordinates in each ordered pair.

- If (a,b) is an ordered pair of a relation **R** then (b,a) is an ordered pair of its inverse - relation denoted by $R^{-1} = \{(y,x): (x,y) \in R\}$

Example

If $R = \{(3,4), (-2,5), (1,7), (8,-1)\}$, then

$R^{-1} = \{(4,3), (5,-2), (7,1), (-1,8)\}$

Note:

(i) The domain of a relation becomes the range of the inverse relation.

(ii) The range of relation becomes the domain of the inverse.

- Domain of R^{-1} = Range of R.
- Range of R^{-1} = Domain of R.

Finding the Inverse of Relation

To find the inverse of the relation that is given by an equation in x and y.

(i) Interchange components of the ordered pairs of the relation, or

(ii) Interchange the role of x and y in the equation.

Illustrative Example

16. Find the inverse of the relation.

a. $R = \{(x,y): y = -3x + 4\}$ b. $R = \{(x,y): y = 4x - 1\}$

Solution

a. $R^{-1} = \{(y,x): y = -3x + 4\}$. Interchange x and y

or $R^{-1} = \{(x,y): x = -3y + 4\}$,

$$R^{-1} = \left\{(x,y): y = \frac{-1}{3}x + \frac{4}{3}\right\}$$ Solve for y.

b. $R^{-1} = \{(y,x): y = 4x - 1\}$. Interchange x and y.

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or $R^{-1} = \{(x,y): x = 4y - 1\}$. solve for y

$$R^{-1} = \left\{(x,y): y = \frac{1}{4}x + \frac{1}{4}\right\}.$$

17. If the relation R is given by $R = \{(x,y): y = 2x - 3$, for the domain $x = -2, -1, 1, 3\}$, then what is the domain of R^{-1}?

A. $\{-2, -1, 1, 3\}$ C. $\{1, -5, -1, -2\}$

B. $\{-7, -4, -1, 1\}$ D. $\{-7, -5, -1, 1\}$

Solution

When $x = -2$, $y = 2(-2) - 3 = -4 - 3 = -7$

$x = -1$, $y = 2(-1) - 3 = -2 - 3 = -5$

$x = 1$, $y = 2(1) - 3 = 2 - 3 = -1$

$x = 3$, $y = 2(3) - 3 = 6 - 3 = 3$

$\therefore$ Domain of R^{-1} = Range of $R = \{-7, -5, -1, 3\}$ **Answer: D**

18. If the relation R is given by $R = \{(x,y): 2x - 3y > 3\}$ then which of the following points belongs to the inverse of R.

A. (-3, -2) B. (2, -1) C. (3,1) D. (-2,-3)

Solution

If $(b,a) \in R^{-1}$, then $(a,b) \in R$

A. If $(-3, -2) \in R^{-1}$, then $(-2, -3) \in R$

$\therefore$ (-2, -3): $2(-2) - 3(-3) = -4 + 9 = 5 > 3$. True

B. If $(2, -1) \in R^{-1}$, then $(-1, 2) \in R$ is false

$\therefore$ (-1, 2): $2(-1) - 3(2) = -2 - 6 = -8 > 3$. false

C. If $(3, 1) \in R^{-1}$, then $(1,3) \in R$ is false

$\therefore$ (1,3): $2(1) - 3(3) = 2 - 9 = -7 > 3$. False

D. If $(-2, -3) \in R^{-1}$, then $(-3, -2) \in R$ is false

$\therefore$ (-3, -2): $2(-3) - 3(-2) = -6 + 6 = 0 > 3$ false

Answer: A

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19. Let $R = \{(x, y): y = \sqrt{x+4}\}$

a. Find the inverse of R

b. Find the domain of R and domain of R^{-1}

c. Find the range of R and range of R^{-1}

Solution

a. $R^{-1} = \{(y,x): y = \sqrt{x+4}\}$ • change ordered pair

$= \{(x,y): x = \sqrt{y+4}\}$ • Interchange x and y.

$= \{(x,y): x^2 = y+4\}$ • squaring both side .

$= \{(x,y): y = x^2 - 4\}$ • solving for y.

$\therefore$ $R^{-1} = \{(y,x): y = \sqrt{x+4}\}$ or

$R^{-1} = \{(x,y): y = x^2 - 4\}$

b. Domain of $R = \{x: x + 4 \geq 0\} = [-4, \infty)$.

Range of $R = \{y: y \geq 0\} = [0, \infty)$.

$\therefore$ Domain of R^{-1} = range of $R = [0, \infty)$

c. Range of R^{-1} = Domain of $R = [-4, \infty)$

20. Let $R = \{(x,y): y = x^2 + 5\}$

a. Find the inverse of R

b. Find the domain of R and domain of R^{-1}

c. Find the range of R and range of R^{-1}

Solution

a. $R^{-1} = \{(y,x): y = x^2 + 5\}$ • changing ordered pair components.

$= \{(x,y): x = y^2 + 5\}$ • Interchanging the role of x and y.

$= \{(x,y): y^2 = x - 5\}$ • solve for y.

$= \{(x,y): y = \pm\sqrt{x-5}\}$

$\therefore$ $R^{-1} = \{(y,x): y = x^2 + 5\}$ or $R^{-1} = \{(x,y): y = \pm\sqrt{x-5}\}$

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b. Domain of $R = \{x: x \in \mathbb{R}\} = (-\infty, \infty)$.

Range of $R = \{y: y \geq 5\} = [5, \infty)$

$\therefore$ Domain of R^{-1} = range of $R = [5, \infty)$

c. Range of R^{-1} = Domain of $R = (-\infty, \infty)$.

21. Let $R = \{(x,y): y \geq 2x + 3 \text{ and } y \leq 2x - 5\}$. Find the inverse of R.

Solution

$R^{-1} = \{(y,x): y \geq 2x + 3 \text{ and } y \leq 2x - 5\}$ or

$= \{(x,y): x \geq 2y + 3 \text{ and } x \leq 2y - 5\}$ • Interchange x and y.

$= \{(x,y): x - 3 \geq 2y \text{ and } x + 5 \leq 2y\}$

$$= \left\{(x,y): y \leq \frac{1}{2}x - \frac{3}{2} \text{ and } y \geq \frac{1}{2}x + \frac{5}{2}\right\}$$

$\therefore R^{-1} = \{(y,x): y \geq 2x + 3 \text{ and } y \leq 2x - 5\}$ or

$$R^{-1} = \left\{(x,y): y \leq \frac{1}{2}x - \frac{3}{2} \text{ and } y \geq \frac{1}{2}x + \frac{5}{2}\right\}$$

Graphs of Inverse Relation

The graph of the inverse relation is the **reflection** of the graph of the relation.

- The mirror of the reflection is the line $y = x$.

Illustrative Example

22. Sketch the graph of the relation and its inverse of each of the following relation.

a. $R = \{(-2,0), (-1,2), (1,3)\}$.

b. $R = \{(x,y): y = 2x - 3\}$

Solution

a. $R^{-1} = \{(0,-2), (2,-1), (3,1)\}$

b. Interchange the role of x and y in the equation.

$R^{-1} = \{(y, x): y = 2x - 3\}$ change x to y, y to x.

$= \{(x,y): x = 2y - 3\}$ • solve for y

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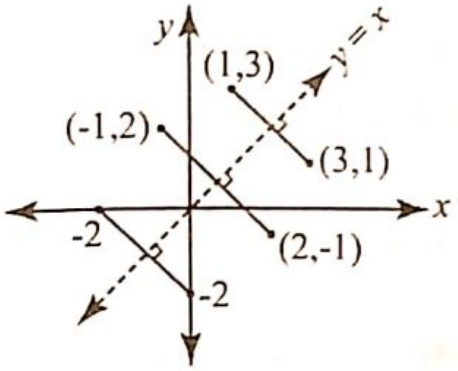
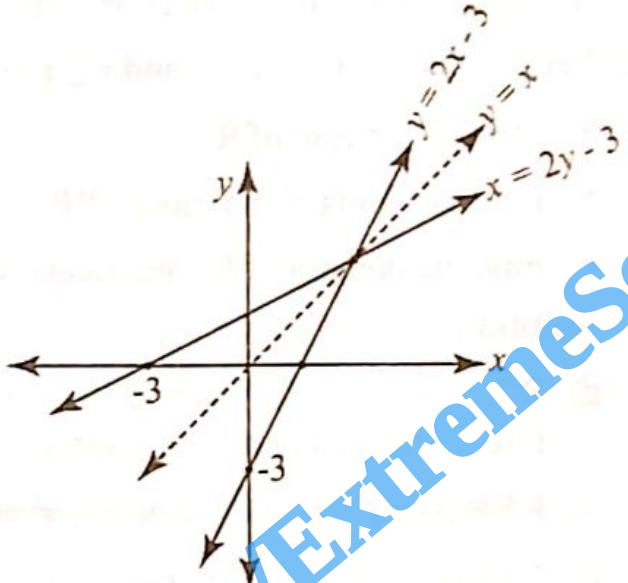
$$= \left\{(x, y) : y = \frac{1}{2}x + \frac{3}{2}\right\}$$

- The graphs of $y = 2x - 3$ and $x = 2y - 3$ are reflection of each other in the line $y = x$.





23. How are related the graph of R and R^{-1}? Explain?

Explanation

The graph of R and the graph of the inverse relation R^{-1} are symmetric with respect to the line given by: $y = x$.

24. Find the image of the relation R after it has been reflected a long the line $y = x$.

a. $R = \{(-3,2),(-2,1),(-1,0),(2,5)\}$

b. $R = \{(x,y): y = 3x - 4\}$

c. $R = \{(x,y): y = 4x^2 - 6\}$

d. $R = \{(x,y): x^2 - 2y^2 = 8\}$

Solution

The image of the relation after the reflection in the line $y = x$ is the inverse of relation.

a. Image of R is $R^{-1} = \{(2, -3), (1, -2), (-1,0), (5,2)\}$

b. Image of R in the line of reflection $y = x$ is

$R^{-1} = \{(y,x): y = 3x - 4\}$ or $R^{-1} = \{(x,y): x = 3y - 4\}$

$$= \left\{(x, y) : y = \frac{1}{3}x + \frac{4}{3}\right\}$$

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c. Image of R is $R^{-1} = \{(y,x): y = 4x^2 - 6\}$ or $R^{-1} = \{(x,y): x = 4y^2 - 6\}$

$$= \left\{(x,y): y = \frac{\pm\sqrt{x+6}}{2}\right\}$$

d. Image of R is $R^{-1} = \{(y,x): x^2 - 2y^2 = 8\}$ or $R^{-1} = \{(x,y): y^2 - 2x^2 = 8\}$

25. Let $R = \{(x,y): y < 2x + 2 \text{ and } y \geq x^2 + 3x - 4\}$

a. Find the inverse of R

b. Find the domain and range of R

c. Find the domain of R^{-1} and range of R-1

Solution

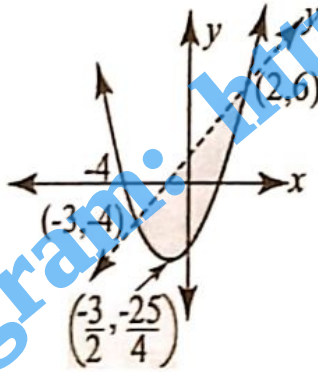
a. $R^{-1} = \{(y,x): y < 2x + 2 \text{ and } y \geq x^2 + 3x - 4\}$ or
Inter - change the role of x and y.

$R^{-1} = \{(x,y): x < 2y + 2 \text{ and } x \geq y^2 + 3y - 4\}$

b. To find the domain and range of R, solve the intersection of

$y = 2x + 2$ and $y = x^2 + 3x - 4$

Solving together $\begin{cases} y = x^2 + 3x - 4 \\ y = 2x + 2 \end{cases}$



$\Rightarrow x^2 + 3x - 4 = 2x + 2$

$\Rightarrow x^2 + 3x - 2x - 4 - 2 = 0$

$\Rightarrow x^2 + x - 6 = 0$

$\Rightarrow (x - 2)(x + 3) = 0$

$\Rightarrow x = 2 \text{ or } x = -3$

To find the corresponding y - value, we substance

$x = 2$ and -3 in the equation $y = 2x + 2$

Thus, when $x = -3$, $y = 2(-3) + 2 = -6 + 2 = -4$, (-3, -4)

when $x = 2$, $y = 2(2) + 2 = 4 + 2 = 6$, (2,6)

There fore, the domain of $R = \{x: -3 < x < 2\}$ and the range of

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$$R = \left\{ y : \frac{-25}{4} \leq y < 6 \right\}$$

c. Domain of R^{-1} = Range of $R = \left\{ x : \frac{-25}{4} \leq x < 6 \right\}$

Range of R^{-1} = Domain of $R = \{y: -3 < y < 2\}$

26. Let $R = \{(x,y): y \geq x^2 \text{ and } y \leq 4\}$

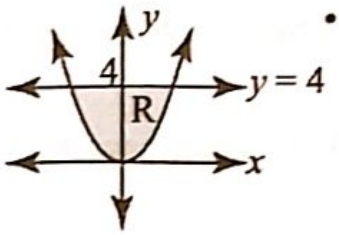
a. Find the inverse of R

b. Find the domain and range of R

c. Find the domain and range of R^{-1}

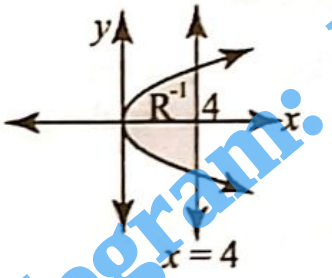
Solution:

- The equation of $y = x^2$ is parabola opens upward with vertex (0,0) and its inverse is $x = y^2$ opens right
- The equation $y = 4$ is horizontal line parallel to x - axis and its inverse is $x = 4$,



• To find the intersection point of $y = x^2$ and $y = 4$, is equating: $x^2 = 4$, this implies

$$x = \pm\sqrt{4} = \pm 2$$



a. $R^{-1} = \{(x,y): x \geq y^2 \text{ and } x \leq 4\}$

b. Domain of $R = \{x: -2 \leq x \leq 2\}$ and range of $R = \{y: 0 \leq y \leq 4\}$.

c. Domain of R^{-1} = range of $R = \{x: 0 \leq x \leq 4\}$

Range of R^{-1} = Domain of $R = \{y: -2 \leq x \leq 2\}$

1.2 What is a Function?

A function is a relation in which each element of the domain is paired with exactly one element of the range.

Example

The ordered pairs (2, -3) and (2,4) do not make up a function because $f(2) = -3$ and $f(2) = 4$, the value of f at 2 is not unique, it

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has different value.

DEFINITION of a Function

Given a relation in x and y, we say that **y is a function of x** if for each value of x in the domain, there is exactly one value of y in the range.

Note: If $(x,y) \in f$, we say that y is the value of f at x, and write $y = f(x)$.

Illustrative Example

27. In each of the following determine whether the relation is a function (y as a function of x)

a. $R_1 = \{(4, 2), (3,1), (-4, 3), (-1, 0), (4, -5)\}$

b. $R_2 = \{(-1, 4), (2, 3), (3, 4), (-4, 5)\}$

c. $R_3 = \{(7, 3), (4, -1), (2,3)\}$

d. $R_4 = \{(-3, 2), (9, 5), (1, 0), (-3, 1)\}$

Solution

a. The relation R_1 is not a function because when $x = 4$, there are two different y - values: $y = 2$ and $y = -5$,

b. The relation R_2 is a function. No two ordered pairs have the same x - value but different y- values.

c. The relation R_3 is a function. No domain is repeated.

d. The relation R_4 is not function because the domain -3 is repeated. When $x = -3$, $y = 2$ and $y = 1$. (-3, 2) and (-3, 1) do not make up a function.

28. Describe in your own words which types of a relations are functions.

Explanation:- A relation is a function if each elements of domain paired with exactly one range.

29. Describe, under what condition the ordered pairs (x,y) and (x,z) is a function?

A. $x = y$ B. $x = z$ C. $y = z$ D. $y > z$

Explanation:- The ordered pair (x,y) and (x,z) is a function if $y = z$

Answer: C

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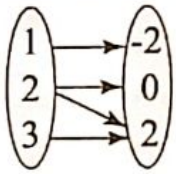
30. Which value of x would cause the relation $\{(-1,6), (x, 8), (2, 4), (3,4)\}$ not to be a function?

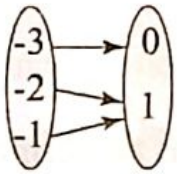
A. 6 B. -1 C. 4 D. 8

Explanation:- A function may not have the same x - value paired with different y - values.

Answer: B

31. Which the following is not function?

A. 

B. 

C. $\{(-1, -1), (0, 0), (1, 1)\}$ D. $\{(-3, 2) (2, -3)\}$

Solution:

A $\{(1, -2) (2, 0), (2, 2), (3, 2)\}$ is not function.

Answer: A

32. Which set of ordered pairs do not describes a function:

A. $R_1 = \{(-5, 1), (0, 1), (5, 3), (2, -1), (3, 2), (4,3)\}$

B. $R_2 = \{(-2, 2), (0, 5), (1, 3), (2, -1), (1, 4)\}$

C. $R_3 = \{(-2, 3), (-1, 1), (0, -5), (5, 4), (1, 5)\}$

D. $R_3 = \{(-3, 2), (-2, 5), (-1, 0), (1, 0), (2, 3)\}$

Solution

R_2 is not a function, because when $x = 1$, there are two different y - values $y = 3$ and $y = 4$.

Answer: B

33. Determine whether which of the following relations are function.

A. $R = \{(x,y)$: y is the father of $x\}$

B. $R = \{(x,y)$: y is the brother of $x\}$

C. $R = \{(x,y)$: y is the wife of $x\}$

D. $R = \{(x,y)$: $y < x\}$

Solution: B,C and D are not function.

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Answer: A

34. Determine whether the relation R' is a function or not in the sense of the definition of a function.

a. $R = \{(x, y): y = x^2 + 4\}$

b. $R = \{(x,y): y^2 - x^2 = 0\}$

c. $R = \{(x,y): y = 3\}$

d. $R = \{(x,y): x = 3\}$

e. $R = \{(x,y): x^2 + y^2 = 4\}$

f. $R = \{(x,y): |y| = |x|\}$

g. $R = \{(x,y): x = |y|\}$

h. $R = \{(x,y): y = |x|\}$

Solution

a. Function

b. Not function, as an example {(-2, 2), (-2, -2)}

c. Function.

d. Not function, as an example: {(3, 2), (3, -4)}

e. Not function, as an example: {(0 -2), (0 2)}

f. Not function, as an example: {(-1, -1), (-1, 1)}

g. Not function, as an example {(3, -3), (3, 3)}

h. Function

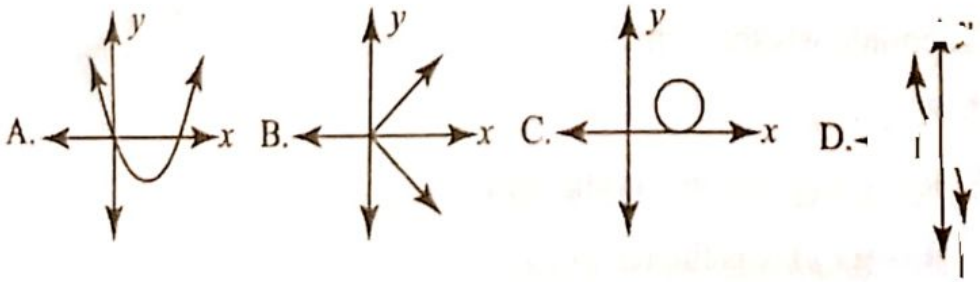
Vertical Line Test

The graph defines y as a function of x if no vertical line intersects the graph in more than one point.

If you can find one vertical line that passes through two points of the relation, then the relation is not function.

Illustrative Example

35. Which graphs describes functions.



Solution

A. a function, because vertical line intersects the graph at one.

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B. Not a function, because one Vertical line intersects two points.

C. Not a function, because one vertical line intersects two points.

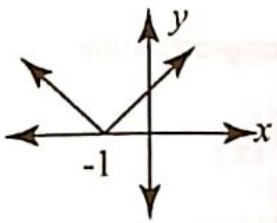
D. a function, because vertical line intersects the gaph at one point.

36. Determine whether if the relation defines y as a function of x.

a. $y = |x + 1|$ c. $y = x^2 + 2$ e. $x^2 + y^2 = 16$

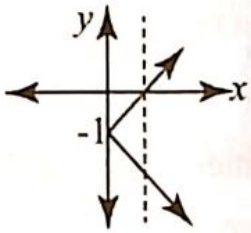
b. $x = |y + 1|$ d. $x = y^2 + 2$ f. $x^3 + y^3 = 1$

Solution



$y = |x + 1|$

a. $y = |x + 1|$ is a function, no two ordered pairs have the same x - value but different y - values.

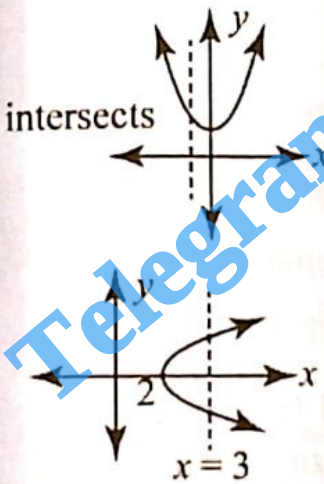


$x = |y + 1|$

b. $x = |y + 1|$ is not a function, because when $x = 3$, $|y + 1| = 3$ implies $y = -4$ and $y = 2$.

(3,2) and (3, -4) when $x = 1$, $y = 0$ and $y = -2$

(1, 0) and (1, -2), do not make up the function.



c. $y = x^2 + 2$ is a function, since vertical line intersects the graph at one point.

$x = 3$

d. $x = y^2 + 2$ is not a function, because vertical line intersects the graph in two point.

Example $x = 3$, $y^2 + 2 = 3$ implies $y^2 = 3 - 2 = 1$

$\therefore$ (3, -1) and (3, 1) not make up function.

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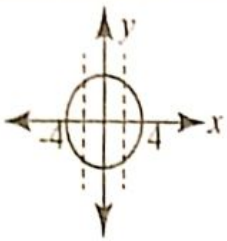

e. $x^2 + y^2 = 16$ is not a function, because the vertical line intersects the circle in two points

f. $x^3 + y^3 = 1$ is a function.

37. Find the domain

a. $f(x) = \sqrt{3x + 1}$

b. $f(x) = \dfrac{1}{\sqrt{3x + 1}}$

c. $f(x) = \sqrt{x + 2} + \sqrt{2 - x}$

d. $f(x) = \sqrt{8 - x^2}$

Solution

a. Domain of $f = \{x: 3x + 1 \geq 0\} = \left\{x : x \geq -\frac{1}{3}\right\}$

b. Domain of $f = \{x: 3x + 1 > 0\} = \left\{x : -x > -\frac{1}{3}\right\}$

c. Domain of $f = \{x: x + 2 \geq 0$ and $2 - x \geq 0\}$ The intersection of the domain $= \{x: -2 \leq x \leq 2\}$

d. Domain of $f = \{x: 8 - x^2 \geq 0\} = \left\{x : -2\sqrt{2} \leq x \leq 2\sqrt{2}\right\}$

38. Let $f(x) = \dfrac{\sqrt{4 - x^2}}{x + 2}$ Find the domain of f

Solution

Domain of $f = \{x: 4 - x^2 \geq 0$ and $x \neq -2\}$

$= \{x: -2 < x \leq 2\}$.

1.2.1 Type of Function

Here, we will going to see some type of special functions such as power function, Absolute value function, signum function and greatest integer function

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1.3 A Family of Power Functions

A family of power function is a group of function with common characteristics.

A function of the form

$f(x) = ax^n$, are called **power function.**

where a is non zero areal number and n is areal number.

39. Determine whether each of the following are power function.

state the domain and range.

a. $f(x) = 3x^{\frac{1}{4}}$

b. $f(x) = -6x$

c. $f(x) = 3^{4x}$

d. $f(x) = 2x^{\frac{4}{3}}$

e. $f(x) = 3x^{\frac{-3}{4}}$

f. $f(x) = \sqrt{x^2}$

g. $f(x) = \sqrt[3]{x^2}$

h. $f(x) = \sqrt[8]{\frac{1}{2x}}$

i. $f(x) = \sqrt[5]{x^3}$

Solution

a. $f(x) = 3x^{\frac{1}{4}}$ is a power function

Domain $= \{x : x \geq 0\}$ and Range $= \{y : y \geq 0\}$

b. $f(x) = -6x$ is a power function

Domain $= \mathbb{R}$ and Range $= \mathbb{R}$

c. $f(x) = 3^{4x}$ is not a power function.

d. $f(x) = 2x^{\frac{4}{3}}$ is a power function

Domain $= \mathbb{R}$ and range $= [0, \infty)$

e. $f(x) = 3x^{\frac{-3}{4}} = \frac{3}{\sqrt[4]{x^3}}$ is a power function with

domain $= (0; \infty)$ and range $= (0, \infty)$

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f. $f(x) = \sqrt{x^2} = |x|$ is not a power function.

g. $f(x) = \sqrt[3]{x^2} = x^{2/3}$ is a power function with domain $= \mathbb{R}$ and range $= [0, \infty)$

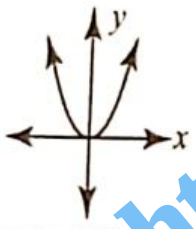
h. $f(x) = \sqrt[8]{\frac{1}{2x}} = \left(\sqrt[8]{\frac{1}{2}}\right) = x^{\frac{-1}{8}}$ is a power function with dc and rang $= (0, \infty)$

i. $f(x) = \sqrt[5]{x^3} = x^{3/5}$ is a power function with domain $= \mathbb{R}$ and range $= \mathbb{R}$

1.3.1 Graph of Power Function

The general shape of the graph of power function $f(x) = x^n$ depends on whether n is even or odd.

(i) If **n is even, positive,** the graph of $f(x) = x^n$ is similar to the parabola $y = x^2$



Graph of $f(x) = x^n$, n = 2, 4, 6...

Example: $f(x) = x^2, x^4, x^6, \ldots$

- **Symmetry:** since $f(-x) = (-x)^4 = x^4 = f($ function, so the graph is symmetric w to y - axis
- The graph contains the point $(-1, 1), (0, 0)$ and $(1, .$
- **Domain** $= \mathbb{R} = (-\infty, \infty)$.
- **Range** $= [0, \infty)$.

$\ldots, x^6 < x^4 < x^2$ in $-1 < x < 0$ and $0 < x < 1$.

$\ldots, x^6 > x^4 > x^2$ in $-\infty < x < -1$ and $1 < x < \infty$.

(ii) If **n is odd** the graph of $f(x)$ is similar in shape to the graph of $y = x$, and $y = x^3$

Example: $f(x) = x^3, x^5, x^7 \ldots$.

- **Symmetry:** since $f(-x) = -x^3 = -f(x)$, f is an odd function, so the graph is symmetric w.r.t Origin

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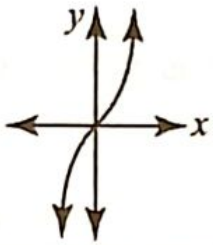
• The graph contains the point (-1, -1) (0, 0) and (1, 1).



Graph of $f(x) = x^n$, n = 3, 5, 7...

..., $x^7 > x^5 > x^3$ if $-1 < x < 0$ and $1 < x < \infty$.

..., $x^7 < x^5 < x^3$ if $0 < x < 1$ and $-\infty < x < -1$

Domain = R = $(-\infty, \infty)$.

Range = R = $(-\infty, \infty)$.

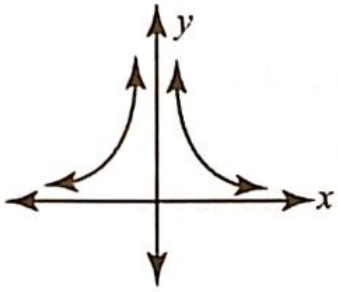
$f(ab) = f(a).\ f(b)$.

(iii) If **n is even** then the graph of $f(x) = x^{-n}$ is similar to

$$y = x^{-2} = \frac{1}{x^2}$$

Example: $f(x) = x^{-2}, x^{-4}, x^{-6}, ...$

Common properties are:



Graph of $f(x) = x^{-n}$

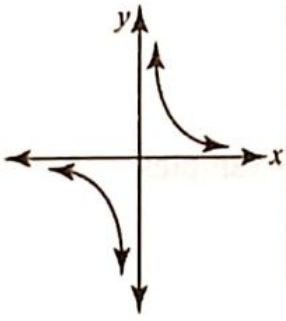
- **Symmetry:** since $f(-x) = f(x)$, then f is even function, so the graph is symmetric with respect to y - axis.
- The graph contains (-1, 1) and (1, 1).
- **Domain** = R/ {0} = $(-\infty, 0)$ and $(\infty, 0)$
- **Range** = $(0, \infty)$
- $f(ab) = f(a).f(b)$.

(iv) If n is **odd**,.... then the graph of $f(x) = x^{-n}$ is similarly to

$$y = x^{-1} = \frac{1}{x}$$

Example: $f(x) = x^{-1}, x^{-3}, x^{-5}, ...$

Common properties are:



- **Symmetry,** since $f(-x) = -f(x)$ f is an odd function, so the graph is symmetric w.r.t origin.
- The graph contains the point (-1, -1) and (1, 1).
- **Domain** = R\{0} and **Range** = R\ {$y = 0$}

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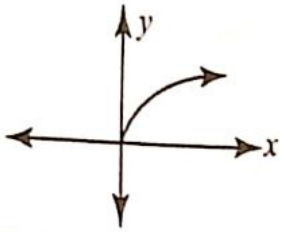
- $f(ab) = f(a) \cdot f(b)$.

(v) The general graph of $f(x) = x^{\frac{m}{n}}$, where m is an odd and n is even number.

$f(x) = x^{\frac{m}{n}} = \sqrt[n]{x^m}$, where, $m = 1, 3, 5, \ldots$ and $n = 2, 4, 6, \ldots$

Common properties are:



$f(x) = x^{\frac{m}{n}}$, m is odd and n is even and m < n

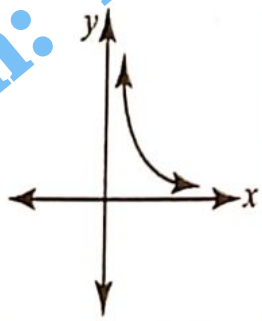
- The graph containing the point (0,0) and (1,1)
- The graph is not symmetric.
- Domain = $[0, \infty)$
- Range = $[0, \infty)$
- Increasing function.

$f(x) = x^{\frac{1}{2}}, x^{\frac{1}{4}}, x^{\frac{1}{4}}, x^{\frac{5}{6}}$... are some of the examples.

(vi) The general of $f(x) = x^{\frac{-m}{n}}$, where m is an odd n is even.

$f(x) = x^{\frac{-m}{n}} = \frac{1}{x^{\frac{m}{n}}} = \frac{1}{\sqrt[n]{x^m}}$, where, m = 1,35,... and n = 2,4,6,.....

Common properties are:



$f(x) = x^{\frac{-m}{n}}$, Where m is odd and n is even

- As $x \to \infty$, $f(x) \to 0^+$
- As $x \to 0^+$, $f(x) \to \infty$
- Not symmetric
- **Domain** = $R^+ = (0, \infty)$
- **Range** = $R^+ = (0, \infty)$

- $f(x) = x^{\frac{-1}{2}}, x^{\frac{-1}{4}}, x^{\frac{-3}{4}}, \ldots$ are some of the examples.
- The graph containing the point (1, 1).

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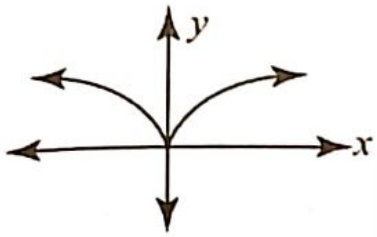
(vii) The general graph $f(x) = x^{\frac{m}{n}}$, where m is even and n is odd.

$f(x) = x^{\frac{m}{n}} = \sqrt[n]{x^m}$, $n = 3, 5, 7, \ldots$ and $m = 2, 4, 6, \ldots$

Example : $f(x) = x^{\frac{2}{3}}, x^{\frac{2}{5}}, x^{\frac{4}{5}}, \ldots = \sqrt[3]{x^2}, \sqrt[5]{x^2}, \sqrt[5]{x}, \ldots$

Common properties are:



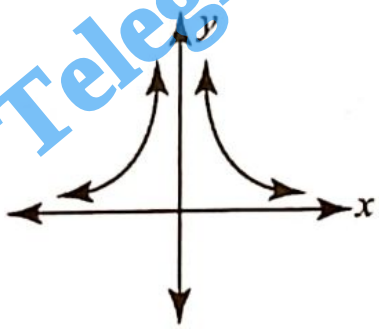
- Symmetric w.r.t y - axis
- The graph containing the point (-1, 1), (0, 0) and (1, 1).
- **Domain** = $\mathbb{R}$
- **Range** = $[0, \infty)$
- Even function : $f(-x) = f(x)$.
- $f(ab) = f(a).f(b)$.

(viii) The general graph of $f(x) = x^{-m/n}$ for m is even and n is odd.

$f(x) = x^{\frac{-m}{n}} = \frac{1}{x^{\frac{m}{n}}} = \frac{1}{\sqrt[n]{x^m}}$, $n = 3, 5, 7, \ldots$ and $m = 2, 4, 6, \ldots$

Example : $f(x) = x^{\frac{-2}{3}}, x^{\frac{-2}{5}}, \ldots = \frac{1}{\sqrt[3]{x^2}}, \frac{1}{\sqrt[5]{x^2}}, \ldots$

Common properties are :



- $f(-x) = f(x)$
- Even function
- symmetric w.r.t y - axis
- Domain = $\mathbb{R} \setminus \{0\}$
- Rang = $(0, \infty)$
- The graph containing (-1, 1) and (1, 1)
- As $x \to -\infty$, $f(x) \to 0^+$
- As $x \to \infty$ $f(x) \to 0^+$

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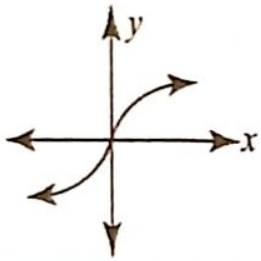
As $x \to 0^-, f(x) \to +\infty$

As $x \to 0^+ f(x) \to +\infty$,

(ix) The general graph of $f(x) = x^{\frac{m}{n}}$, for both m and n are odd.

$f(x) = x^{\frac{1}{3}}, x^{\frac{3}{5}}, \ldots\ldots = \sqrt[3]{x}, \sqrt[5]{x^3} \ldots$

Common Properties are:



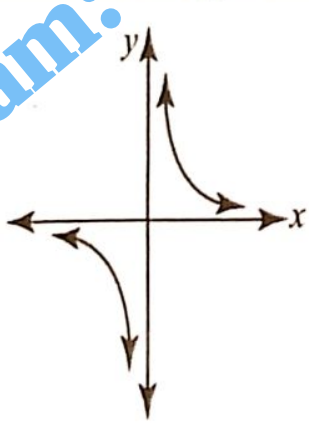
$f(x) = x^{\frac{m}{n}}$, for both m and n are odd

- $f(-x) = -f(x)$
- Symmetric w.r.t origin
- An odd function.
- (-1, -1), (0,0) and (1, 1) are points on the gr
- Domain = $\mathbb{R}$
- Range = $\mathbb{R}$
- Increasing function.
- $f(ab) = f(a).f(b)$

(x) The general graph of $f(x) = x^{\frac{-m}{n}}$ for both m and n are odd.

$f(x) = x^{-3}, x^{\frac{-5}{7}}, \ldots\ldots = \frac{1}{\sqrt[3]{x}}, \frac{1}{\sqrt[5]{x^3}}, \ldots$

Common Properties are:



- $f(-x) = -f(x)$
- symmetric w.r.t origin.
- An odd - function.
- (-1, -1) and (1, 1) are points on the graph.
- Domain = $\mathbb{R} \setminus \{0\}$
- Range = $\mathbb{R} \setminus \{0\}$
- As $x \to \infty$, $f(x) \to 0^+$
- As $x \to \infty$, $f(x) \to 0^-$

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• As $x \to 0^-$, $f(x) \to -\infty$

• As $x \to 0^+$, $f(x) \to \infty$

. Which of following power function does not satisfies the condition $f(xy) = f(x).f(y)$.

A. $f(x) = \sqrt[3]{\frac{1}{xy}}$

C. $f(x) = x^{\frac{2}{3}}$

B. $f(x) = x^4$

D. $f(x) = 2x^6$

Solution

A. $f(x) = \sqrt[3]{\frac{1}{xy}} = \sqrt[3]{\frac{1}{x}} \cdot \sqrt[3]{\frac{1}{y}} = f(x) \cdot f(y)$

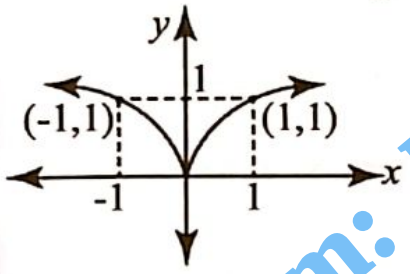
B. $f(xy) = (xy)^4 = x^4 \cdot y^4 = f(x) \cdot f(y)$

C. $f(xy) = (xy)^{2/3} = x^{2/3} \cdot y^{2/3} = f(x).f(y).$

D. $f(xy) = 2(xy)^6 = 2x^6 y^6 = 2f(xy)$

Answer: D

. Which of the following function best describes the graph at the left?



A. $f(x) = x^{\frac{-2}{3}}$

B. $f(x) = x^{\frac{-3}{2}}$

C. $f(x) = x^{2/3}$

D. $f(x) = x^{3/2}$

Solution

A. $f(x) = x^{-2/3} = \frac{1}{x^{2/3}} = \frac{1}{\sqrt[3]{x^2}}$

- Domain = $\mathbb{R} / \{0\}$
- Range = $(0, \infty)$

B. $f(x) = x^{-3/2} = \frac{1}{x^{3/2}} = \frac{1}{\sqrt{x^3}}$

- Domain = $(0, \infty)$
- Range = $[0, \infty)$

C. $f(x) = x^{2/3} = \sqrt[3]{x^2}$

- Domain = $\mathbb{R} = (0, \infty)$
- Range = $[0, \infty)$

D. $f(x) = x^{3/2} = \sqrt{x^3}$

- Domain = $[0, \infty)$
- Range = $[0, \infty)$

Answer: C

12. Which of the following function best describes the graph at the left?

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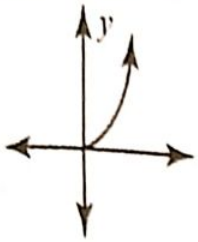


A. $f(x) = x^{2/3}$

B. $f(x) = x^{3/2}$

C. $f(x) = x^{-3/4}$

D. $f(x) = x^{-4/3}$

Solution

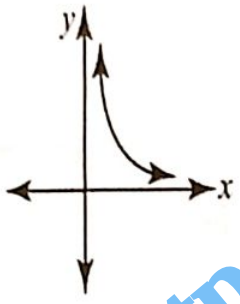
From the graph, the domain (set of x - value) are $\{x: x \geq 0\}$ and th range (set of y - value) are $\{y: y \geq 0\}$.

A. $f(x) = x^{2/3} = \sqrt[3]{x^2}$, Domain = $\mathbb{R}$ and range = $[0, \infty)$

B. $f(x) = x^{3/2} = \sqrt{x^3}$, *Domain* = $[0, \infty)$ *and range* = $[0, \infty)$

Answer:

43. Which of the following function best describes the graph at the left?



A. $f(x) = x^{\frac{-2}{3}}$

B. $f(x) = x^{\frac{-3}{2}}$

C. $f(x) = x^{2/3}$

D. $f(x) = x^{3/2}$

Solution:

Refer, ≠ (VI), $f(x) = x^{-m/n}$

A. $f(x) = x^{\frac{-2}{3}} = \frac{1}{x^{2/3}} = \frac{1}{\sqrt[3]{x^2}}$

Domain = $\mathbb{R} \setminus \{0\}$, Range = $\mathbb{R}^+ = (0, \infty)$

B. $f(x) = x^{-3/2} = \frac{1}{x^{3/2}} = \frac{1}{\sqrt{x^3}}$

Domain = $\{x: x^3 > 0\} = (0, \infty)$ and range = $(0, \infty)$

As $x \to \infty, f(x) \to 0^+$ and as $x \to 0^+, f(x) \to \infty$

Answer:

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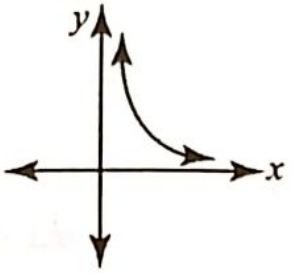
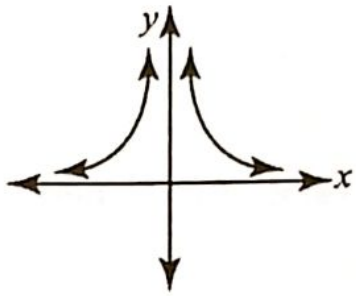
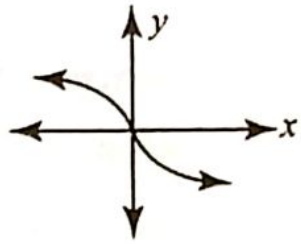
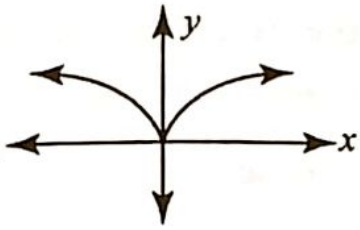
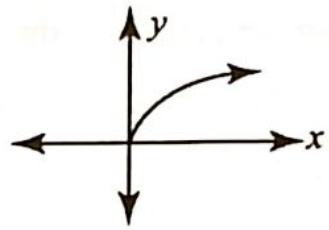
44. Match each graph with its corresponding function.

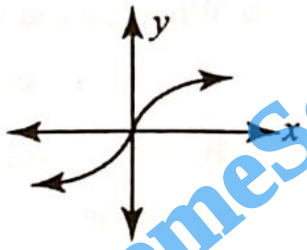
1. 

2. 

3. 

4. 

5. 

6. 

A. $f(x) = x^{\frac{4}{3}}$

B. $f(x) = -x^{\frac{1}{3}}$

C. $f(x) = x^{\frac{-3}{4}}$

D. $f(x) = x^{\frac{-4}{3}}$

E. $f(x) = x^{\frac{1}{7}}$

F. $f(x) = x^{\frac{3}{4}}$

Solution:

Refer all the classes of $(i - x)$ $f(X) = x^{\frac{m}{n}}$

1. C. Refer ≠ (VI) $f(x) = \frac{1}{\sqrt[4]{x^3}}$
2. D, for detail refer (≠ iii and viii)

   $f(x) = x^{\frac{-n}{m}}$ for n is even and m is odd.
3. B, Explanation, refer (≠ X).
4. A, Explanation, refer (v ii).
5. F, Explanation, refer (v).
6. E, Explanation, refer. (X).

45. What are the domain and range of the function $f(x) = 8x^{-3/4}$ respectively?

A. $(0, \infty)$ and $[0, \infty)$

B. $[0, \infty)$ and $(0, \infty)$

C. $(0, \infty)$ and $(0, \infty)$

D. $[0, \infty)$ and $[0, \infty)$

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Solution

$$f(x) = 8x^{\frac{-3}{4}} = \frac{8}{\sqrt[4]{x^3}}$$

Domain = $\{x: x > 0\} = (0, \infty) = R^+$

when $x > 0$, the value of $y > 0$, range = $(0, \infty)$

Answer: C

46. What are the domain and range of the function $f(x) = 8x^{-3/4}$ respective

A. $(-\infty, \infty)$ and $(-\infty, \infty)$

B. $(0, \infty)$ and $(0, \infty)$

C. $R\setminus\{0\}$ and $R\setminus\{0\}$

D. $R\setminus\{0\}$ and R^+

Solution

$$f(x) = 8x^{\frac{-1}{3}} = \frac{8}{\sqrt[3]{x^4}}$$

Domain = $R\setminus\{0\}$

For all x in the domain, the value of y is $\{y: y > 0\} = R^+$

Answer: D

47. Find the domain and range of

a. $f(x) = 2x^{3/4}$

b. $f(x) = 2x^{4/3}$

Solution

a. Domain = $\{x : x \geq 0\} = [0, \infty)$ and range = $[0, \infty)$

b. Domain = R and range = $[0, \infty)$

48. The graph below is the graph of the function $f(x) = x^{m/n}$, where m and n are natural numbers which of the following is true bout m and n?

A. m is odd, n is even and m < n

B. m is even, n is odd and m > n.

C. m is odd, n is even and m > n.

D. m is even, n is odd and m < n.

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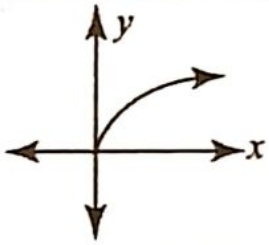
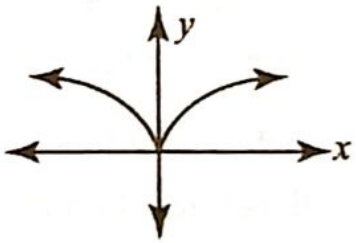
Solution

A. As an example, is $x^{\frac{3}{4}}, x^{\frac{5}{6}}, x^{\frac{7}{8}}$ ....



B. As an example is $x^{\frac{4}{3}}, x^{\frac{6}{5}}, x^{\frac{8}{7}}$

Domain = $\mathbb{R}$ and range = $[0, \infty)$,



C. Refer example ≠ 44.

Answer: C

49. If the graph of the power function $f(x) = x^{\frac{-m}{n}}$ is containing the point (-1, 1) and (1, 1), then which of the following is true about m and n?

A. m is odd and n is even.
C. Both m and n are odd.
B. m is even and n is odd
D. Both m and n are even.

Solution

Since $f(-1) = f(1) = 1$ and $f(-x) = f(x)$

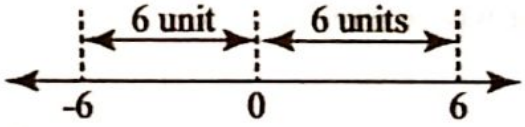
As an example: $x^2, x^{2/3}, x^{-4/3}$ .....

Answer: B

1.4 Absolute Value (Modulus) Function

- How fer is 6 from zero?
- How far is -6 from Zero ?

The numbers 6 and -6 are both the same distance 6 units from zero on the number line.



The **absolute value** of a number x is the distance from x to 0 in either direction on the number line.

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Definition: Absolute value of any real number x is given by:

$$|x| = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -x & \text{if } x < 0 \end{cases}$$

- Distance between two points x and y on a number line is given by:

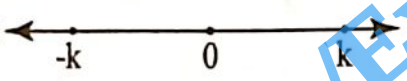
  $|x - y| = |y - x|$.

- $|x| \geq 0$, because distance is always positive or zero.

Geometrically absolute value equation interpreted as follows:

(i) $|x| = k$ means "the distance from x to zero is exactly k units".

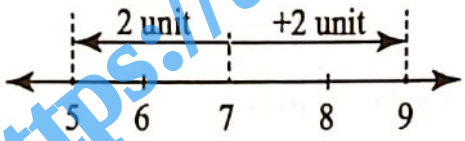
Graph:



Solution: $x = -k$ or $x = k$

(ii) $|x - 7| = 2$ means "the distance from x to 7 is 2 units".

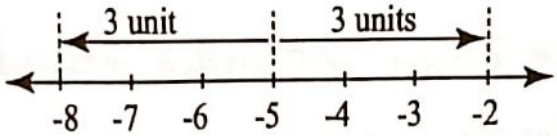
Graph:



Solution: $x = 5$ or $x = 9$

(iii) $|x + 5| = 3$ means " the distance from x to -5 is 3 units".

Graph:



Solution: $x = -8$ or $x = -2$

Illustrative Example

50. (a) Name all real number whose distance from zero on the number line is 8

(b). Name all real number whose absolute is 5.

Solution

a. Let x be the number, then $|x| = 8 \quad \therefore \quad x = 8$ or $x = -8$

b. $|x| = 5$, that is $x = -5$ or $x = 5$

51. Find the set of real number whose absolute value is

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a. 3 b. -8 c. $\frac{2}{5}$ d. $\frac{-2}{5}$

Solution: Let x - the set of real number.

a. $|x| = 3$, then $x = 3$, or $x = -3$

$\therefore$ The set of real number is $\{-3, 3\}$

b. $|x| = -8$ is impossible, because absolute value measures distance and distance is never negative . $\therefore$ The solution set is empty set.

c. $|x| = \frac{2}{5}$, then the solution set is $\left\{\frac{-2}{5}, \frac{2}{5}\right\}$

d. $|x| = \frac{-2}{5}$, is impossible, so that the solution set is $\varnothing$.

52. Find the number whose distance between a number and 4 on the number line is 6.

Solution: Let x the number. Then, $|x - 4| = 6$

$x - 4 = 6$ or $x - 4 = -6$

$x = 6 + 4 = 10$ or $x = -6 + 4 = -2$

$\therefore$ The requires numbers is $\{2, 10\}$.

53. If $|x + y| < |x| + |y|$, what can you conclude about the real number x and y?

A. $x < y < 0$ B. $x > y > 0$ C. $xy < 0$ D. $y < x$

Solution

A. When $x < y < 0$, $|x + y| = |x| + |y|$

B. When $x > y > 0$, $|x + y| = |x| + |y|$

C. $xy < 0$ means x and y are opposite sign.

Then $|x + y| < |x| + |y|$, Let $x = -8$, $y = 3$

Answer: C

54. For what real number x and y is each equation (expression) satisfied?

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a. $|x - y| = |y - x|$ b. $|xy| = |x|\,|y|$ c. $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$

Solution

a. $|x - y| = |y - x|$ is true for all real numbers. $(-\infty, \infty)$

b. $|xy| = |x||y|$ is true for all real numbers. $(-\infty, \infty)$

c. $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$ is true for all real numbers except $y = 0$.

55. Evaluate each of the following absolute value expression:

a. $|\sqrt{2} - 2|$ c. $|\sqrt{6} - 3|$ e. $|a - b|$

b. $|\pi - 4|$ d. $|\sqrt{7} - 3|$

Solution

a. $|\sqrt{2} - 2| = -(\sqrt{2} - 2) = 2 - \sqrt{2}$, because $\sqrt{2} < 2$.

b. $|\pi - 4| = -(\pi - 4) = 4 - \pi$,

c. $|\sqrt{6} - 3| = |\sqrt{6} - \sqrt{9}| = -(\sqrt{6} - \sqrt{9}) = \sqrt{9} - \sqrt{6} = 3 - \sqrt{6}$.

d. $|\sqrt{7} - 3| = (\sqrt{7} - 3) = 3 - \sqrt{7}$

e. $|a - b| = \begin{cases} a - b \text{ if } a \geq b \\ -(a - b) = b - a \text{ if } a < b \end{cases}$

56. For what real values of x is it true that $|x| > -x$?

Solution

Since $|x| \geq 0$ for all real number x.

$\Rightarrow |x| \geq 0 > -x$, that is $-x < 0 < |x|$

Hence, $-x < 0$ means $x > 0$

$\therefore$ $|x| > -x$ is true for $x \in (0, \infty)$.

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1.4.1 Absolute Value Equation

Absolute value equation can be solved algebraically by using the three sets of equivalent equation.

(i) $|ax + b| = c$ is equivalent to $ax + b = c$ or $ax + b = -c$

(ii) $|ax + b| = |kx + c|$ is equivalent to $ax + b = \pm (kx + c)$

(iii) $|ax + b| < c$ is equivalent to $-c < ax + b < c$

(iv) $|ax + b| > c$ is equivalent to $ax + b < -c$ or $ax + b > c$.

Illustrative Example

57. Find the solution set of each of the following equation.

a. $|x - 2| = \pi$ c. $|3 - 5x| = 6$ e. $2|3x - 7| + 7 = 9$

b. $|4x + 3| = 0$ d. $|x^2 - 3| = 2$ f. $|9x - 3| + 7 = 6$

Solution

a. $|x - 2| = \pi$ means the value of $x - 2$ is π or $-\pi$

$x - 2 = \pi$ or $x - 2 = -\pi$

$\therefore x = 2 + \pi$ or $x = 2 - \pi$

b. $|4x + 3| = 0$ equivalent to $4x + 3 = 0$

$$\therefore \quad x = \frac{-4}{3}$$

c. $|3 - 5x| = 6$ is equivalent to $3 - 5x = -6$ or $3 - 5x = 6$

$3 - 5x = -6$ or $3 - 5x = 6$

$-5x = -6 - 3 = -9$ or $-5x = 6 - 3 = 3$

$$\therefore \quad x = \frac{9}{5} \quad \text{or} \quad x = \frac{-3}{5}$$

d. $|x^2 - 3| = 2$ is equivalent to $x^2 - 3 = -2$ or $x^2 - 3 = 2$

$x^2 = -2 + 3 = 1$ or $x^2 = 2 + 3 = 5$

$x = \pm\sqrt{1}$ or $x = \pm\sqrt{5}$

$\therefore$ The solution set is $\{-1, 1, -\sqrt{5}, \sqrt{5}\}$

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e. $2|3x - 7| + 7 = 9$ implies $2|3x - 7| = 9 - 7 = 2$

$$|3x - 7| = \frac{2}{2} = 1$$

$3x - 7 = -1$ or $3x - 7 = 1$

$3x = -1 + 7 = 6$ or $3x = 1 + 7 = 8$

$\therefore$ The solution set is $\left\{2, \frac{8}{3}\right\}$

f. $|9x - 3| + 7 = 6$ is equivalent to $|9x - 3| = 6 - 7 = -1$

$|9x - 3| = -1$ has no solution because absolute value measures distance and distance is never negative.

$\therefore$ The solution set is $\{\}$.

Illustrative Example

58. Find the solution set of each of the following equation.

a. $|3x + 5| = |x + 1|$

b. $|3 - 2x| = |x + 5|$

c. $|x - 5| = |2x + 1|$

d. $|x - 3| = |3 - x|$

Solution

We use $|x| = |k|$ equivalent to $x = \pm k$

a. $|3x + 5| = |x + 1|$ is equivalent to $3x + 5 = \pm (x + 1)$

$3x + 5 = -(x + 1)$ or $3x + 5 = x + 1$

$3x + 5 = -x - 1$ or $3x - x = 1 - 5 = -4$

$4x = -1 - 5 = -6$ or $2x = -4$

$\therefore$ The solution set is $\left\{\frac{-3}{2}, -2\right\}$

b. $|x - 3| = |3 - x|$ is equivalent to $x - 3 = \pm (3 - x)$

$x - 3 = 3 - x$ or $x - 3 = -(3 - x) = -3 + x$

$2x = 3 + 3 = 6$ or $x - x = -3 + 3$

$x = \frac{6}{2} = 3$ or $0 = 0$

$\therefore$ The solution set is $\mathbb{R}$

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59. What is the solution set of $|x - 4| = 4 - x$?

A. $\mathbb{R}$ B. $(-\infty, 4]$ C. $[4, \infty)$ D. 4

Solution

Since absolute value is positive or zero

Domain $= \{x: 4 - x \geq 0\} = \{x: x \leq 4\} = (-\infty, 4]$

$\therefore |x - 4| = 4 - x$ if and only if $x - 4 = \pm (4 - x)$

$x - 4 = 4 - x$ or $x - 4 = -4 + x$

$2x = 8$ or $x - x = -4 + 4$

$x = 4$ or $0 = 0$. True for all x truth set must in the domain.

$\therefore$ Solution set is $\{x: x \leq 4\}$

Answer: B

60. What is the solution set of $|x - 6| = 3 - 2x$?

A. $\{3, -3\}$ B. $\{3\}$ C. $\{\}$ D. $\{-3\}$

Solution

The solution set should lie in the domain $= \{x: 3 - 2x \geq 0\} = \{x: x \leq 3/2\}$

$|x - 6| = 3 - 2x$ if $x - 6 = \pm (3 - 2x)$

$x - 6 = 3 - 2x$ or $x - 6 = -3 + 2x$

$3x = 9$ or $2x - x = -6 + 3$

$x = 3$ or $x = -3$

$x = 3$ is not in the domain $\left(-\infty, \frac{3}{2}\right]$

$x = 3$ makes the equation $3 - 2x$ negative.

Answer: D

61. What is the solution set of the equation $|x + 6| = 2x + 4$?

A. $\left\{\frac{-10}{3}, 2\right\}$ B. $\{2\}$ C. $\left\{\frac{-10}{3}\right\}$ D. $\{\ \}$

Answer: B

62. What is the solution set of the equation $x^3 + |x| = 0$?

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A. {1, 0, -1} B. {1, -1} C. {0, -1} D. {0, 1}

Solution: $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$.

Case 1: if $x \geq 0$, then $x^3 + |x| = x^3 + x = 0$

$\Rightarrow x(x^2 + 1) = 0$, so that $x = 0$

Case 2: If $x < 0$, them $x^3 + |x| = x^3 - x = 0$

$x(x^2 - 1) = 0$, so that $x = -1, 0, 1$.

But $x < 0$, so that $x = -1$,

Answer: (

1.4.2 Graph of Absolute Value Function

DEFINITION: The function $f: R \to R^+$ defined by $f(x) = |x|$ is called **the absolute value function.**

$f(x) = |x|$, where $f(x) \geq 0$ for all values of x.

- The graph of $f(x) = |x|$ is V - shaped and opens up. The corner point o the graph is its Vertex.
- The vertex of absolute value a function is a point where the functio reaches a maximum or minimum.
- The vertex of $f(x) = |ax + b| + c$ is $\left(\frac{-b}{a}, c\right)$. To find the vertex, set

$ax + b = 0$ then $x = \frac{-b}{a}$, and at $x = \frac{-b}{a}$, $y = c$

- The symmetry line is $x = \frac{-b}{a}$
- The range is $\{y : y \geq c\}$

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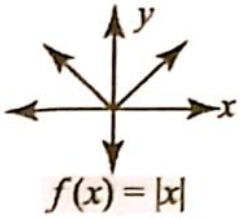
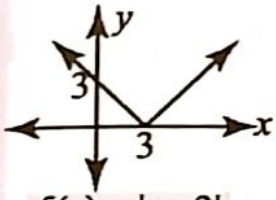
(i) Let $f(x) = |x| = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x < 0 \end{cases}$



$f(x) = |x|$

- The graph of $f(x) = |x|$ is graph of
  $y = x$ if $x \geq 0$ and $y = -x$ if $x < 0$.
- V - shaped with its vertex (0, 0)
- Domain = $\mathbb{R}$ and its range = $[0, \infty)$

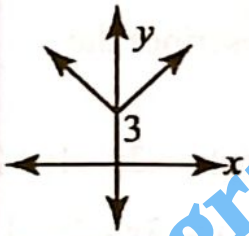
(ii) Let $f(x) = |x - 3| = \begin{cases} x - 3 \text{ if } x \geq 3 \\ -(x - 3) \text{ if } x < 3 \end{cases}$



$f(x) = |x - 3|$

- Then the graph of $f(x) = |x - 3|$ is the graph of
  $y = x - 3$ if $x \geq 3$ and $y = -x + 3$ if $x < 3$.
- V - shaped with its vertex (3, 0)
- Domain = $\mathbb{R}$ and its range = $[0, \infty)$.

(iii) Let $f(x) = |x| + 3 = \begin{cases} x + 3 & \text{if } x \geq 0 \\ -x + 3 & \text{if } x < 0 \end{cases}$



$f(x) = |x| + 3$

- Then the graph of $f(x) = |x| + 3$ is the graph of
  $y = x + 3$ if $x \geq 0$.
  $y = -x + 3$ if $x < 0$.
- V - shaped with its vertex (0, 3)
- Domain = $\mathbb{R}$ and its range = $[3, \infty)$.
- Symmetric w.r,t y - axis ($x = 0$)

(iv) Let $f(x) = |x + 3| + 1 = \begin{cases} (x + 3) + 1 \text{ if } x \geq -3 \\ -(x + 3) + 1 \text{ if } x < -3 \end{cases}$

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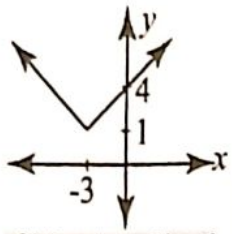


$f(x) = |x + 3| + 1$

- Then the graph of $f(x) = |x + 3| + 1$ is the graph o

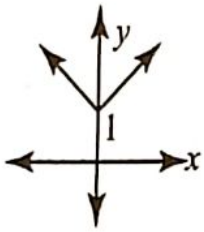
  $y = x + 4$ if $x \geq -3$

  $y = -x - 2$ if $x < -3$

- V - shaped with its vertex (-3, 1)
- Domain = R and rang = $[1, \infty)$
- Symmetric w.r.t the line $x = -3$

Illustrative Example

63. Which of the following absolute value function best describes the graph at left



A. $f(x) = |2x + 1|$ C. $f(x) = |2x| +$

B. $f(x) = |2x - 1|$ D. $f(x) = |2x| -$

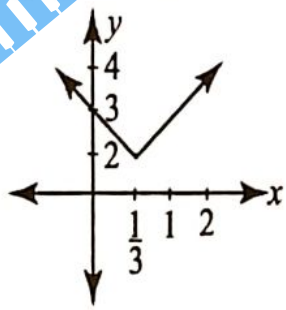
Solution: For detail, refer (iii)

- The vertex of the graph is (0,1)
- The domain is R and the range is $\{y: y \geq 1\}$

C. The graph of $f(x) = |2x| + 1$ is V - shaped with vertex (0, 1) and range is $\{y: y \geq 1\}$

∴ **Answer is**

64. Which of the following absolute value function best describes the graph at left?



A. $f(x) = |3x + 1| + 2$ C. $f(x) = |3x - 1| +$

B. $f(x) = |3x - 1| + 2$ D. $f(x) = |3x - 2| + 1$

Solution

- The vertex of the graph is $\left(\frac{1}{3}, 2\right)$
- The range is $\{y | y \geq 2\}$

A. $f(x) = |3x + 1| + 2 = \begin{cases} (3x + 1) + 2 = 3x + 3 \text{ if } x \geq \frac{-1}{3} \\ -(3x + 1) + 2 = -3x + 1 \text{ if } x < \frac{-1}{3} \end{cases}$

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- Vertex $= \left(\frac{-1}{3}, 2\right)$ and Range $= [2, \infty)$

B. $f(x) = |3x - 1| + 2 = \begin{cases} (3x - 1) + 2 = 3x + 1 & \text{if } x \geq \frac{1}{3} \\ -(3x - 1) + 2 = -3x + 3 & \text{if } x < \frac{1}{3} \end{cases}$

- Vertex $= \left(\frac{1}{3}, 2\right)$ and Range $= [2, \infty)$

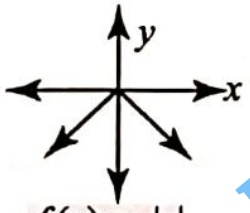
C. $f(x) = |3x + 2| + 1$, vertex $= \left(\frac{-2}{3}, 1\right)$ and range $= [1, \infty)$

D. $f(x) = |3x - 2| + 1$, vertex $= \left(\frac{2}{3}, 1\right)$ and range $= [1, \infty)$

Answer: B

Graph of $f(x) = C - |ax + b|$

(i) Let $f(x) = -|x|$

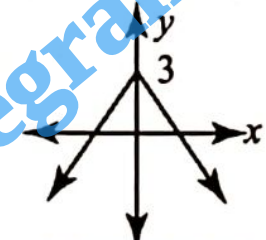


$f(x) = -|x|$

The graph of $f(x) = -|x|$ is inverted V - shaped (opens down ward)

- The vertex is (0,0) with axis $x = 0$
- Domain = R and range $= (-\infty, 0)$

(ii) Let $f(x) = 3 - |x|$, the graph opens down ward



$f(x) = 3 - |x|$

- Vertex (0, 3).
- Domain = R
- Range $= (-\infty, 3]$

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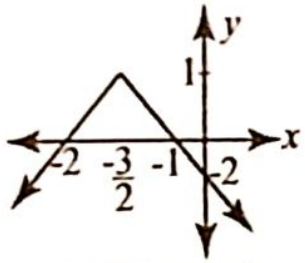
(iii) Let $f(x) = 1 - |3 + 2x|$, the graph opens down ward.



$f(x) = 1 - |3 + 2x|$

- The vertex is $\left(\frac{-3}{2}, 1\right]$
- Domain = R
- Range = $(-\infty, 3]$

1.5 Signum Function

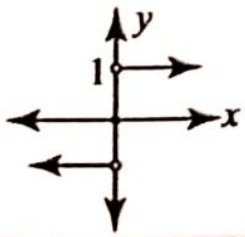
We now look at special category or piece wise - defined functions called **signum function.**

Signum function is defined by :

$$f(x) = \text{Sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

- Domain of signum function is R
- The range is {-1, 0, 1}

Note : $f(x) = \text{Sgn}(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$



Graph of $f(x) = \text{sign}(x)$

- Domain = R
- Range = {-1, 0, 1}
- sgn (-x) = -sgn x

Illustrative Example

65. Evaluate $f(x) = \text{Sgn}(x)$ for the given value of x

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a. $f(10)$ c. $f(-\pi)$ e. $f\left(\frac{-4}{3}\right)$

b. $f\left(\frac{3}{5}\right)$ d. $f\left(\log 2^{\frac{1}{8}}\right)$ f. $f\left(\sqrt{6}-3\right)$

Solution

Using definition $f(x) = \operatorname{sgn} x = \begin{cases} 1, \text{ if } x > 0 \\ 0, \text{ if } x = 0 \\ -1 \text{ if } x < 0 \end{cases}$

a. $f(10) = \operatorname{sgn}(10) = 1$, since $10 > 0$

b. $f\left(\frac{3}{5}\right) = \operatorname{sgn}\left(\frac{3}{5}\right) = 1$, since $\frac{3}{5} > 0$

c. $f(-\pi) = \operatorname{sgn}(-\pi) = -1$, since $-\pi < 0$.

d. $f\left(\log 2^{\frac{1}{8}}\right) = \operatorname{sgn}\left(\log 2^{\frac{1}{8}}\right) = \operatorname{sgn}(-3) = -1$

e. $f\left(\frac{-4}{3}\right) = \operatorname{sgn}\left(\frac{-4}{3}\right) = -1$, since $\frac{-4}{3} < 0$.

f. $f\left(\sqrt{6} - 3\right) = \operatorname{sgn}\left(\sqrt{6} - 3\right) = -1$, since $\sqrt{6} - 3 < 0$.

66. Let $f(x) = 3 + \operatorname{sgn}(x)$.

a. Find the domain and range b. Sketch the graph.

Solution

$$f(x) = 3 + \operatorname{sgn}(x) = 3 + \begin{cases} 1 \text{ if } x > 0 \\ 0 \text{ if } x = 0 \\ -1 \text{ if } x < 0 \end{cases} = \begin{cases} 4 & \text{if } x > 0 \\ 3 & \text{if } x = 0 \\ 2 & \text{if } x < 0 \end{cases}$$

a. Domain = R and range = {2, 3, 4}

b. To sketch the graph.

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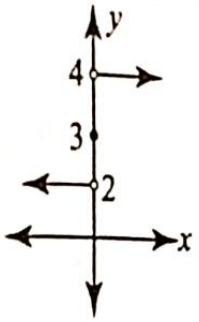


For all $x > 0$, $f(x) = 4$

For all $x < 0$, $f(x) = 2$

For $x = 0$, $f(x) = 3$

67. Let $f(x) = x + \text{sgn}(x)$

a. Find the domain and range

b. Sketch the graph.

Solution

$$f(x) = x + \text{sgn}(x) = x + \begin{cases} 1 \text{ if } x > 0 \\ 0 \text{ if } x = 0 \\ -1 \text{ if } x < 0 \end{cases}$$

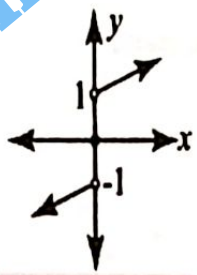
$$\therefore f(x) = \begin{cases} x + 1 \text{ if } x > 0 \\ x \text{ if } x = 0 \\ x - 1 \text{ if } x < 0 \end{cases}$$

a. Domain = R Range = $(-\infty, -1) \cup \{0\} \cup (1, \infty)$.

b. To sketch the graph, we have

$y = x + 1$ for all $x > 0$

$y = x - 1$ for all $x < 0$



$f(x) = x - \text{sgn}(x)$

x	$\frac{1}{4}$	1	2	3	4	-1	
$y = x + \text{sgn}(x)$	$\frac{5}{4}$	2	3	4	5	-2	-9

68. Find the solution set of each of the following equation

a. $\text{sgn}(x) = 1$

b. $\text{sgn}(x - 4) = 1$

c. $\text{sgn}(4 - x^2) = 1$

d. $\text{sgn}(9 - x^2) = -1$

e. $\text{sgn}(4 - x^2) = 0$

f. $\text{sgn}(x^2 - 6) = 0$

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Solution

a. $\text{sgn}(x) = 1$ if $x > 0$ $\therefore$ The solution set is $\{x : x > 0\}$

b. $\text{sgn}(x - 4) = 1$ if $x - 4 > 0$ $\therefore$ The solution set $\{x : x > 4\}$

c. $\text{sgn}(4 - x^2) = 1$ if $4 - x^2 > 0$ $\therefore$ The solution set $\{x : -2 < x < 2\}$

d. $\text{sgn}(9 - x^2) = -1$ if $9 - x^2 < 0$

$\therefore$ The solution set is $\{x : x < -3 \text{ or } x > 3\}$

e. $\text{sgn}(4 - x^2) = 0$ if $4x^2 = 0$

$\therefore$ The solution set is $\{x : x = -2 \text{ or } x = 2\}$

f. $\text{sgn}(x^2 - 6) = 0$ if $x^2 - 6 = 0$

$\therefore$ The solution set is $\left\{x : x = \pm\sqrt{6}\right\}$

69. Let $f(x) = \text{sgn}(x) + 1$, then find:

a. $f(-3) + f(3)$ b. $f(-x) + f(x)$

Solution

a. $f(-3) + f(3) = [\text{sgn}(-3) + 1] + [\text{sgn}(3) + 1]$

$= [-1 + 1] + [1 + 1] = 2$

b. $+ \begin{cases} f(-x) = \text{sgn}(-x) + 1 = -\text{sgn}(x) + 1 \\ f(x) = \text{sgn}(x) + 1 = \text{sgn}(x) + 1 \end{cases}$

$f(-x) + f(x) = -\text{sgn}(x) + 1 + \text{sgn}(x) + 1$

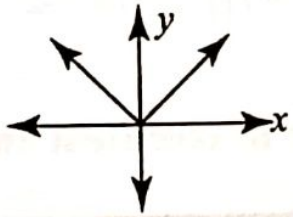
$= -\text{sgn}(x) + \text{sgn}(x) + 1 + 1 = 2$

70. Let $f(x) = x\,\text{sgn}(x)$

a. Sketch the graph b. Find the domain and range

Solution

a. $f(x) = x\,\text{sgn}(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -x & \text{if } x < 0 \end{cases}$



$f(x) = x\text{sgn}(x) = |x|$

b. Domain = R and range = $[0, \infty)$.

71. Let $f(x) = x^2\,\text{sgn}(x)$

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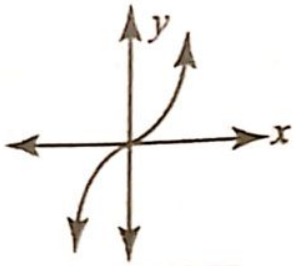
a. Sketch the graph b. Find the domain and range

Solution

a. $f(x) = x^2 \operatorname{sgn}(x) = \begin{cases} x^2 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -x^2 & \text{if } < 0 \end{cases}$



$f(x) = x^2 \operatorname{sgn}(x)$

b. Domain = R and rang = R

1.6 The Greatest Integer Function

If x is a real number, we define the symbol $\lfloor x \rfloor$ as follows:

$\lfloor x \rfloor = m$, where m is greatest inter such that $m \leq x$

Example : $\lfloor 0 \cdot 6 \rfloor = 0$, $\lfloor 2 \cdot 3 \rfloor = 2$, $\lfloor -2 \cdot 3 \rfloor = -3$

$\lfloor \sqrt{2} \rfloor = 1$, $\lfloor -\sqrt{2} \rfloor = -2$ $\lfloor \pi \rfloor = 3$

Illustrative Example

72. Evaluate $f(x) = \lfloor x \rfloor$ for the given value of x.

a. $f(2 \cdot 47)$ c. $f(1 \cdot 24)$ e. $f(0 \cdot 25)$

b. $f(-2 \cdot 47)$ d. $f(-1 \cdot 24)$ f. $f(-0 \cdot 25)$

Solution

$f(x) = \lfloor x \rfloor$

a. $f(2 \cdot 47) = \lfloor 2 \cdot 47 \rfloor$

b. $f(-2 \cdot 47) = \lfloor -2 \cdot 47 \rfloor = -3$

c. $f(1 \cdot 24) = \lfloor -2 \cdot 24 \rfloor = 1$

d. $f(-1 \cdot 24) = \lfloor -1 \cdot 24 \rfloor = -2$

e. $f(0 \cdot 25) = \lfloor 0 \cdot 25 \rfloor = 0$

f. $f(-0 \cdot 25) = \lfloor -0 \cdot 25 \rfloor = -1$

Graph of Greatest Integer

The graph of **greatest integer function** (floor function) is a series of discontinuous "step function". It is defined by

$f(x) = \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x.

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Note: The graph of the greatest integer function $y = \lfloor x \rfloor$ lies on below the line $y = x$, so it provides an integer **floor** for x.

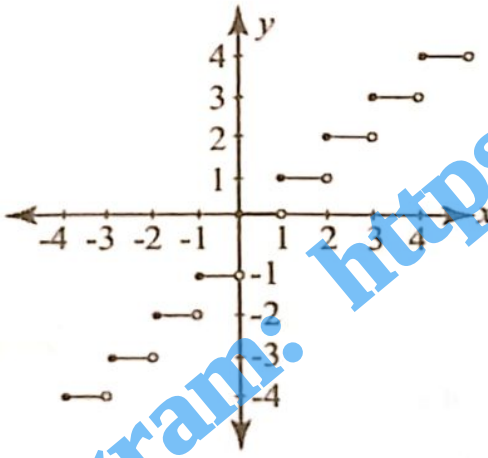
The Domain of $f(x) = \lfloor x \rfloor$ is $\mathbb{R}$ and the range is the set of integer ($\mathbb{Z}$).

To graph $f(x) = \lfloor x \rfloor$, consider the table.

x	-3 · 7	-2 · 7	-1· 7	-1	0	0 · 6	1 · 6	2 · 6	3 · 6	4 · 8
$f(x) = \lfloor x \rfloor$	-4	-3	-2	-1	0	0	1	2	3	4

- Greatest integer less than or equal to -3 · 7 is - 4.
- Greatest integer less than or equal to -1 · 7 is - 2.
- Greatest integer less than or equal to 1 · 6 is 1.
- Greatest integer less than or equal to 3 · 6 is 3.

From the table, we observe a pattern and from the pattern, we form the graph.



- If $-4 \le x < -3$, then $\lfloor x \rfloor = -4$
- If $-3 \le x < -2$, then $\lfloor x \rfloor = -3$.
- If $-2 \le x < -1$, then $\lfloor x \rfloor = -2$
- If $-1 \le x\ 0$, then $\lfloor x \rfloor = -1$
- If $0 \le x < 1$, then $\lfloor x \rfloor = 0$.
- If $1 \le x < 2$, then $\lfloor x \rfloor = 1$.
- If $2 \le x < 3$, then $\lfloor x \rfloor = 2$.

73. Let $f(x) = \lfloor x + 3 \rfloor$, sketch the graph.

Solution: If $\lfloor x \rfloor = a$, $a \in \mathbb{Z}$, then $a \le x < a + 1$.

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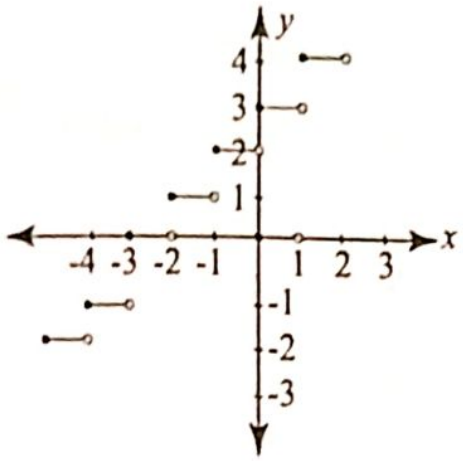


If $-5 \leq x < -4$, then $\lfloor x + 3 \rfloor = -2$

If $-4 \leq x < -3$, then $\lfloor x + 3 \rfloor = -1$

If $-3 \leq x < -2$, then $\lfloor x + 3 \rfloor = 0$

If $-2 \leq x < -1$, then $\lfloor x + 3 \rfloor = 1$

If $-1 \leq x < 0$, then $\lfloor x + 3 \rfloor = 2$

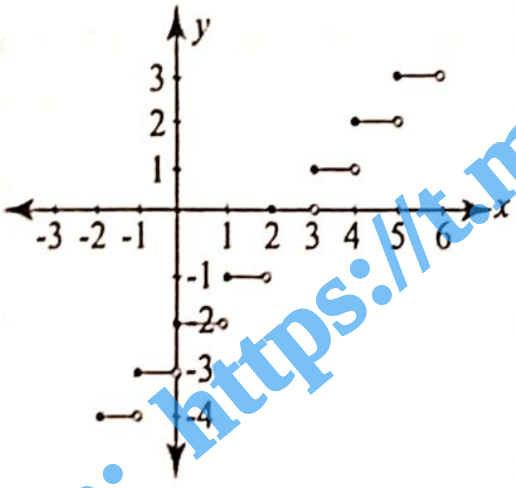
If $0 \leq x < 1$, then $\lfloor x + 3 \rfloor = 3$

If $1 \leq x < 2$, then $\lfloor x + 3 \rfloor = 4$

Note: The graph of $y = \lfloor x + 3 \rfloor$ lies below the line $y = x + 3$.

74. Let $f(x) = \lfloor x - 2 \rfloor$, Sketch the graph.

Solution: If $\lfloor x \rfloor = a$, $a \in Z$, then $a \leq x < a + 1$



If $-2 \leq x < -1$, then $\lfloor x - 2 \rfloor = -4$

If $-1 \leq x < 0$, then $\lfloor x - 2 \rfloor = -3$

If $0 \leq x < 1$, then $\lfloor x - 2 \rfloor = -2$

If $1 \leq x < 2$, then $\lfloor x - 2 \rfloor = -1$

If $2 \leq x < 3$, then $\lfloor x - 2 \rfloor = 0$

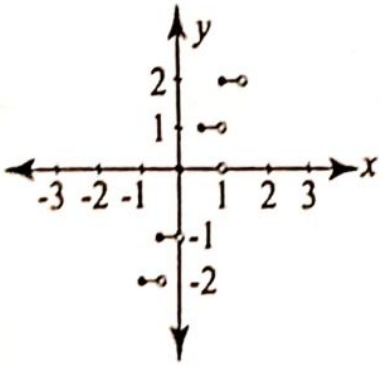
If $3 \leq x < 4$, then $\lfloor x - 2 \rfloor = 1$

If $4 \leq x < 5$, then $\lfloor x - 2 \rfloor = 2$

75. Let $f(x) = \lfloor 2x \rfloor$, then for what values of x

$f(x) = -2$, $f(x) = -1$, $f(x) = 0$, and $f(x) = 1$

Solution: If $\lfloor x \rfloor = m$, then $m \leq 2x < m + 1$



If $-1 \leq x < \frac{-1}{2}$, then $\lfloor 2x \rfloor = -2$

If $\frac{-1}{2} \leq x < 0$, then $\lfloor 2x \rfloor = -1$

If $0 \leq x < \frac{1}{2}$, then $\lfloor 2x \rfloor = 0$

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If $\frac{1}{2} \le x < 1$, then $\lfloor 2x \rfloor = 1$

If $1 \le x < \frac{3}{2}$, then $\lfloor 2x \rfloor = 2$

Note: The graph of $y = \lfloor 2x \rfloor$ lies below the line $y = 2x$.

76. Let $f(x) = x - \lfloor x \rfloor$, then

a. find the domain b. find the range

- Let $x = -2 \cdot 9$, $f(-2 \cdot 9) = -2 \cdot 9 - \lfloor 2 \cdot 9 \rfloor = -2 \cdot 9 + 3 = 0 \cdot 1$
- Let $x = -1 \cdot 98$, $f(-1 \cdot 98) = -1 \cdot 98 - \lfloor -1 \cdot 98 \rfloor = -1 \cdot 98 + 2 = 0 \cdot 02$
- Let $x = -1 \cdot 01$ $f(-1 \cdot 01) = -1 \cdot 01 - \lfloor -1 \cdot 01 \rfloor = -1 \cdot 01 + 2 = 0 \cdot 99$
- Let $x = 2 \cdot 7$, $f(2 \cdot 7) = 2 \cdot 7 - \lfloor 2 \cdot 7 \rfloor = 2 \cdot 7 - 2 = 0 \cdot 7$
- Let $x = 4 \cdot 8$, $f(4 \cdot 8) = 4 \cdot 8 - \lfloor 4 \cdot 8 \rfloor = 4 \cdot 8 - 4 = 0 \cdot 8$
- Let $x = 3$ $f(3) = 3 - \lfloor 3 \rfloor = 3 - 3 = 0$

Let $x = a$, a is an integer, $f(a) = a - \lfloor a \rfloor = a - a = 0$

a. The domain is $\mathbb{R}$.

b. The range is $[0, 1) = 0 \le f(x) < 1$ for every value of x.

77. Let $y = 2\lfloor x \rfloor + 3$ and $y = 3\lfloor x - 2 \rfloor + 5$

Find the value of $\lfloor x + y \rfloor$

Solution

since $y = 2\lfloor x \rfloor + 3$, substitute $2\lfloor x \rfloor + 3$ for y in the second equation.

$\Rightarrow 2\lfloor x \rfloor + 3 = 3\lfloor x - 2 \rfloor + 5 = 3\lfloor x \rfloor - 6 + 5$

$\Rightarrow 3\lfloor x \rfloor - 2\lfloor x \rfloor = 3 + 1 = 4$

$\therefore \quad \lfloor x \rfloor = 4$, if and only if $4 \le x < 5$.

Also, to find y, put $\lfloor x \rfloor = 4$ in $y = 2\lfloor x \rfloor + 3$.

$\therefore \quad y = 2(4) + 3 = 8 + 3 = 11$

$\therefore \quad \lfloor x + y \rfloor = \lfloor 4 + 11 \rfloor = 15$

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Equation Involving Greatest Integer.

An equation of the form $\lfloor x \rfloor$ where a is an integer is equa... involving greatest integer.

$\lfloor x \rfloor = a$ if and only if $a \leq x < a + 1$

Illustrative Example

78. Find the value of x:

a. $\lfloor x \rfloor = 4$

b. $[x] = -3$

c. $\lfloor x + 3 \rfloor 8$

d. $[x + 2] = 10$

e. $[2 + [x]] = 7$

f. $[[x]-1] = 3$

Solution

a. $\lfloor x \rfloor = 4$ if and only $4 \leq x < 5$

$\therefore$ The solution set is $[4, 5)$

b. $\lfloor x \rfloor = -3$ if and only if $-3 \leq x < -2$

c. $\lfloor x + 2 \rfloor = 8$ if and only if $8 \leq x + 2 < 9$

$\therefore$ The solution set is $[6, 7)$

d. $\lfloor 2x + 1 \rfloor = 10$ if and only if $10 \leq 2x + 1 < 11$.

$9 \leq 2x < 10$

$\frac{9}{2} \leq x < 5$

$\therefore$ The solution set is $[4{\cdot}5, 5)$

e. $\lfloor 2 + [x] \rfloor = 7$ and only $7 \leq 2 + \lfloor x \rfloor < 8$

$\Rightarrow 5 \leq \lfloor x \rfloor < 6$

$\Rightarrow 5 \leq x < 6$

$\therefore$ The solution set $[5,6)$

f. $\lfloor \lfloor x \rfloor - 1 \rfloor = 3$ if and only if $3 \leq \lfloor x \rfloor - 1 < 4$

$\Rightarrow 3 + 1 \leq \lfloor x \rfloor < 4 + 1$

$\Rightarrow 4 \leq \lfloor x \rfloor < 5$ the same as $4 \leq x < 5$.

$\therefore$ The solution set $[4,5)$

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9. Let $f(x) = |2x - 3|$. For what values of x, $f(x) = 6$

A. $\left\{x : \frac{9}{2} \leq x < 5\right\}$ B. $\left\{x : x = \frac{9}{2}\right\}$ C. $\{x : x = 5\}$ D. $\left\{x : \frac{9}{2} < x \leq 5\right\}$

Solution

$\Rightarrow \lfloor 2x - 3 \rfloor = 6$ if and only if $6 \leq 2x - 3 < 7$

$\Rightarrow 6 + 3 \leq 2x < 7 + 3$

$9 \leq 2x < 10$, then $\frac{9}{2} \leq x < 5$

$\therefore$ The solution set is $\left\{x : \frac{9}{2} \leq x < 5\right\}$

Answer: A

0. Determine whether each of the following statement is true or false for all real number x and y. If false give a counter example:

a. $[x + y] = [x] + [y]$

b. $\frac{[x]}{2} = \left[\frac{x}{2}\right]$

c. $[xy] = [x][y]$

d. $[x]^2 = [x^2]$

e. $[-x] = -[x]$

f. $[x] + \frac{1}{2} = \left[x + \frac{1}{2}\right]$

Solution

a. False: as an example

Let $x = 3{\cdot}6$, and $y = 2{\cdot}8$

$\lfloor 3{\cdot}6 + 2{\cdot}8 \rfloor \neq \lfloor 3{\cdot}6 \rfloor + \lfloor 2{\cdot}8 \rfloor$

$\lfloor 6{\cdot}4 \rfloor \neq 3 + 2$

$6 \neq 5$

b. False, as an example

Let $x = 5{\cdot}4$

$\frac{\lfloor 5{\cdot}4 \rfloor}{2} \neq \left\lfloor \frac{5{\cdot}4}{2} \right\rfloor$

$\frac{5}{2} \neq 2$

c. False ; as an example

Let $x = 2{\cdot}4, y = 3{\cdot}4$

$\lfloor (2.4)\ (3.4) \rfloor \neq \lfloor 2.4 \rfloor \lfloor 3.4 \rfloor$

$\lfloor 8.16 \rfloor \neq 2 \times 3$

$8 \neq 6$

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d. False: Let $x = 3{\cdot}6$

$$\lfloor 3\cdot 6 \rfloor^2 \neq \lfloor 3\cdot 6^2 \rfloor$$

$$3^2 \neq \lfloor 12\cdot 96 \rfloor$$

$$9 \neq 12$$

e. False as an example

let $x = -4{\cdot}3$

$$\lfloor -4\cdot 3 \rfloor \neq -\lfloor 4\cdot 3 \rfloor$$

$$-5 \neq -4$$

f. False as an example

Let $x = 7{\cdot}6$

$$\lfloor 7\cdot 6 \rfloor + \frac{1}{2} \neq \left\lfloor 7\cdot 6 + \frac{1}{2} \right\rfloor$$

$$7 + \frac{1}{2} \neq \lfloor 8\cdot 1 \rfloor$$

$$7\cdot 5 \neq 8$$

Note: Let $f(x) = \lfloor x \rfloor$

- $\lfloor x + y \rfloor \geq \lfloor x \rfloor + \lfloor y \rfloor$
- $\lfloor xy \rfloor \geq \lfloor x \rfloor \lfloor y \rfloor$
- $\lfloor x \rfloor^2 \leq \lfloor x^2 \rfloor$
- $f(x) \leq x < f(x)+$

81. Determine whether each of the following is true or false

a. $\text{sgn}(x) = \pm |x|$, for all real number x

b. $\text{sgn} \leq \lfloor x \rfloor$, for all $x \leq 0$.

c. $|x| = x \text{ sgn}(x)$, for all real number x

d. $\text{sgn}(x) > \lfloor x \rfloor$, for all $x \geq 0$

Solution

a. False as an example, let $x = 4$.

b. False, as an example, let $x = -1{\cdot}6$

$\text{sgn}(-1{\cdot}6) \leq \lfloor -1{\cdot}6 \rfloor \Rightarrow -1 \leq -2$ is false

c. True, $\text{sgn}(x) = \dfrac{|x|}{x}$, so that $|x| = x \text{ sgn}(x)$

d. False

82. For what values of x:

a. $\text{sgn} = \lfloor x \rfloor$, for $x > 0$

b. $\text{sgn}\, x = \lfloor x \rfloor$, for $x > 0$

Solution

a. We know that, $x > 0$, $\text{sgn}\, x = 1$

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$\Rightarrow \lfloor x \rfloor = 1$, if and only if $1 \leq x < 2$.

$\therefore$ sgn $x = \lfloor x \rfloor$, for $x > 0$ is $\{x: 1 \leq x < 2\}$

b. For the fact that sgn $x = -1$, if $x < 0$

$\Rightarrow -1 = \lfloor x \rfloor$, if and only if $-1 \leq x < 0\}$.

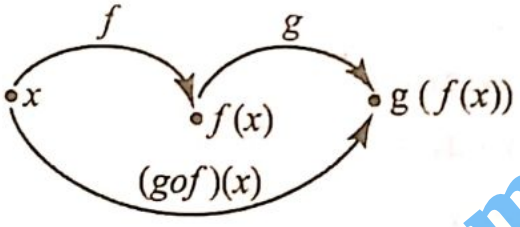
$\therefore$ sgn $x = \lfloor x \rfloor$, for $x < 0$ $\{x: -1 \leq x < 0\}$.

1.7 Composition Function

(i) The composition of function g with function f is denoted as

$(gof)(x) = g(f(x))$ ⟵ g at $f(x)$

(ii) The **domain** of $(gof)(x)$ is consists of x in the domain of f and the value of $f(x)$ is the domain of g.



- x is in the domain of f.
- $f(x)$ sis in the domain of g.
- Domain of $gof \subseteq$ Domain of f.
- Range of $gof \subseteq$ Range of g.

(iii) $(fog)(x) = f(g(x))$ ← f at g (x). Domain of (fog) x is x in the domain of g and g (x) is in the domain of f.

Illustrative Example

83. Let A = {a, b, c}; B. = {-1, 0, 1} and C = {0, 1, 2, 3}

If f: A ⟶ B, given by f = {(a, - 1), (b, 0), (c,1)} and

g: B ⟶ C, given by g = {(-1,0), (0,1), (1, 2)}

Then find gof and domain of gof

Solution

(i) $(gof)(a) = g(f(a)) = g(-1) = 0$, $\therefore$ (a, 0).

$(gof)(b) = g(f(b)) = g(0) = 1$ $\therefore$ (b,1).

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$(gof)(c) = g(f(c)) = g(1) = 2 \quad \therefore \quad (c,2)$

$\therefore \quad gof = \{(a,0), (b,1), (c,2)\}$.

(ii) Domain of gof is $\{a,b,c\}$. Range of gof is $\{0, 1, 2\}$

84. The following table lists values of two functions f and g.

x	-2	-1	0	3	4	7	10
$f(x)$	1	3	5	11	5	3	-4

x	-3	-2	-1	1	2	3
g(x)	-2	1	0	10	9	5

Find the value of the following table.

a. $(fog)(1)$ b. $(gof)(-2)$ c. $(fof)(-1)$ d. $(gog)(-3)$

Solution

a. $(fog)(1) = f(g(1)) = f(10) = -4$

b. $(gof)(-2) = g(f(-2)) = g(f(-2)) = g(1) = 10.$

c. $(fof)(-1) = f(f(-1)) = f(3) = 11.$

d. $(gog)(-3) = g(g(-3)) = g(-2) = 1$

85. Let $f(x) = 3x + 7$ and $g(x) = 4x-5$

find a. $(fog)(x)$ b. $(gof)(x)$

Solution: f at x is $3x + 7 = f(x)$

a. $f(g(x)) = 3g(x) + 7 \longleftarrow f$ at $g(x)$

$= 3(4x - 5) + 7$

$= 12x - 8$

b. $g(f(x))$ is g at $f(x)$

$g(f(x)) = 4f(x) - 5 = 4(3x + 7) - 5 = 12x - 23$

86. Let $f(x) = 3x - 4$ and $g(x) = x^2 + 3x$ Find

a. $(fog)(1)$ c. $(fog)(x)$ e. $fof(2)$

b. $(gof)(1)$ d. $(gof)(x)$ f. $(fof)(x)$

Solution

a. $(fog)(1) = f(g(1)) \longleftarrow f$ at $g(1)$

$= f(4) \longleftarrow g(1) = 1 + 3 = 4$

$= 3(4) - 4 = 8$

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b. $(gof)(1) = g(f(1)) \longleftarrow$ g at $f(1)$

$= g(-1) \longleftarrow f(1) = 3 - 4 = -1$

$= 4$

c. $(fog)(x) = f(g(x)) \longleftarrow$ f at $g(x)$

$= f(x^2 + 3x) \longleftarrow g(x) = x^2 + 3x$

$= 3(x^2 + 3x) - 4 \longleftarrow$ Definition

$= 3x^2 + 9x - 4$

d. $(gof)(x) = g(f(x)) \longleftarrow$ Definition of gof.

$= g(3x - 4) \longleftarrow f(x) = 3x - 4$

$= (3x - 4)^2 + 3(3x - 4) \longleftarrow$ Definition of g.

$= 9x^2 - 24 + 16 + 9x - 12$

$= 9x^2 - 15x + 4$

e. $(fof)(3) = f(f(3)) = f(5) = 11$

f. $(fof)(x) = f(f(x)) \longleftarrow$ f at $f(x)$

$f(3x - 4) \longleftarrow f(x) = 3x - 4$

$3(3x - 4) - 4 = 9x - 12 - 4 = 9x - 16$

87. Suppose $f(3x + 1) = -6x - 9$, then find $f(x)$?

Solution: Let $g(x) = 3x + 1$ and $f(x) = ax + b$.

Then $f(g(x)) = a\,g(x) + b = -6x - 9$

$\Rightarrow a(3x + 1) + b = -6x - 9$

$3ax + a + b = -6x - 9$

$\Rightarrow 3ax = -6x$ and $a + b = -9$

$\Rightarrow 3a = -6$ and $b = -9 - a$

$\Rightarrow a = -2$ and $b = -9 - (-2) = -7$

$\therefore \quad f(x) = -2x - 7$

88. Let $f(x) = \sqrt{x - 2}$ and $g(x) = x + 3$

Find $(fog)(x)$ and domain of $(fog)(x)$.

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Solution

(i) $(fog)(x) = f(g(x)) = f(x+3) = \sqrt{x+3-2} = \sqrt{x+1}$

(ii) The function $f(x)$ is defined for: $x - 2 \geq 0 = \{x; x \geq 2\}$.

• $f(g(x))$ is defined for: $g(x) \geq 2$

$\Rightarrow x + 3 \geq 2$ and $x \geq 2 - 3$

$\Rightarrow x \geq -1$

$\therefore$ Domain of (for) (x) is $\{x: x \geq -1\}$

89. If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

Find the following functions and their domains.

a. $fog(x)$ b. $gof(x)$ c. $fof(x)$ d. $gog(x)$

Solution

a. $(fog)(x) = f(g(x)) = f\left(\sqrt{3-x}\right) = \sqrt{\sqrt{3-x}} = \sqrt[4]{3-x}$

The domain fog is $\{x: 3 - x \geq 0\} = \{x: x \leq 3\} = (-\infty, 3]$

b. $(gof)(x) = g(f(x)) = \sqrt{3 - f(x)} = \sqrt{3-\sqrt{x}}$

To find domain of gof

For $\sqrt{x}$ to be defined, we must have $x \geq 0$.

For $\sqrt{3-\sqrt{x}}$ to be defined, we have

$3 - \sqrt{x} \geq 0$, that is, $\sqrt{x} \leq 3$, or $x \leq 9$

$\therefore$ Domain of gof is $\{x: 0 \leq x \leq 9\} = [0, 9]$

c. $(fof)(x) = f(f(x)) = f\left(\sqrt{x}\right) = \sqrt{\sqrt{x}}, \sqrt[4]{x}$

$\therefore$ Domain of fof is $[0, \infty)$

d. $(gog)(x) = g(g(x)) = \sqrt{3 - g(x)} = \sqrt{3 - \left(\sqrt{3-x}\right)}$

This expression is defined when $3 - x \geq 0$ and $3 - \left(\sqrt{3-x}\right) \geq 0$

$\therefore$ Domain of gog is $[-6, 3]$

90. Let $f(x) = \sqrt{4 - x^2}$ and $g(x) = 3x + 7$. What is the domain of $(fog)(x)$?

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A. $\{x: -2 \leq x \leq 2\}$ C. $\left\{x: -3 \leq x \leq \frac{-5}{3}\right\}$

B. $\{x; x \in \mathbb{R}$ D. $\{x: x \leq -1\}$

Solution: $(fog)(x) = f(g(x))$.

- The function $f(x)$ is defined for: $\{x: 4 - x^2 \geq 0\} = \{x: -2 \leq x \leq 2\}$

$f(g(x)) = \{x: -2 \leq g(x) \leq 2\} = \{x: -2 \leq 3x + 7 \leq 2\}$

$= \{x: -2 - 7 \leq 3x \leq -5\}$

$= \left\{x: -3 \leq x \leq \frac{-5}{3}\right\}$

$\therefore$ Domain of $(fog)(x) = \left\{x: -3 \leq x \leq \frac{-5}{3}\right\}$ **Answer: C**

91. Let $f(x) = 3x - 1$ and $f(g(x)) = 9x^2 + 12x + 8$. The find $g(x)$

Solution

$f(x) = 3x - 1$

$f(g(x)) = 3g(x) - 1 = 9x^2 + 12x + 8$

$3g(x) = 9x^2 + 12x + 9$

$\therefore$ $g(x) = 3x^2 + 4x + 3$.

92. Let $f(x) = \sqrt{4 - x^2}$ and $g(x) = \sqrt{x - 1}$ .what is the domain of $(fog)(x)$?

A. [1 4] B. [-2 2] C. [1, 5] D. [1 2]

Solution

Domain of $f = \{x: 4 - x^2 \geq 0\} = \{x: -2 \leq x \leq 2\}$

Domain of $(fog)(x) = f(g(x)) = \{x: -2 \leq g(x) \leq 2\}$

$= \left\{x: -2 \leq \sqrt{x - 1} \leq 2\right\}$

$= \left\{x: -2 \leq \sqrt{x - 1}$ and $\sqrt{x - 1} \leq 2\right\}$

$= \{x: x - 1 \geq 0$ and $x - 1 \leq 4\}$

$= \{x: x \geq 1$ and $x \leq 5\}$

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$= \{x: 1 \leq x \leq 5\}$

Answer: (

93. If $f(x) = 3x^2 - 7$ and $g(x) = 4x + k$, for what value of k, so that the graph of $(fog)(x)$ crosses the y - axis at 5?

A. $k = 0$, B. $k = \pm 3$ C. $k = \pm 2$ D. $k = \pm\sqrt{2}$

Solution

The graph of $(fog)(x)$ crosses the y - axis when $x = 0$

$\Rightarrow f(g(0)) = f(0 + k) = f(k) = 3k^2 - 7 = 5$

$\therefore \quad 3k^2 = 12$ and $k^2 = \frac{12}{3} = 4$

$\therefore \quad k = \pm\sqrt{4} = \pm 2,$

Answer: (

94. If $f(x) = \frac{1}{x - 2}$ and $g(x) = \frac{1}{x + 4}$

a. Find $(fog)(x)$ and domain of $(fog)(x)$

b. Find $(gof)(x)$ and domain of $(gof)(x)$.

a. (i) Domain of $(fog)(x)$ a $= f(g(x))$, should be

- x is in the domain g and $g(x)$ is in the domain of f

$\Rightarrow x \neq -4$ and $\frac{1}{x + 4} \neq 2$

$\Rightarrow x \neq -4$ and $2x + 8 \neq 1$, thus $x \neq \frac{-7}{2}$

$\therefore$ Domain of fog is $\mathbb{R} / \left\{-4, \frac{-7}{2}\right\}$

(ii) $(fog)(x) = f(g(x)) = \frac{1}{g(x) - 2} = \frac{1}{\frac{1}{x + 4} - 2}$

$= \frac{1}{\frac{1}{x + 4} - \frac{2(x + 4)}{x + 4}} = \frac{1}{\frac{1 - 2x - 8}{x + 4}} = \frac{x + 4}{-2x - 7}$

b. (i) Domain of $(gof)(x) = g(f(x))$

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• x is defined in f and $f(x)$ is defined in g.

$\Rightarrow x \neq 2$ and $\frac{1}{x-2} \neq -4$

$\Rightarrow x \neq 2$ and $1 \neq -4x + 8$

$\Rightarrow x \neq 2$ and $x \neq \frac{7}{4}$

$\therefore$ Domain of gof is $\mathbb{R} / \left\{2, \frac{7}{4}\right\}$

(ii). $(gof)(x) = g(f(x)) = \frac{1}{f(x)+4} = \frac{1}{\frac{1}{x-2}+4}$

$= \frac{1}{\frac{1}{x-2} + \frac{4(x-2)}{x-2}} = \frac{1}{\frac{4x-7}{x-2}} = \frac{x-2}{4x-7}$

1.8 Inverse Function and Its Graph

The inverse of any relation is found by interchanging the value of x and y in the relation. However, the inverse of a function may not itself be a function.

As **an example** let $f = \{(1,4), (2,3), (-2,4)\}$ and its inverse will be $\{(4,1), (3,2), (4,-2)\}$. In this case, the relation defining the inverse of f in not a function, because it fails because the domain $x = 4$ has two different y- values 1 and -2.

What is a condition for a function to have an inverse function?

A function f has an inverse function if and only if it is a one - to - one function.

- If the inverses of a function is also a function, it is called an **invertible** and is denoted by f^{-1}.

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Inverse Function and Notation

The inverse of a one - to - one function f is the fun f^{-1}, which is obtained from f by interchanging the x a coordinate in each ordered pair of f.

If $f(a) = b$, then $f^{-1}(b) = a$.

95. Determine whether each function is invertible. If it is invertible, find the inverse.

a. $f = \{(-3,3), (-2,2), (0,0), (1,2)\}$

b. $g = \{(-1,-1), (0, 1), (2, 3)\}$

c. $h = \{(2, -2), (1, -1), (3,$

d. $k = \{(3, 1), (2, 1), (0, 4)\}$

Solution

a. It is not invertible function, because (-2, 2) and (1, 2) have the y - coordinate, this function is not one to one.

b. It is invertible, because this function is one to one.

$\therefore g^{-1} = \{(1, -1), (1, 0), (3,2)\}$

c. It is invertible function, because this function is one to one.

$\therefore h^{-1} = \{(-2, 2), (-1, 1) (4, 3)\}$

d. It is not invertible, because (3, 1) and (2, 1) have same y - coordi this function is one to one.

96. Is the function $f = \{(x,y): y \text{ is the father of } x\}$ invertible?

Solution: No, because one father may have more than one child.

97. Determine whether each of the following function is invertible.

a. $f = \{(x,y): y = 4x - 7\}$

b. $g = \{(x,y): y = x^2\}$

c. $h = \{(x,y): y = x^3 + 8\}$

d. $k = \{(x,y): y = x^3 - x - 1\}$

Solution

a. f is invertible, because linear function are one - to - one functi

b. g is not invertible, because g is not - one to one, as an example

g = {(-2, 4), (-1, 1), (1, 1), (2,4)}, the inverse of this ordered p is {(4, -2), (1, -1), (1, 1), (4, 2)} not function.

c. h is an invertible function, because h is one - to - one.

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d. k is not invertible, because the function is not one - to - one As an example, k = {(1, -1), (-1, -1)}, the inverse of this function is not function.

98. Which of the following functions have inverses?

A. $f(x) = |x - 2|$ C. $f(x) = x^3 + 1$

B. $f(x) = x^4 + 1$ D. $f(x) = x^3 + x^2$

Solution

A. $f(x) = |x - 2|$ is not one - to - one, because

$f(3) = |3 - 2| = 1$ and $f(1) = |1 - 2| = 1$

This shows $f(3) = f(1) = 1$, but $3 \neq 1$

$\Rightarrow$ So that f has no inverse.

B. $f(x) = x^4 + 1$ is not one - to - one, because

$f(-1) = 2$ and $f(1) = 2$

$\Rightarrow f(-1) = f(1)$ but $-1 \neq 1$

$\Rightarrow$ So that f has no inverse.

C. $f(x) = x^3 + 1$ has an inverse because f is one - to - one.

D. $f(x) = x^3 + x^2 = x^2 (x + 1)$ is not one - to - one function, as an example $f(0) = 0$ and $f(-1) = 0$ but $0 \neq -1$

$\Rightarrow$ So that f has no inverse

Answer: C

99. If g is a function with domain A and range B,

a. What is the domain of g^{-1} b. What is the range of g^{-1}

Solution

a. Domain of g^{-1} = Range of g = B

b. Range of g^{-1} = Domain of g = A

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How To Find the Inverse of Function

To find the inverse of $y = f(x)$ (when it exists):

1. Replace $f(x)$ by y
2. Interchange x and y
3. Solve for y
4. Replace y by $f^{-1}(x)$

Illustrative Example

100. Find the inverse of each of the following function.

a. $f(x) = 4x - 7$

b. $h(x) = -3x + 2$

Solution

a. $y = 4x - 7$

$x = 4y - 7$

$4y = x + 7$

$$y = \frac{x + 7}{4}$$

$$\therefore f^{-1}(x) = \frac{x + 7}{4}$$

- Replace $f(x)$ by y
- Interchange x and y
- Solve for y
- Replace y by $f^{-1}(x)$

b. $y = -3x + 2$

$x = -3y + 2$

$-3y = x - 2$

$$y = \frac{x - 2}{-3} = \frac{-x + 2}{3}$$

$$h^{-1}(x) = \frac{-x + 2}{3}$$

- Replace $h(x)$ by y.
- Interchange x and y
- Solve for y
- Replace y by $h^{-1}(x)$

101. Find the inverse of each of the following function and domain and range of f^{-1}

a. $f(x) = \frac{3}{x - 4}$

b. $f(x) = \frac{2x}{x + 1}$

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c. $f(x) = \frac{4x+3}{5x-2}$

d. $f(x) = \frac{x+1}{3x-4}$

Solution

a. $y = \frac{3}{x-4}$

$x = \frac{3}{y-4}$

$xy - 4x = 3$

$xy = 4x + 3$

$y = \frac{4x+3}{x}$

$f^{-1}(x) = \frac{4x+3}{x}$

Domain of f^{-1} is $\mathbb{R}\setminus\{0\}$

Range of f^{-1} is $\mathbb{R}\setminus\{4\}$

b. $y = \frac{2x}{x+1}$

$x = \frac{2y}{y+1}$

$xy + x = 2y$

$xy - 2y = -x$

$y(x-2) = -x$

$y = \frac{-x}{x-2}$

$f^{-1}(x) = \frac{-x}{x-2}$

Domain of f^{-1} is $\mathbb{R}\setminus\{2\}$

Range of f^{-1} is $\mathbb{R}\setminus\{-1\}$

c. $y = \frac{4x+3}{5x-2}$,

$x = \frac{4y+3}{5y-2}$,

$5xy - 2x = 4y + 3$

$5xy - 4y = 2x + 3$

$y(5x-4) = 2x + 3$

$y = \frac{2x+3}{5x-4}$

$\therefore f^{-1}(x) = \frac{2x+3}{5x-4}$

Domain of f^{-1} is $\mathbb{R}/\left\{\frac{4}{5}\right\}$

Range of f^{-1} is $\mathbb{R}/\left\{\frac{2}{5}\right\}$

d. $y = \frac{x+1}{3x-4}$

$x = \frac{y+1}{3y-4}$

$3xy - 4x = y + 1$

$3xy - y = 4x + 1$

$y(3x-1) = 4x + 1$

$y = \frac{4x+1}{3x-1}$

$\therefore f^{-1}(x) = \frac{4x+1}{3x-1}$

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Domain of f^{-1} is $\mathbb{R}/\left\{\frac{1}{3}\right\}$

Range of f^{-1} is $\mathbb{R}/\left\{\frac{4}{3}\right\}$

102. Find the inverse of each of the following function

a. $f(x) = 2x^3 + 5$ b. $f(x) = 3x^5 - 4$

Solution

a. $y = 2x^3 + 5$

$x = 2y^3 + 5$

$2y^3 = x - 5$

$y^3 = \frac{x-5}{2}$

$y = \sqrt[3]{\frac{x-5}{2}}$

$\therefore f^{-1}(x) = \sqrt[3]{\frac{x-5}{2}}$

b. $y = 3x^5 - 4$

$x = 3y^5 - 4$

$3y^5 = x + 4$

$y^5 = \frac{x+4}{3}$

$y = \sqrt[5]{\frac{x+4}{3}}$

$\therefore f^{-1}(x) = \sqrt[5]{\frac{x+4}{3}}$

103. Find the inverse of each of the following function

a. $f(x) = \sqrt[3]{2x - 5}$ b. $f(x) = \sqrt[3]{4x + 1}$

Solution

a. $y = \sqrt[3]{2x - 5}$

$x = \sqrt[3]{2y - 5}$

$x^3 = 2y - 5$

$2y = x^3 + 5$

$y = \frac{x^3 + 5}{2}$

$f^{-1}(x) = \frac{x^3 + 5}{2}$

b. $y = \sqrt[3]{4x + 1}$

$x = \sqrt[3]{4y + 1}$

$x^3 = 4y + 1$

$4y = x^3 - 1$

$y = \frac{x^3 - 1}{4}$

$f^{-1}(x) = \frac{x^3 - 1}{4}$

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Place A Restriction on A Domain

For a function that is not one - to - one, sometimes we restrict its domain to create a new function that is one to one.

The graph of the function defined by $f(x) = ax^2 + bx + c$, $a \neq 0$, is a parabola that opens upward.

It is not a one - to - one function and therefore does not have an inverse function. However the function.

$f(x) = ax^2 + bx + c$, $a \neq 0$ with domain restricted to

$\left\{x \middle| x \geq \frac{-b}{2a}\right\}$ or $\left\{x \middle| x \leq \frac{-b}{2a}\right\}$ is a one to one function.

It has an inverse function denoted by f^{-1}.

104. Find the inverse of each of the following function

a. $f(x) = x^2 + 3$, for $x \geq 0$.

b. $f(x) = x^2 - 4$, for $x \leq 0$.

c. $f(x) = (x - 4)^2$, for $x \geq 4$

d. $f(x) = x^2 + 4x + 1$

Solution

a. $f(x) = x^2 + 3, \quad x \geq 0,$

$y = x^2 + 3, \quad x \geq 0$ • Replace f (x) by y.

$x = y^2 + 3, \quad y \geq 0$ • Interchange x and y.

$y^2 = x - 3, \quad y \geq 0$

$y = \pm\sqrt{x - 3},$ • since $y \geq 0$, then we choose positive square root

$y = \sqrt{x - 3}$

$f^{-1}(x) = \sqrt{x - 3},$ • Replace y by $f^{-1}(x)$

Domain of f^{-1} is $\{x: x \geq 3\}$ and Range of f^{-1} is $\{y: y \geq 0\}$

b. $f(x) = x^2 - 4$, for $x \leq 0$

$y = x^2 - 4; \ x \leq 0$

$x = y^2 - 4; \ y \leq 0$ • Interchange x and y.

$y^2 = x + 4;$

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$y = \pm\sqrt{x+4}$ • since $y \le 0$, we take negative square root

$y = -\sqrt{x+4}$

$f^{-1}(x) = -\sqrt{x+4}$, for $y \le 0$ Domain of f^{-1} is $\{x : x \ge -4\}$

c. $f(x) = (x-4)^2$ for $x \ge 4$

$y = (x-4)^2$, • Replace y for $f(x)$

$x = (y-4)^2 ; y \ge 4$ • Interchange x and y.

$y - 4 = \pm\sqrt{x}$ • since $y \ge 4$, we take the positive square root

$y - 4 = \sqrt{x}$

$y = \sqrt{x} + 4$

$\therefore \quad f^{-1}(x) = 4 + \sqrt{x}$

• Domain of f^{-1} is $\{x: x \ge 0\}$

• Range of f^{-1} is $\{y: y \ge 4\}$

105. Let $f(x) = (x-4)^2 - 1$ for $x \ge 4$. Find $f^{-1}(x)$

Solution

$y = (x-4)^2 - 1$, for $x \ge 4$

$x = (y-4)^2 - 1, y \ge 4$ • Interchange x and y.

$(y-4)^2 = x + 1$ • solve for y.

$y - 4 = \sqrt{x+1}, \ y \ge 4$

$y = 4 + \sqrt{x+1}$,

$\therefore f^{-1}(x) = 4 + \sqrt{x+1}$,

Domain of $f^{-1} = \{x : x \ge -1\}$ and range $= \{y : y \ge 4\}$.

106. Let $f(x) = x^2 + 6x + 4$, for $x \ge -3$. Find $f^{-1}(x)$

Solution

To find f^{-1}, we uses the completing the square method.

$y = x^2 + 6x + 4$, for $x \ge -3$

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$y = x^2 + 6x + 9 + 4 - 9$, for $x \geq -3$ • completing the square

$y = (x + 3)^2 - 5, x \geq -3$

$x = (y + 3)^2 - 5, y \geq -3$ • interchange x and y

$(y + 3)^2 = x + 5$

$y + 3 = \pm\sqrt{x + 5}$,

$y = -3 + \sqrt{x + 5}$, since $y \geq -3$

$\therefore f^{-1}(x) = -3 + \sqrt{x + 5}$,

Domain $f^{-1} = \{x | x \geq -5\}$ and range $= \{y | y \geq -3\}$.

07. Find the inverse function

a. $f(x) = \sqrt{x - 2}$ b. $f(x) = \sqrt{2x - 1}$ c. $f(x) = 4 + \sqrt{x}$

Solution

a. $f(x) = \sqrt{x - 2}$ with domain $\{x : x \geq 2\}$ and range is $\{y : y \geq 0\}$

$y = \sqrt{x - 2}$ $x \geq 2$ and $y \geq 0$

$x = \sqrt{y - 2}$, $y \geq 2$, $x \geq 0$ • Interchange x and y.

$x^2 = y - 2$ • squaring both side

$y = x^2 + 2$ • solve for y

$\therefore f^{-1}(x) = x^2 + 2$, $x \geq 0$ and $y \geq 2$

b. $f(x) = \sqrt{2x - 1}$ Domain is $\left\{x : x \geq \frac{1}{2}\right\}$ and range is $\{y \geq 0\}$.

$y = \sqrt{2x - 1}$; $x \geq \frac{1}{2}$ and $y \geq 0$

$x = \sqrt{2y - 1}$, $y \geq \frac{1}{2}$ and $x \geq 0$ • Interchange x and y.

$x^2 = 2y - 1$ • squaring both side

$y = \frac{x^2 + 1}{2}$, $y \geq \frac{1}{2}, x \geq 0$

$\therefore f^{-1}(x) = \frac{x^2 + 1}{2}$, $y \geq \frac{1}{2}$ and $x \geq 0$

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c. $y = 4 + \sqrt[3]{x}$ • Interchange x and y

$x = 4 + \sqrt[3]{y}$

$\sqrt[3]{y} = x - 4$

$y = (x - 4)^3$

$\therefore \quad f^{-1}(x) = (x - 4)^3$

108. Let $f(x) = |x| + 5$, for $x \leq 0$. Find $f^{-1}(x)$

Solution

$y = |x| + 5$

$y = -x + 5, \ x \leq 0$

$x = -y + 5, \ y \leq 0$

$-y = x - 5$

$y = -x + 5 = 5 - x$

$\therefore \quad f^{-1}(x) = (x - 4)^3$

109. Let $f(x) = |x - 3| + 1$, for $x \leq 3$.

Solution

$y = |x - 3| + 1$, for $x \leq 3$

$y = -(x - 3) + 1, \ x \leq 3$

$y = -y + 4, \ y \leq 3,$ • Interchange x and y

$-y = x + 4$ • solve for y.

$y = -x + 4$

$f^{-1}(x) = -x + 4$

Important: Let f be invertible (one - to - one) function with domain A and range B. Then its **inverse function** f^{-1} has domain B and range A and is defined by $f^{-1}(y) = x$ and $f(x) = y$ for any y in B

Symbols: Let f be one - to - one function; then

(i) $f: A \longrightarrow B$ and defined $f(x) = y$

(ii) $f^{-1}: B \longrightarrow A$, and defined by $f^{-1}(y) = x$ for any y in B

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Example: If $f(1) = 3, f(2) = 6$ and $f(5) = -7$, then $f^{-1}(3) = 1$, $f^{-1}(6) = 2$ and $f^{-1}(7) = 5$. These shows f^{-1} reverses the effect of f.

Illustrative Example

110. Let $f(x) = x^2 + 7, x \leq 0$, then what is the value of $f^{-1}(8)$?

A. 15 B. -1 C. 1 D. $\sqrt{15}$

Solution

Let $f^{-1}(8) = a$, with $a \leq 0$

$\Rightarrow f(a) = 8$ and $f(a) = a^2 + 7$

$\Rightarrow a^2 + 7 = 8$ implies $a^2 = 8 - 7 = 1$

$\therefore \quad a = \pm\sqrt{1} = \pm 1$, but $a \leq 0$, so that $a = -1$

Answer: B

111. Let $f(x) = 2x^3 + 5x + 8$ and $f^{-1}(2x) = 1$. Find the value of x

Solution

If $f^{-1}(2x) = -1$, then $f(-1) = 2x$

$\Rightarrow 2(-1)^3 + 5(-1) + 8 = 2x$, and $2x = 1$

$\therefore \quad x = \frac{1}{2}$

112. Let $f^{-1}(3x - 1) = -2$, and $f(-2) = 9$; Find x

Solution

$f^{-1}(3x - 2) = -2$ if $f(-2) = 3x - 1$

$\Rightarrow 3x - 1 = 9$, and $3x = 10$

$\therefore \quad x = \frac{10}{3}$

113. Let $f(x) = \sqrt{x - 3}$, then what is the value of $f^{-1}(4)$?

A. 7 B. 1 C. 19 D. 13

Solution

Let $f^{-1}(4) = a$, then $f(a) = 4$,

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$\Rightarrow \sqrt{a-3} = 4$, • squaring both side

$\Rightarrow a - 3 = 16$

$a = 16 + 3 = 19$ and $f^{-1}(4) = 19$ Answer: (

114. Let $f(x) = x^2 + 6x$, $x \leq -3$, then what is the equation of $f^{-1}(x)$?

A. $f^{-1}(x) = 3 \pm \sqrt{x+9}$ C. $f^{-1}(x) = -3 + \sqrt{x-9}$

B. $f^{-1}(x) = -3 \pm \sqrt{x+9}$ D. $f^{-1}(x) = -3 - \sqrt{x+9}$

Solution

First apply completing square method

$y = x^2 + 6x = x^2 + 6x + 9 - 9$

$y = (x+3)^2 - 9, \quad x \leq -3$

$x = (y+3)^2 - 9, \quad y \leq -3$ • Interchange x and y.

$(y+3)^2 = x + 9$

$y + 3 = \pm\sqrt{x+9}$, since $y \leq -3$: then $y + 3 = -\sqrt{x+9}$

$\therefore \quad y = -3 - \sqrt{x+9}$ and $f^{-1}(x) = -3 - \sqrt{x+9}$

115. Let $f(x) = ax - b$ and $f^{-1}\left(\frac{1}{2}x + 3\right) = x + 1$ then the value of a and b is equal to

A. $a = 2, b = 5$ C. $a = \frac{1}{2}, b = \frac{-5}{2}$

B. $a = \frac{1}{2}, b = \frac{5}{2}$ D. $a = \frac{-1}{2}, b = 5$

Solution

The fact that $f^{-1}(a) = b$ is the same as $f(b) = a$

So that $f^{-1}\left(\frac{1}{2}x + 3\right) = x + 1$ is the same as $f(x+1) = \frac{1}{2}x + 3$

Since $f(x) = ax - b$, then $f(x+1) = a(x+1) - b = \frac{1}{2}x + 3$

$ax + a - b = \frac{1}{2}x + 3$. This implies $a = \frac{1}{2}$ and $a - b = 3$

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$\frac{1}{2}$ - b = 3 we get b = $\frac{1}{2}$ - 3 = $\frac{-5}{2}$ **Answer: C**

116. Determine whether each of the following statement are true or false. If false, give example

a. Every function are invertible.

b. Every greatest integer function are invertible.

c. Every linear function are invertible.

d. Every absolute value function has an inverse .

e. Every exponential function has an inverse.

f. Every quadratic function are invertible.

g. Every constant function are invertible.

Solution

a. False, as an example $f = \{(-1, 1), (0, 0), (1, 1)\}$ is not invertible.

b. False, as an example, $f(x) = [x], f = \{(1{\cdot}2, 1), (1{\cdot}2, 1), (1{\cdot}9, 1)\}$ is not invertible.

c. True

d. False, as an example, $f\ [x] = |x|, f = \{(-2, 2), (2, 2)\}$ is not invertible.

e. True,

f. False, as an example, $f(x) = x^2, f = \{(-2, 4), (2, 4)\}$, is not invertible.

g. False, as an example, $f = \{(3, 1), (2, 1)\}$ is not invertible.

How to Determine whether Two Given Functions Are Inverses of Each other

Functions f and g are inverses of each other if and only if:

1. $(fog)(x) = x$ for all x in the domain of g
2. $(gof)(x) = x$, for all x in the domain of f

Note: Let f be one - to - one function with domain A and range B. The inverse function f^{-1} satisfies.

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$f^{-1}(f(x)) = x$, for every x in A

$f^{-1}(f(x)) = x$, for all x in the domain of f.

$f(f^{-1}(x)) = x$, for all x in the domain of f^{-1}.

Illustrative Example

117. Determine whether each pair of function f and g are inverses of ea other

a. $f(x) = 3x - 5$ and $g(x) = \dfrac{x+5}{3}$

b. $f(x) = \sqrt[3]{x-1}$ and $g(x) = x^3 + 1$

c. $f(x) = x^3 - 4$ and $g(x) = \sqrt[3]{x+4}$

d. $f(x) = x^2 + 5$ and $g(x) = \sqrt{x-5}$

e. $f(x) = x^2 + 5,\ x \geq 0$ and $g(x) = \sqrt{x-5},\ x \geq 5$

Solution

We need to show that $(fog)(x) = x$ and $(gof)(x) = x$

a. $(fog)(x) = f(g(x)) = 3\,g(x) - 5 = 3\left(\dfrac{x+5}{3}\right) - 5 = x$

$(gof)(x) = g(f(x)) = \dfrac{f(x)+5}{3} = \dfrac{3x-5+5}{3} = x$

$\therefore$ f and g are inverses of each other.

b. $f(g(x)) = \sqrt[3]{g(x)-1} = \sqrt[3]{x^3 + 1 - 1} = \sqrt[3]{x^3} = x$

$g(f(x)) = (f(x))^3 + 1 = \left(\sqrt[3]{x-1}\right)^3 + 1 = x - 1 + 1 = x$

$\therefore$ f and g are inverses of each other.

c. $f(g(x)) = (g(x))^3 - 4 = \left(\sqrt[3]{x+4}\right)^3 - 4 = x + 4 - 4 = x$

$g(f(x)) = \sqrt[3]{f(x)+4} = \sqrt[3]{x^3 - 4 + 4} = \sqrt[3]{x^3} = x$

$\therefore$ f and g are inverses of each other.

d. $f(g(x)) = (g(x))^2 + 5 = \left(\sqrt{x-5}\right)^2 + 5 = x - 5 + 5 = x$

$g(f(x)) = \sqrt{f(x)-5} = \sqrt{x^2 + 5 - 5} = \sqrt{x^2} = |x|$

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Since $(fog)(x) \neq (gof)(x)$ then $x \neq |x|$

So that f and g are not inverse of each other

e. $f(g(x)) = x$ and $g(f(x)) = \sqrt{x^2} = |x| = x,\ x \geq 0$

$\therefore$ f and g are inverses of each other.

118. If $f(0) = -1$ and $f(-1) = 2$, then find

a. $f^{-1}(-1)$ b. $f^{-1}(f(0))$ c. $f(f^{-1}(2))$

Solution

a. $f^{-1}(-1) = 0$ because $f(0) = -1$ c. $f(f^{-1}(2)) = f(-1) = 2$

b. $f^{-1}(f(0)) = f^{-1}(-1) = 0$

119. Let $f(x) = x^2 + 4, x \geq 0$, then which of the following statement is true.

A. $f^{-1}(f(1)) = f(f^{-1}(1))$ C. $f(f^{-1}(6)) = f^{-1}(f(6))$

B. $f^{-1}(f(3)) = f(f^{-1}(3))$ D. $f(f^{-1}(2)) = f^{-1}(f(2))$

Solution

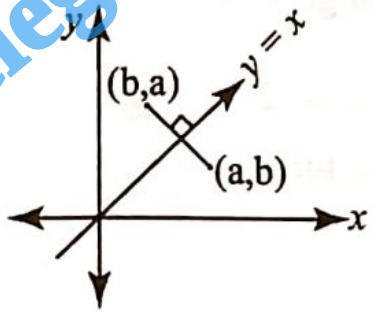
- Domain of $f = [0, \infty)$ and Range of $f = [4, \infty)$.
- Domain of $f^{-1} = [4, \infty)$ and range of $f^{-1} = [0, \infty)$

Since 1, 2, and 3 are not in the domain of f^{-1}, then $f(f^{-1}(1)), f(f^{-1}(2))$ and $f(f^{-1}(3))$ are not defined.

Answer: C

1.8.1 Graphs of Inverse Functions

How are the graphs f and f^{-1} related?



The graphs f and f^{-1} are symmetric with respect to the line $y = x$.

If a point (a, b) is on the graph of f then point (b, a) must be on the graph of f^{-1}

Let $f(x) = x^2 - 1$, with $x \geq 0$, then $f^{-1}(x) = \sqrt{x+1}$

Their graphs are shown below.

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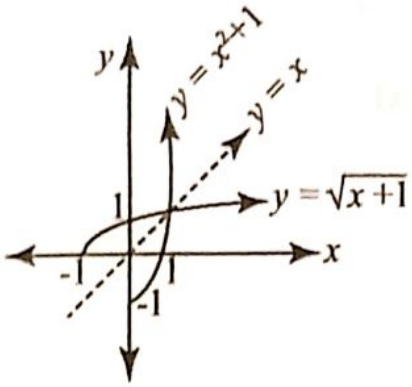

- $(0, -1) \in f$, then $(-1, 0) \in f^{-1}$
- $(1, 0) \in f$, then $(0, 1) \in f^{-1}$
- $(2, 3) \in f$, then $(3, 2) \in f^{-1}$
- Domain of $f = [0, \infty)$ and range of $f = [-1, \infty)$
- Domain of $f^{-1} = [-1, \infty)$ and range of $f = [0, \infty)$

For Fast learner skill

120. Find the inverse of the function:

a. $f(x) = \ell n(2x + 1)$ b. $f(x) = \ell n\left(\frac{x}{x-1} + 2\right)$

Solution: we use the face that $\log e^x = \ell n x$

$a. y = \ell n(2x + 1)$, this implies $e^y = 2x + 1$

$e^x = 2y + 1$ • Interchanging x and y

$y = \frac{1}{2}(e^x - 1)$

$\therefore \quad f^{-1}(x) = \frac{1}{2}(e^x - 1)$

b. $y = \ell n\left(\frac{x}{x-1} + 2\right)$, this implies $e^y = \frac{x}{x-1} + 2$

$e^y = \frac{x}{x-1} + 2\left(\frac{x-1}{x-1}\right) = \frac{x + 2x - 2}{x - 1} = \frac{3x - 2}{x - 1}$

$e^x = \frac{3y - 2}{y - 1}$, interchange x to y

$e^x y - e^x = 3y - 2$, implies $e^x y - 3y = e^x - 2$

$y(e^x - 3) = e^x - 2$, solving for y

$y = \frac{e^x - 2}{e^x - 3}$ and $f^{-1}(x) = \frac{e^x - 2}{e^x - 3}$

121. Find the inverse of the following function

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a. $f(x) = \dfrac{1}{e^x - 1}$ b. $f(x) = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

Solution

a. $y = \dfrac{1}{e^x - 1}$, interchange x to y:

$x = \dfrac{1}{e^y - 1}$, this implies $xe^y - x = 1$

$\Rightarrow xe^y = x + 1$, implies that $e^y = \dfrac{x+1}{x}$

$\therefore \quad y = \ell n\left(\dfrac{x-1}{x}\right)$, hence, $f^{-1}(x) = \ell n\left(\dfrac{x+1}{x}\right)$

b. $y = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$, then $x = \dfrac{e^y - e^{-y}}{e^y + e^{-y}}$

$\Rightarrow xe^y + xe^{-y} = e^y - e^{-y}$, multiply both side by (e^y)

$\Rightarrow xe^{2y} + x = e^{2y} - 1$,

$\Rightarrow xe^{2y} - e^{2y} = -1 - x = -x - 1 = -(x+1)$

$\Rightarrow e^{2y}(x-1) = -(x+1)$, then $e^{2y} = \dfrac{x+1}{1-x}$

$\therefore f^{-1}(x) = \dfrac{1}{2}\ell n\left(\dfrac{x+1}{1-x}\right)$

122. Let $f(x) = \log_5^{3x-4}$, then find $f^{-1}(x)$

Solution

$y = \log_5^{3x-4}$, change x *to* y

$x = \log_5^{3y-4}$ this implie $3y - 4 = 5^x$

$y = \dfrac{1}{3}(5^x + 4)$

$\therefore \quad f^{-1}(x) = \dfrac{1}{3}(5^x + 4)$

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1.9 Applications of Relation and Function

Here, we will going to see application based on business and economics.

Cost Analysis:

The cost of manufacturing an item commonly consists of two parts.

(i) **Fixed cost** are costs like rent, machinery training workers, and so on. The fixed cost is constant for a particular product and does not change as more items are made.

(ii) **Variable costs** are costs per item for labor, materials, packing, transporting, and so on. This cost vary according to the amount being produced.

- The sum of the fixed cost and the variable cost gives the **total cost** of producing a product.

Break - Even Analysis : When a company manufacture x - units of a product, it inverts money. This is **total cost** and can be thought of as a function C, where C (x) is the total cost of producing x - units.

The revenue R(x) - when the company sells x units of the product, it takes money.

$$R(x) = Px,$$

Where P is the selling price of the product (item) and x is the number of units sold (demand).

The profit P (x) is the difference between revenue R (x) and cost C (x).

That is $P(x) = R(x) - C(x)$

Note:

(i) A company can make profit only if the revenue revived from its customers exceeds the cost of producing and selling its goods and service. That is $R(x) > C(x)$

(ii) The number of units at which revenue just equals cost is the break - even quantity $R(x) = C(x)$

This is called break – even point.

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Illustrative Example

123. A Company is planning to make a new kind of radio. Fixed cost will be Birr 180,000 and it will cost Birr 240 to produce each radio. Each radio sells for Birr 270

a. Find the total cost C (x) of producing x radio.

b. Find total revenue R (x)

c. Find the total profit P (x)

d. What profit or loss will the company from the production of 400 radios? of 900 radio

e. The break - even point.

Solution

a. Total cost is given by: C (x) = (Fixed cost) + (Variable costs)

C $(x) = 18,000 + 240x$, where x is the number of radios produced.

b. Total revenue is given by R $(x) = 270x$ Birr 270 times the number of radios sold.

c. Total profit is given by : P (x) = R (x) - C (x)

$= 270x - (18,000 + 240x) = 30x - 18,000$

d. P $(400) = 30 \times 400 - 18,000 = -6000$ Birr when 400 radios are produced and sold, the company loses money.

P $(900) = 30 \times 9000 - 18,000 = 27000 - 18,000 = 9000$. When 900 radios are produced and sold the company makes profit of Birr 9000.

e. The break - even point is given by R (x) = C (x)

$270x = 18000 + 240x \Rightarrow 270x - 240x = 18,000$

$$30x = 18,000, \text{and } x = \frac{18,000}{30} = 600$$

$\therefore$ The break - even point $x = 600$. If the company sells more than 600 radio, it makes a profit. If it sells fewer than 600 radio is loss money.

124. A firm producing poultry feed whose production cost in Birr is given by: C $(x) = 32x + 200$

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Management plans to charge Birr 36 per unit for the feed.

a. How many units must be sold for the firm to ensure no loss

b. How many units must be sold for the firm to break even.

c. What is the profit if 80 units of feed are sold?

d. How may units must be sold to produce a profit of Birr 1400.

Solution

We have $C(x) = 32x + 200$ and $R(x) = 36x$

a. $R(x) > C(x)$ implies $36x > 32x + 200$

$\Rightarrow 36x - 32x > 200$ and $x > \frac{200}{4}$, $x > 50$

If $x > 50$, it makes profit

b. Break - even point is $R(x) = C(x)$

$\Rightarrow 36x = 32x + 200$ and $4x = 200$

$$x = \frac{200}{4} = 50$$

c. $P(x) = R(x) - C(x) = 36x - (32x + 200) = 4x - 200$

$P(80) = 4 \times 80 - 200 = 320 - 200 = 120$

The firm will make a profit of Birr 120 form the sale of 80 units of feed.

d. Let $P(x) = 1400$ and $P(x) = 4x - 200$

$\Rightarrow 4x - 200 = 1400$

$4x = 1400 + 200 = 1600$

$$x = \frac{1600}{4} = 40$$

Sales of 400 units will produce Birr 1400 profit.

125. At a parking garage in a city the charge for parking consists of a flat fee of Birr 10 and an additional Birr 2.5 per hour.

a. Write the cost of parting for t hours.

b. How much cost to park for 2 hour.

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Solution

a. $C(t) = 10 + 2{\cdot}5t$

b. $C(2) = 10 + 2{\cdot}5(2) = 15$ birr

Review Exercise 1

1. What is a relation?
2. What is the domain of a relation?
3. What is the range of a relation?
4. What is a function?
5. What is the difference between relation and function?
6. What is the inverse of a function?
7. What is the domain of f^{-1}?
8. What is the range of f^{-1}?
9. How are the graph of f and f^{-1} related?
10. What is the domain of composition of f and g is denoted by $(fog)(x)$?
11. Let $R = \{(x, y): y = 2x - 1\}$
    a. Find the ordered pairs belonging to R where $x \in \left\{-2, -1, \frac{-1}{2}, 0, \frac{1}{2}, 1\right\}$
    b. Find the ordered pairs belonging to R where $y \in \{-2, -1, 0, 1, 3\}$.
12. Let $A = \{1, 2, 3, 4\}$. If R is the relation from A to A such that $R = \{(x, y): y \leq x$, where $x, y \in A\}$. List all elements of R.
13. Let R is a relation from N to N defined by $R = \{(x, y): x + y = 12\}$.
    a. List all the elements of R
    b. List the domain and range of R.
14. Let $R = \{(x,y): y = 3x - 1\}$. Find R^{-1}
15. Let $R = \{(x, y): y < x + 5$ and $y \geq x^2 - 1\}$
    a. Find the domain and range of R.
    b. Find the inverse of R.

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c. Find the domain and rang R^{-1}

16. Let $R = \{(x,y): x - y < 1 \text{ and } x < -2\}$

a. Find domain and range of R b. Find the inverse of R

c. Find domain and range of R^{-1}

17. Let $R = \{(x,y): x + y > 1 \text{ and } x > y + 2\}$

a. Find the domain and range of R

b. Find the inverse of R

c. Find the domain and range of R^{-1}

18. Let $R = \{(x,y): 3x + y > 6\}$, then determine which of the following points belongs to R^{-1}?

a. (1, 3) b. (1, 2) c. (2, 1) d. (-1, 5)

19. Determine whether each relation is a function. If not, explain.

a. $R = \{(x,y): x \text{ is the husband of } y\}$.

b. $R = \{(x,y): y \text{ is the husband of } x\}$.

c. $R = \{(x,y): y \text{ is the wife of } x\}$.

d. $R = \{(x,y): y \text{ is the sister of } x\}$

e. $R = \{(-2, 2), (-1, -4), (0, -6), (1, -4), (2, 2)\}$

f. $R = \{(-1, 0), (1, 1), (7, 2)\}$

g. $R = \{(x, y): |x| + |y| = 1\}$

h. $R = \{(x,y): y = |x|\}$

i. $R = \{(x,y): x = |y|\}$

20. Describe the common characteristics of the graph.

a. $f(x) = x^{\frac{m}{n}}$, where m is odd and n is even positive integer.

b. $f(x) = x^{\frac{-m}{n}}$, Where m is odd and n is even positive integer.

c. $f(x) = x^{\frac{m}{n}}$, Where m is even and n is odd positive integer.

d. $f(x) = x^{\frac{-m}{n}}$, Where m is even and n is odd positive integer.

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e. $f(x) = x^{\frac{m}{n}}$, Where both m and n are odd.

f. $f(x) = x^{\frac{-m}{n}}$, Where both m and n are odd.

21. a. Name all rational number whose absolute value is 6.

b. Find the set of rational numbers whose distance from 5 and a number is 8.

22. Solve

a. $|x| + |x - 2| = 6$

b. $|x - 4| = 3x + 1$

c. $|2x + 1| = |3x - 4|$

d. $|x^2 - 3| = |2x|$

23. Determine whether each statement is true or false. Explain

a. $|x| > x$

b. $|x| + |y| > 0$

c. $|x + y| \leq |x| + |y|$

d. $|x + y| = |x| + |y|$

e. $|x\,y| = |x|\,|y|$

f. $|x| = x$

g. $|-x| = x$

h. $|x - y| = |x| - |y|$

24. An expression $|x + y| < |x| + |y|$ is always true if:

A. $x > 0$ and $y > 0$

B. $x < 0$ and $y \geq 0$

C. $xy \geq 0$

D. $xy < 0$

25. Write the vertex of each of the graph of absolute value

a. $y = |x| - 3$

b. $y = |x - 1| + 5$

c. $y = |x + 2| - 3$

d. $y = |3x - 1| + 2$

26. Solve

a. $\lfloor x + 2 \rfloor = 7$

b. $\lfloor 3 + \lfloor x \rfloor \rfloor = 4$

c. $\lfloor 2x - 3 \rfloor = 6$

d. $\lfloor x + 2 \rfloor = -4$

27. Given $f(x) = x^2 + 3x$ and $g(x) = x - 2$ Find

a. $f(g(1))$ b. $g(f(1))$ c. $f(g(0))$ d. $g(f(0))$

28. Let $f(x) = \dfrac{x}{x - 1}$ and , $g(x) = \dfrac{9}{x^2 - 4}$.

Find $(fog)(x)$ and domain of $(fog)(x)$

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29. Suppose $f(2x - 1) = 6x - 2$, then find $f(x)$ 1

30. Let $f(x) = \sqrt{x + 3}$ and , $g(x) = x - 2$.

Find $(fog)(x)$ and domain of $(fog)(x)$ 1

31. Determine whether each of the following function is invertible If it invertible find the inverse.

a. $\{(-2, 5), (0, 1), (2,5)\}$

b. $\{(-1, 3), (0, 1), (1, -1)\}$

c. $\{(-3, 1), (-2, 1), (0, 1)\}$ 1

d. $\{(1, 2), (2, 1), (3, 1)\}$ 1

32. Find the inverse of the function

a. $f(x) = 3x - 4$

b. $f(x) = \dfrac{1 - x}{x + 3}$

c. $f(x) = \dfrac{x + 1}{3x - 4}$

d. $f(x) = 2x^2 + 1,\ x \geq 0$

e. $f(x) = 3x^2 - 2,\ x \leq 0$

f. $f(x) = (x + 5)^2,\ x \leq -5$ 1

g. $f(x) = (x - 2)^2,\ x \geq 2$ 1

h. $f(x) = \sqrt{x - 3}$

i. $f(x) = x^3 - 4$

j. $f(x) = \sqrt[3]{x + 2}$

Answer and Explanation

1. A relation is any set of ordered pairs.
2. The domain of a relation is the set of all first coordinates.
3. The range of a relation is the set of all second coordinate. 1
4. A function is a set of ordered pairs in which no two pairs have the same first coordinate and different second coordinates.
5. A relation is any set of ordered pairs, but function is a special type relation in which no domain is repeated.
6. The inverse of a function is a function with the same ordered pairs except that the coordinates are revered.
7. The domain of f^{-1} is the range of f.
8. The range of f^{-1} is the domain of f.
9. The graphs of f and f^{-1} are symmetric with respect to the line $y = x$.

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0. The domain of fog is the set of real numbers x in the domain of g such that $g(x)$ is on the domain of f

1. a. $R = \left\{(-2,-5), (-1,-3), \left(\frac{-1}{2}, -2\right), (0,-1), \left(\frac{1}{2}, 0\right), (1,1)\right\}$

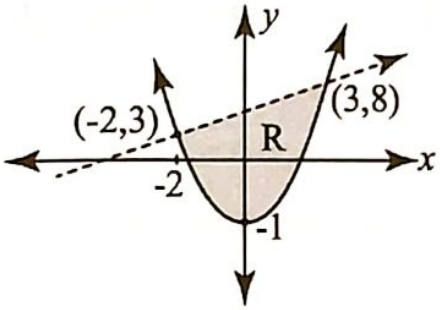
b. $R = \left\{\left(\frac{-1}{2}, -2\right), (0,-1), \left(\frac{1}{2}, 0\right), (1,1), (2,3)\right\}$

2. $R = \{(1,1), (2,2), (3,3), (4,4), (2,1), (3,1), (3,2), (4,1), (4,2), (4,3),\}$

3. $R = \{(1,11),(2,10),(3,9),(4,8),(5,7),(6,6),(7,5),(8,4),(9,3),(10,2),(11,1)\}$

4. $R^{-1} = \{(y,x): y = 3x - 1\}$ or $R^{-1} = \left\{(x,y): y = \frac{x+1}{3}\right\}$

5. Solve the intersection of $y = x + 5$ and $y = x^2 - 1$



$\Rightarrow x^2 - 1 = x + 5$

$\Rightarrow x^2 - x - 6 = 0$

$\Rightarrow (x - 3)(x + 2) = 0$

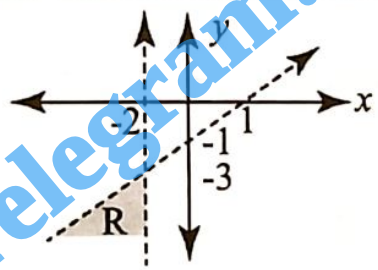
$\therefore \quad x = 3$ or $x = -2$

a. Domain of $R = \{x: -2 < x \leq 3\}$ Range of $R = \{y: -1 \leq y < 8\}$

b. $R^{-1} = \{(y,x): y < x + 5$ and $y \geq x^2 - 1\}$

c. Domain of $R^{-1} = \{x: -1 \leq x < 8\}$ Range of $R^{-1} = \{y: -2 < y < 3\}$

6. Solve the intersection of $x - y$ and $x = -2$



$\Rightarrow -2 - y = 1$

$\Rightarrow -y = 1 + 2 = 3$

$\Rightarrow y = -3$

a. Domain of $R = (-\infty, -2)$ Range of $R = (-\infty, -3)$

b. $R^{-1} = \{(y,x): x - y < 1$ and $x < -2\}$

$= \{(x,y): y - x < 1$ and $y < -2\}$

c. Domain of $R^{-1} = (-\infty, -3)$ Range of $R^{-1} = (-\infty, -2)$

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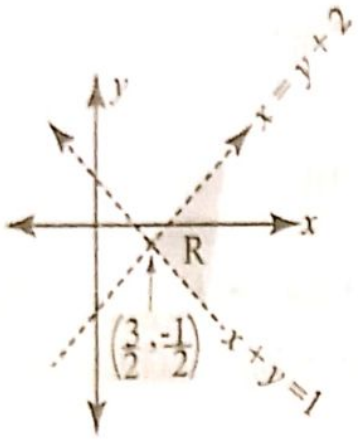
17. Solve the intersection of $x + y = 1$ and $x = y + 2$.



$(y + 2) + y = 1 \Rightarrow 2y + 2 = 1$

$\therefore \quad y = \frac{-1}{2}$, and $x = \frac{-1}{2} + 2 = \frac{3}{2}$

a. Domain of $R = \left(\frac{3}{2}, \infty\right)$

Range of $R = (-\infty, \infty)$

d. $R^{-1} = \{(y,x) : x + y > 1$ and $x > y +$

or $R^{-1} = \{(x, y) : y + x > 1$ and $y >$

c. Domain of $R^{-1} = (-\infty, \infty)$

Range of $R^{-1} = \left(\frac{3}{2}, \infty\right)$

18. Note: If $(a, b) \in R^{-1}$, then $(b,a) \in R$

a. If $(1, 3) \in R^{-1}$ then $(3, 1) \in R$

$\Rightarrow (3, 1): 3(3) + 1 = 10 > 6$ True

b. If $(1, 2) \in R^{-1}$, then $(2, 1) \in R$

$\Rightarrow (2, 1) : 3(2) + 1 > 6$ True

c. $(2, 1) \notin R^{-1}$, because $(1, 3) \notin R$

$\Rightarrow (1, 2): 3(1) + 2 > 6$ is False

$\therefore$ The ordered pairs $(2, 1) \notin R$

d. If $(-1, 5)\ R^{-1}$ then $(5, -1) \in R$

$\Rightarrow (5, -1): 3\ (5) + 1 > 6$ is True

19. a. Not function, x may have one or two

b. Function

c. Not function,

d. Not function, x may have two or more sisters.

e. Function

f. Not function because the number 7 have two values -2 and 2

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g. Not function: because $\left\{\left(\frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{-1}{2}\right)\right\}$ an element of R and the number $\frac{1}{2}$ have two different y - values $\frac{1}{2}$ and $\frac{-1}{2}$

h. Function

i. Not function, because (2, -2) and (2,2) are an element of R and the x - value 2 have two different y - value.

20.a. Domain = $\{x: x \geq 0\}$ and range = $\{y: y \geq 0\}$

- The graph containing the point (0,0) and (1, 1)
- Increasing function, Not symmetric

b. Domain = $(0, \infty)$ and range = $(0, \infty)$

- The graph containing the point (1, 1).

c. Domain = $(-\infty, \infty)$ and rang = $[0, \infty)$

- The graph containing the point (-1, 1), (0, 0) and (1, 1).
- Symmetric W. r. t y - axis

d. The graph containing the point (-1, 1) and (1, 1).

- Domain = $\mathbb{R} \setminus \{0\}$ and range = $(0, \infty)$.
- Symmetric w.r.t y - axis

e. Domain = R and range = R

- The graph containing the point (-1, -1), (0, 0) and (1, 1)
- Symmetric w.r.e origin

f. Domain = $\mathbb{R} \setminus \{0\}$ and range = $\mathbb{R} \setminus \{0\}$

- The graph containing the point (-1, -1) and (1, 1).
- Symmetric w.r.t origin.

21.a. $|x| = 6$ if $x = -6$ or $x = 6$

b. $|5 - x| = 8$ if $x = -3$ or $x = 13$

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22. a. $\{-2, 4\}$ b. $\left\{\frac{3}{4}\right\}$ c. $\left\{5, \frac{3}{5}\right\}$ d. $\{-1, 1, -3, 3\}$

23.

a. True	e. True
b. False, let $x = 0, y = 0$	f. False, let $x = -3$
c. True	g. False
d. False, let $x = -3, y = 5$	h. False

24. $|x + y| < |x| + |y|$ is true if x and y have opposite sign

Answer:

25. a. $(0, -3)$ b. $(1, 5)$ c. $(-2, -3)$ d. $\left(\frac{1}{3}, 2\right)$

26. a. $7 \le x + 2 < 8$, so that $5 \le x < 6$

b. $4 \le 3 + |x| < 5$, then $1 \le |x| < 2$, $1 \le x < 2$

c. $6 \le 2x - 3 < 7 \Rightarrow 9 \le 2x < 10 \Rightarrow \frac{9}{2} \le x < 5$

d. $-4 \le x + 3 < -3 \Rightarrow -7 \le x < -6$

27. a. $f(g(1)) = f(-1) = 1 - 3 = -2$ c. $f(g(0)) = f(-2) = 4 + 3(-2) =$

b. $g(f(1)) = g(4) = 4 - 2 = 2$ d. $g(f(0)) = g(0) = 0 - 2 = -2$

28. $(fog)(x) = f(g(x)) = \dfrac{g(x)}{g(x) - 1} = \dfrac{\frac{9}{x^2 - 4}}{\frac{9}{x^2 - 4} - 1} = \dfrac{9}{13 - x^2}$

$$\text{Domain} = \left\{\mathbb{R} / \left\{2, -2, -\sqrt{13}, \sqrt{13}\right\}\right\}$$

29. Let $g(x) = 2x - 1$, let $f(x) = ax + b$

$f(g(x)) = a\,g(x) + b = 6x - 2$

$\Rightarrow a(2x - 1) + b = 6x - 2$

$2ax - a + a + b = 6x - 2$

$\therefore \quad 2ax = 6x, 2a = 6, a = 3$ and $-a + b = -2$

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$\Rightarrow$ $-3 + b = -2$

$\therefore$ $b = -2 + 3 = 1$

$\therefore$ $f(x) = 3x + 1$

30. $f(g(x)) = \sqrt{g(x) + 3} = \sqrt{x - 2 + 3} = \sqrt{x + 1}$

Domain $(fog)(x) = [-1, \infty)$

31. a. Not invertible b. invertible, $\{(3, -1), (1, 0), (-1, 1)\}$

c. Not invertible d. Not invertible

32.

a. $y = 3x - 4$

$x = 3y - 4$

$y = \frac{x + 4}{3}$

$f^{-1}(x) = \frac{x + 4}{3}$

b. $y = \frac{1 - x}{x + 3}$

$x = \frac{1 - x}{y + 3}$

$xy + 3x = 1 - y$

$xy + y = 1 - 3x$

$y(x + 1) = 1 - 3x$

$y = \frac{1 - 3x}{x + 1}$

$f^{-1}(x) = \frac{1 - 3x}{x + 1}$

c. $f^{-1}(x) = \frac{4x + 1}{3x - 1}$

d. $f^{-1}(x) = \sqrt{\frac{x - 1}{2}}, x \geq 1,$

e. $y = 3x^2 - 2, x \leq 0$

$x = 3y^2 - 2, y \leq 0$

$f^{-1}(x) = \sqrt{\frac{x + 2}{3}}, x \geq -2$

f. $y = (x + 5)^2, x \leq -5$

$x = (y + 5)^2, y \leq -5$

$f^{-1}(x) = -5 - \sqrt{x}, x \geq 0$

g. $y = (x - 2)^2, x \geq 2$

$x = (y - 2)^2, x \geq 2$

$f^{-1}(x) = 2 + \sqrt{x}, x \geq 0$

h. $y = \sqrt{x - 3} \Rightarrow x = \sqrt{y - 3}$

$f^{-1}(x) = x^2 + 3,$

i. $f^{-1}(x) = \sqrt[3]{x + 4}$

j. $f^{-1}(x) = (x - 2)^3$

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UNIT 2

2. Rational Expressions and Rational Function

2.1 Rational Expression

A rational number is a fraction in which the numerator a denominator are integers.

- Rational numbers = $\left\{\frac{a}{b}\middle|\ a \text{ and } b \text{ are integers, } b \neq 0\right\}$

What is a Rational Expression?

A **rational expression** is a fraction in which the numerator a denominator are polynomials.

DEFINITION: A rational expression is defined as any expressio that can be expressed as the ratio of two polynomial expression.

Symbols:

Rational expression = $\left\{\frac{p(x)}{q(x)}\middle|\ p(x) \text{ and } q(x) \text{ are polynomials, } q(x) \neq 0\right\}$

For example : 4, $\frac{5x}{x-1}$, $\frac{x^2+1}{x+4}$ and $\frac{x-4}{x+3}$ are rational expressions.

Illustrative Example

1. Determine whether each of the following are rational expression or n

a. $\frac{3}{5}$ b. $1.4\dot{}$ c. $x^{-1}+x^3$ d. π e. $\sqrt{2}$ f. $\sqrt{x}+6$

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g. $4x + 1$ h. $\frac{3x^2 + 4\sqrt{x}}{x + 1}$ i. $4x^{2/3} - 1$ j. $\frac{x - 2}{3x^2 - 4}$ k. $\frac{x^3 - y^3}{x - y}$

Solution

a. $\frac{3}{5}$ is a rational expression, because $\frac{3}{5} = \frac{3x^0}{5x^0}$ is a ratio of two polynomials.

b. $1.\dot{4} = \frac{13}{9}$ is rational expression.

c. $x^{-1} + x^{-3} = \frac{1}{x} + \frac{1}{x^3}$ is a rational expression.

d. π is a rational expression.

e. $\sqrt{2}$ is a rational expression because $\sqrt{2} = \frac{\sqrt{2}\,x^0}{1x^0}$ is a ratio of two polynomials.

f. $\sqrt{x} + 6$ is not a rational expression because $\sqrt{x}$ is not a polynomial.

g. $4x + 1$ is a rational expression.

h. Not a rational expression because $3x^2 + 4\sqrt{x}$ is not a polynomial, expression.

i. $4x^{2/3} - 1$ is not a rational expression because $x^{2/3}$ is not a polynomial; the exponent of x is not integer

j. A rational expression.

k. A rational expression.

2. Which of the following is not a rational expression.

A. $\sqrt{2}\ x^{-1} + \pi x^{-2}$ C. $\sqrt{(x^2 + 1)^2}$

B. $x^{\sqrt{2}} + x$ D. $\sqrt{(x^2 - 1)^2}$

Solution

A. $\sqrt{2}\ x^{-1} + \pi x^{-2} = \frac{\sqrt{2}}{x} + \frac{\pi}{x^2}$ is a rational expression

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B. $x^{\sqrt{2}} + x$ is not a rational expression because the exponent of $x^{\sqrt{2}}$ is not interger, so that it is not polynomial.

C. $\sqrt{(x^2+1)^2} = |x^2+1| = x^2+1$, for all x. It is a rational expression

D. $\sqrt{(x^2-1)^2} = |x^2-1|$, for all x, this is not a rational expression.

Answer: B and D

Domain of Rational Expression

What is the domain of a rational expression?

The domain of a rational expression is the set of all real numbers except those that cause the denominator to be zero.

Illustrative Example

3. Find the domain of each rational expression.

a. $\frac{4x}{x-3}$

b. $\frac{x+2}{x(x+4)(x-1)}$

c. $\frac{2x-1}{x^2-9}$

d. $\frac{y+3}{y^2+9}$

e. $\frac{2x-3}{x^2+5x+6}$

f. $\frac{3x+1}{x^2-3x-4}$

g. $\frac{x-1}{x^2-4x+9}$

Solution

a. The denominator is zero if $x - 3 = 0$ or $x = 3$. The domain is $\{x|x \neq 3\}$ or $\mathbb{R} \setminus \{3\}$. In interval notation as $(-\infty, 3) \cup (3, \infty)$

b. The denominator is zero if $x = 0$, $x = -4$ or $x = 1$

∴ The domain is $\{x|x \neq 0,\ x \neq -4$, and $x \neq 1\}$ or $\mathbb{R} \setminus \{0, -4, 1\}$

In interval notation as : $(-\infty, -4) \cup (-4, 1) \cup (1, \infty)$

c. The denominator is zero if $x^2 - 9 = 0$ or $x = -3$ and $x = 3$

∴ The domain is $\{x|x \neq -3$ and $x \neq 3\}$ or $\mathbb{R} \setminus \{-3, 3\}$

In interval notation as : $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$

d. The denominator can not be zero, because $y^2 + 9 > 0$

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$\therefore$ The domain is R

e. The denominator is zero if $x^2 + 5x + 6 = 0$

or $(x + 2)(x + 3) = 0$ or $x = -2$ and $x = -3$

$\therefore$ The domain is $\{x|x \neq -2 \text{ and } x \neq -3\}$

Interval notation as : $(-\infty, -3) \cup (-3, -2) \cup (-2, \infty)$

f. The denominator is zero if $x^2 - 3x - 4 = 0$ or

$(x - 4)(x + 1) = 0$, $x = 4$ and $x = -1$

$\therefore$ The domain is $\{x|x \neq -1 \text{ and } x \neq 4\}$

In interval notation as : $(-\infty, -1) \cup (-1\ 4) \cup (4, \infty)$

g. The denominator can not be zero because

$x^2 - 4x + 9 = (x - 2)^2 + 5 > 0$ for all real number x.

$\therefore$ The domain is R

2.1.1 Simplification of Rational Expression

A rational expression is simplified (in lowest terms) when the numerator and denominator have no common factors other than 1

To simplify a rational expression, factor the numerator and denominator, then apply the property of equivalent algebraic fractions.

Properties of Rational Expression

For all rational expression P, q and r:

$$\frac{P \cdot r}{q \cdot r} = \frac{P}{q} \text{ for } r \neq 0 \text{ and } q \neq 0$$

$$-\frac{P}{q} = \frac{-P}{q} = \frac{P}{-q}$$

The steps for simplification of a rational expression:

STEP - 1 : Factor numerator and denominator.

STEP - 2: Divide out common factor. (Cancel out the common factor)

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Illustrative Example

4. Simplify each of the following rational expression.

a. $\dfrac{x^2 + x - 12}{x^3 - 8x^2 + 15x}$ b. $\dfrac{4x^2 - 7x - 2}{2x^2 - x - 6}$ c. $\dfrac{-6x^2 + 11x + 2}{18x^2 + 3x}$

Solution

a. $\dfrac{x^2 + x - 12}{x^3 - 8x^2 + 15x} = \dfrac{(x - 3)(x + 4)}{x(x^2 - 8x + 15)}$ ⟵ Factoring

$= \dfrac{(x-3)(x+4)}{x(x-3)(x-5)}$ • Cancel out common factor

$= \dfrac{x+4}{x(x-5)}, \quad x \neq 3, \; x \neq 0 \quad \text{and} \quad x \neq 5$

b. $\dfrac{4x^2 - 7x - 2}{2x^2 - x - 6} = \dfrac{(x - 2)(4x + 1)}{(x - 2)(2x + 3)}$ • Factoring

$= \dfrac{4x+1}{2x+3}, \; x \neq 2, \text{ and } x \neq -\dfrac{3}{2}$

c. $\dfrac{-6x^2 + 11x + 2}{18x^2 + 3x} = \dfrac{-6x^2 + 12x - x + 2}{3x(6x + 1)}$ • Factoring

$= \dfrac{-6x(x - 2) - 1(x - 2)}{3x(6x + 1)}$

$= \dfrac{(-6x - 1)(x - 2)}{3x(6x + 1)}$

$= \dfrac{-(6x + 1)(x - 2)}{(6x + 1)(3x)}$

$= \dfrac{-(x - 2)}{3x}$

Simplify to lowest terms: $\dfrac{x^2 - 4x + ax - 4a}{x^2 - ax - 4x + 4a}$

$= \dfrac{x(x - 4) + a(x - 4)}{x(x - a) - 4(x - a)}$ • Factoring

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$$= \frac{(x + a)(x - 4)}{(x - a)(x - 4)}$$ • Divide out common factor.

$$= \frac{x + a}{x - a}$$

2.1.2 Operations on Rational Expression

Arithmetic operations are defined on rational expressions just as they are on rational numbers.

Addition: $\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b}$, $b \neq 0$

Substraction: $\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b}$, $b \neq 0$

Multiplication: $\frac{a}{b} \times \frac{c}{b} = \frac{ac}{b^2}$

Addition and Subtraction with the Same denomination

As with rational number, the procedure used to add (or subtract) two rational expressions depends upon whether the expression have **like** or **unlike** denominators.

To add (or subtract) two rational expressions with like denominator simply add (or subtract) their numerators and place the result over the common denominator.

As an example :

a. $\frac{3}{4} + \frac{2}{4} = \frac{3 + 2}{4} = \frac{5}{4}$

b. $\frac{5}{6x} + \frac{7}{6x} = \frac{5 + 7}{6x} = \frac{12}{6x} = \frac{2}{x}$

c. $\frac{2}{x + 3} - \frac{x}{x + 3} = \frac{2 - x}{x + 3}$

Illustrative Example

5. In each case of the following perform the operation and simplify

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a. $\frac{3x-2}{x-1}+\frac{4-5x}{x-1}$

b. $\frac{x^2+8x-7}{x^2-9}-\frac{x^2+4x+5}{x^2-9}$

c. $\frac{5x^2-2x-5}{2x^2+3x-2}-\frac{x^2-9x-3}{2x^2+3x-2}$

d. $\frac{x}{x^2-4}+\frac{2}{x^2-4}$

Solution

a. Since the denominators are the same, we simply add numerator.

$$\frac{3x-2}{x-1}+\frac{4-5x}{x-1}=\frac{3x-2+4-5x}{x-1}=\frac{-2x+2}{x-1}$$

$$=\frac{-2(x-1)}{x-1}=-2,\ x\neq 1$$

b. Since the denominators are the same, we simply subtract numerator.

$$\frac{x^2+8x-7}{x^2-9}-\frac{x^2+4x+5}{x^2-9}=\frac{x^2+8x-7-(x^2+4x+5)}{x^2-9}$$

$$=\frac{x^2+8x-7-x^2-4x-5}{x^2-9}$$

$$=\frac{4x-12}{x^2-9}=\frac{4(x-3)}{(x-3)(x+3)}=\frac{4}{x+3},\ x\neq 3,\ x\neq -3$$

c. $\frac{5x^2-2x-5}{2x^2+3x-2}-\frac{x^2-9x-3}{2x^2+3x-2}=\frac{5x^2-2x-5-(x^2-9x-3)}{2x^2+3x-2}$

$$=\frac{5x^2-2x-5-x^2+9x+3}{2x^2+3x-2}$$

$$=\frac{4x^2+7x-2}{2x^2+3x-2},$$

$$=\frac{4x^2+8x-x-2}{2x^2+4x-x-2}$$

$$=\frac{4x(x+2)-1(x+2)}{2x(x+2)-1(x+2)}$$

$$=\frac{(4x-1)(x+2)}{(2x-1)(x+2)},\ x\neq\frac{1}{2},\ x\neq -2$$

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$$= \frac{4x - 1}{2x - 1}, \quad x \neq \frac{1}{2}, x \neq -2$$

d. $$\frac{x}{x^2 - 4} + \frac{2}{x^2 - 4} = \frac{x + 2}{x^2 - 4} = \frac{x + 2}{(x - 2)(x + 2)} = \frac{1}{x - 2}$$

Adding and Subtracting with Different Denominators.

To add or subtract rational expressions with different denominators, we must **build up** each rational expression to equivalent forms with like denominators using the least common denominator, (L C D)

a. As an Example: Add $\frac{11}{14} + \frac{5}{6}$

First building up the denominator to identical denominator (L C D)

Step 1: Factor both 14 and 6 : 14 = 2 × 7 and 6 = 2 × 3

STEP: 2 Identify the missing factor.

The first denominator 14 = 2 × 7 is missing factor 3.

The second denominator 6 = 2 × 3 is missing factor 7.

STEP 3: Multiply the numerator and denominator by missing factor by 3 and 7, then add or subtract.

$$\frac{1}{2 \cdot 7} \cdot \left(\frac{3}{3}\right) + \frac{5}{2 \cdot 3}\left(\frac{7}{7}\right) = \frac{(1 \times 3) + 5 \times 7}{2 \cdot 3 \cdot 7} = \frac{38}{42}$$

b. Add: $\frac{1}{x + 2} + \frac{2}{x - 3}$

To build up identical denominator to get L C D.

We multiply the numerator and denominator of $\frac{1}{x + 2}$

by missing factor $x - 3$: $\frac{1}{x + 2} = \frac{1}{x + 2} \cdot \left(\frac{x - 3}{x - 3}\right)$

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And we multiply the numerator and denominator of $\frac{2}{x-3}$ by missing factor $x+2$: $\frac{2}{x-3} = \frac{2}{x-3}\left(\frac{x+2}{x+2}\right)$

$$\therefore \frac{1}{x+2}\left(\frac{x-3}{x-3}\right) + \frac{2}{x-3}\left(\frac{x+2}{x+2}\right) = \frac{x-3+2(x+2)}{(x-3)(x+2)} = \frac{3x+1}{(x-3)(x+2)}$$

Illustrative Example

6. Simplify : $\frac{1}{x^2+xy} - \frac{2}{x^2-y^2}$

Solution

Factor: $x^2 + xy = x(x+y)$, $x^2 - y^2 = (x-y)(x+y)$.

Then build up to L C D: Multiply the numerator and denominator of

(i) $\frac{1}{x(x+y)}$ by missing factor $x-y$: $\frac{1}{x(x+y)} = \frac{1}{x(x+y)} \cdot \left(\frac{x-y}{x-y}\right)$

(ii) $\frac{2}{(x-y)(x+y)}$ by missing factor x : $\frac{2}{(x-y)(x+y)}$

$$= \frac{2}{(x-y)(x+y)} \cdot \left(\frac{x}{x}\right)$$

$$\therefore \frac{1}{x^2+xy} - \frac{2}{x^2-y^2} = \frac{1}{x(x+y)} - \frac{2}{(x-y)(x+y)}$$

$$= \frac{1}{x(x+y)} \cdot \left(\frac{x-y}{x-y}\right) - \frac{2}{(x-y)(x+y)}\left(\frac{x}{x}\right)$$

$$= \frac{x-y-2x}{(x-y)(x+y)x} = \frac{-x-y}{(x-y)(x+y)(x)}$$

$$= \frac{-(x+y)}{(x+y)(x-y)x} = \frac{-1}{x(x-y)}$$

7. Add : $\frac{2}{x^2+5x+6} - \frac{4}{x^2+4x+3} + \frac{3}{x^2+3x+2}$

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Solution

$$\frac{2}{(x+2)(x+3)} - \frac{4}{(x+3)(x+1)} + \frac{3}{(x+2)(x+1)}$$

Building to L C D by multiplying missing factor

$$\frac{2}{(x+2)(x+3)} \cdot \left(\frac{x+1}{x+1}\right) - \frac{4}{(x+3)(x+1)} \cdot \left(\frac{x+2}{x+2}\right) + \frac{3}{(x+2)(x+1)}\left(\frac{x+3}{x+3}\right)$$

$$= \frac{2x+2-4x-8+3x+9}{(x+3)(x+1)(x+2)}$$

$$= \frac{x+3}{(x+3)(x+1)(x+2)} = \frac{1}{(x+1)(x+2)}$$

8. Add: $\frac{x+7}{2x+12} + \frac{6}{x^2-36}$

Solution

$$\frac{x+7}{2(x+6)} + \frac{6}{(x+6)(x-6)}$$

$$\frac{x+7}{2(x+6)} \cdot \left(\frac{x-6}{x-6}\right) + \frac{6}{(x+6)(x-6)} \cdot \left(\frac{2}{2}\right)$$

$$\frac{x^2+x-42+12}{2(x+6)(x-6)} = \frac{x^2+x-30}{2(x+6)(x-6)} = \frac{(x-5)(x+6)}{2(x+6)(x-6)}$$

$$= \frac{x-5}{2(x-6)}, \; x \neq 6 \text{ and } x \neq -6$$

9. Add: $\frac{2x-8}{3x^2+8x+4} + \frac{x+3}{3x^2+5x+2}$ ... Exercise

Answer : $\frac{x-1}{(x+1)(x+2)}$

10. **Simplify :** $\frac{1}{x} - \frac{3}{x^2+4x} - \frac{12}{x^3+4x^2}$

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We build the denominator to L C D : $x^3 + 4x^2 = x^2(x+4)$
and $x^2 + 4x = x(x+4)$

$$\frac{1}{x}\cdot\left(\frac{x(x+4)}{x(x+4)}\right) - \frac{3}{x^2+4x}\cdot\left(\frac{x}{x}\right) - \frac{12}{x^2(x+4)}$$

$$= \frac{x^2+4x-3x-12}{x^3+4x} = \frac{x^2+x-12}{x^2(x+4)}$$

$$= \frac{(x-3)(x+4)}{x^2(x+4)} = \frac{x-3}{x^2}$$

Multiplying and Dividing rational Expression

We multiply rational expressions in the same way that we multiply rational numbers:

$$\frac{a}{b}\cdot\frac{c}{b} = \frac{ac}{bd}$$

To divide one rational expressions by another, multiply the first by the reciprocal of the second.

$$\frac{a}{b} \div \frac{c}{b} = \frac{a}{b}\cdot\frac{d}{c}$$

How do you multiply rational expression?

To multiply rational expressions, multiply the numerators and the denominators.

How do we divide rational expression?

To divide rational expressions invert the divisor and multiply.

Illustrative Example

11. Multiply: $\frac{x^2-4x+3}{x^2-x-6}\cdot\frac{x^2-5x+6}{x^2-1}$

Solution

$$\frac{\cancel{(x-3)}\,\cancel{(x-1)}}{\cancel{(x+2)}\,\cancel{(x-3)}}\,\frac{(x+3)\,\cancel{(x+2)}}{\cancel{(x-1)}\,(x+1)} = \frac{x+3}{x+1}$$

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12. Multiply: $\frac{6xy}{x^2y - 4xy} \cdot \frac{x^2 - 8x + 6}{4x - 16}$

Solution

$$\frac{6xy}{xy(x-4)} \cdot \frac{(x-4)^2}{4(x-4)} = \frac{\cancel{6xy} \cdot \cancel{(x-4)}\ \cancel{(x-4)}}{\cancel{xy}\ \cancel{(x-4)} \cdot 4 \cdot \cancel{(x-4)}} = \frac{6}{4} = \frac{3}{2}$$

13. Divide: $\frac{5x - 5y}{x^2 - 2xy + y^2} \div \frac{15x}{x^2 - xy}$

Solution

Multiply the first faction by the reciprocal of the second fraction.

$$\frac{5x - 5y}{x^2 - 2xy + y^2} \cdot \frac{x^2 - xy}{15x}$$

$$\frac{5\cancel{(x-y)}}{\cancel{(x-y)}^2} \cdot \frac{x\cancel{(x-y)}}{15x} = \frac{5x}{15x} = \frac{1}{3}$$

14. Divide: $\frac{2x^2 - 5x - 12}{4x^2 + 8x + 3} \div \frac{x^2 - 16}{2x^2 + 7x + 3}$

Solution: First factorize, then multiply the first fraction by the reciprocal of the second fraction.

$$2x^2 - 5x - 12 = 2x^2 - 8x + 3x - 12$$
$$= 2x(x - 4) + 3(x - 4)$$
$$= (2x + 3)(x - 4)$$
$$4x^2 + 8x + 3 = 4x^2 + 2x + 6x + 3$$
$$= 2x(2x + 1) + 3(2x + 1)$$
$$= (2x + 3)(2x + 1)$$
$$2x^2 + 7x + 3 = 2x^2 + 6x + x + 3$$
$$= 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$$

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Therefore,

$$\frac{\cancel{(2x+3)}\ \cancel{(x-4)}}{\cancel{(2x+3)}\ \cancel{(2x+1)}} \cdot \frac{\cancel{(2x+1)}\ (x+3)}{\cancel{(x-4)}\ (x-4)} = \frac{x+3}{x-4}$$

15. Divide: $\frac{9x^2 - 64}{3x^2 - 5x - 8} \div \frac{25x^2 - 4}{5x^2 + 3x - 2}$ · **Exercise**

Answer : $\frac{3x+8}{5x+2}$

Simplifying a Complex Fraction

A complex fraction is a fraction that has rational expressions in the numerator, the denominator, or both.

Methods to simplify a complex fraction

STEP - 1: Multiply the numerator and denominator of the complex fraction by the L C D of all individual fraction.

STEP - 2: Simplify the numerator and denominator.

STEP - 3: Simplify the resulting expression if possible.

Illustrative Example

16. Simplify : $\frac{1 - \frac{9}{x^2}}{1 + \frac{1}{x} - \frac{12}{x^2}}$

Solution

Multiply numerator and denominator by the L C D: x^2

$$\frac{x^2\left(1 - \frac{9}{x^2}\right)}{x^2\left(1 + \frac{1}{x} - \frac{12}{x^2}\right)} = \frac{x^2 - 9}{x^2 + x - 12} = \frac{\cancel{(x-3)}\ (x+3)}{\cancel{(x-3)}\ (x+4)} = \frac{x+3}{x+4}$$

17. Simplify : $\frac{\frac{1}{x+4} + \frac{1}{x-4}}{\frac{1}{x+4} - \frac{1}{x-4}}$

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Solution

Multiply numerator and denominator by the L C D: $(x+4)(x-4)$

$$\frac{(x+4)(x-4)\left(\frac{1}{x-4}+\frac{1}{x-4}\right)}{(x+4)(x-4)\left(\frac{1}{x+4}-\frac{1}{x-4}\right)} = \frac{x-4+x+4}{x-4-(x+4)}$$

$$= \frac{2x}{-8} = \frac{-x}{4}$$

18. Simplify : $\dfrac{1-\frac{1}{x+1}}{1+\frac{1}{x-1}}$

Solution

$$\frac{\frac{x+1-1}{x+1}}{\frac{x-1+1}{x-1}} = \frac{\frac{x}{x+1}}{\frac{x}{x-1}} = \frac{x}{x+1}\cdot\frac{x-1}{x} = \frac{x-1}{x+1}$$

19. Simplify: $\dfrac{\frac{3}{x-2}-\frac{1}{x+2}}{\frac{5}{x^2-4}}$

Solution

Multiply numerator and denominator by the L C D: $(x-2)(x+2)$

$$\frac{(x-2)(x+2)\left(\frac{3}{x-2}-\frac{1}{x+2}\right)}{(x-2)(x+2)\left(\frac{5}{x^2-4}\right)}$$

$$= \frac{3(x+2)-1(x-2)}{5}$$

$$= \frac{3x+6-x+2}{5} = \frac{2x+8}{5} = \frac{2}{5}(x+4)$$

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20. Check your skill: What is the simplified form of

$$\frac{4-(a-2)^2}{a^4+2a+1} \div \frac{1-\frac{1}{a^2+1}}{a}$$

A. $\frac{4-a}{a}$ B. $\frac{a}{a^2+1}$ C. $\frac{-1}{a^2(a^2+1)}$ D. $\frac{4-a}{a^2+1}$

Solution

The expression is equivalent to

$$\frac{[2-(a-2)][2+(a-2)]}{(a^2+1)(a^2+1)} \div \frac{a^2+1-1}{a^2+1} \cdot \frac{1}{a}$$

$$\frac{(4-a)\cancel{a}}{(a^2+1)\cancel{(a^2+1)}} \cdot \frac{\cancel{a^2+1}}{\cancel{a}} = \frac{4-a}{a^2+1}$$

Answer: D

21. The simplified form of $\dfrac{y-\frac{1}{y}}{1+\frac{y}{1-y}} \div \dfrac{y^2-2y+1}{y}$

A. $y+1$ B. $y-1$ C. $1-y$ D. $-(y+1)$

Solution: The expression will be

$$\left(\frac{\frac{y^2-1}{y}}{\frac{1-y+y}{1-y}}\right)\left(\frac{y}{(y-1)(y-1)}\right) = \left(\frac{(y-1)(y+1)}{y}(1-y)\right)\left(\frac{y}{(y-1)(y-1)}\right)$$

$$= -(y+1)$$

Answer: D

2.2 Decomposition of Rational Expression

The method by which a rational expression is written as sum of rational expressions is called **paratial fraction decomposition**. For example:

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$$\frac{4x+1}{(x+2)(x-5)} = \frac{1}{x+2} + \frac{3}{x-5}$$

$$\frac{2x-2}{(x-3)(x+1)} = \frac{1}{x-3} + \frac{1}{x+1}$$

The procedure for finding a partial fraction decomposition of a rational expression depends on factorization of the denominator of the rational expression. There are four cases :

(Assume that the degree of the numerator is less than the degree of denominator)

Case 1: Non repeated Linear factor denominators.

Example : $\frac{1}{(2x+3)(x-1)} = \frac{A}{2x+3} + \frac{B}{x-1}$

Case 2: Repeated Linear factor denominator.

Example : $\frac{2x-7}{x(x-3)^2} = \frac{A}{x} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$

$$\frac{4x-3}{(x-1)(x+2)^3} = \frac{A}{x-1} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2} + \frac{D}{(x+2)^3}$$

Case 3: Non repeated irreducible Quadratic Factor denominator.

Example : $\frac{3x+1}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$

$$\frac{4x-1}{(x-2)(x^2+x+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+x+1}$$

Note: A quadratic function $f(x) = ax^2 + bx + c$

Then $f(x)$ is irreducible if and only if $b^2 - 4ac < 0$

Example : (a) $x^2 + 4$ is irreducible,

since $b^2 - 4ac = 0 - 4(1)(4) = -16 < 0$

(b) $x^2 + 2x + 6$ is irreducible,

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since $b^2 - 4ac = 4 - 4(1)(6) = -20 < 0$

Case 4: Repeated irreducible quadratic Factor denominator

Example: $\frac{2x - 5}{(x^2 + 3)^2} = \frac{A}{x^2 + 3} + \frac{Bx + C}{(x^2 + 3)^2}$

(i) Finding a partial Fraction decomposition

(Non - repeated linear factor case)

Illustrative Example

22. Find a partial fraction decomposition of non - repeated linear factor denominator.

a. $\frac{-11x - 13}{(3x + 1)(x - 2)}$ b. $\frac{4x + 1}{x^2 - 3x - 10}$ c. $\frac{12}{x^2 - 9}$

Solution

a. The denominator has two non - repeated linear factor. The partial fraction decomposition will have the form:

$$\frac{-11x - 13}{(3x + 1)(x - 2)} = \frac{A}{3x + 1} + \frac{B}{x - 2}$$ • To solve for A and B

Multiplying each side by the L C D: $(3x + 1)(x - 2)$

$$-11x - 13 = A(x - 2) + B(3x + 1)$$

$$= Ax - 2A + 3Bx + B$$

$$= (A + 3B)x + (-2A + B)$$

Using the equality of polynomials, equate coefficients of like powers. The result will be the system of equation:

$$\begin{cases} A + 3B = -11 \\ -2A + B = -13 \end{cases} \Rightarrow \begin{cases} 2A + 6B = -22 \\ -2A + B = -13 \end{cases}$$

$$7B = -35$$

Solving the system, we have $A = 4$ and $B = -5$ we obtain

$$\frac{-11x - 13}{(3x + 1)(x - 2)} = \frac{4}{3x + 1} - \frac{5}{x - 2}$$

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b. First factor the denominator: $x^2 - 3x - 10 = (x - 5)(x + 2)$

The factors are non - repeated linear factor. Therefore, the partial fraction decomposition will have the form:

$$\frac{4x + 1}{(x - 5)(x + 2)} = \frac{A}{(x - 5)} + \frac{B}{(x + 2)}$$

To solve for A and B, multiply each side by L C D: $(x - 5)(x + 2)$.

$4x + 1 = A(x + 2) + B(x - 5) = Ax + 2A + Bx - 5B$

$4x + 1 = (A + B)x + (2A - 5B)$

Equate coefficients of like power

$$\begin{cases} A + B = 4 \\ 2A - 5B = 1 \end{cases} \Rightarrow \begin{cases} -2A - 2B = -8 \\ 2A - 5B = 1 \end{cases}$$

$$-7B = -7$$

Solving the system we have $A = 3$ and $B = 1$

Thus the partial fraction decomposition is

$$\frac{4x + 1}{x^2 - 3x - 10} = \frac{3}{x - 5} + \frac{1}{x + 2}$$

c. $$\frac{12}{x^2 - 9} = \frac{12}{(x - 3)(x + 3)} = \frac{A}{x - 3} + \frac{B}{x + 3}$$

To find A and B, multiply each side by L C D $(x - 3)(x + 3)$

$12 = A(x + 3) + B(x - 3) = Ax + 3A + Bx - 3B$

$12 = (A + B)x + 3A - 3B$

Equate coefficients of like power:

$A + B = 0$ and $3A - 3B = 12$

$A = -B$ and $-3B - 3B = 12$, $B = -2$ and $A = 2$

Thus the partial fraction decomposition is

$$\frac{12}{(x - 3)(x + 3)} = \frac{2}{x - 3} - \frac{2}{x + 3}$$

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(ii) Finding a partial fraction decomposition.

(Repeated linear factor case)

23. Find a partial fraction decomposition: of repeated linear factor denominator:

a. $\frac{-x^2 + 3x + 8}{x^3 + 4x^2 + 4x}$ b. $\frac{3x^2 - 6x - 25}{(x + 1)(x^2 - 6x + 9)}$

Solution

Factor the denominator:

$$x^3 + 4x^2 + 4x = x(x^2 + 4x + 4) = x(x + 2)^2$$

$$\frac{-x^2 + 3x + 8}{x(x + 2)^2} = \frac{A}{x} + \frac{B}{(x + 2)} + \frac{C}{(x + 2)^2}$$

To find A, B, and C, multiply each side by L.C.D: $x(x + 2)^2$.

$$-x^2 + 3x + 8 = A(x + 2)^2 + Bx(x + 2) + Cx.$$

- Let $x = -2$. Then:

$$-(-2)^2 + 3(-2) + 8 = A \cdot 0 + B \cdot 0 + C(-2)$$

$$-2C = -4 - 6 + 8 = -2, \quad C = 1$$

- Let $x = 0$. Then $0 + 0 + 8 = A(0 + 2)^2 + 0 + 0$

$$4A = 8, \quad A = \frac{8}{4} = 2$$

- To find B, let $x = 1$

$$-1 + 3 + 8 = A(1 + 2)^2 + B \cdot 1 \cdot (1 + 2) + C \cdot 1$$

$$9A + 3B + C = 10$$

$$9(2) + 3B + 1 = 10, \quad B = \frac{-9}{3} = -3$$

Therefore, the partial decomposition is

$$\frac{-x^2 + 3x + 8}{x(x + 2)^2} = \frac{2}{x} - \frac{3}{x + 2} + \frac{1}{(x + 2)^2}$$

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b. First factor the denominator: $x^2 - 6x + 9 = (x - 3)^2$

$$\frac{3x^2 - 6x - 25}{(x + 1)(x - 3)^2} = \frac{A}{x + 1} + \frac{B}{(x - 3)} + \frac{C}{(x - 3)^2}$$

To find A, B, and C, multiply each side by L C D $(x + 1)(x - 3)^2$.

$3x^2 - 6x - 25 = A(x - 3)^2 + B(x + 1)(x - 3) + C(x + 1)$

Let $x = 3$: Then:

$3(3)^2 - 6(3) - 25 = A \cdot 0 + B \cdot 0 + C(3 + 1)$ and $C = -4$

Let $x = -1$ Then :

$3(-1)^2 - 6(-1) - 25 = A(-4)^2 + B \cdot 0 + C \cdot 0$

$-16 = 16A, \quad A = -1$

Let $x = 0$. Then $0 - 0 - 25 = 9A - 3B + C$

$9A - 3B + C = -25$ and $-9 - 3B + -4 = -25$

$-3B = -25 + 13 = -12, B = 4,$

Hence,

$$\frac{3x^2 - 6x - 25}{(x + 1)(x - 3)^2} = \frac{-1}{x + 1} + \frac{4}{(x - 3)} - \frac{4}{(x - 3)^2}$$

(iii) Finding a partial fraction decomposition of

Non - repeated irreducible quadratic factor denominator.

24. Find the partial decomposition:

a. $\dfrac{x^2 + x - 15}{x^3 + 5x}$ b. $\dfrac{3x + 16}{(x - 2)(x^2 + 7)}$

Solution

a. Factor $x^3 + 5x = x(x^2 + 5)$

Since x is a non repeated linear factor and $x^2 + 5$ is an irreducible quadratic factor, the decomposition will have form:

$$\frac{x^2 + x - 15}{x(x^2 + 5)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 5}$$

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To find A, B and C, multiply each side by the L C D $x(x^2 + 5)$ you will get

$$x^2 + x - 15 = A(x^2 + 5) + (Bx + C)x$$

Let $x = 0$. Then : $0 + 0 - 15 = A(5) + 0$, $A = -3$

$$x^2 + x - 15 = Ax^2 + 5A + Bx^2 + Cx = (A + B)x^2 + C\,x$$

By comparing coefficient of x^2: $A + B = 1$, $B = 4$

By comparing coefficient of x : $C = 1$

$$\therefore \frac{x^2 + x - 15}{x^3 + 5x} = \frac{-3}{x} + \frac{4x + 1}{x^2 + 5}$$

b. $$\frac{3x + 16}{(x - 4)(x^2 + 7)} = \frac{A}{x - 4} + \frac{Bx + C}{x^2 + 7}$$

Multiply each side by the L C D $(x - 2)(x^2 + 7)$

$$3x + 16 = A(x^2 + 7) + (Bx + C)(x - 2)$$

• Let $x = 2$. Then : $3(2) + 16 = A(2^2 + 7) + 0$

$$22 = 11A, \quad A = 2$$

Expanding the right side :

$$3x + 16 = (A + B)x^2 + (-2B + C)x + (7A - 2C)$$

Comparing coefficient of x^2: $A + B = 0$, $B = -2$

Comparing constant term: $7A - 2\ C = 16$

$$7(2) - 2C = 16, C = -1$$

The decomposition is

$$\frac{3x + 16}{(x - 2)(x^2 + 7)} = \frac{2}{x - 2} + \frac{-2x - 1}{x^2 + 7}$$

If the degree of numerator is greater than or equal to the degree of denominator, use long division to first write the rational expression as a polynomial plus a remainder over the denominator.

Illustrative Example

25. Find the partial decomposition

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a. $\dfrac{2x^3-15x+1}{x^2-9}$ b. $\dfrac{3x^2+6x-8}{x^2-4}$

Solution

Since the degree of the numerator is greater than the degree of denominator, we use long division to obtain

a. $\dfrac{2x^3-15x+1}{x^2-9} = 2x + \dfrac{3x+1}{x^2-9} = 2x + \dfrac{A}{x-3} + \dfrac{B}{x+3}$

The remaining term now satisfies the requirement that the degree of the numerator is less than the degree of the denominator.

At this point we proceed as in case (1 up to 4) to obtain the decomposition.

Then: $\dfrac{3x+1}{x^2-9} = \dfrac{A}{x-3} + \dfrac{B}{x+3}$

Remaining multiply **each side by L C D** $(x-3)(x+3)$

$3x+1 = A(x+3) + B(x-3)$

Let $x = 3$, then $3(3)+1 = A(3+3) + B\cdot 0$, $A = \dfrac{10}{6} = \dfrac{5}{3}$

Let $x = -3$, then $3(-3)+1 = A\cdot 0 + B(-3-3)$, $B = \dfrac{-8}{6} = \dfrac{-4}{3}$

$\therefore \dfrac{2x^2-15x+1}{x^2-9} = 2x + \dfrac{5}{3(x-3)} - \dfrac{4}{3(x+3)}$

b. We use long division to obtain:

$\dfrac{3x^2+6x-8}{x^2-4} = 3 + \dfrac{6x+4}{x^2-4}$

Also, $\dfrac{6x+4}{x^2-4} = \dfrac{6x+4}{(x-2)(x+2)} = \dfrac{A}{x-2} + \dfrac{B}{x+2}$

Multiply each side by the L C D : $(x-2)(x+2)$.

$6x+4 = A(x+2) + B(x-2)$

Let $x = -2$, then $6(-2)+4 = A\cdot 0 + B(-2-2)$, $-8 = -4B$, $B = 2$

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Let $x = 2$, then $6(2) + 4 = A(2 + 2) + B \cdot 0$, $4A = 16$, $A = 4$

$$\therefore \quad \frac{3x^2 + 6x - 8}{x^2 - 4} = 3 + \frac{6x + 4}{(x - 2)(x + 2)} = 3 + \frac{4}{x - 2} + \frac{2}{x + 2}$$

26. Test your skill: What are the respective value of A and B, so that

$$\frac{1}{(2x + 3)(x - 1)} = \frac{A}{2x + 3} + \frac{B}{x - 1}$$

A. $\frac{2}{5}, \frac{1}{5}$ B. $\frac{-2}{5}, \frac{-1}{5}$ C. $\frac{-2}{5}, \frac{1}{5}$ D. $\frac{2}{5}, \frac{-1}{5}$

Solution

To clear the denominator Multiply each side by L C D $(2x + 3)(x - 1)$

$1 = A(x - 1) + B(2x + 3)$

Let $x = 1$, then $1 = A \cdot 0 + B \cdot 5$, $5B = 1$, $B = \frac{1}{5}$.

Let $x = 0$, then $1 = -A + 3B$, $-A = 1 - 3B = 1 - \frac{3}{5} = \frac{2}{5}$, $A = \frac{-2}{5}$

Answer: C

27. What are the respective value of A, B and C, so that

$$\frac{2x - 7}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}?$$

A. -1, -1, -1 B. 1, 1, -1 C. -1, 1, 1 D. -1, 1, -1

Solution

To clear the denominator multiply each side by L C D, we obtain

$2x - 7 = A(x - 2)^2 + B(x + 1)(x - 2) + C(x + 1)$

Let $x = 2$, then $4 - 7 = A \cdot 0 + B \cdot 0 + C(2 + 1)$, $C = -1$

Let $x = -1$, then $-2 - 7 = A(9) + B \cdot 0 + C \cdot 0$, $A = -1$

Let $x = 0$, then $0 - 7 = 4A + -2B + C$

$4A - 2B + C = -7$, and $4(-1) - 2B + (-1) = -7$

$-2B = -7 + 5 = -2$, $B = 1$,

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Answer: D

28. What are the respective value of A, B, and C, So that

$$\frac{x-1}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}?$$

A. -1, 1, 1 B. -1, 1, 2 C. 1, -2, 0 D. 3, 1, 2

Solution

To clear the denominator multiply each side by L C D $(x)(x^2+1)$, we obtain

$x - 1 = A(x^2+1) + (Bx+c)x$ By expanding the right side

$x - 1 = (A+B)x^2 + A + Cx$

- Comparing coefficient of x^2, $A + B = 0$, $B = 1$
- Comparing coefficient of x, $C = 1$
- Comparing constant term $A = -1$

Answer: A

29. If $\frac{x^2+14}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, then

What are the values of A, B and C respectively?... UEE

A. 3, -2, 6 B. 1, -1, 3 C. 2, -1, 5 D. 2, -2 -5

Solution

Multiply both side by $(x+2)(x-1)^2$:

$x^2 + 14 = A(x-1)^2 + B(x+2)(x-1) + C(x+2)$

Let $x = 1$, the $1 + 14 = 3C$, $C = 5$

Let $x = -2$, then $18 = 9A$, $A, = 2$

Let $x = 0$, then $14 = A - 2B + 2C$, $C = -1$

Answer: C

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2.3 Solving Rational Equation

What is a rational equation?

A **rational equation** is an equation in which each term contains a rational expression.

How do you solve a rational equation?

To solve a rational equation, our first step is to clear the fraction of the denominator. We multiply both sides of the equation by the **L C M** of all the denominators. Then we carry out the equation - solving process.

Note: If a variable appears in the denominator of a fraction, we must restrict the values of the variable to avoid division by zero.

Illustrative example

30. Solve each of the following rational equation.

a. $\frac{x+1}{2} - \frac{x-3}{3} = 3$ b. $\frac{x-2}{4} - \frac{x+3}{6} = \frac{-7}{12}$

Solution

a. First clear fractions by multiplying both sides by the L C D, 6

$$6\left(\frac{x+1}{2} - \frac{x-3}{3}\right) = 3(6)$$

$3x + 3 - (2x - 6) = 18$

$x = 18 - 9 = 9$ $\therefore$ The solution set is $\{9\}$

b. Multiplying both sides by the LCD, 12

$$12\left(\frac{x-2}{2} - \frac{x+3}{6}\right) = 12\left(\frac{-7}{12}\right)$$

$3(x-2) - 2(x+3) = -7$

$3x - 6 - 2x - 6 = -7$

$x = -7 + 12 = 5$

31. Solve each of the following rational equation.

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a. $\frac{2}{x-1} = \frac{3}{x+1}$

b. $\frac{8}{x^2-9} = \frac{2}{x+3}$

c. $\frac{2}{x-5} - \frac{1}{x+5} = \frac{11}{x^2-25}$

d. $\frac{2}{x-3} - \frac{3}{x+3} = \frac{12}{x^2-9}$

Solution

a. $\frac{2}{x-1} = \frac{3}{x+1}$ • Multiply both sides by LCD $(x-1)(x+1)$

$(x-1)(x+1)\left[\frac{2}{x-1}\right] = \left[\frac{3}{x+1}\right](x+1)(x-1)$, $x \neq$ and $x \neq -1$

$2(x+1) = 3(x-1)$

$2x + 2 = 3x - 3$ implies that $3x - 2x = 2 + 3 = 5$

$\therefore$ The solution set is $\{5\}$

b. $\frac{8}{x^2-9} = \frac{2}{x+3}$ • Domain is $\{x | x \neq 3$ and $x \neq -3\}$

Multiply both sides by the L C D $(x-3)(x+3)$, we obtain:

$8 = 2(x-3)$, this implies $2x - 6 = 8$, $x = 7$

$\therefore$ The solution set is $\{7\}$

c. $\frac{2}{x-5} - \frac{1}{x+5} = \frac{11}{x^2-25}$, $x \neq 5$ and $x \neq -5$

Multiply both sides by the L C D $(x-5)(x+5)$ we get

$2(x+5) - 1(x-5) = 11$ this implies $x + 15 = 11$,

$\therefore$ The solution set is $\{-4\}$

d. $\frac{2}{x-3} - \frac{3}{x+3} = \frac{12}{x^2-9}$, $x \neq 3$ and $x \neq -3$

Multiply both sides by the L C D $(x-3)(x+3)$:

$2(x+3) - 3(x-3) = 12$

$2x + 6 - 3x + 9 = 12$

$-x + 15 = 12$

$x = 3$

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Since $x \neq 3$, and $x = 3$ makes the rational expression undefined.

$\therefore$ The solution set is $\{\}$ or $\emptyset$.

32. Check your skill: What is the solution set of

$$1 - \frac{5}{x^2+1} = \frac{x-2}{x^2-2x} - \frac{1}{x}\ ?$$

A. $\{-2, -2\}$ B. $\{2\}$ C. $\{-2\}$ D. $\{-1, -2, 2\}$

Solution

$$\frac{x^2+1}{x^2+1} - \frac{5}{x^2+1} = \frac{x-2}{x(x-2)} - \frac{1}{x},\ x \neq 2 \text{ and } x \neq 0.$$

$$\frac{x^2+1-5}{x^2+1} = \frac{1}{x} - \frac{1}{x} = \frac{1-1}{x} = \frac{0}{x} = 0$$

$$\frac{x^2-4}{x^2+1} = 0 \text{ if } x^2 - 4 = 0, \text{ then } x = 2, \text{ or } x = -2$$

But $x = 2$ is not in the domain

$\therefore$ The solution set is $\{-2\}$ **Answer: C**

33. What is the solution set of $\frac{2}{3x+1} = \frac{1}{x} - \frac{6x}{3x+1}$?

A. $\left\{\frac{1}{2}\right\}$ B. $\left\{\frac{-1}{3}, \frac{1}{2}\right\}$ C. $\left\{\frac{-1}{2}, \frac{1}{3}\right\}$ D. $\left\{\frac{-1}{2}\right\}$

Solution : $x \neq \frac{-1}{3}$ and $x \neq 0$

Multiplying both sides by the L C D $x(3x + 1)$, We obtain:

$2x = 3x + 1 - 6x^2$ implies that $-6x^2 + 3x + 1 = 2x$

$-6x^2 + 3x - 2x + 1 = 0$, factorizing

$-3x(2x - 1) - 1(2x - 1) = 0$

$(2x - 1)(-3x - 1) = 0$, in this case, $x = \frac{1}{2}$ and $x = \frac{-1}{3}$

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Since $x = \frac{-1}{3}$ causing the denominator $3x + 1 = 0$.

$\therefore$ The solution set is $\left\{\frac{1}{2}\right\}$

Answer: A

34. What is the solution set of $\frac{5x - 3}{x^2 - 9} = \frac{2}{x - 3} + \frac{3}{x + 3}$

A. $\{3, -3\}$ B. R C. $R \setminus \{3, -3\}$ D. $\{\}$

Solution

Domain $x \neq 3$ and $x \neq -3$ Multiplying both sides by the L C D

$(x - 3)$ $(x + 3)$, we obtain:

$5x - 3 = 2(x + 3) + 3(x - 3) = 5x + 6 - 9$

$5x - 3 = 5x - 3$

$5x - 5x = -3 + 3$

$0 = 0 \longleftarrow$ True for all x, $x \neq 3, -3$

Answer: C

35. What is the solution set of $\frac{2x + 3}{x - 1} = \frac{10}{x^2 - 1} + \frac{2x - 3}{x + 1}$

A. $\{1, -1\}$ B. $\{-1\}$ C. $\{1\}$ D. $\{\}$

Solution

Multiplying both sides by the L C D $(x - 1)(x + 1)$: We get

$(2x + 3)(x + 1) = 10 + (2x - 3)(x - 1)$

$2x^2 + 5x + 3 = 10 + 2x^2 - 5x + 3$

$5x + 5x = 13 - 3 = 10$

$10x = 10,\ x = 1$

Since $x = 1$ is not in the domain of expression

$\therefore$ The solution set is $\{\}$

Answer: D

36. If $q(x) = 4x^3$ and $P(x) = 3x^2 - 3x$, then

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what is the solution set of $\frac{q(x)}{(4x)P(x)} + \frac{1}{x} = \frac{1}{P(x)}$?

A. {1, -4} B. {1} C. {-4} D. {-1, 4}

Solution

Multiply both sides by $4x\,P(x)$, will get :

$$q(x) + 4P(x) + 4P(x) = 4x$$

$$4x^3 + 4(3x^2 - 3x) + 4(3x^2 - 3x) = 4x$$

$$4x^3 + 12x^2 - 12x - 4x = 0 \Rightarrow 4x(x^2 + 3x - 4) = 0$$

$$\Rightarrow 4x(x - 1)(x + 4) = 0, \quad \therefore \quad x = 0,\ x = 1,\ x = -4$$

since $x \neq 0, x \neq 1$

Answer: C

Finding the zeros of Rational Expression

37. Solve

a. $\frac{x^2 - 4}{x^2 - 9} = 0$ c. $\frac{x^2 + 5x + 6}{x^2 - 2x + 8} = 0$ e. $\frac{x - 1}{x^2 - 1} = 0$

b. $\frac{x + 2}{x^2 - 3x + 2} = 0$ d. $\frac{x + 1}{x - 3} = 0$ f. $\frac{x - 3}{x^2 + x - 12} = 0$

Solution

a. $\frac{x^2 - 4}{x^2 - 9} = 0$ if $x^2 - 4 = 0$, provided that $x \neq 3, x \neq -3$

$\therefore \quad x = 2$ and $x = -2$ are the zeros.

b. $\frac{x + 2}{(x - 2)(x - 1)} = 0$, if $x + 2 = 0$, and $x \neq 2$, and $x \neq -3$

$\therefore \quad x = -2$ is the zeros of expression.

c. $\frac{(x + 3)(x + 2)}{(x - 4)(x + 2)} = 0$ if $\frac{(x + 3)}{x - 4} = 0,\ x \neq 4,\ x \neq -2$

$\therefore \quad x = -3$ is the zeros of expression.

d. $x = -1$ is the zeros of expression

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e. $\frac{x-1}{x^2-1} = \frac{x-1}{(x-1)(x+1)} = \frac{1}{x+1}$

$\therefore \frac{x-1}{x^2-1}$ has no zeros

f. $\frac{x-3}{(x+4)(x-3)} = \frac{1}{x+4}$ has no zeros.

2.4 Solving Rational Inequality

What is a rational inequality?

An inequality involving a rational expression. The inequalities :

$\frac{x+1}{x-4} \leq 3$, $\frac{2x-3}{x+5} \leq 0$ and $\frac{2}{x+1} > \frac{1}{x-2}$

are called rational inequalities.

Critical Vales of a Rational Expression

The critical Values of a rational expression are the numbers that cause the numerator of the rational expression to equal zero or the denominator of the rational expression to equal zero.

To Solve a rational inequality by the Critical Value Method.

STEP 1: Move all non – zero terms to one side and the other side is zero.

STEP 2: Find the zeros of numerator and the zeros of denominator (They are the critical values of the inequality)

STEP 3: Determine the intervals formed by the critical values.

STEP 4: Multiply the signed numbers to determine which intervals satisfy the original inequality.

STEP 5: Write the solution set using interval.

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Illustrative Example

38. Solve the inequalities

a. $\frac{2x-3}{x+1} \geq 1$ b. $\frac{2x-3}{x+1} > 1$ c. $\frac{2x-3}{x+1} = 1$

c. $\frac{2x-3}{x+1} < 1$ d. $\frac{2x-3}{x+1} \leq 1$

Solution

We do not multiply each side by $x + 1$. Instead, subtract 1 from each side to get zero on the right.

$$\frac{2x-3}{x+1} \geq 1$$

$$\frac{2x-3}{x+1} - 1 \geq 0 \quad \bullet \text{ Set one side to zero.}$$

$$\frac{2x-3}{x+1} - 1\left(\frac{x+1}{x+1}\right) \geq 0 \quad \bullet \text{ Get a common denominator}$$

$$\frac{2x-3-x-1}{x+1} \geq 1 \quad \bullet \text{ Simplify}$$

$$\frac{x-4}{x+1} \geq 0$$

- The zeros of numerator is $x = 4$ and the zeros of denominator is $x = -1$. The number 4 and -1 are critical number, we divide the intervals and make sign graph.

	$-\infty$		-1		4		∞
Sign of $(x-4)$	- - -		- - -		+ + +		
Sign of $(x+1)$	- - -		+ + +		+ + +		
Sign of $\left(\frac{x-4}{x+1}\right)$	+ + +	∄	- - -	0	+ + +		

Therefore;

a. $\frac{2x-3}{x+1} \geq 1$ the solution set is $(-\infty, -1) \cup [4, \infty)$.

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b. $\frac{x-4}{x+1} > 0$, the solution set is $(-\infty, -1) \cup (4, \infty)$

c. $\frac{2x-3}{x+1} < 1, \frac{x-4}{x+1} < 0$, the solution set is $(-1\ 4)$

d. $\frac{x-4}{x+1} \leq 0$, the solution set is $(-1, 4]$

e. $\frac{x-4}{x+1} = 0$, at $x = 4$

39. Solve the inequalities:

a. $\frac{3}{4-x} \leq \frac{6}{1-x}$

b. $\frac{3}{4-x} < \frac{6}{1-x}$

c. $\frac{3}{4-x} \geq \frac{6}{1-x}$

d. $\frac{3}{4-x} > \frac{6}{1-x}$

e. $\frac{3}{4-x} = \frac{6}{1-x}$

Solution

We do not multiply the L C D as we do in solving equations. In stead, subtract $\frac{6}{1-x}$ from each side:

$\frac{3}{4-x} - \frac{6}{1-x} \leq 0$ • Set one side to zero.

$\left(\frac{1-x}{1-x}\right)\left(\frac{3}{4-x}\right) - \frac{6}{1-x}\left(\frac{4-x}{4-x}\right) \leq 0$ • Get a common denominator

$\frac{3-3x-24+6x}{(1-x)(4-x)} \leq 0$ • Simplify

$\frac{-21+3x}{(1-x)(4-x)} \leq 0$ • Note: $-21 + 3x = 3(x - 7)$

$\frac{3(x-7)}{(1-x)(4-x)} \leq 0$

The critical numbers are 7, 1, and 4. Now, we divide these numbers on

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intervals, and make sign chart.

	$-\infty$		1		4		7	∞
Sign of $3(x-7)$	- - -		- - -		- - -	0	+ + +	
Sign of $(1-x)$	+ + +	0	- - -		- - -		- - -	
Sign of $(4-x)$	+ + +		+ + +	0	- - -		- - -	
$\frac{3(x-7)}{(1-x)(4-x)}$	- - -	∄	+ + +	∄	- - -	0	+ + +	

Therefore

a. $\frac{3(x-7)}{(1-x)(4-x)} \leq 0$, on interval, $(-\infty,1)\cup(4, 7]$

b. $\frac{3(x-7)}{(1-x)(4-x)} < 0$, on interval, $(-\infty,1)\cup(4, 7)$

c. $\frac{3(x-7)}{(1-x)(4-x)} \geq 0$, on interval, $(1\ \ 4)\cup[7, \infty)$

d. $\frac{3(x-7)}{(1-x)(4-x)} > 0$, on interval, $(1\ \ 4)\cup(7, \infty)$

e. $\frac{3(x-7)}{(1-x)(4-x)} = 0$, at $x = 7$ only.

40. Solve the inequalitie

a. $\frac{5}{4x}+\frac{1}{4}\leq\frac{3}{2x^2}$ c. $\frac{5}{4x}+\frac{1}{4}\geq\frac{3}{2x^2}$ e. $\frac{5}{4x}+\frac{1}{4}=\frac{3}{2x^2}$

b. $\frac{5}{4x}+\frac{1}{4}<\frac{3}{2x^2}$ d. $\frac{5}{4x}+\frac{1}{4}>\frac{3}{2x^2}$

Solution

We do not multiply by the L C D as we do in solving equation.

Instead, subtract $\frac{3}{2x^2}$ from each side:

$\frac{5}{4x}+\frac{1}{4}-\frac{3}{2x^2}\leq 0$ • Set one side to zero

$\left(\frac{x}{x}\right)\frac{5}{4x}+\left(\frac{x^2}{x^2}\right)\frac{1}{4}-\frac{3}{2x^2}\left(\frac{2}{2}\right)\leq 0$ • Get a common denominator.

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$\frac{x^2 + 5x - 6}{4x^2} \leq 0$ implies that $\frac{(x-1)(x+6)}{4x^2} \leq 0$

$x = 1$, $x = -6$ and $x = 0$ are critical number

	$-\infty$	-6	0	1	∞
$x+6$	- - -	+ + +	+ + +	+ + +	
$x-1$	- - -	- - -	- - -	+ + +	
$4x^2$	+ + +	+ + +	+ + +	+ + +	
$\frac{(x-1)(x+6)}{4x^2}$	+ + +	- - -	- - -	+ + +	

Therefore

a. $\frac{(x-1)(x+6)}{4x^2} \leq 0$, on interval, $[-6, 0) \cup (0\ 1]$

b. $\frac{(x-1)(x+6)}{4x^2} < 0$, on interval, $(-6, 0) \cup (0, 1)$

c. $\frac{(x-1)(x+6)}{4x^2} \geq 0$, on interval, $(-\infty, -6] \cup [1, \infty)$

d. $\frac{(x-1)(x+6)}{4x^2} > 0$, on interval, $(-\infty, -6) \cup (1, \infty)$

e. $\frac{(x-1)(x+6)}{4x^2} = 0$, at $x = 1$ and $x = -6$

41. Solve the inequalities

a. $\frac{1}{x-3} = \frac{1}{x+4}$

b. $\frac{1}{x-3} \geq \frac{1}{x+4}$

c. $\frac{1}{x-3} \leq \frac{1}{x+4}$

d. $\frac{1}{x-3} < \frac{1}{x+4}$

Solution

Subtract $\frac{1}{x+4}$ from each side:

$\frac{1}{x-3} - \frac{1}{x+4} = 0$ • set one side to zero.

$\left(\frac{x+4}{x+4}\right)\left(\frac{1}{x-3}\right) - \left(\frac{1}{x+4}\right)\left(\frac{x-3}{x-3}\right) = 0$ • Get a common denominator

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$$\frac{x+4-(x-3)}{(x+4)(x-3)}=0$$

$$\frac{7}{(x+4)(x-3)}=0$$

$x=-4$ and $x=3$ are critical numbers.

	$-\infty$	-4	3	∞
7	+ + +	+ + +	+ + +	
$x+4$	- - -	0 + + +	+ + +	
$x-3$	- - -	- - -	0 + + +	
$\frac{7}{(x+4)(x-3)}$	+ + +	∄ - - -	∄ + + +	

Therefore

a. $\frac{1}{x-3}=\frac{1}{x+4}$ implies $\frac{7}{(x+4)(x-3)}=0$ has no solution

b. $\frac{7}{(x+4)(x-3)}\geq 0$, on interval, $(-\infty,-4)\cup(3,\infty)$

c. $\frac{7}{(x+4)(x-3)}\leq 0$, on interval, $(-4,3)$

d. $\frac{7}{(x+4)(x-3)}<0$, on interval, $(-4,3)$

42. For what values of x, $\frac{1}{x^4}\geq\frac{1}{x^2}$?

A. $(-\infty,-1]$ B. $[-1,1]$ C. $[-1,0]\cup(0,1]$ D. $(-1,1)$

Solution : $\frac{1}{x^4}-\frac{1}{x^2}\geq 0\Rightarrow\frac{1-x^2}{x^4}\geq 0$

	$-\infty$	-1	0	1	∞
x^4	+	+	+	+	
$1-x$	+	+	+	-	
$1+x$	-	+	+	+	
$\frac{(1-x)(1+x)}{x^4}$	-	+	+	-	

Answer: C

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43. **Mixed skill.** What is the solution set of $\left|\frac{3x-4}{8-x}\right| \leq 1$?

A. $x \leq -2, x \geq 3$ C. $-2 \leq x \leq 3$

B. $2 \leq x \leq 3$ D. $x < 8$

Solution: Recall that, if $|x| \leq a$, then $-a \leq x \leq a$

If $\left|\frac{3x-4}{8-x}\right| \leq 1$, then $-1 \leq \frac{3x-4}{8-x} \leq 1$

$\frac{3x-4}{8-x} \geq -1$ and $\frac{3x-4}{8-x} \leq 1$

$\frac{3x-4}{8-x} + 1 \geq 0$ and $\frac{3x-4}{8-x} - 1 \leq 0$

$\frac{3x-4}{8-x} + \frac{8-x}{8-x} \geq 0$ and $\frac{3x-4}{8-x} - \left(\frac{8-x}{8-x}\right) \leq 0$

$\frac{2x+4}{8-x} \geq 0$ and $\frac{4x-12}{8-x} \leq 0$

	$-\infty$		-2		8		∞
$2x+4$	- - -	0	+ + +	+ + +			
$8-x$	+ + +	+ + +		+ + +	0	- - -	
$\left(\frac{2x+4}{8-x}\right)$	- - -	0	+ + +	0		- - -	

and

	$-\infty$		3		8		∞
$4x-12$	- - -	0	+ + +	+ + +			
$8-x$	+ + +	+ + +		+ + +	0	- - -	
$\left(\frac{4x-12}{8-x}\right)$	- - -	0	+ + +	0		- - -	

$S.S_1 = \{x: -2 \leq x < 8\}$ and $S.S_2 = \{x | -x \leq 3 \text{ and } x > 8\}$

$S.S = \{S.S_1\} \cap \{S.S_2\} = \{x : -2 \leq x \leq 3\}$

Answer: C

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2.5 Rational Function and Their Graphs

What is a rational function?

A ratio of two polynomial functions are called **a rational function.**

Definition: Let $P(x)$ and $q(x)$ be a polynomial, $q(x) \neq 0$.

A function f defined by

$$f(x) = \frac{P(x)}{q(x)}$$

is called a **rational function**. The domain of a rational function is all re numbers except the real zero of q(x).

Illustrative Example

44. Find the domain of a rational function

a. $f(x) = \frac{3x^2 - 9}{x + 4}$ b. $f(x) = \frac{x^2 - 4}{x - 4}$

Solution

a. Domain of f = R/ $\{x - 4\}$ or $(-\infty, -4) \cup (-4, \infty)$

b. Domain of f = R/ $\{x = 2\}$

2.5.1 Asymptotes of a Rational function

Here, we go to see, the three asymptotes **Vertical asymptote, horizontal asymptotes and oblique asymptote.**

- If a function value gets larger and larger in magnitude without bound as x approaches the number 'a ; then the line x = **a** is called a **vertical asymptote.**
- If the value of y approaches a number 'b' as $|x|$ gets larger and larger, the line y = **b** is called a **horizontal asymptote.**

(i) Vertical Asymptotes

A vertical asymptote, when it occurs, describes a certain behaviour of the graph when x close to some number a, **where** x = a is the zeros of the denominator. What happens to the value of f as x closer and closer to a, where a is the zeros of the denominator.

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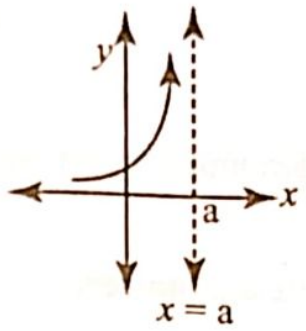
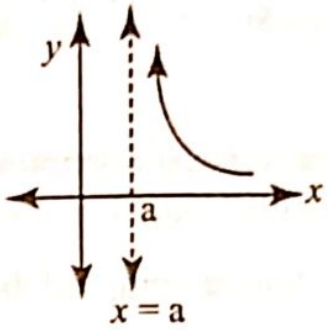
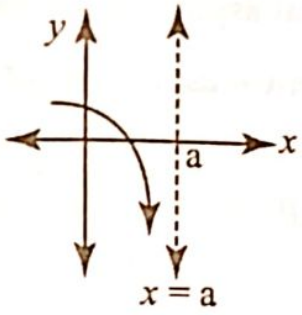
DEFINITION:- The line $x = a$ is a **Vertical asymptote** of the graph of function. if $f(x) \to \infty$ or $f(x) \to -\infty$ as x closer and closer to a from either the left or the right.



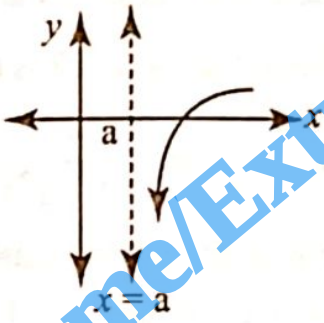
As $x \to a^-$, $f(x) \to \infty$



As $x \to a^+$, $f(x) \to \infty$



As $x \to a^-$, $f(x) \to -\infty$



As $x \to a^+$, $f(x) \to -\infty$

Locating Vertical Asymptote:

How do you locate a vertical asymptote?

A rational function

$$f(x) = \frac{P(x)}{q(x)}$$

has a vertical asymptote at the line $x = a$ if $q(a) = 0$ and $P(a) \neq 0$.

Note: The graph of the function will never intersect a vertical asymptote.

Illustrative Example

45. Find the vertical asymptote, if any, of the graph of each rational function

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a. $f(x) = \frac{4x}{x^2 - 9}$ c. $f(x) = \frac{x - 1}{x - x^3}$ e. $f(x) = \frac{x}{x^2 + 4}$

b. $f(x) = \frac{x + 4}{x - 5}$ d. $f(x) = \frac{x^2 - 5x + 6}{x^2 + x - 12}$ f. $f(x) = \frac{x^3 + 8}{x^2 + x - 2}$

Solution

To find the vertical asymptote, set the denominator equal to zero solve, provided that f is in lowers term.

a. f is in lowest term, and the zeros of the denominator:

$x^2 - 9 = 0$, implies $x = 3$ and $x = -3$

$\therefore$ The line $x = 3$ and $x = -3$ are the vertical asymptote of the graph

b. f is in lowest term, $x = 5$ is the only vertical asymptote of the gra

c. $f(x) = \frac{x - 1}{x - x^3} = \frac{x - 1}{x(1 - x^2)} = \frac{x - 1}{x(1 - x)(1 + x)} = \frac{-1}{x(x + x)}$

The zeros of the denominator is $x = 0$ or $x = -1$

$\therefore$ The line $x = 0$ and $x = -1$ are the vertical asymptotes of the graph.

Note: $x = 1$ is not a vertical asymptote because $x - 1$ is common factor of f and the graph has hole at $x = 1$

d. $f(x) = \frac{x^2 - 5x + 6}{x^2 + x - 12} = \frac{(x - 3)(x - 2)}{(x - 3)(x + 4)} = \frac{x - 2}{x + 4}$

$\therefore$ The line $x = -4$ is the only vertical asymptote of the graph of

e. The denominator $x^2 + 4$ has no real zeros, so that graph of f has n vertical asymptote

f. $f(x) = \frac{x^3 + 8}{x^2 + x - 2} = \frac{(x + 2)(x^2 - 2x + 4)}{(x + 2)(x - 1)} = \frac{x^2 - 2x + 4}{x - 4}$

$\therefore$ The line $x = 1$ is the only vertical asymptote of the graph of f.

46. Find the vertical asymptotes of the function $f(x) = \frac{x^2 + 3}{x^2 + 4x + 1}$

Solution

Set the denomination equal to zero $x^2 + 4x + 1 = 0$,

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Using completing square: $x^2 + 4x + 4 + 1 - 4 = 0$

$(x + 2)^2 - 3 = 0$ is the same as $x + 2 = \pm\sqrt{3}$

$\therefore$ The line $x = -2 + \sqrt{3}$ and $x = -2 - \sqrt{3}$ are the vertical asymptotes of the graph.

47. What one of the following has a graph that does not have a vertical asymptote?

A. $f(x) = \dfrac{2x + 3}{x^3 + 1}$ B. $f(x) = \dfrac{1}{x^2 - 2}$

C. $f(x) = \dfrac{1}{x^2 + 1}$ D. $f(x) = \dfrac{5}{x^2}$

Solution

A, B and D have vertical asymptote $x = -1$, $x = \pm\sqrt{2}$, and $x = 0$ respectively

Answer: C

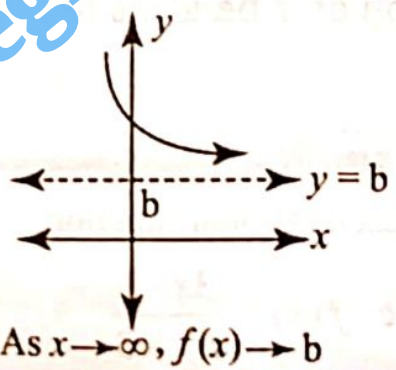
(ii) Horizontal Asymptote

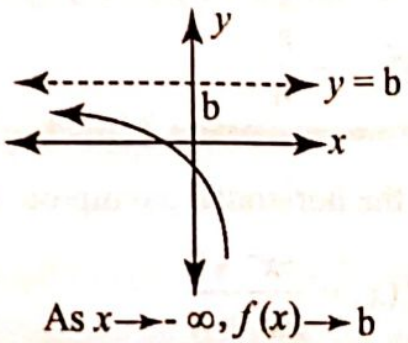
A horizontal asymptote, when it occurs, describes a certain behaviour of the graph as $x \to \infty$ or as $x \to -\infty$ that is its end behaviour.

What happens to the value of f as $x \to \infty$ or as $x \to -\infty$?

DEFINITION: The line $y = b$ is a **horizontal asymptote** of the graph of a function f if

$f(x) \to b$ as $x \to \infty$ or $x \to -\infty$





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Locating Horizontal Asymptote:

The horizontal asymptote is determined by the leadi coefficients of the numerator and denominator of the functi This determines the end behaviour of a rational function.

Let f be a rational function defined by

$$f(x) = \frac{P(x)}{q(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \ldots + b_1 x + b_0}$$

Where n is the degree of the numerator and m is the degree of th denominator.

1. If $n < m$, then f has horizontal asymptote $y = 0$ (the x - axis)
   - As an exampe: $\frac{4x}{x^2 - 1}$, $\frac{2x^2}{x^3 - 2x - 1}$, $\frac{x+1}{x^2 + x - 4}$, e.t.c

     each of them has H. A, $y = 0$
2. If $n = m$, then f has horizontal asymptote

$$y = \frac{a_n}{b_m} = \frac{\text{leading coefficient of } P(x)}{\text{leading coefficient of } q(x)}$$

3. If $n = m + 1$, then f has an oblique asymptote $y = ax + b$. To find an equation of oblique asymptote, divide the numerator by the denominator.
4. If $n > m + 1$, the graph of $f(x)$ has no horizontal asymptote and no oblique asymptote

Note: For large value of $|x|$ the graph of f behaves like the graph of $y = \frac{a_n}{b_n}$

Illustrative Example

48. Find the horizontal asymptote (if any) each rational function

a. $f(x) = \frac{3x^2 - 1}{5x^2 + 4}$

b. $f(x) = \frac{x^3 + 2}{x^2 - 6}$

c. $f(x) = \frac{3x - 1}{x^2 + 5}$

d. $f(x) = \frac{2x + 3}{4x + 7}$

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Solution

a. $f(x) = \dfrac{3x^2 - 1}{5x^2 + 4}$, we have degree: n = 2 and m = 2

since n = m,

then line $y = \dfrac{3}{5}$ is a horizontal asymptotes of f.

b. The degree of numerator: n = 3 and the degree of denominator m = 2. Since n > m, then the function has no horizontal asymptote.

c. We have n = 1 and m = 2

Since n < m, then the line $y = 0$ (the x - axis) is a horizontal asymptote.

d. We have n = 1 and m = 1, since n = m then the line $y = \dfrac{2}{4} = \dfrac{1}{2}$ is a horizontal asymptote of the function.

49. Find any horizontal and vertical asymptote for each rational function.

a. $f(x) = \dfrac{-4}{x+3}$

b. $f(x) = \dfrac{-4x}{x+2}$

c. $f(x) = \dfrac{3-2x}{4x+1}$

d. $f(x) = \dfrac{x^2 + 7x + 12}{x+3}$

Solution

a. Vertical asymptote : $x = -3$ and horizontal asymptote : $y = 0$

b. Vertical asymptote : $x = -2$ and horizontal asymptote : $y = -4$

c. Vertical asymptote: $x = -\dfrac{1}{4}$ and horizontal asymptote : $y = \dfrac{-2}{4}$

d. The value $x = -3$ makes the numerator zero, and simplify the function :

$$\frac{x^2 + 7x + 12}{x+3} = \frac{(x+3)(x+4)}{x+3} = x + 4, \text{ for } x \neq -3$$

$\therefore$ The graph of this function is the graph of $f(x) = x + 4$ with a hole at $x = -3$

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Determining where a Graph of a Function crosses a Horizontal Asymptote

Suppose that the line $y = b$ is a horizontal asymptote of a rational function $y = f(x)$.

- Solving for the equation $f(x) = b$ are the values of x where the graph of f crosses its horizontal asymptote.
- If the equation has no real solution, then the graph does not cross its horizontal asymptotes.

Illustrative Example

50. Let $f(x) = \dfrac{3x^2 - 2x - 1}{2x^2 + 3x - 2}$

Find a. the vertical asymptote

b. the horizontal asymptote

c. the point where the graph of f crosses its horizontal asymptote.

Solution

a. We first factor : $f(x) = \dfrac{3x^2 - 2x - 1}{(2x - 1)(x + 2)}$

$\therefore$ The line $x = \dfrac{1}{2}$ and $x = -2$ are the vertical asymptotes of f.

b. The degrees of numerator anddenominator are the same (m = n), then

Horizontal asymptote: $y = \dfrac{\text{leading coefficient of numerator}}{\text{leading coefficient of denominator}} = \dfrac{3}{2}$

c. Since horizontal asymptotes is $y = \dfrac{3}{2}$, then we set $f(x) = \dfrac{3}{2}$

$\dfrac{3x^2 - 2x - 1}{2x^2 + 3x - 2} = \dfrac{3}{2}$, this implies $3x^2 - 2x - 1 = \dfrac{3}{2}(2x^2 + 3x - 2)$

$3x^2 - 2x - 1 = 3x^2 + \dfrac{9}{2}x - 3$, it gives $-2x - 1 = \dfrac{9}{2}x - 3$

$-4x - 2 = 9x - 6$, solving for x, $x = \dfrac{13}{4}$

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$\therefore$ The function crosses its horizontal asymptote at $\left(\frac{13}{4}, \frac{3}{2}\right)$

51. Which one of the following has a graph that does not have a horizontal asymptote?

A. $f(x) = \frac{3x - 1}{x^2 + 1}$ B. $f(x) = \frac{4x}{x^2 - 16}$

C. $f(x) = \frac{x + 2}{(x + 1)(x - 1)}$ D. $f(x) = \frac{x^2 - 4}{x + 2}$

Solution

A, B and C have horizontal asymptote $y = 0$, but

$f(x) = \frac{x^2 - 4}{x + 2} = \frac{(x - 2)(x + 2)}{x + 2} = x - 2$ has no HA

Answer: D

2.6 Analyzing the Graph of a Rational Function

We now turn our attention to graphing rational functions.

Since any value of x such that $q(x) = 0$ are excluded from the domain of rational function : $\frac{p(x)}{q(x)}$

Illustrative Example

52. Sketch and analyze the graph of $f(x) = \frac{1}{x - 3}$.

a. Evaluate f at $x = 2.5, 2.9, 2.99, 2.999$ and 2.9999.

What happens to the values of f as x approaches 3 and $x < 3$?

b. Evaluate f at $x = 3.5, 3.1, 3.01, 3.001$ and 3.0001 What happens to the values of f as x approaches 3, and $x > 3$?

c. Evaluate f at $x = 13, 103, 1003, 10003$ and 100003.

What happens to the values of f as $|x|$ approaches very large number ∞?

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d. Sketch the graph.

Solution

This function is undefined at $x = 3$. Since x can take on any value except 3, the value of x can approach 3 as closely as desired from either side of 3

To evaluate the function at given x - values, we use the following table

x	2.5	2.9	2.99	2.999	2.9999
$f(x) = \frac{1}{x-3}$	-2	-10	-100	-1000	-10,000

The table above suggests that, as x gets closer and closer to 3 from the left $|x|$ gets larger and larger. We may describe this behaviour of $f(x)$, denoted by : $f(x) \to -\infty$, as $x \to 3^-$

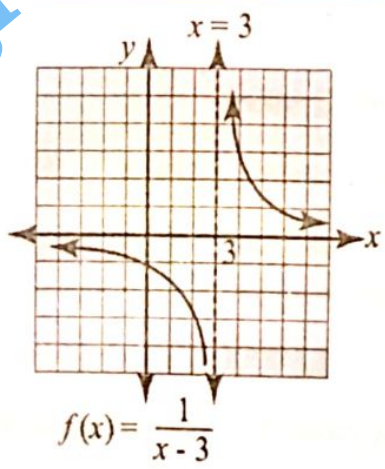
x	3.5	3.1	3.01	3.001	3.0001
$f(x) = \frac{1}{x-3}$	2	10	100	1000	10,000

As x close to 3, and $x > 3$, the value of $f(x)$ approaches to very large Positive number. We may describe this behaviour of $f(x)$ by writing:

$f(x) \to -\infty$, as $x \to 3^+$

c

x	13	103	1003	10003	100003
$f(x) = \frac{1}{x-3}$	0.1	0.01	0.001	0.0001	0.00001



As $|x|$ gets larger and larger

$f(x) = \frac{1}{x-3}$ gets closer and closer to 0.

$f(x) \to 0$, as $x \to \infty$.

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Analysing the Graph of a Rational Function

Let $f(x) = \frac{P(x)}{q(x)}$ rational function in lowest term.

STEP 1: Find the domain and factor. Factor the numerator and denominator

STEP 2: Intercepts

Find the x - intercepts by determining the zeros of numerator.

STEP 3: Locate Vertical Asymptote.

The vertical asymptote occurs at the values of x for which the denominator is zero and the numerator is non zero and then if $y \longrightarrow \infty$ or $y \longrightarrow -\infty$ on each side of each vertical asymptote by using test value.

STEP 4: Locate horizontal asymptote.

Find horizontal asymptote (if any)by dividing both numerator and denominator by highest power of x that appears in the denominator, and then evaluating $x \longrightarrow \pm \infty$, then see, $f(x) \longrightarrow b$.

STEP 5: Determine whether the graph will intersect its horizontal asymptote.

STEP 6: Determine the behaviour of f on each interval

Determine where the graph is above x - axis, below x - axis using the zeros of numerator and denominator to divide the x - axis into intervals.

Notation Describing Behaviour of a Function

Let $x = a$, the zeros of the denominator of rational function $f(x) = \frac{P(x)}{q(x)}$.

We will use the following notation table:

Notation	Meaning
$x \longrightarrow a^-$	◆ x approaches a from the left ($x < a$) or x close to a from the left.
$x \longrightarrow a^+$	◆ x approaches a from the right ($x > a$) or x close to a from the right.
$x \longrightarrow \infty$	◆ x approaches infinity or (x - increases with out bound)

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$x \longrightarrow -\infty$	• x approaches negative infinity or (x - decrease without bound)
$f(x) \longrightarrow \infty$	• $f(x)$ increases with out bound.
$f(x) \longrightarrow -\infty$	• $f(x)$ decreases with out bound.

Remark: For any real number, except zero.

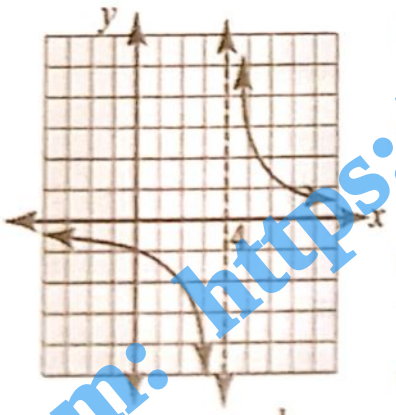
- As the denominator gets larger, the fraction gets smaller.
- As the denominator gets smaller, the fraction gets larger.

Investigating the Behaviour of a Rational Function

We now turn, describing and investigating the infinite behavio of a graph of rational function. The infinite behaviour of a functi usually called the **end behaviour** of a function.

Illustrative Example

53. Investigating the behaviour of the graph of $f(x) = \dfrac{1}{x-4}$ is given



a. What happens to the value of $f(x)$ as x g closer and closer to 4 from the **left** side?

b. What happens to the value of $f(x)$ as x g closer and closer to 4 from the **right** side?

c. What happens to the value of $f(x)$ as $|x|$ g larger and larger

d. For what values of x, $f(x) > 0$, $f(x) < 0$, $f(x) =$

Solution

a. As $x \longrightarrow 4^-$, $f(x) \longrightarrow -\infty$

b. As $x \longrightarrow 4^+$, $f(x) \longrightarrow \infty$

c. As $x \longrightarrow \infty$, $f(x) \longrightarrow 0$

d. $f(x) > 0$, if $x > 4$, and $f(x) < 0$, if $x < 4$

But no x - value that makes $f(x) = 0$

- The graph is decreasing over the interval $(-\infty, 4) \cup (4, \infty)$.

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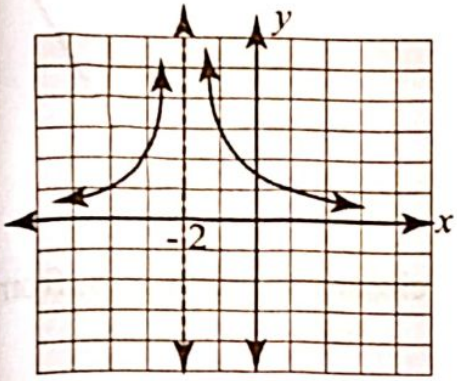
4. Analyze The graph of $f(x) = \frac{1}{(x+2)^2}$ is given



$f(x) = \frac{1}{(x+2)^2}$

a. What happens to the value of $f(x)$ as x close to -2 from the left side?

b. What happens to the value of $f(x)$ as x close to -2 form the right side?

c. What happens to the value of $f(x)$ as x is larger positive?

d. What happens to the value of $f(x)$ as x is large negative?

e. For what values of x, the graph crosses the y - axis?

f. For what values of x, the graph of $f(x)$ is increasing and decreasing.

g. What is the range of $f(x)$?

Solution

a. As $x \longrightarrow -2^-, f(x) \longrightarrow \infty$

b. As $x \longrightarrow -2^+, f(x) \longrightarrow \infty$

c. As $x \longrightarrow \infty, f(x) \longrightarrow 0$

d. As $x \longrightarrow -\infty, f(x) \longrightarrow 0$

e. When $x = 0$, $y = \frac{1}{4}$

f. $f(x)$ is increasing on interval $(-\infty, -2)$ and $f(x)$ is decreasing on interval $(-2, \infty)$

g. Range = $\{y: y > 0\}$

5. Let $f(x) = \frac{1}{(x+2)^2} + 1$

- The graph of f is the graph of $f(x) = \frac{1}{x^2}$ with a shift to the left 2 units and shift upward 1 units.

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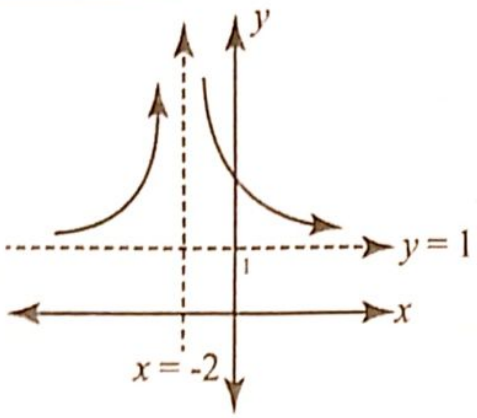

- Vertical asymptote: $x = -2$
- Horizontal asymptote: $y = 1$
- y - intercept: $f(0) = \frac{1}{4} + 1 =$
- No - x - intercept.

56. Analyze the graph of $f(x) = \frac{x}{x - 3}$

a. What happens to the values of $f(x)$ as x closer and closer to 3, from the right and from the left?

b. What happens the values of $f(x)$ as $|x|$ gets larger and larger number?

c. on what values of x, the graph is increasing or decreasing.

Solution

Domain is $\{x | x \neq 3\}$, x - intercept, (0, 0) vertical asymptote, is the line $x = 3$ and Horizontal asymptote, $y = 1$

a. As x gets closer and closer to 3 : from the right

- $x \longrightarrow 3^+$, $f(x) = \frac{3}{3^+ - 3} = \frac{6}{0^+} \longrightarrow \infty$.
- from the left: $x \longrightarrow 3^-$, $f(x) = \frac{3}{3^- - 3} = \frac{6}{0^-} \longrightarrow -\infty$.

Note: 0^+ doesn't mean 0, "slightly greater than 0"

0^- doesn't mean 0, "slight less than 0".

b. Divide the numerator and denominator by x :

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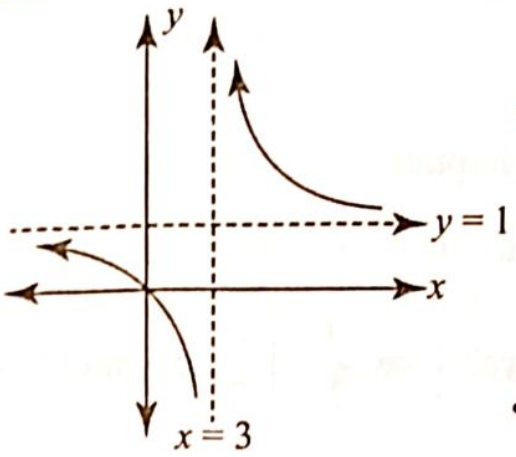

$$f(x) = \frac{x}{x-3} = \frac{\frac{x}{x}}{\frac{x-3}{x}} = \frac{1}{1-\frac{3}{x}}$$

as $x \longrightarrow \infty, f(x) \longrightarrow 1$,

as $x \longrightarrow -\infty, f(x) \longrightarrow 1$.

- The graph of f is never increasing
- The graph of f is decreasing over the interval $(-\infty, 3) \cup (3, \infty)$.

57. Let $f(x) = \dfrac{2x+1}{x-4}$

a. Find the domain, the intercept and asymptote.

b. What happens the values of $f(x)$ as x close to 4.

c. What happens the values of $f(x)$ as $|x| \longrightarrow \infty$.

d. For what values of x the graph is below the x - axis and above the x - axis.

Solution

a. The domain of f is $\{x | x \neq 4\}$

- The x-intercept is the point $\left(\dfrac{-1}{2}, 0\right)$
- The y-intercept is $f(0) = \dfrac{0+1}{0-4} = \dfrac{-1}{4}$
- The line $x = 4$ is vertical asymptote
- Since the degree of numerator is equal to the degree of denominator $(n = m)$ then horizontal asymptote is $y = \dfrac{2}{1} = 2$

b. As x close to 4 :

- from the right side: $x \longrightarrow 4^+, f(x) = \dfrac{8+1}{4^+ - 4} = \dfrac{9}{0^+} \to \infty$
- from the left side: $x \longrightarrow 4^-, f(x) = \dfrac{8+1}{4^- - 4} = \dfrac{9}{0^-} \to -\infty$

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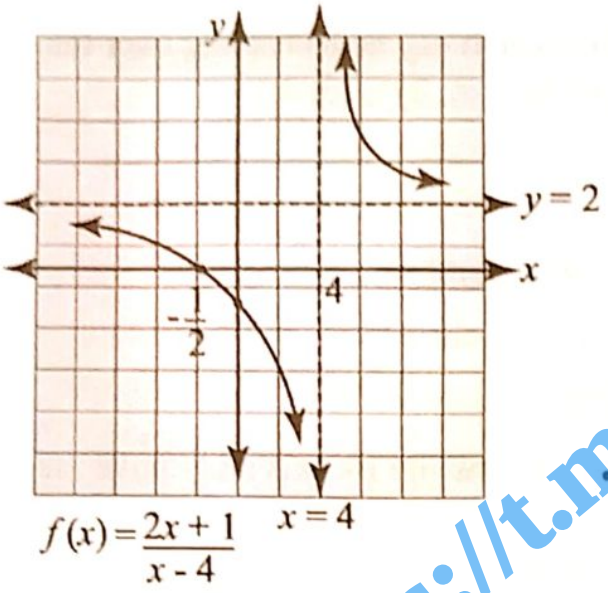
c. The line $y = \frac{a_n}{b_m} = \frac{2}{1} = 1$

$\therefore$ The line $y = 2$ is a horizontal asymptote

d. The zeros of numerator, $x = \frac{-1}{2}$ and the zeros of denominator $x = 4$. Now, we divide into intervals $\left(-\infty, \frac{-1}{2}\right)$, $\left(\frac{-1}{2}, 4\right)$ and $(4, \infty)$



$f(x) = \frac{2x+1}{x-4}$

	$-\infty$	$\frac{-1}{2}$		4	∞
$2x+1$	- -	0	+ +	+ +	
$x-4$	- -	- -	- -	0	+ +
$\frac{2x+1}{x-4}$	+ +	0	- -	∄	+ +
Graph Location	above x - axis		below x - axis		above x - axis

- On interval: $(-\infty, \frac{-1}{2}) \cup (4, \infty)$, the graph lies above the x - axis
- On interval $(\frac{-1}{2}, 4)$ the graph lies below the x - axis

58. Let $f(x) = \frac{x^2}{x^2 + 2x - 8}$

a. Find the domain, the range, the intercept, and the asymptote.

b. Analyze the behaviour of $f(x)$ near the vertical asymptote.

c. What happens the value of $f(x)$ as $|x| \to \infty$?

d. Determine whether the graph will intersect its horizontal asymptote

e. For what values of x, the graphs is above the x- axis, below the x - axis

f. On what values of f increasing or decreasing

Solution

First, we factor the denominator.

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- $f(x) = \dfrac{x^2}{x^2 + 2x - 8} = \dfrac{x^2}{(x-2)(x+4)}$

a. The domain of f is $\{x | x \neq 2, x \neq -4\}$

- The x - intercept is $x = 0$.
- The y - intercept is $f(0) = 0$
- The line $x = 2$ and $x = -4$ are vertical asymptote.
- Since n = m, then the ratio of the leading coef. $y = \dfrac{1}{1} = 1$ is horizontal asymptote.

b. The line $x = 2$ and $x = -4$ are vertical asymptotes

(i) As x close to 2:

- from the right side: $x \to 2^+$, $f(x) = \dfrac{4}{(2^+ - 2)(2^+ + 4)} = \dfrac{4}{0^+} \to -\infty$
- from the left side: $x \to 2^-$, $f(x) = \dfrac{4}{(2^- - 2)(2^- + 4)} = \dfrac{4}{0^-} \to -\infty$

That means as $x \to 2^+, y \to \infty$ and as $x \to 2^-, y \to -\infty$

(ii) As x close to -4:

- from the right side: $x \to -4^+$,

$$f(x) = \frac{16}{(-4 - 2)(-4^+ + 4)} = \frac{16}{(-6)(0^+)} = \frac{16}{0^-} \to -\infty$$

- from the left side: $x \to -4^-$,

$$f(x) = \frac{16}{(-4^- - 2)(-4^- + 4)} \quad \frac{16}{(-6)(0^-)} = \frac{16}{0^+} = \to \infty$$

That means, as $x \to 4^+, y \to -\infty$ and as $x \to 4^-, y \to -\infty$.

c. As $|x| \longrightarrow \infty, f(x) \longrightarrow 1$

$\therefore$ $y=1$ is horizontal asymptote

d. We set $f(x) = 1$, this implies $\dfrac{x^2}{x^2 + 2x - 8} = 1$

$x^2 = x^2 + 2x - 8$, therefore, $2x - 8 = 0, x = 4$

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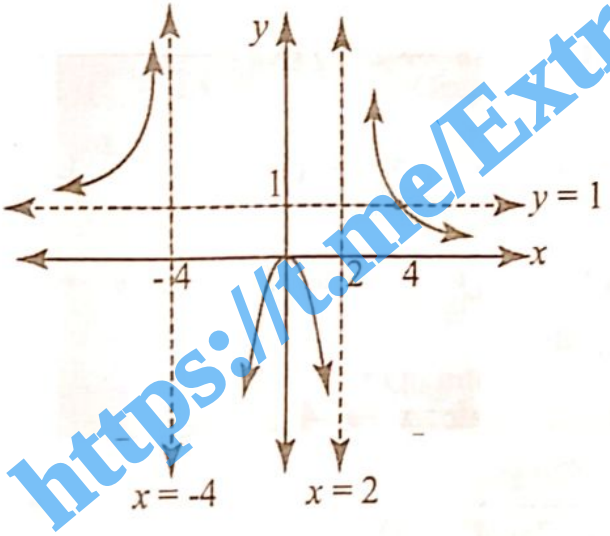
The graph will intersect its horizontal asymptote at the point (4, 1)

e. The zeros of numerator is $x = 0$, the zeros of denominator are $x = 2$ and $x = -4$.

Now we divide into interval : $(-\infty, -4)$, $(-4, 0)$, $(0, 2)$ and $(2, \infty)$

	$-\infty$	-4	0	2	∞
x^2	+ +	0 + +	+ +	+ +	+ +
$x - 2$	- -	- -	- -	0 + +	
$x + 4$	- -	0 + +	+ +	+ +	
$\frac{x^2}{(x-2)(x+4)}$	+ +	∄ - -	0 - -	∄ + +	
Location of Graph	above x - axis	below x - axis	below x - axis	above x - axis	



$$f(x) = \frac{x^2}{x^2 + 2x - 8}$$

- The graph of f is increasing over the interval $(-\infty, -4) \cup (-4, 0)$
- The graph of f is decreasing over the interval $(0, 2) \cup (2, \infty)$.

59. Let $f(x) = \dfrac{x^2 + x - 6}{x^2 - 4}$

a. Find the domain, the intercept and the asymptote.

b. Analyze, the behavior of $f(x)$ near the vertical asymptote.

c. What happens the values of $f(x)$ as $|x| \to \infty$

d. Determine where f crosses its horizontal asymptote (if at all).

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e. For what values of x, the graph is above x - axis, below x - axis

Solution

First, faction $f(x) = \dfrac{(2x-3)(x+2)}{(x-2)(x+2)} = \dfrac{2x-3}{x-2}$

a. The domain is $\{x | x \neq 2, x \neq -2\}$

- x - intercept is $\left(\frac{3}{2}, 0\right)$ and y - intercept is $\left(0, \frac{3}{2}\right)$

- The line $x = 2$ is the only vertical asymptote, the value $x = -2$ makes the numerator zero. The graph will have a 'hole' at $x = -2$ rather than a vertical asymptote.

b. The line $x = 2$ is a vertical asymptote

As x close to 2:

- from the right side, $x \to 2^+$, $f(x) = \dfrac{4-3}{2^+ - 2} = \dfrac{1}{0^+} \to \infty$

- from the left side, $x \to 2^-$, $f(x) = \dfrac{4-3}{2^- - 2} = \dfrac{1}{0^-} \to -\infty$

That means, as $x \to 2^+$, $f(x) \to \infty$ and as $x \to 2^-$, $f(x) \to -\infty$.

Note: 0^+ and 0^- doesn't mean zero, but near to zero.

c. As $x \to \infty$, $y \to 2$ and as $x \to -\infty$, $y \to 2$

d. Solve the equation $f(x) = 2$,

$\dfrac{2x-3}{x-2} = 2 \Rightarrow 2x - 3 = 2x - 4 \Rightarrow 3 \neq -2$

$\therefore$ The graph of f does not cross its horizontal asymptote.

e. The zero of the numerator is $\dfrac{3}{2}$ and the zeros of the denominator is 2, divide the x - axis into 3 intervals. $\left(-\infty, \frac{3}{2}\right)$, $\left(\frac{3}{2}, 2\right)$ and $(2, \infty)$

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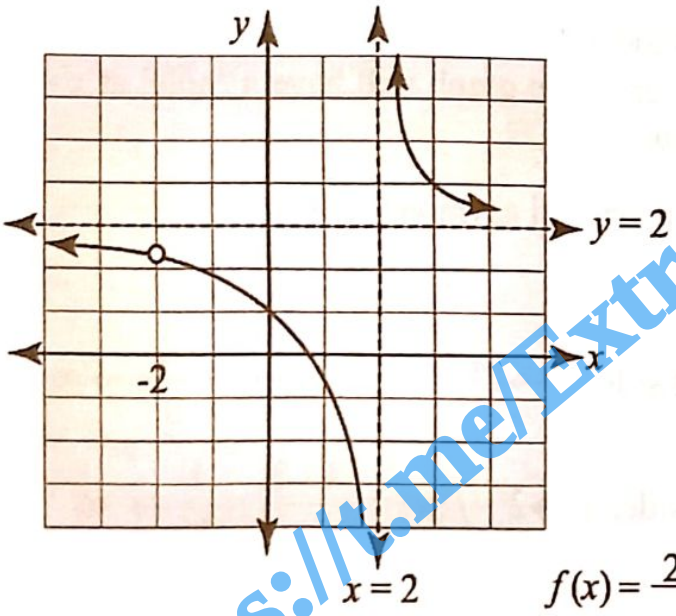
	$-\infty$	$\frac{3}{2}$	2	∞
$2x - 3$	- -	0	+ +	+ +
$x - 2$	- -		- -	+ +
$\frac{2x-3}{x-2}$	+ +	0	- - ∄	+ +
Location of Graph	above x - axis		below x - axis	above x - axis



$f(x) = \frac{2x^2 + x - 6}{x^2 - 4}$

60. Analyze the graph of $f(x) = \frac{-4x}{x^2 - 9}$

Solution

1. Determine the domain

   Domain is $\{x | x \neq 3, x \neq -3\}$

2. The x - intercept is the point $(0, 0)$

   y - intercept is the point $(0, 0)$

3. The line $x = -3$ and $x = 3$ are vertical asymptote.

- Since n < m, then f is a proper rational function, and f will have horizontal asymptote $y = 0$ (x - axis).

4. $f(x) = \frac{-4x}{x^2 - 9} = \frac{-4x}{(x - 3)(x + 3)}$

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(i) As $x \to 3^+$, $f(x) \to \dfrac{-12}{(3^+ - 3)(3^+ + 3)} = \dfrac{-12}{0^+} \to -\infty$

As $x \to 3^-$, $f(x) \to \dfrac{-12}{0^+} \to \infty$

$\therefore$ $y \to -\infty$, as $x \to 3^+$ and $y \to \infty$, as $x \to 3^-$

(ii) As $x \to -3^+$, $f(x) \to \dfrac{12}{(-6)(0^+)} = \dfrac{12}{0^-} \to -\infty$

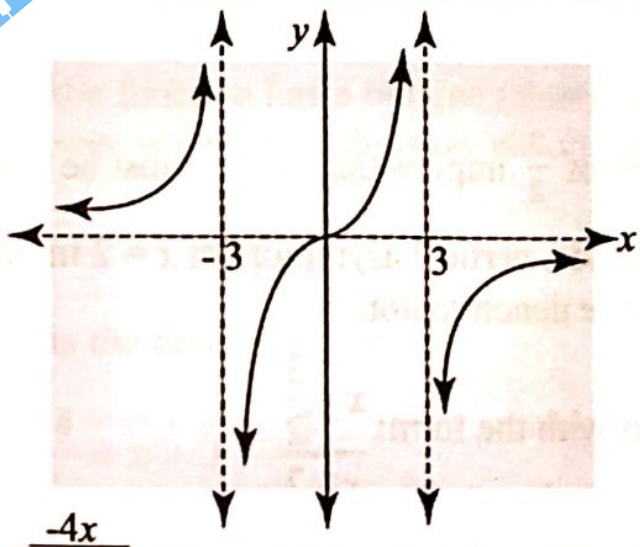
As $x \to -3^-$, $f(x) \to \dfrac{-12}{(-6)(0^-)} = \dfrac{12}{0^+} \to \infty$

$\therefore$ $y \to -\infty$, as $x \to -3^+$ and $y \to \infty$, as $x \to -3^-$

5. $y \to 0^-$, as $x \to \infty$ and $y \to 0^+$, as $x \to -\infty$

6. The zero of the numerator 0, the zeros of denominator, -3 and 3 divide the x - axis into four intervals: $(-\infty, -3)$, $(-3, 0)$, $(0, 3)$ and $(3, \infty)$.

	$-\infty$		-3		0		3	∞
$-4x$	+ +	0	+ +		- -		- -	
$x+3$	- -		+ +		+ +		+ +	
$x-3$	- -		- -		- -	0	+ +	
$\dfrac{-4x}{(x+3)(x-3)}$	+ +	∄	- -	0	+ +	∄	- -	
Location of Graph	above x - axis		below x - axis		above x - axis		below x - axis	



$f(x) = \dfrac{-4x}{x^2 - 9}$

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- The graph of a function is increasing over the interval $(-\infty,-3)\cup(-3,3)\cup(3,\infty)$

Finding an equation of a rational function satisfying prescribed condition.

61. Find an equation of a rational function that have:

x - intercept 3; vertical asymptote: $x = -4$; and horizontal asymptote : $y = \frac{-2}{3}$.

Solution

- An x - intercept of 3 implies that $x - 3$ must be a factor in the numerator.
- A vertical asymptote of $x = -4$ implies that $x + 4$ is a factor of in the denominator. So we can start with the from: $\frac{x-3}{x+4}$
- The horizontal asymptote is $y = \frac{-2}{3}$

We can multiply the numerator by -2 and the denominator by 3 to get the form

$$f(x) = \frac{-2(x-3)}{3(x+4)} = \frac{-2x + 6}{3x + 12}$$

62. Find an equation of a rational function that have an x - intercept: $\frac{3}{2}$, vertical asymptote: $x = 2$, horizontal asymptote: $y = 2$, and a hole at $x = -2$.

Solution

- An x - intercept of $\frac{3}{2}$ implies that $x - \frac{3}{2}$ must be a factor in thenumerator, and a vertical asymptote of $x = 2$ implies that $x - 2$ isa functor in the denominator.

So we can start with the form: $\frac{x - \frac{3}{2}}{x - 2}$

- The horizontal asymptote is $y = 2$. We can multiply the numerator by 2 and the denominator by 1 to get the form:

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$$\frac{2\left(x - \frac{3}{2}\right)}{x - 2} = \frac{2x - 3}{x - 2}$$

Lastly, since there is a hole at $x = -2$, we must have a factor of $x + 2$ in both the numerator and denominator. Thus, an equator for f is

$$f(x) = \frac{(2x - 3)(x + 2)}{(x - 2)(x + 2)} = \frac{2x^2 + x - 6}{x^2 - 4}$$

Finding the oblique asymptotes of a rational function

If a rational function f is **improper**, if the degree of the numerator is greater than or equal to the degree of the denominator, we use the division algorithm to express the function in the form :

$$f(x) = ax + b + \frac{r(x)}{q(x)},$$

Where

The degree of r (x) is less than the degree of q (x) and a ≠ 0.

Illustrative Example

63. Find an oblique asymptote, if any, of the graph:

a. $f(x) = \frac{x^2 - 4x - 5}{x - 3}$ b. $f(x) = \frac{x^2 + 4}{x - 1}$

Solution

a. Since the degree of the numerator is one more than the degree of the denominator, the function has a oblique (slant) asymptote. To find oblique asymptote we use long division, we obtain.

$$f(x) = x - 1 - \frac{8}{x - 3}$$

$\therefore$ $y = x - 1$ is the oblique asymptote.

b. $f(x) = \frac{x^2 + 4}{x - 1} = x + 1 + \frac{5}{x - 1}$

$\therefore$ The line $y = x + 1$ is an oblique asymptote

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64. **Test your skill** : For what values of x, the graph of $f(x) = \frac{3x^4 + 4}{x^3 + 3x}$ inter sects the oblique asymptote?

A. $x = \frac{3}{4}$ B. $x = \pm\frac{4}{3}$ C. $x = \pm\frac{2}{3}$ D. $x = \frac{3}{2}$

Solution: By using long division:

$$f(x) = \frac{3x^4 + 4}{x^3 + 3x} = 3x + \frac{4 - 9x^2}{x^3 + 3x}$$

The line $y = 3x$ is an oblique asymptote. Then we solve $f(x) = 3x$

$\frac{3x^4 + 4}{x^3 + 3x} = 3x$, implies that $3x^4 + 4 = 3x(x^3 + 3x)$

$\Rightarrow 3x^4 + 4 = 3x^4 + 9x^2$, implies that $9x^2 = 4$

$\Rightarrow x^2 = \frac{4}{9}$, implies that $x = \pm\sqrt{\frac{4}{9}} = \pm\frac{2}{3}$

Answer: (

65. Analyze the graph of $f(x) = \frac{2x^2 - 7x + 3}{x - 2}$

Solution

1. The x -intercept : $\frac{2x^2 - 7x + 3}{x - 2} = 0$

$\Rightarrow 2x^2 - 6x - x + 3 = 0 \Rightarrow 2x(x - 3) - 1(x - 3) = 0$

$(2x - 1)(x - 3) = 0$, $x = \frac{1}{2}$, *or* $x = 3$

- The y- intercept is $f(0) = \frac{0 - 0 + 3}{0 - 2} = \frac{-3}{2}$

2. The line $x = 2$ is a vertical asymptote.

3. Identify the oblique asymptote. To find the oblique asymptote, divide:

$$\frac{2x^2 - 7x + 3}{x - 2} = 2x - 3 + \frac{-3}{x - 2}$$

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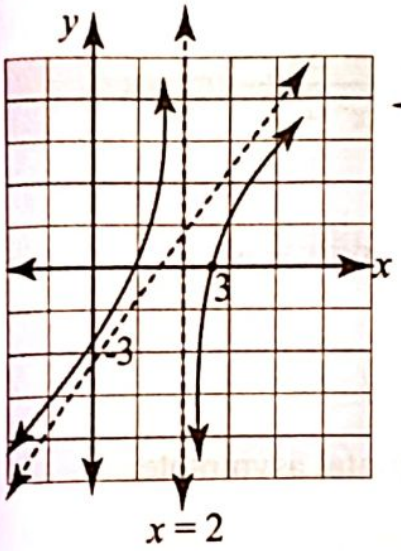
$\therefore$ The oblique (slant) asymptote is $y = 2x - 3$

4. As $x \to 2^+$, $f(x) \to \dfrac{-3}{2^+ - 2} = \dfrac{-3}{0^+} \to -\infty$

As $x \to 2^-$, $f(x) \to \dfrac{-3}{2^- - 2} = \dfrac{-3}{0^-} \to +\infty$

5. Determine the location of the graph of f on each interval

$$\left(-\infty, \frac{1}{2}\right), \left(\frac{1}{2}, 2\right), (2, 3), (3, \infty)$$



	$-\infty$	$\frac{1}{2}$	2	3	∞
$2x - 1$	- -	+ +	+ +	+ +	
$x - 3$	- -	- -	- -	+ +	
$x - 2$	- -	- -	+ +	+ +	
$\frac{(2x - 1)(x - 3)}{(x - 2)}$	- -	+ +	- -	+ +	
Location of Graph	below x - axis	above x - axis	below x - axis	above x - axis	

$$f(x) = \frac{2x^2 - 7x + 3}{x - 2}$$

- The graph of the function is increasing over the interval $(-\infty, 2) \cup (2, \infty)$.

66. The graph of $f(x) = \dfrac{1}{(x + 2)^2} + 3$ will behave which function for large values of $|x|$?

A. $y = 0$ B. $y = \dfrac{1}{x^2}$ C. $y = 3$ D. $y = \dfrac{13}{4}$

Solution

For large values of $|x|$ mean that : As $|x| \to \infty, f(x) \to 0 + 3 = 3$

Answer: C

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67. The graph of $f(x) = \dfrac{-x^3 + 2x + 1}{4x^3}$ will be have like which function for large values of $|x|$?

A. $y = \dfrac{-1}{4}$ B. $y = \dfrac{-1}{4}x$ C. $y = \dfrac{1}{3}$ D. $y = \dfrac{-x^3}{4}$

Solution

$$y = \frac{\text{leading coefficient of numerator}}{\text{leading coefficient of denominator}} = \frac{-1}{4}$$

Answer: A

68. For what value (s) of x, the graph of $f(x) = \dfrac{x^2 + 3x - 4}{x^3 + 1}$ crosses its horizontal asymptote?

A. $x = 2$ C. $x = 1$ and $x = -4$

B. $x = 3$ and $x = 1$ D. $x = 0$

Solution

Since f is proper, then the line $y = 0$ is horizontal asymptote.

set $f(x) = 0$ and $\dfrac{x^2 + 3x - 4}{x^3 + 1} = 0$

$\Rightarrow x^2 + 3x - 4 = 0$, this implies $(x - 1)(x + 4) = 0$

$\therefore x = 1$ and $x = -4$ **Answer: C**

69. If $f(x) = \dfrac{x - 2}{(x - 3)^2 (x + 1)}$, then which of the following is true about f.

A. It has an oblique asymptote

B. The graph does not cross its asymptote

C. As $x \to 3^-, f(x) \to -\infty$

D. As $x \to -1^+, f(x) \to -\infty$

Solution

- It has horizontal asymptote $y = 0$
- The graph crosses horizontal asymptote at $x = 2$

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- As $x \to 3^-$, $f(x) = \dfrac{3 - 2}{(3^- - 3)^2 (3 + 1)} = \dfrac{1}{0^+ \cdot 4} \to \infty$
- As $x \to -1^+$, $f(x) = \dfrac{1 - 2}{(1 - 3)^2 (-1^+ + 1)} = \dfrac{-1}{4 \cdot 0^+} = \dfrac{-1}{0^+} \to -\infty$

Answer: D

70. Let $f(x) = \dfrac{x^2 - 2x}{x^2 + x - 6}$, then which of the following is **Not** true about f and its graph?

A. The graph of f has hole at $\left(2, \frac{2}{5}\right)$

B. The line $y = 1$ is a horizontal asymptote.

C. The line $x = -3$ is a vertical asymptote.

D. The graph of f is decreasing over the interval $(-\infty, -3) \cup (-3, \infty)$

Solution

$$f(x) = \frac{x^2 - 2x}{x^2 + x - 6} = \frac{x(x - 2)}{(x + 3)(x - 2)} = \frac{x}{x + 3}$$

- The graph has hole at $x = 2, \left(2, \frac{2}{5}\right)$
- The line $y = 1$ is a H.A and the line $x = -3$ is a. V. A.
- The graph is increasing over the intervals $(-\infty, -3) \cup (-3, \infty)$

Answer: D

71. If $f(x) = \dfrac{3x + 2}{2x - 1}$ then which of the following is not true about f.

A. $f(x) > \frac{3}{2}$, *if* $x > \frac{1}{2}$

B. $f(x) < 0$, *if* $\frac{-2}{3} < x < \frac{1}{2}$

C. $0 \le f(x) < \frac{3}{2}$, *if* $-\infty < x < \frac{-2}{3}$

D. $f(x) < 0$, *if* $x < \frac{1}{2}$

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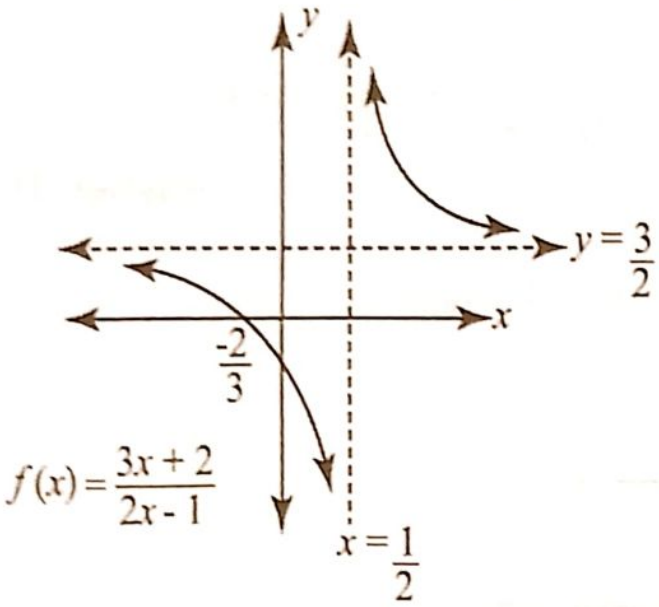
Solution



Let us analyze the graph of

$$f(x) = \frac{3x + 2}{2x - 1}$$

Domain $\left\{x \middle| x \neq \frac{1}{2}\right\}$, vertical asymptote the line $x = \frac{1}{2}$ and horizontal asymptote is the line $y = \frac{3}{2}$

- x - intercept, $x = \frac{-2}{3}$
- y - intercept, $y = -2$

Based on the graph of

$$f(x) = \frac{3x + 2}{2x - 1}$$

A. $f(x) > \frac{3}{2}$, for all x interval $\left(\frac{1}{2}, \infty\right)$

B. $f(x) < 0$, for all x in the interval $\left(\frac{-2}{3}, \frac{1}{2}\right)$

C. $0 \leq f(x) < \frac{3}{2}$, for all x in the interval $\left(-\infty, \frac{-2}{3}\right]$

D. $f(x) < 0$, *for* $x < \frac{1}{2}$ is false as an example, let $x = -6$, then $y = \frac{16}{13} > 0$.

Answer: D

72. Which rational function beast describes the graph are the right?

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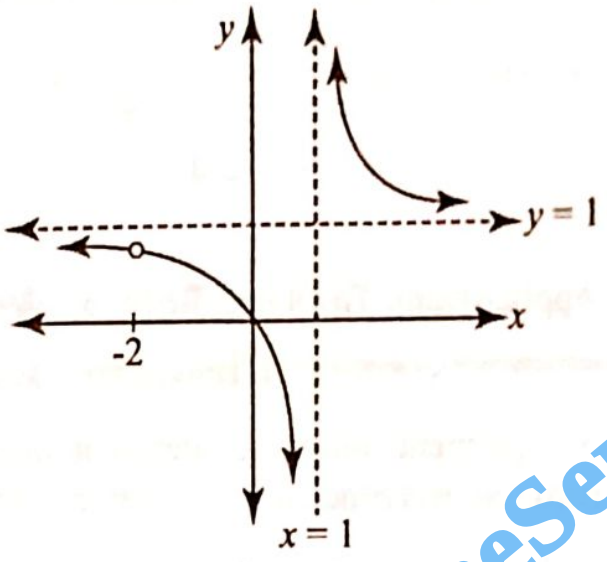
A. $f(x) = \dfrac{x}{x-1}$

B. $f(x) = \dfrac{x^2+2x}{x^2+x-2}$

C. $f(x) = \dfrac{x+1}{x-1}$

D. $f(x) = \dfrac{x^2+2x}{x^3+4x}$



Solution

The graph has an x - intercept of 0 implies that 'x - 0' must a factor in the numerator, and a vertical asymptote of $x = 1$ implies that $x - 1$ is a factor in the denominator. So we can write in the form: $\dfrac{x}{x-1}$

Since the graph has a hole at $x = -2$, we must have a factor of $x + 2$ in both numerator and the denominator.

Therefore, $f(x) = \dfrac{x(x+2)}{(x-1)(x+2)} = \dfrac{x^2+2x}{x^2+x-2}$

Answer: B

73. Which one of the following is **Not** true about the graph of $f(x) = 5 + \dfrac{|x|}{x+1}$?

A. $x = -1$ is its vertical asymptote of f.

B. $y = 4$ is its horizontal asymptote.

C. $y = 6$ is its horizontal asymptote of f.

D. $y = 5 + x$ is its oblique asymptote.

Solution

We know that $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$. Then

- $f(x) = 5 + \dfrac{|x|}{x+1} = 5 + \dfrac{x}{x+1}$, $x \geq 0$, then

  V · A: $x = -1$ and H · A: $y = 5 + 1 = 6$

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- $f(x) = 5 + \dfrac{|x|}{x+1} = 5 - \dfrac{x}{x+1}$, $x < 0$, then

  $H \cdot A: y = 5 - 1 = 4$

Answer: D

Applications Involving Rate of Work Done

Illustrative Example

74. At a garment company, Abeba can process 70-orders in 6 hours. Birtukan can process 70 - order in 4 hours.

a. How long would it take them to process 70 - orders if they work together.

b. How long would it take them to process 280-orders if they work together.

Solution

Let t = the amount of time required to process 70 - orders working together

a. $\begin{pmatrix}\text{Abeba's}\\ \text{speed}\end{pmatrix} + \begin{pmatrix}\text{Birtukan's}\\ \text{speed}\end{pmatrix} = \begin{pmatrix}\text{Speed working}\\ \text{together}\end{pmatrix}$

$$\frac{70}{6\text{hr}} + \frac{70}{4\text{hr}} = \frac{70}{t}$$

$\dfrac{1}{6} + \dfrac{1}{4} = \dfrac{1}{t}$ implies that $t = \dfrac{6 \times 4}{6+4} = \dfrac{24}{10} = \dfrac{12}{5}$

$\therefore$ Abeba and Birtukan can process 70- orders in $\dfrac{12}{5}$ hrs

b. $\dfrac{70}{6\text{hr}} + \dfrac{70}{4\text{hr}} = \dfrac{280}{\text{t-hr}}$ this implies that $\dfrac{1}{6} + \dfrac{1}{4} = \dfrac{4}{t}$

$$\therefore \quad t = \left(\frac{6 \times 4}{6+4}\right)4 = \frac{96}{10} = 9 \cdot 6 \text{ hr}$$

75. Suppose, person A and person B can do a job alone in 6hr and 8hr respectively. How long would take, if A and B working together?

Solution

To translate by considering how much of the job is finished

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in 1hr, 2hr, 3hr, and so on.

(i) It takes person A 6hr to do the job alone.

Then in 1hr, he can do $= \frac{1}{6}$ of the job.

(ii) It takes person B 8hr to do the job alone. Then in 1hr, he can do $= \frac{1}{8}$ of the job.

Working together they can do $\frac{1}{6} + \frac{1}{8}$ or $\frac{7}{24}$ of the job in 1hr.

From this reasoning, we can see the table of the following

Time	Parts of the job completed		
	Person A	Person B	Together
1 hr	$\frac{1}{6}$	$\frac{1}{8}$	$\frac{1}{6}+\frac{1}{8}$ or $\frac{7}{24}$ of the job
2 hr	$2\left(\frac{1}{6}\right)$	$2\left(\frac{1}{8}\right)$	$2\left(\frac{1}{6}\right)+2\left(\frac{1}{8}\right)$ or $2\left(\frac{7}{24}\right)$ of the job
3 hr	$3\left(\frac{1}{6}\right)$	$3\left(\frac{1}{8}\right)$	$3\left(\frac{1}{6}\right)+3\left(\frac{1}{8}\right)=\frac{7}{8}$ of the job
t - hr	$t\left(\frac{1}{6}\right)$	$t\left(\frac{1}{8}\right)$	$t\left(\frac{1}{6}\right)+t\left(\frac{1}{8}\right)=t\left(\frac{7}{24}\right)$

What we need is a number 't' such that the fraction of the job the gets completed is 1.

That is $t\left(\frac{1}{6}\right)+t\left(\frac{1}{8}\right)=1$

$\therefore \quad t=\frac{8\times 6}{6+8}=\frac{48}{14}=\frac{24}{7}$

$\therefore$ It takes $\frac{24}{7}$ hr for them to do the job working together.

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The Work principle

Let t_1 = the time takes person A to do a job.

t_2 = the time it takes person B to do a job.

and t = the time it takes to do the same job working togeth

$$\text{Then } \frac{t}{t_1} + \frac{t}{t_2} = 1 \text{ implies } t = \frac{t_1 \times t_2}{t_1 + t_2}$$

76. Zinabu can plants 200 coffee trees in 4 days, Taddes Can plants 180 coffee tress in 6 days.

How many days would it take them to plant 240 coffee trees if they work together.

Solution: Let d = Number of days for working together.

$$\begin{pmatrix}\text{Zenabu}\\ \text{speed}\end{pmatrix} + \begin{pmatrix}\text{Taddes}\\ \text{speed}\end{pmatrix} = \begin{pmatrix}\text{Speed working}\\ \text{together}\end{pmatrix}$$

$$\left(\frac{200 \text{ coffee tree}}{4 \text{ days}}\right) + \left(\frac{180 \text{ coffee trees}}{6 \text{ days}}\right) = \frac{240 \text{ coffee tre}}{\text{d - days}}$$

$$50 + 30 = \frac{240}{d}, \text{ implies that } d = \frac{240}{80} = 3 \text{ days}$$

$\therefore$ 3 days are need to plant 240 coffee when they work together.

77. A and B together can do a job in 12 days. B alone can finish it in 2 days. In how many days can A alone finish the work?

Solution: Let d = Number of working day for A.

$$\frac{1 \text{ job}}{28} + \frac{1 \text{ job}}{d} = \frac{1 \text{ job}}{12} \text{ implies that}$$

$$\frac{1}{28} + \frac{1}{d} = \frac{1}{12} \Rightarrow \frac{1}{d} = \frac{1}{12} - \frac{1}{28}$$

$$\therefore \quad d = \frac{28 \times 12}{28 - 12} = 21 \text{ days.}$$

$\therefore$ A alone can finish in 21 days.

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78. Two pipes carry water to the same tank. Pipe A, working alone, can fill the tank three times as fast as pipe B. Together the pipes can fill the tank in 9hr. Find the time each would take to fill the tank alone.

Solution: let $t_2 = 3t_1$, then by work principle:

$$\frac{t}{t_1} + \frac{t}{3t_1} = 1 \text{ implies that } t = \frac{(3t_1)(t_1)}{3t_1 + t_1} = 9$$

$$\Rightarrow \frac{3t_1^2}{4t_1} = \frac{3}{4}t_1 = 9 \text{ and } t_1 = \frac{4}{3}(9) = 12\,\text{hr}$$

$\therefore$ Pipe A, 12hr; and pipe B takes 36 hr.

79. In a class there are 8 more boys than girls. If the ratio of boys to girls is 8 to 7. Find the number of boys and girls in a class.

Solution

Let x = The number of girl

Then $x + 8$ = The number of boys.

$$\frac{x+8}{x} = \frac{8}{7} \text{ implies that } 7(x+8) = 8x$$

$\therefore$ $7x + 56 = 8x$ and $8x - 7x = 56$

$\therefore$ $x = 56$

$\therefore$ The number of girl is 56 and the number of boy is 64.

80. The ratio of the number of boys and girls in a school is 3:4. If there are 420 students in the school. Find the number of boys and girls in a school.

Solution

Let the numbers are $3x$ and $4x$

$$3x + 4x = 420, \Rightarrow x = \frac{420}{7} = 60$$

Then $3x = 3 \times 60 = 180$ are boys and $4x = 4(60) = 240$ are girls

81. UEE. A 20m high building casts a shadow of 4m, at the same time a 35m tree casts a shadow of x -meter. What is the value of x.

A. 11m B. 8m C. 15m D. 7m

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Solution

$$\frac{20m}{4m} = \frac{35m}{x} \text{ implies } x = \frac{35 \times 4}{20} = 7$$

Answer:

82. If $P(x) = \frac{x+1}{x-1}$ and $q(x) = \frac{x-1}{x+1}$, then the simplified form of $\frac{P(x) - q(x)}{P(x) + q(x)}$

A. $\frac{x}{x-1}$ B. $\frac{x}{x^2-1}$ C. $\frac{2x}{x^2+1}$ D. $\frac{2x}{x^2-1}$

Solution

Substituting the given values of P(x) and q(x), we get: $\frac{2x}{x^2+1}$

Answer:

83. What is the simplified form of the expression:

$$\left(\frac{2}{2-x} - \frac{x}{2+x} - \frac{5}{4-x^2}\right) \div \frac{x+1}{4-x^2}$$

A. $-(x+1)$ B. $x-1$ C. $-(x-1)$ D. $2-x$

Solution

The given expression is equal to:

$$\frac{2(2+x) - x(2-x) - 5}{4-x^2} \div \frac{x+1}{4-x^2}$$

$$\frac{4+2x-2x+x^2-5}{4-x^2} \cdot \frac{4-x^2}{x+1} = \left(\frac{x^2-1}{4-x^2}\right)\left(\frac{4-x^2}{x+1}\right)$$

$$\frac{x^2-1}{x+1} = \frac{(x-1)(x+1)}{x+1} = x-1$$

Answer: B

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Review Exercise 2

1. What is are a rational expression.
2. What is the domain of rational expression.
3. What is a rational function.
4. What does it mean to say that $x = a$ is a vertical asymptote of $y = f(x)$?
5. What does it mean to say that $y = b$ is a horizontal asymptote of $y = f(x)$?
6. Under what circumstances does a rational function have an oblique asymptote?
7. Perform the indicated operation

a. $\frac{4x+7}{5x-10} - \frac{x+1}{x-2}$

b. $\frac{x+1}{2x-2} - \frac{2}{x^2-1}$

c. $\frac{2x+1}{x^2+x-6} - \frac{x+2}{x^2+5x+6}$

d. $\frac{-3}{x^2+x-2} + \frac{5}{x^2-x-6}$

8. Perform the indicated operation

a. $(x+2) \div \frac{x^2-4}{x-1}$

b. $\frac{3-x}{x-1} \div \frac{x^2-x-6}{x^2-1}$

c. $\frac{4x-2}{x^2-5x} \cdot \frac{x^2-25}{2x^2+9x-5}$

d. $\frac{5-10x}{x^2-2x} \cdot \frac{x^2+2x-8}{2x^2+7x-4}$

9. Find the solution set to each equation:

a. $\frac{5}{2x+4} - \frac{1}{x-1} = \frac{3}{x+2}$

b. $\frac{2}{x+2} + \frac{x}{x-3} = \frac{-1}{x^2-x-6}$

c. $\frac{1}{x-2} = \frac{3}{x+2} - \frac{6x}{x^2-4}$

d. $\frac{2}{x+4} + \frac{2x-1}{x^2+2x-8} = \frac{1}{x-2}$

10. Find the solution set to each equation.

a. $\frac{5}{x-4} - \frac{5}{x} = \frac{20}{x^2-4x}$

b. $\frac{4}{x+4} - \frac{3}{x+3} = \frac{x}{x^2+7x+12}$

c. $\frac{2x+3}{x-1} = \frac{10}{x^2-1} + \frac{2x-3}{x+1}$

d. $\frac{3x+5}{x^2+4x-3} + \frac{x}{x+1} = \frac{2}{x+3}$

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11. Find the solution set to each inequality

a. $\frac{x+4}{x-3} \leq 3$

b. $\frac{5}{x+3} > \frac{3}{x+1}$

c. $\frac{x}{x-3} \leq \frac{-8}{x-6}$

d. $\left|\frac{x+2}{x-1}\right| \leq 3$

12. Find the partial decomposition to each expression.

a. $\frac{8x+4}{x^2+x-6}$

b. $\frac{x}{x^2-3x+2}$

c. $\frac{x+1}{x^3-4x^2+4x}$

d. $\frac{x-3}{(x+2)(x+1)^2}$

13. Find the vertical asymptotes of the function

a. $f(x) = \frac{4}{x^2-1}$

b. $f(x) = \frac{x}{x^2+1}$

c. $f(x) = \frac{x-2}{2x^2-9x-5}$

d. $f(x) = \frac{x+3}{9x^2-6x+1}$

e. $f(x) = \frac{x-3}{2x^2-7x+3}$

f. $f(x) = \frac{x^2-4}{x^2+x-6}$

14. Find the horizontal asymptote (if any). Determine the point where t graph crosses the horizontal asymptote

a. $f(x) = \frac{3-2x}{5x+1}$

b. $f(x) = \frac{5}{x^2+2x+1}$

c. $f(x) = \frac{x^2-4}{x+1}$

d. $f(x) = \frac{-4x^2+3x+2}{x^2-4}$

e. $f(x) = \frac{x^2+4x+1}{x^3+2x+3}$

f. $f(x) = \frac{1}{(x-2)^2} - 3$

15. Find an equation of a rational function f that satisfies the given conditions.

a. an x - intercept: 3, vertical asymptote: $x = 4$ horizontal asympto

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$y = -\frac{1}{2}$

b. an x - intercept: -2, vertical asymptote: $x = 3$

horizontal asymptote: $y = -1$ and has hole as $x = 4$

16. Let $f(x) = \frac{5x - 8}{x^2 - 4}$

a. As $x \longrightarrow 2^+$, the values of $f(x) \longrightarrow$ _____

b. As $x \longrightarrow 2^-$, the values of $f(x) \longrightarrow$ _____

c. As $x \longrightarrow -2^+$, the values of $f(x) \longrightarrow$ _____

d. As $x \longrightarrow -2^-$, the values of $f(x) \longrightarrow$ _____

e. As $x \longrightarrow \infty$, the values of $f(x) \longrightarrow$ _____

f. As $x \longrightarrow -\infty$, the values of $f(x) \longrightarrow$ _____

17. Let $f(x) = \frac{1 - x^2}{x^2 - 7x + 12}$

a. As $x \longrightarrow 3^+$, the values of $f(x) \longrightarrow$ _____

b. As $x \longrightarrow 3^-$, the values of $f(x) \longrightarrow$ _____

c. As $x \longrightarrow 4^+$, the values of $f(x) \longrightarrow$ _____

d. As $x \longrightarrow 4^-$, the values of $f(x) \longrightarrow$ _____

e. As $x \longrightarrow -\infty$, the values of $f(x) \longrightarrow$ _____

f. As $x \longrightarrow \infty$, the values of $f(x) \longrightarrow$ _____

18. Two pipes can fill a tank in $1\frac{1}{2}$ hr . One pipe requires 4hr longer running alone to fill the tank than the other. How long would it take for the faster pipe, working alone, to fill the tank ?

19. Takele and Obsa can complete a job in 16 days. Obsa alone can do it in 24 days. How long will Takele alone take to finish the whole work?

20. Person A can deliver a job 3 times as fast as person B. If they work together, it takes them 6hr. How long would it take each to deliver a job a lone?

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Answer to Review Exercise 2

1. Rational expression is the ratio of two polynomial expression.
2. The domain of a rational expression is the set of all real number except that cause the denominator zero.
3. The ratio of two polynomial function is a rational function.
4. The line $x = a$ is a vertical asymptote of f, if $y \to \pm\infty$ as $x \to a$ from the right or left.
5. The line $y = b$ is a horizontal asymptote of a function $y = f(x)$ if $y \to b$ as $|x| \to \infty$
6. If the degree of the numerator is one more than the degree of the denominator.

7. a. $-\frac{1}{5}$ b. $\frac{x+3}{2(x+1)}$ c. $\frac{1}{x-2}$ d. $\frac{2}{(x-3)(x+2)}$

8. a. $\frac{x-1}{x-2}$ b. $-(x-1)$ c. $\frac{2}{x}$ d. $\frac{-5}{x}$

9. a. $\{-1\}$ b. $\{1, -5\}$ c. $\varnothing$ d. $\{3\}$

10. a. $\mathbb{R}/\{0, 4\}$ b. $\mathbb{R}/\{-3, -4\}$ c. $\varnothing$ d. $\varnothing$

11. a. $\left\{x : \frac{-13}{2} \le x < 3\right\}$ c. $[-6, 3) \cup [4\ 6)$

b. $(-\infty, -3) \cup (-1, 2)$ d. $\left(-\infty, \frac{1}{4}\right] \cup \left[\frac{5}{2}, \infty\right)$

12. a. $\frac{8x+4}{(x-2)(x+3)} = \frac{A}{x-2} + \frac{B}{x+3} = \frac{4}{x-2} + \frac{4}{x+3}$

b. $\frac{x}{x^2-3x+2} = \frac{x}{(x-2)(x-1)} = \frac{A}{x-2} + \frac{B}{x-1} = \frac{2}{x-2} - \frac{1}{x-1}$

c. $\frac{x+1}{x(x-2)^2} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2} = \frac{1}{4x} + \frac{-11}{4(x-2)} + \frac{3}{2(x-2)^2}$

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d. $\dfrac{x-3}{(x+2)(x+1)^2} = \dfrac{A}{x+2} + \dfrac{B}{x+1} + \dfrac{C}{(x+1)^2} = \dfrac{-5}{x+2} + \dfrac{5}{x+1} + \dfrac{-4}{(x+1)^2}$

13. a. $x = 1$ and $x = -1$ c. $x = \dfrac{-1}{2}$, $x = 5$ e. $x = \dfrac{1}{2}$

b. No- vertical asymptote d. $x = \dfrac{1}{3}$ f. $x = -3$

14.

a. $y = \dfrac{-2}{5}$, Graph does not cross $y = \dfrac{-2}{5}$

b. $y = 0$, graph does not cross $y = 0$

c. No horizontal asymptote.

d. $y = -4$, graph cross H. A at $(-6, -4)$.

e. $y = 0$, graph cross H. A at $\left(-2+\sqrt{3},0\right)$ and $\left(-2-\sqrt{3},0\right)$

f. $y = -3$, graph does not cross, $y = -3$

15. a. $f(x) = \dfrac{-1(x-3)}{2(x-4)} = \dfrac{-x+3}{2x-8}$

b. $f(x) = \dfrac{-(x-2)(x-4)}{x-3(x-4)} = \dfrac{-(x^2-2x-8)}{x^2-7x+12} = \dfrac{-x^2+2x+8}{x^2-7x+12}$

16. a. As $x \longrightarrow 2^+, f(x) \longrightarrow \infty$

b. As $x \longrightarrow 2^-, f(x) \longrightarrow -\infty$

c. As $x \longrightarrow -2^+, f(x) \longrightarrow \infty$

d. As $x \longrightarrow -2^-, f(x) \longrightarrow -\infty$

e. As $x \longrightarrow \infty$, $f(x) \longrightarrow 0$

f. As $x \longrightarrow -\infty, f(x) \longrightarrow 0$

17. a. As $x \longrightarrow 3^+$, $f(x) \longrightarrow -\infty$

b. As $x \longrightarrow 3^-$, $f(x) \longrightarrow \infty$

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c. As $x \longrightarrow 4^{+}$, $f(x) \longrightarrow -\infty$

d. As $x \longrightarrow 4^{-}$, $f(x) \longrightarrow \infty$

e. As $x \longrightarrow -\infty$, $f(x) \longrightarrow -1$

f. As $x \longrightarrow \infty$, $f(x) \longrightarrow 1$

18. $\frac{t_1(t_1+1)}{2t_1+4} = \frac{3}{2}$ implies that $t_1^2 + t - 6 = 0 \quad \therefore \; t = 2$

19. $\frac{t_1 \cdot t_2}{t_1+t_2} = 16 \Rightarrow \frac{24t_2}{24+t_2} = 16, \; t_2 = 48$ days

20. $\frac{t_1(3t_2)}{4t_1} = 6 \Rightarrow t_1 = \frac{6 \times 4}{3} = 8$ hr

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UNIT 3

3. Matrices

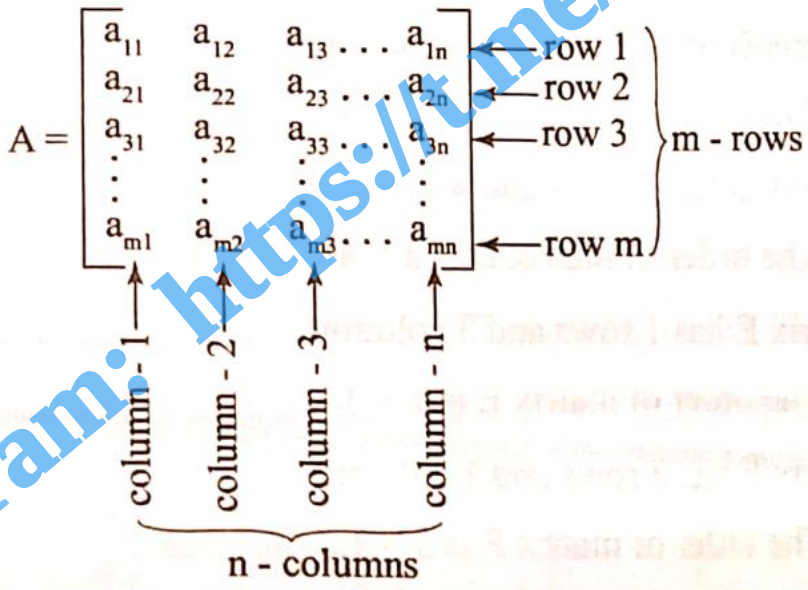
If you have ever used a spreadsheet program on the computer, you have worked with matrices.

What is a Matrix?

A matrix is a rectangular arrangement of numbers in rows and columns. It is usually described by the number of rows and columns called the **dimensions or order of matrix.**

DEFINITION of Matrix

An m × n **matrix** is a rectangular array of a numbers with m **rows** and n **columns.**



- Matrix A has m × n dimension (order) or we can say that the **order of matrix** is m × n.
- Each entry in a matrix is called an **element** of the matrix.
- Elements in the **horizontal line** are called rows.
- Elements in the **vertical line** are called column.

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- The number (a_{ij}) are the **entries** or element of the matrix.
- The subscript on the entry (a_{ij}) indicates element in the i^{th} **row** and j^{th} **column.**

Illustrative Example

1. Determine the order of each matrix.

a. $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \\ 0 & -1 \end{bmatrix}$ b. $B \begin{bmatrix} 1 & \sqrt{2} & 3 \\ 1 & 4 & 3 \end{bmatrix}$ c. $D. \begin{bmatrix} a & 1 & 3 & 8 \\ 2 & 0 & 5 & 4 \\ 7 & 1 & b & c \end{bmatrix}$

d. $E = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix}$ e. $F = \begin{bmatrix} 3 \\ 8 \\ 1 \end{bmatrix}$ f. $G = [2]$

Solution

a. Matrix A has 3 rows and 2 columns

∴ The order of matrix A is 3×2

b. Matrix B has 2 rows and 3 columns

∴ The order of matrix B is 2×3

c. Matrix D has 3 rows and 4 columns

∴ The order of matrix D is 3×4.

d. Matrix E has 1 rows and 3 columns.

∴ The order of matrix E is 1×3.

e. Matrix F has 3 rows and 1 column

∴ The order of matrix F is 3×1.

f. Matrix G has 1 row and 1 column.

∴ The order of matrix G is 1×1.

2. Let $A = \begin{bmatrix} -1 & 1 & 0 & \pi \\ \sqrt{2} & -1 & 3 & 8 \\ 2 & 7 & 4 & 5 \end{bmatrix}$, where $A = [a_{ij}]$

Determine the value of each element

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a. a_{23} b. a_{32} c. a_{13} d. a_{31}

Solution

a. a_{23} represents the element in the 2nd row, 3rd column; $a_{23} = 3$.

b. a_{32} represents the element in the 3rd row, 2nd column: $a_{32} = 7$.

c. a_{13} represents the element in the 1st row, 3rd column: $a_{13} = 0$.

d. a_{31} represent the element in the 3rd row, 1st column: $a_{31} = 2$

3. Which element in matrix: $A = \begin{bmatrix} \frac{1}{3} & 2 & 5 & 0 \end{bmatrix}$ is the number 0.

A. a_{41} B. a_{14} C. a_{13} D. a_{31}

Solution

0 is the element in the 1st row, 4th column

Answer: B

Equality of Matrices

Two matrices are equal if and only if they have the same element in the same positions with same order (or the same dimension).

Example

$$\begin{bmatrix} 2 & -3 \\ a & b \end{bmatrix} = \begin{bmatrix} 2 & c \\ 7 & 1 \end{bmatrix}$$

$a = 7$, $b = 1$ and $c = -3$

Illustrative Example

4. Given $A = \begin{bmatrix} 3 & a \\ b & -4 \end{bmatrix}$ and $B = \begin{bmatrix} c & 7 \\ 5 & -4 \end{bmatrix}$

For what values of a, b and c will the matrix A = B?

Solution

$a = 7$, $b = 5$ and $c = 3$

5. Let $A = [a_{ij}]_{3\times 2}$ and $a_{ij} = i + j$

Find the elements of matrix A

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$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} 1+1 & 1+2 \\ 2+1 & 2+2 \\ 3+1 & 3+2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 4 & 5 \end{bmatrix}$$

6. Let $A = \begin{bmatrix} 2x-5 & 4 \\ 3 & 3y-8 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 4 \\ 3 & y-2 \end{bmatrix}$

for what values of x, y, and z will the matrix A = B.

Solution

$A = B$ if and only if $2x - 5 = 7$ and $3y - 8 = y - 2$

$\Rightarrow 2x = 5 + 7$ and $3y - y = -2 + 8$

$\therefore \quad x = \frac{12}{2} = 6$ and $2y = 6$, $y = 3$

7. Let $A = \begin{bmatrix} 3 & x & w+3 \\ 4y & 2 & 5z \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2w-1 & 2x-9 \\ 20 & 2 & z-12 \end{bmatrix}$

For what value of x, y, w and z will matrix A = B?

Solution

Since A = B, then $a_{ij} = b_{ij}$. i,e

$a_{11} = b_{11}$, $a_{12} = b_{12}$ and $a_{13} = b_{13}$

$x = 2w - 1$ and $w + 3 = 2x - 19$

By substitution: $w + 3 = 2(2w - 1) - 19$

$-3w = -21 - 3 = -24$, $w = \frac{24}{3} = 8$

$\therefore \quad x = 2w - 1 = 2(8) - 1 = 15$

$\therefore \quad x = 15$, $w = 8$, $y = 5$ and $z = -3$

3.1 Addition and Subtraction of Matrices

Two matrices can be added or subtracted if they have the same dimension. Other wise, their sum or difference is undefined.

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Sum and difference of matrices

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be matrices of the same dimension $m \times n$.

- The **sum** $A + B$ is the $m \times n$ matrix obtained by adding corresponding entries of A and B.

i. $A + B = [a_{ij} + b_{ij}]$ ii. $A - B = [a_{ij} - b_{ij}]$

- The sum (difference) of two matrices is found by adding their corresponding elements.

Scalar Multiplication of Matrices

To multiply a matrix by a number K, we multiply every element of the matrix by K.

The scalar product KA is the $m \times n$ matrix obtained by multiplying each entry A by K.

$KA = [Ka_{ij}]$ where $A = [a_{ij}]$

Illustrative Example

1. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ -1 & 4 \end{pmatrix}$. Then which of the following is equal to $3A + 2B$?

A. $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ B. $\begin{pmatrix} 3 & 4 \\ 2 & 18 \end{pmatrix}$ C. $\begin{pmatrix} 7 & 4 \\ 7 & 18 \end{pmatrix}$ D. $\begin{pmatrix} 7 & 1 \\ 7 & 7 \end{pmatrix}$

Solution

$$3A + 2B = 3\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 2\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix} + \begin{pmatrix} 4 & -2 \\ -2 & 6 \end{pmatrix} = \begin{pmatrix} 7 & 4 \\ 7 & 18 \end{pmatrix}$$

Answer: C

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Solving Matrix Equations

BY using the properties of matrix addition, subtraction and of scalar multiplication we can solve matrix equations.

Illustrative Example

9. Let $A = \begin{bmatrix} 1 & 3 \\ 2 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 3 & 6 \end{bmatrix}$ Solve for X

a. $2X - B = A$

b. $3X + A = B$

c. $B - A = 5X$

d. $B + 10X = 2A$

Solution

a. $2X - B = A$ implies that $2X = A + B$

$$\therefore \quad 2X = A + B = \begin{pmatrix} 1 & 3 \\ 2 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 4 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 7 & 4 \end{pmatrix}$$

$$\therefore \quad X = \frac{1}{2}\begin{pmatrix} 2 & 7 \\ 5 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 3.5 \\ 2.5 & 2 \end{pmatrix}$$

b. $3X + A = B$ implies that $3X = B - A$

$$\therefore \quad 3X = \begin{pmatrix} 1 & 4 \\ 3 & 6 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 8 \end{pmatrix}$$

$$\therefore \quad X = \frac{1}{3}\begin{pmatrix} 0 & 1 \\ 1 & 8 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & \frac{8}{3} \end{pmatrix}$$

c. $5X = B - A = \begin{pmatrix} 0 & 1 \\ 1 & 8 \end{pmatrix}$

$$\therefore \quad X = \frac{1}{5}\begin{pmatrix} 0 & 1 \\ 1 & 8 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{5} \\ \frac{1}{5} & \frac{8}{5} \end{pmatrix}$$

d. $B + 10X = 2A$, implies that $10X = 2A - B$

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$$10X = 2\begin{pmatrix}1 & 3\\ 2 & -2\end{pmatrix} - \begin{pmatrix}1 & 4\\ 3 & 6\end{pmatrix} = \begin{pmatrix}2 & 6\\ 4 & -4\end{pmatrix} - \begin{pmatrix}1 & 4\\ 3 & 6\end{pmatrix}$$

$$X = \frac{1}{10}\begin{pmatrix}1 & 2\\ 2 & -10\end{pmatrix} = \begin{pmatrix}0\cdot1 & 0\cdot2\\ 0\cdot1 & -1\end{pmatrix}$$

10. Let $A = \begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}$, $B = \begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix}$, $C = \begin{pmatrix}-1 & 0\\ 1 & 1\end{pmatrix}$ and $D = \begin{bmatrix}3 & 3\\ 0 & -2\end{bmatrix}$

for what values of a,b and c, the matrix D is a linear combination (aA + bB + cC) of matrix A, B and C.

A. 1, -2, 1 B. 2, 2, -2 C. 1, 1, -1 D. 2, 1, -1

Solution

we have given: aA + bB + cC = D

$$a\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix} + b\begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix} + c\begin{pmatrix}-1 & 0\\ 1 & 1\end{pmatrix} = \begin{bmatrix}3 & 3\\ 0 & -2\end{bmatrix}$$

$$\begin{pmatrix}a & a\\ 0 & -a\end{pmatrix} + \begin{pmatrix}0 & b\\ b & b\end{pmatrix} + \begin{pmatrix}-c & 0\\ c & c\end{pmatrix} = \begin{pmatrix}3 & 3\\ 0 & -2\end{pmatrix}$$

a + 0 - c = 3 and a + b + 0 = 3

0 + b + c = 0 and -a + b + c = -2

(i) Now b + c = 0, substitute in -a + b + c = -2

$\Rightarrow$ -a + 0 = -2 implies that a = 2.

(ii) Put a = 2 in a + b = 3

$\Rightarrow$ 2 + b = 3 implies that b = 3 - 2 = 1

(iii) Put a = 2 in a + 0 - c = 3

$\Rightarrow$ 2 - c = 3 implies that c = -1

Answer: D

11. Find matrix A and B if $2A + B = \begin{pmatrix}2 & 3\\ -1 & 4\end{pmatrix}$ and $A - 2B = \begin{pmatrix}4 & 1\\ -2 & 3\end{pmatrix}$

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Solution

$$2(2A + B) = 4A + 2B = \begin{pmatrix} 4 & 6 \\ -2 & 8 \end{pmatrix}$$

$$\therefore \quad (4A + 2B) + (A - 2B) = \begin{pmatrix} 4 & 6 \\ -2 & 8 \end{pmatrix} + \begin{pmatrix} 4 & 1 \\ -2 & 3 \end{pmatrix}$$

$$5A = \begin{pmatrix} 8 & 7 \\ -4 & 11 \end{pmatrix}, \text{ then } A = \frac{1}{5}\begin{pmatrix} 8 & 7 \\ -4 & 11 \end{pmatrix}$$

$$\text{Also, } 2A + B = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}, \text{ and } B = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix} - 2A$$

$$\therefore \quad B = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix} - \frac{2}{5}\begin{pmatrix} 8 & 7 \\ -4 & 11 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -4 & 8 \\ 3 & -2 \end{pmatrix}$$

Properties of Addition and Scalar Multiplication of Matrices

Let A, B and C be m x n matrices and let c and d be scalars.

(i) $A + B = B + A$ • Commutative property of matrix Addition

(ii) $(A + B) + C = A + (B + C)$ • Associative property of matrix addition

(iii) $c(dA) = (cd)A$ • Asso. prop. of scalar multiplication

(iv) $(c + d)A = cA + dA$ • Distributive properties of scalar.

(v) $c(A + B) = cA + cB$ • Multiplication

3.2 Multiplication of Two Matrices

To perform **matrix multiplication,** multiply the elements of each row of the first matrix by the elements of each column of the second matrix then add the products.

DEFINITION: Product of Two Matrices

Let A be an $m \times n$ and let B be an $n \times k$ matrix. To find element C_{ij} of the product matrix AB, multiply each element in the **ith row** of A by the corresponding element in the **jth column** of B, and adding the results. The order of product matrix AB is an $m \times k$ matrix.

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In symbols: C_{ij} is the sum of the product of the element in the i^{th} row of A and j^{th} column of B

Illustrative Example

12. Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3\times2}$

Given product matrix: $AB = [C_{ij}]$, explain how to find:

a. C_{11} c. C_{21} e. C_{23} g. C_{32}

b. C_{12} d. C_{22} f. C_{31} h. C_{33}

Solution

Since A is a 3 × 3 matrix and B is a 3 × 2 matrix, then the product AB will be a matrix of order 3 × 2.

a. C_{11} is sum of the product of the elements in the 1st row of A and 1st column of B.

$$\therefore \quad c_{11} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \cdot \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = a_{11}\ b_{11} + a_{12}\ b_{21} + a_{13}\ b_{31}$$

b. C_{12} is the sum the product of the elements in the 1st row of A and 2nd column B.

$$c_{12} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \cdot \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = a_{11}\ b_{12} + a_{12}\ b_{22} + a_{13}\ b_{32}$$

c. C_{21} is the sum of the product of the elements in 2nd row of A and 1st column of B

$$\therefore \quad c_{21} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \end{bmatrix} \cdot \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix} = a_{21}\ b_{11} + a_{22}\ b_{12} + a_{23}\ b_{13}$$

d. C_{22} exercise left for you.

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e. C_{23} is does not exist (undefined) because C_{23} is the sum of the product of elements of 2nd row of A and 3rd column of B (but 3rd column of B does not exist).

f. C_{31} is the sum of the product of the elements of 3rd row of A and 1st column of B.

$$\therefore \quad c_{31} = \begin{bmatrix} a_{31} & a_{32} & a_{33} \end{bmatrix} \cdot \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix} = a_{31}\ b_{11} + a_{32}\ b_{12} + a_{33}\ b_{13}$$

g. C_{32} Exercise left for you

h. C_{33} does not exist (not defined).

13. Let $A = \begin{pmatrix} 1 & -2 & 4 \\ 2 & 3 & 1 \end{pmatrix}_{2 \times 3}$ and $B = \begin{pmatrix} 2 & 3 \\ -1 & 2 \\ 2 & -4 \end{pmatrix}_{3 \times 2}$.

Then find a. AB b. BA

Solution

a. Since A is a 2×3 matrix and B is a 3×2 matrix, then the product AB will be a matrix of order 2×2

$$C_{11} = \begin{bmatrix} 1 & -2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 1 \times 2 + (-2) \times (-1) + 4 \times 3 = 16$$

$$C_{12} = \begin{bmatrix} 1 & -2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} = 1 \times 3 + (-2) \times 2 + 4(-4) = -17$$

$$C_{21} = \begin{bmatrix} 2 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 2 \times 2 + 3 \times (-1) + 1 \times 3 = 4$$

$$C_{22} = \begin{bmatrix} 2 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} = 2 \times 3 + 3 \times 2 + 1 \times (-4) = 8$$

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$$\therefore \quad C = AB = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} 16 & -17 \\ 4 & 8 \end{bmatrix}$$

b. $BA = [C_{ij}]$ means that C_{ij} is the sum of the product of the elements of i^{th} row of B and j^{th} column of B.

Since B is 3×2 matrix and A is a 2×3 matrix, then the product BA will be a matrix of order 3×3.

C_{11} = sum of the product of elements of 1st row in B and 1st column of A

$$C_{11} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2 \times 2 + 3 \times 2 = 10$$

$$C_{12} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = 2 \times (-2) + 3 \times 3 = -4 + 9 = 5$$

$$C_{13} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = 2 \times 4 + 3 \times 1 = 8 + 3 = 1$$

$$C_{21} = \begin{bmatrix} -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = (-1)(1) + 2 \times 2 = -1 + 4 = 3$$

$$C_{22} = \begin{bmatrix} -1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = (-1)(-2) + 2 \times 3 = 8$$

$$C_{23} = \begin{bmatrix} -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = (-1)(4) + 2 \times 1 = -4 + 2 = -2$$

$$C_{31} = \begin{bmatrix} 3 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 \times 1 + (-4) \times 2 = 3 - 8 = -5$$

$$C_{32} = \begin{bmatrix} 3 & -4 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = 3 \times (-2) + (-4) \times 3 = -6 - 12 = -18$$

$$C_{33} = \begin{bmatrix} 3 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = 3 \times 4 + (-4) \times 1 = 12 - 4 = 8$$

$$\therefore \quad C = BA = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} 10 & 5 & 11 \\ 3 & 8 & -2 \\ -5 & -18 & 8 \end{bmatrix}$$

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If $A = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ 4 & 2 \end{bmatrix}$ then $3A+7B-2AB$ is equation

A. $\begin{bmatrix} -1 & 5 \\ -1 & 8 \end{bmatrix}$ B. $\begin{bmatrix} 1 & -5 \\ 1 & -8 \end{bmatrix}$ C. $\begin{bmatrix} 1 & 5 \\ 1 & 8 \end{bmatrix}$ D. $\begin{bmatrix} 1 & 5 \\ -1 & -8 \end{bmatrix}$

Solution: Exercise left for you

Note: The product AB is not always possible.

- The product AB of two matrices A and B exists only if the number of **columns** of A is equal to the number of **rows of** B. That is columns of A = rows of **B**.

The order (size) of product matrix.

If A is an m × k matrix and B is a k × n matrix, then the product AB is an m × n matrix.

Example

Size of A	Size of B	Size of AB
• 2 × 3	3 × 4	2 × 4
• 3 × 5	5 × 2	3 × 2
• 4 × 1	1 × 3	4 × 3
• 1 × 4	4 × 1	1 × 1
• 3 × 2	2 × 3	3 × 3
• 3 × 5	3 × 5	AB is not defined

Conceptual Example

14. What is a matrix?

Explanation:- A matrix is a rectangular array of number.

15. What is the difference between a row and a column of a matrix?

Explanation:- A row runs horizontal and a column runs vertically.

16. What is the order of a matrix?

Explanation:- The order of a matrix is the number of rows and

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columns of matrix.

17. If A is an m × n, what do the m and n represent?

Explanation

- m is the number of rows of matrix.
- n is the number of columns of matrix.

18. What are the requirement needed to multiply two matrices A and B?

Explanation

(i) The product AB is possible if the number of columns in A be equal to the number of rows in B.

(ii) The product BA is possible if the number of columns in B be equal to the number of rows in A.

19. If A is 4 × 7 matrix and B is 7 × 6 then what will be the order of the product AB?

Explanation

The resulting matrix will have dimensions equal to the numbers of rows of A by the number of columns of B.

△ So that the product AB will be 4 × 6 matrix.

20. Given that A is a 4 × 3 matrix and B is a 3 × 2 matrix.

a. Is AB defined? If so what is the order of AB?

b. Is BA defined? If so what is the order of BA?

Solution

a. Yes, 4 × 2 b. No

21. Given that A is a 3 × 1 matrix and B is a 1 × 3 matrix.

a. Is AB defined? If so what is the order of AB?

b. Is BA defined? If so what is the order of BA?

Solution

a. Yes, size: 3 × 3 b. Yes, size: 1 × 1.

22. If A is an m × n and B is a p × n are two matrices, state the requirements needed for the existence of

a. A + B b. AB c. BA

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Solution

a. A + B is defined if matrix A and matrix B both have the same order.

Δ So that A + B is defined if m = P and n = q

b. AB is defined if number of columns of A is equal to the number of rows in B.

Δ So that AB is defined if n = P

c. BA is defined if number of columns of B is equal to the number of rows in A.

Δ So that BA is defined if q = m

23. Let $A = \begin{pmatrix} 2 & 3 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 5 & -4 \\ 3 & 2 & 6 \end{pmatrix}$, then AB is equal to

A. $\begin{pmatrix} 16 & 8 \\ 10 & -14 \end{pmatrix}$

C. $\begin{pmatrix} 8 & 13 & -18 \\ -1 & 8 & -2 \end{pmatrix}$

B. $\begin{pmatrix} 13 & 16 & 10 \\ 1 & 8 & -14 \end{pmatrix}$

D. $\begin{pmatrix} 8 & -1 \\ 13 & 8 \\ 18 & -2 \end{pmatrix}$

Solution

Since A is a 2 × 2 matrix and B is a 2 × 3 matrix, the resulting matrix AB is order of 2 × 3 matrix. Then:

$$C_{11} = (2 \quad 3)\begin{pmatrix} 2 \\ 3 \end{pmatrix} = (2)(2) + (3)(3) = 13$$

$$C_{12} = (2 \quad 3)\begin{pmatrix} 5 \\ 2 \end{pmatrix} = 2 \times 5 + 3 \times 2 = 16$$

$$C_{13} = (2 \quad 3)\begin{pmatrix} -4 \\ 6 \end{pmatrix} = 2 \times (-4) + 3 \times 6 = 10$$

$$C_{21} = (2 \quad -1)\begin{pmatrix} 2 \\ 3 \end{pmatrix} = 4 + (-3) = 1$$

$$C_{22} = (2 \quad -1)\begin{pmatrix} 5 \\ 2 \end{pmatrix} = 2 \times 5 + (-1) \times 2 = 8$$

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$$C_{23} = (2 \quad -1)\begin{pmatrix} -4 \\ 6 \end{pmatrix} = 2 \times (-4) + (-1) \times 6 = -14$$

$$\therefore \quad C = AB = \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \end{pmatrix} = \begin{pmatrix} 13 & 16 & 10 \\ 1 & 8 & -14 \end{pmatrix}$$

Answer: B

24. Let $A = \begin{pmatrix} 4 & 4 & -2 \\ 2 & 6 & 0 \\ 3x & 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 2 \\ 3 & 0 \\ -1 & 5 \end{pmatrix}$ are given if

$C_{ij} = AB$ and $C_{32} = 11$, then what is the value of x?

A. $\frac{3}{5}$ B. $\frac{-4}{3}$ C. $\frac{-2}{3}$ D. -8

Solution

C_{32} is the sum of the product of the element in 3rd row of A and 2nd column of B.

$$\therefore \quad C_{32} = \begin{pmatrix} 3x & 1 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} = (3x)(2) + 1 \times 0 + 3 \times 5 = 11$$

$$\Rightarrow 6x + 15 = 11 \text{ implies that } 6x = -4$$

$$\therefore \quad x = \frac{-4}{6} = \frac{-2}{3}$$

Answer: C

25. Let $A = \begin{pmatrix} -2 & x+1 \\ y & 0 \end{pmatrix}, B = \begin{pmatrix} 3 & z-1 \\ 2 & 1 \end{pmatrix}$ and $= \begin{pmatrix} 0 & -7 \\ -3 & -5 \end{pmatrix}$

for what values of x, y, and z respectively, so that $AB = C$

A. -2, -1, 6 B. 2, -1, 6 C. 2, -1, -6 D. 2, 1, 6

Solution

We have given $C_{11} = 0$, $C_{12} = -7$, $C_{21} = -3$, $C_{22} = -5$

$$C_{11} = (-2 \quad x+1)\begin{pmatrix} 3 \\ 2 \end{pmatrix} = -2 \times 3 + 2(x+1) = 0 \Rightarrow 2x + 2 - 6 = 0$$

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$2x - 4 = 0$ implies $x = 2$

$$C_{12} = (-2 \quad x+1)\begin{pmatrix} z-1 \\ 1 \end{pmatrix} = (-2z + 2) + (x + 1) = -7$$

$-2z + x + 3 = -7$ and $-2z + x = -10$

Put $x = 2$ in $-2z + x = -10$

$\Rightarrow$ $-2z + 2 = -10$ implies $z = \dfrac{-12}{-2} = 6$

$$C_{21} = (y \quad 0)\begin{pmatrix} 3 \\ 2 \end{pmatrix} = 3y + 0 = -3, \text{ implies } y = -1$$

$\therefore$ $x = 2$, $y = -1$ and $z = 6$ **Answer**

26. For what values x, y, z and W the matrix

$$\begin{pmatrix} 2 & 1 \\ x & y \end{pmatrix}\cdot\begin{pmatrix} 8 & -3 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} Z & W \\ -16 & 8 \end{pmatrix}$$

Solution

We have $C_{11} = Z$, $C_{12} = W$, $C_{21} = -16$

$$C_{11} = (2 \quad 1)\begin{pmatrix} 8 \\ 0 \end{pmatrix} = 2 \times 8 + 1 \times 0 = 16$$

$$C_{12} = (2 \quad 1)\begin{pmatrix} -3 \\ -2 \end{pmatrix} = 2 \times (-3) + 1 \times (-2) = -6 - 2 = -8$$

$$C_{21} = (x \quad y)\begin{pmatrix} 8 \\ 0 \end{pmatrix} = 8x$$

$$C_{22} = (x, \quad y)\begin{pmatrix} -3 \\ -2 \end{pmatrix} = -3x - 2y$$

$$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}\cdot\begin{pmatrix} 16 & -8 \\ 8x & -3x - 2y \end{pmatrix} = \begin{pmatrix} Z & W \\ -16 & 8 \end{pmatrix}$$

$\therefore$ $Z = 16$, $8x = -16$, $x = -2$, $W = -8$

Also $-3x - 2y = 8$, put $x = -2$

$\Rightarrow$ $-3(-2) - 2y = 8$ implies $-2y = 2$, $y = -1$

$\therefore$ $x = -2$, $y = -1$, $Z = 16$ and $W = -1$

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27. What are the values of x and y so that

$$\begin{pmatrix} x & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 4 & 1 \end{pmatrix}=\begin{pmatrix} y & y \\ 2 & 1 \end{pmatrix}?$$

A. $x=0,\ y=-1$ C. $x=4,\ y=5$

B. $x=3,\ y=2$ D. $x=-2,\ y=2$

Solution

$$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}=\begin{pmatrix} 2x-4 & x-1 \\ 2 & 1 \end{pmatrix}=\begin{pmatrix} y & y \\ 2 & 1 \end{pmatrix}$$

$\Rightarrow$ $2x - 4 = y$ and $x - 1 = y$. solving together

$2x - 4 = x - 1 \Rightarrow x = -1 + 4 = 3$

Also, $y = x - 1 = 3 - 1 = 2$

Answer: B

28. If $\begin{pmatrix} 2 & -1 \\ a & 4 \end{pmatrix}\begin{pmatrix} 1 & b \\ -1 & 3 \end{pmatrix}=\begin{pmatrix} 3 & 1 \\ -1 & 18 \end{pmatrix}$.

Then what are the values of a and b respectively

A. 3, 2 B. 2, -2 C. -1, 2 D. 0, 1

Solution

$$C_{12}=\begin{pmatrix} 2 & -1 \end{pmatrix}\begin{pmatrix} b \\ 3 \end{pmatrix}=2b-3=1,\ b=2$$

$$C_{21}=\begin{pmatrix} a & 4 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix}=a-4=-1,\ a=-3$$

Answer: A

29. Let $A=\begin{pmatrix} -3 & 7 & 2 \end{pmatrix}$ and $B=\begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix}$. Then find

a. AB b. BA

Solution

a. A is a 1×3 matrix and B is a 3×1 matrix, then the resulting matrix AB is 1×1 matrix.

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$$\therefore \quad C_{11} = \begin{pmatrix} -3 & 7 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix} = -3 \times 1 + 7 \times 4 + 2 \times (-5) =$$

b. Since the rows of B is 3 and columns of A is 3, then the result matrix BA is 3×3 matrix.

$$\begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix} = \begin{pmatrix} -3 & 7 & 2 \\ -12 & 28 & 8 \\ 15 & -35 & -10 \end{pmatrix}$$

Properties of Matrix Multiplication

(i) Matrix multiplication is not commutative : $AB \neq BA$

(ii) Matrix multiplication is associative: $(AB)C = A(BC)$

(iii) Matrix multiplicative is distributive with respect to Addition: $A(B + C) = AB + AC$

Illustrative Example

30. True of false? $\begin{bmatrix} 5 & 3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -3 & 5 \end{bmatrix}$

Solution False

$$\begin{pmatrix} 5 \times 2 + 3 \times 0 & 5 \times 0 + 3 \times 1 \\ -3 \times 2 + 5 \times 0 & -3 \times 0 + 5 \times 1 \end{pmatrix} ? \begin{pmatrix} 2 \times 5 + 0 \times (-3) & 2 \times 3 + 0 \times 5 \\ 0 \times 5 + 1 \times (-3) & 0 \times 3 + 1 \times 5 \end{pmatrix}$$

$$\begin{pmatrix} 10 & 3 \\ -6 & 5 \end{pmatrix} \neq \begin{pmatrix} 10 & 6 \\ -3 & 5 \end{pmatrix}$$

$\therefore$ Matrix multiplication is not commutative.

31. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, then what is the value of A^n.

A. $\begin{bmatrix} n & 1 \\ 0 & 1 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 1 \\ 0 & n \end{bmatrix}$ C. $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 2 & 1 \\ 0 & \end{bmatrix}$

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Solution

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = A^2 . A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$$

Answer: C

32. Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$, what is a matrix B if AB = A + B.

A. $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 3 & -3 \end{pmatrix}$

B. $\frac{1}{2}\begin{pmatrix} 1 & 2 \\ -3 & 3 \end{pmatrix}$

C. $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ -3 & 3 \end{pmatrix}$

D. $\frac{1}{2}\begin{pmatrix} -1 & -1 \\ 3 & -3 \end{pmatrix}$

Solution : We have given AB = A + B

Let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $AB = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 2a-c & 2b-d \\ 3a & 3b \end{pmatrix}$

$$A + B = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix} + \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a+2 & b-1 \\ c+3 & d \end{pmatrix}$$

Since AB = A + B, then $\begin{pmatrix} 2a-c & 2d-d \\ 3a & 3b \end{pmatrix} = \begin{pmatrix} a+2 & b-1 \\ c+3 & d \end{pmatrix}$

Equality of matrix property:

i. $2a - c = a + 2$ and $3a = c + 3$

$c = a - 2$ by substitution: $3a = a - 2 + 3$,

$\therefore \quad a = \frac{1}{2}$ and $c = \frac{1}{2} - 2 = -\frac{3}{2}$

ii. $2b - d = b - 1$ and $3b = d$

By substitution for d; $2b - 3b = b - 1$

$\therefore \quad b = \frac{1}{2}$ and $d = 3b = 3\left(\frac{1}{2}\right) = \frac{3}{2}$

Answer:C

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3.3 Types of Matrices

I. **Row matrix** is a matrix having exactly single row

Example: (3 0 1), (-2 1 4, 7),.....

II. **Column matrix** is a matrix having exactly single column.

III. **Square matrix** is a matrix with the same number of rows and columns.

Example : $\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}_{2\times 2}$, $\begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 4 \\ 6 & -2 & 0 \end{bmatrix}_{3\times 3}$ ... etc

IV. **Identity matrix** is a square matrix in which all diagonal elements are 1' while all other element are 0.

Example

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},\ I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The real number 1 is the multiplicative identity for real numbers. Here, we define a multiplicative identity matrix I that has properties similar to the real number 1.

a.1 = a, b.1 = b and c.1 = c.

Multiplication with an identity matrix.

Identity Property

(i) If A is an $m \times n$ matrix, then $I_m A = A$ and $AI_n = A$

(ii) If A is an $n \times n$ matrix, then $IA = AI = A$.

33. Let $A = \begin{bmatrix} -1 & 2 & 1 \\ 3 & 0 & 4 \end{bmatrix}$, and $B = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 6 & 4 \end{bmatrix}$

Find a. AI_3 b. I_2A c. BI_2

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Solution

a. $AI_3 = \begin{bmatrix} -1 & 2 & 1 \\ 3 & 0 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 1 \\ 3 & 0 & 4 \end{bmatrix} = A$

b. $I_2A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 & 1 \\ 3 & 0 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 1 \\ 3 & 0 & 4 \end{bmatrix} = A$

c. $BI_2 = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 6 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 6 & 4 \end{bmatrix} = B$

Vi. **Triangular matrix** is a square matrix which is either upper triangular or lower triangular matrix.

a. **Upper triangular matrix** is a square matrix in which all the elements below the diagonal elements are zero.

Example : $\begin{pmatrix} 2 & 3 \\ 0 & 5 \end{pmatrix}, \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$ are upper triangular matrix

b. **Lower triangular matrix** is a square matrix in which all the elements above the diagonal elements are zero.

Example : $\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 & 0 \\ 5 & 1 & 0 \\ 3 & 4 & 6 \end{pmatrix}$ are lower triangular matrix

IV. Diagonal Matrices

A square matrix in which all non - diagonal elements are 0 is called a **diagonal matrix.**

Let $A = [a_{ij}]_{n\times n}$ be a square diagonal matrix of order n and if $a_{ij} = 0$, for all $i \neq j$.

- The elements $a_{11}, a_{22}, a_{33}, \ldots a_{nn}$ are called a **diagonal elements of A**
- The sum of the diagonal elements of A is called the **trace** of A, denoted by tr (A). That is, $tr(A) = a_{11} + a_{22} + a_{33} + \ldots.. + a_{nn}$

Example

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$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}, \begin{pmatrix} a_{11} & 0 & 0 & \dots & 0 \\ 0 & a_{22} & 0 & & 0 \\ 0 & 0 & a_{33} & & 0 \\ 0 & 0 & 0 & & a_{nn} \end{pmatrix}$$

V. Scalar Matrix is a diagonal matrix in which all the diagonal elemen are equal

Example

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \dots \begin{pmatrix} x & 0 & 0 & \dots & 0 \\ 0 & x & 0 & \dots & 0 \\ 0 & 0 & x & & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & x \end{pmatrix}_{n \times n}$$, for some $x \in \mathbb{R}$

3.4 Transpose of a Matrix

If we interchange rows into columns and columns into rows of an $m \times n$ matrix A we get an $n \times m$ matrix, which is called the **transpose** of A, denoted by A^t.

Example

i. If $A = \begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix}$, then $A^t = \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix}$

ii. If $A = \begin{pmatrix} -1 & 0 & 2 \\ 3 & 5 & 4 \end{pmatrix}_{2\times3}$, then $A^t = \begin{pmatrix} -1 & 3 \\ 0 & 6 \\ 2 & 4 \end{pmatrix}_{3\times2}$

iii. If $A = \begin{pmatrix} -2 & 0 & 2 \end{pmatrix}_{1\times3}$, then $A^t = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}_{3\times1}$

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Properties of transpose matrix

Len A and B denotes matrices and let k denote a scalar.

(i) If A is an m × n matrix, then At is an n × m matrix.

(ii) $(A^t)^t = A$ (iV) $(A + B)^t = A^t + B^t$

(iii) $(kA)^t = kA^t$ (V) $(AB)^t = B^t A^t$

Symmetric and skew symmetric matrix.

(i) A square matrix A is called symmetric if $A^t = A$

Example

a. Let $A = \begin{bmatrix} 3 & 2 \\ 2 & -1 \end{bmatrix}$, then $A^t = \begin{bmatrix} 3 & 2 \\ 2 & -1 \end{bmatrix}$, $A = A^t$

b. Let $A = \begin{bmatrix} a & x \\ x & b \end{bmatrix}$, then $A^t = \begin{bmatrix} a & x \\ x & b \end{bmatrix}$, $A = A^t$

c. Let $A = \begin{pmatrix} -2 & 3 & 5 \\ 3 & -1 & 4 \\ 5 & 4 & 8 \end{pmatrix}$, then $A^t = \begin{pmatrix} -2 & 3 & 5 \\ 3 & -1 & 4 \\ 5 & 4 & 8 \end{pmatrix}$

(ii) A square matrix A is called a **skew symmetric** matrix if $A^t = -A$

For skew - symmetric matrix $A = [a_{ij}]$, $a_{ij} = -a_{ji}$ i.e $a_{12} = -a_{21}$, $a_{13} = -a_{31}$ ... Putting $i = j$, $a_{ii} = -a_{ii}$ this implies $2a_{ii} = 0$ or $a_{ii} = 0$ for $i = j$

Thus, all diagonal elements of a skew -symmetric matrix are zero.

Example : $\begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 & -4 \\ -1 & 0 & 2 \\ 4 & -2 & 0 \end{bmatrix}$. are skew - symetric

Refer example - 36

Note: Every square matrix A can uniquely be expressed as the sum of symmetric matrix and a skew - symmetric matrix

Refer example - 38

For any square matrix A:

i. $\frac{1}{2}\left(A + A^t\right)$ is symmetric matrix.

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ii. $\frac{1}{2}\left(A - A^t\right)$ is a skew - symmetric matrix.

Illustrative Example

34. In each case find the matrix A, if:

a. $2A - \begin{bmatrix} 1 & 0 & -2 \\ 4 & 7 & 3 \end{bmatrix}^t = \begin{bmatrix} 2 & 0 \\ -3 & 4 \\ 0 & 8 \end{bmatrix}$

b. $\left\{A^t + 2\begin{bmatrix} 1 & -2 \\ 5 & 0 \end{bmatrix}\right\}^t = 5\begin{bmatrix} 3 & -1 \\ 1 & 7 \end{bmatrix}$

Solution

a. $2A = \begin{pmatrix} 2 & 0 \\ -3 & 4 \\ 0 & 8 \end{pmatrix} + \begin{pmatrix} 1 & 4 \\ 0 & 7 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ -3 & 11 \\ -2 & 11 \end{pmatrix}$

$\therefore A = \frac{1}{2}\begin{pmatrix} 3 & 4 \\ -3 & 11 \\ -2 & 11 \end{pmatrix} = \begin{pmatrix} 1\cdot 5 & 2 \\ 1\cdot 5 & 5\cdot 5 \\ -1 & 5\cdot 5 \end{pmatrix}$

b. we use: $(A + B)^t = A^t + B^t$

$\left(A^t\right)^t + 2\begin{bmatrix} 1 & -2 \\ 5 & 0 \end{bmatrix}^t = 5\begin{bmatrix} 3 & -1 \\ 1 & 7 \end{bmatrix}$

$A + 2\begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 15 & -5 \\ 5 & 35 \end{bmatrix}$

$\therefore A = \begin{bmatrix} 15 & -5 \\ 5 & 35 \end{bmatrix} - \begin{bmatrix} 2 & 10 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 13 & -15 \\ 9 & 35 \end{bmatrix}$

35. Let $A = \begin{pmatrix} -2 & o & x \\ 2y & x+y & -4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$ such that $A + 2B^t = 0$, then the value of y?

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A. 0 B. $\frac{-13}{2}$ C. -8 D. $\mathbb{R}$

Solution

$$B^t = \begin{pmatrix} 1 & 0 & 1-x \\ -y & 3 & 2 \end{pmatrix} \text{ and } 2B^t = \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix}$$

$$A + 2B^t = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix}$$

$$\begin{pmatrix} -2+2 & 0+0 & x+2-2x \\ 2y-2y & (x+y+6) & -4+4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

(i) $-x + 2 = 0 \Rightarrow x = 2$

(ii) $x + y + 6 = 0 \Rightarrow y + 6 = -2,\ y = -8$

Answer: C

36. If $\begin{pmatrix} 0 & -5 & 6 \\ 2m-1 & 0 & 8 \\ -6 & 1-k^2 & 0 \end{pmatrix}$ is skew symmetric matrix then find value of m and k.

Solution

$A^t = -A$

If $A = [a_{ij}]_{m\times m}$ is skew symmetric then $a_{ij} = -a_{ij}$ for $i \neq j$

(i) $a_{21} = -a_{12} \Rightarrow 2m - 1 = -(-5) = 5$ and $m = 3$

(ii) $a_{32} = -a_{23} \Rightarrow 1 - k^2 = -8$

$\therefore \quad k^2 = 9$, and $k = \pm 3$,

37. Let $A = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 3 & 0 \\ 0 & 6 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 4 & -2 \\ 1 & -3 & 0 \end{pmatrix}$

then what is the value of $C_{21} = AB^t$?

A. -10 B. 0 C. 6 D. not defined

Solution

$$C_{21} = AB^t = (\text{Row 2 of A})\,(\text{column 1 of } B^t)$$

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$$= \begin{pmatrix} -2 & 3 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} = -2 \times 3 + 3 \times 4 + 0 \cdot (-2) = 6$$

Answer: C

38. Let $A = \begin{pmatrix} 3 & 5 \\ -2 & 4 \end{pmatrix}$, express matrix A as sum of a symmetric and skew symmetric matrix.

Solution

(i) symmertic matrix: $S = \frac{1}{2}(A + A^t) = \frac{1}{2}\left[\begin{pmatrix} 3 & 5 \\ -2 & 4 \end{pmatrix} + \begin{pmatrix} 3 & -2 \\ 5 & 4 \end{pmatrix}\right]$

$$\therefore \quad S = \frac{1}{2}\begin{pmatrix} 6 & 3 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 3 & \frac{3}{2} \\ \frac{3}{2} & 4 \end{pmatrix}$$

(ii) Skew symmetric: $T = \frac{1}{2}(A - A^t)$

$$= \frac{1}{2}\left[\begin{pmatrix} 3 & 5 \\ -2 & 4 \end{pmatrix} - \begin{pmatrix} 3 & -2 \\ 5 & 4 \end{pmatrix}\right] = \frac{1}{2}\begin{bmatrix} 0 & 7 \\ -7 & 0 \end{bmatrix}$$

$$\therefore \quad S + T = \begin{pmatrix} 3 & \frac{3}{2} \\ \frac{3}{2} & 4 \end{pmatrix} + \begin{pmatrix} 0 & \frac{7}{2} \\ \frac{-7}{2} & 0 \end{pmatrix} = \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} = A$$

39. Suppose A,B and C are square matrices of the same size.

Determine whether each of the following is always true?

a. $(A + B)^2 = A^2 + 2AB + B^2$

b. $(A + B)^2 = A^2 + AB + BA + B^2$

c. $A^2 - B^2 = (A - B)(A + B)$

d. If $AC = BC$ then $A = B$

e. If $AB = 0$, then $A = 0$ or $B = 0$

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Solution

a. False, matrix multiplication is not commutative, so that $AB \neq BA$

b. True

c. False, $(A - B)(A + B) = A(A + B) - B(A + B) = A^2 + AB - BA - B^2$, and $AB \neq BA$

d. False because cancellation low does not hold for matrices multiplication

e. False, as example let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ we have $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

3.5 Elementary Row Operations

In this lesson you will learn an operation for transforming a matrix into a equivalent **triangular matrix** of the form:

$$\begin{bmatrix} 1 & -3 & 4 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

What is an Elementary row operations?

An elementary row operation is a process to transform the given matrix to an equivalent triangular matrix form.

DEFINITION: **Elementary Row operations**

Row operations producing row-equivalent matrices. Any of the following elementary row operations performed on a given matrix will produce a matrix that is **row - equivalent** to the original matrix.

1. **Swapping:** Interchange of any two rows $(R_i \leftrightarrow R_j)$
2. **Scaling:** Multiplying element of a row by a non - zero constant $(KR_i \rightarrow R_i)$

   Symbol: $KR_i \rightarrow R_i$

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3. Pivoting: Add a multiple of the elements of one row to another row.

Symbol: $KR_i + R_j \rightarrow R_j$

Note: The arrow " $\longrightarrow$ " may be read as replaces.

Meaning for arrow symbol:

(i) $3R_2 \rightarrow R_2$ $3R_2$ is replacing new row 2 (R_2)

(ii) $3R_2 + R_1 \rightarrow R_1$ $3R_2 + R_1$ is replacing new row 1 (R_1)

40. Let $A = \begin{bmatrix} -3 & -5 & -13 \\ 1 & 1 & 5 \end{bmatrix}$, Find the matrix B such that B is obtained by performing the series of the following elementary row operations:
$R_1 \leftrightarrow R_2$, $3R_1 + R_2 \rightarrow R_2$, $-2R_2 + R_1 \rightarrow R_1$

Solution

A. $\begin{bmatrix} -3 & -5 & -13 \\ 1 & 2 & 5 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 5 \\ -3 & -5 & -13 \end{bmatrix} \rightarrow \begin{matrix} 3R_1 \\ +R_2 \\ \hline R_2 \end{matrix} \begin{matrix} \begin{bmatrix} 3 & 6 & 15 \\ -3 & -5 & -13 \end{bmatrix} \\ \hline 0 \quad 1 \quad 2 \end{matrix}$

$\xrightarrow{3R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{-2R_2 + R_1 \rightarrow R_1} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{pmatrix}$

$\therefore B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{pmatrix}$

Illustrative Example

41. Explain the meaning of the notation in elementary row operation.

a. $R_1 \leftrightarrow R_3$ b. $-R_2 \rightarrow R_2$ c. $3R_1 + R_2 \rightarrow R_2$

Explanation

a. Interchange row 1 and row 3.

b. Multiply row 2 by (-1) and replace the original row 2 with the result.

c. Add 3 times row 1 to row 2 and replace the original row 2 with the result $(3R_1 + R_2)$.

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42. Let $A = \begin{bmatrix} 3 & 1 & 4 \\ 1 & 5 & -3 \\ -2 & 4 & 2 \end{bmatrix}$. By performing a series of elementary row operations change matrix A into upper triangular form of row equivalent matrix B, in the form $B = \begin{bmatrix} 1 & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$

Solution

Our goal is to transform $a_{11} = 1$, $a_{21} = 0$ $a_{31} = 0$ and $a_{32} = 0$.

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 1 & 5 & -3 \\ -2 & 4 & 2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 5 & -3 \\ 3 & 1 & 4 \\ -2 & 4 & 2 \end{bmatrix}$$

our goal is to change $a_{21} = 0$, $a_{31} = 0$

$$\begin{bmatrix} 1 & 5 & -3 \\ 3 & 1 & 4 \\ -2 & 4 & 2 \end{bmatrix} \xrightarrow[2R_1 + R_3 \to R_3]{-3R_1 + R_2 \to R_2} \begin{bmatrix} 1 & 5 & -3 \\ 0 & -14 & 13 \\ 0 & 14 & -4 \end{bmatrix}$$

our goal is to change $a_{32} = 0$

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -14 & 13 \\ 0 & 14 & -4 \end{bmatrix} \xrightarrow{R_2 + R_3 \to R_3} \begin{bmatrix} 1 & 5 & -3 \\ 0 & -14 & 13 \\ 0 & 0 & -9 \end{bmatrix}$$

3.5.1 Row - Echelon Form and Reduced Row - Echelon Form of a Matrix

A matrix is in **row - echelon form** if it satisfies the following condition.

1. The first non - zero number in any row is 1. This called the leading 1. (The entry in row 1, column 1 is 1, and 0's appear below it.)
2. All zero rows are at the bottom.
3. Each leading 1 is to the right of all leading 1' S in the row above it.

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A row - echelon matrix is said to be **reduced row echelon form** if, in addition, it satisfies the following condition.

4. Every number above and below each leading number 1 is zero (Above and below 1 must zero).

The following is an illustration of matrices in echelon form.

Row echelon form

$$\begin{bmatrix} 1 & a_{12} & a_{13} & a_{14} \\ 0 & 1 & a_{23} & a_{24} \\ 0 & 0 & 1 & a_{34} \end{bmatrix}$$

Row - echelon form

$$\begin{bmatrix} 1 & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} \\ 0 & 1 & a_{23} & a_{24} & a_{24} & a_{24} \\ 0 & 0 & 0 & 1 & a_{35} & a_{36} \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

In a reduced echelon matrix the additional requirement is that all entries above leading 1 must also be zero.

Row - echelon form

$$\begin{bmatrix} 1 & * & * & * & * \\ 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Reduced - echelon form

$$\longrightarrow \begin{bmatrix} 1 & * & 0 & * & 0 \\ 0 & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Hare is a systematic way to put a matrix in row - echelon form using elementary row operations

- Start by obtaining 1 in the top left corner. Then obtain zeros below 1 by setting: $KR_i + R_j \rightarrow R_j$
- Next obtain leading 1 in the next row, and then obtain zeros below that 1.
- Continue this process until you get matrix in row - echelon form.

Illustrative Example

43. Determine whether each of the following matrices are row echelon form. Explain why the matrix is not in row - echelon for form.

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a $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ b. $B = \begin{bmatrix} 1 & 2 & -3 & 6 & 0 \\ 0 & 0 & 1 & 4 & 8 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

c. $E = \begin{bmatrix} 0 & 1 & -2 & 0 \\ 1 & 0 & 3 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 1 & 1 & 0 \end{bmatrix}$ d. $F = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

Solution

- Matrix A and B are row - echelon form of matrix.
- Matrix E is not row - echelon form because row 2, column 1 (a_{21}, the leading 1' S do not shift to the right in successive rows.

d. Matrix F is not row - echelon form because by condition # 2, all zero rows are not at the bottom.

44. Determine whether each of the matrix are reduced row - echelon form. If not, explain why the matrix is not reduced row - echelon form.

a. $E = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ c. $F = \begin{bmatrix} 1 & 4 & 0 & 3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ d. $B = \begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$

e. $G = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

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Solution

Let Us consider the reduced - row echelon form matrix illustration is as follow:

$$\begin{bmatrix} 1 & * & 0 & * & 0 \\ 0 & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

a. Matrix E is reduced echelon from

b. Matrix A is not reduced - echelon form because the elements above the leading entry 1 in second column is not zero. That is the elements in row 1, column 2 should be zero.

c. Matrix F is reduced echelon form

d. Matrix B is not reduced - echelon form because the elements above the leading 1 in third column are not all zero. That is the 4 in a_{13} should be zero.

e. Matrix G is reduced - echelon form

45. Determine whether each of the matrix are reduced row echelon form. If not state the row operation (s) required and transform to reduced form.

a. $\left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 4 & -8 \end{array}\right]$

b. $\left[\begin{array}{ccc|c} 0 & 1 & 0 & -4 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$

c. $\left[\begin{array}{ccc|c} 1 & 5 & 4 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$

d. $\left[\begin{array}{ccc|c} 1 & 4 & 0 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$

Solution

a. Not reduced, condition 1 is violated: the 4 in a_{22} should be 1. Because the first non - zero number in any row is 1, $\frac{1}{4}\ R_2 \longrightarrow R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & -8 \end{array}\right]$$

b. Not reduced row echelon from, because condition 3 is violated: the leading 1 in second row is not to the right of the leading 1 in the firs row.

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Perform: $R_1 \leftrightarrow R_2$ to obtain $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 2 \end{array}\right]$

c. Not reduced echelon form, condition 4 is violated. Above and below leading element should be zero.

$$\xrightarrow{-5R_2 + R_1 \longrightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & -6 & 7 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

d. Not reduced echelon form because condition 2 is violated.

$$R_2 \leftrightarrow R_3 . \left[\begin{array}{ccc|c} 1 & 4 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

STEP 4: Go to column - n obtain a leading of in row Then obtain 0's above this element.

Illustrative Example

46. Let $A = \left[\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 2 & -1 & 2 & 2 \\ 1 & -2 & 3 & -1 \end{array}\right]$

a. Reduce A to echelon form

b. Reduce A to row - reduced echelon form

Solution

a. Goal: change a_{11} to 1: go to column 1: obtain leading element $a_{11} = 1$

$$R_1 \leftrightarrow R_3 \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{array}\right]$$

Goal: To change a_{21} to 0, a_{31} to 0. Add - 2 times the first row to the second row and -3 times the first row to the third row.

$$\xrightarrow[-3R_1 + R_3 \longrightarrow R_3]{-2R_1 + R_2 \longrightarrow R_2} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 7 & -7 & 6 \end{array}\right]$$

Goal: To change a_{22} to 1 and a_{32} to 0: multiply the second row by 7 and third row by 3. Then, add - 1 times the second row to the third row.

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$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 21 & -28 & 28 \\ 0 & 21 & -21 & 18 \end{array}\right] \xrightarrow{-R_2 + R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 21 & -28 & 28 \\ 0 & 0 & 7 & -10 \end{array}\right]$$

Multiply row 2 by $\frac{1}{21}$ and row 3 by $\frac{1}{7}$

$$\xrightarrow[\frac{1}{7}R_3 \to R_3]{\frac{1}{21}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 1 & -\frac{4}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -\frac{10}{7} \end{array}\right]$$

. This is a matrix in echelon form.

b. To obtain the reduced echelon form;

Goal: change $a_{12} = -2$ to 0. Add 2 times the second row to the first row

$$2R_2 + R_1 \to R_1 \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & \frac{5}{3} \\ 0 & 1 & -\frac{4}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -\frac{10}{7} \end{array}\right]$$

Goal: To change, a_{23} to 0, a_{13} to 0.

$$\frac{-1}{3} R_3 + R_1 \longrightarrow R_1$$

and $\frac{4}{3} R_3 + R_2 \longrightarrow R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{15}{7} \\ 0 & 1 & 0 & -\frac{4}{7} \\ 0 & 0 & 1 & -\frac{10}{7} \end{array}\right]$$

resulting in the row reduced echelon form.

42. Which one of the following is a reduced row echelon form of the matrix

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & 1 \\ 4 & -1 & 4 \end{bmatrix} ?$$

A. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ C. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ D. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

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Solution

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & 1 \\ 4 & -1 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & -1 & 4 \end{bmatrix}$$

our Goal to make $a_{21} = 0$ and $a_{31} = 0$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & -1 & 4 \end{bmatrix} \xrightarrow[-4R_1 + R_3 \to R_3]{-2R_1 + R_2 \to R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & -5 & 0 \end{bmatrix}$$

We need to make $a_{32} = 0$ and $a_{33} = 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & -5 & 0 \end{bmatrix} \xrightarrow{5R_2 + R_3 \to R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{-\frac{1}{5} R_3 \to R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

We need to make $a_{12} = 0$, and $a_{23} = 0$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[1R_3 + R_2 \to R_2]{-R_2 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We need to make $a_{13} = 0$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2R_3 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Answer: B

48. Explain why the matrix is not in reduced row echelon form:

$$\left[\begin{array}{ccc|c} 1 & 0 & 6 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

Explanation: The elements above the leading element 1 in third column are not all zero. Specifically the elements in row 1 column 3 should be zero.

49. Explain the difference between a matrix in row - echelon form and reduced row - echelon form.

Explanation:- reduced row - echelon form is the same formate row-echelon form with the added condition that all elements above the leading 1's must be 0's.

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50. Which of the matrices given below is the reduced row - echelon form of the following matrix? $\left[\begin{array}{ccc|c} 1 & 0 & 6 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right]$

A. $\left[\begin{array}{ccc|c} 0 & 0 & 1 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 0 & 1 \end{array}\right]$

C. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & 0 \end{array}\right]$

B. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right]$

D. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{2} \end{array}\right]$

Answer: C

3.6 Augmented Matrices of a System of Equation

What is a system of equation?

A system of equation is two or more equations considered together is called a **system of equation.**

A **solution** of a system of equation in two or more variable is an ordered pair (triple) that satisfies each equation of the system.

* Equivalent systems of equations have the same solution set.

Consider the system of linear equation

a. $\begin{cases} 3x - 4y = 10 \\ 2x + 5y = -1 \end{cases} \xrightarrow{\text{equivalent to}} \begin{cases} 3x - 4y = 10 \\ y = -1 \end{cases}$

Triangular form

b. $\begin{cases} 2x - 3y + z = 10 \\ -x + 3y + 3z = 4 \\ x - 2y - z = 1 \end{cases} \longrightarrow \begin{cases} x - 2y - z = 1 \\ y + 2z = 5 \\ z = 3 \end{cases}$

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What is an Augmented matrix

A matrix created from a system of linear equation an **augmented matrix**. Where the entries in the first column are the coefficients of x, the entries in the 2nd column are the coefficients of y and the entries in the 3rd columns are coefficient of z and the entries in the 4th columns are the constants from system of equation

- An augmented matrix contains the **coefficient** of the variable and the **constants** from a system of equation.

Example

Linear system → Augmented matrix

a. $\begin{cases} 2x - 3y + z = 10 \\ -x + 3y + 3z = 4 \\ x - 2y - z = 1 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 2 & -3 & 1 & 10 \\ -1 & 3 & 3 & 4 \\ 1 & -2 & -1 & 1 \end{array}\right]$

b. $\begin{cases} 2x + z = 2 \\ 3y - 5z = 4 \\ 4x - y + 3z = 0 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 2 & 0 & 1 & 2 \\ 0 & 3 & -5 & 4 \\ 4 & -1 & 3 & 0 \end{array}\right]$

a. coefficient matrix: $\begin{bmatrix} 2 & -3 & 1 \\ -1 & 3 & 3 \\ 1 & -2 & -1 \end{bmatrix}$, constant matrix $\begin{bmatrix} 10 \\ 4 \\ 1 \end{bmatrix}$

b. coefficient matrix: $\begin{bmatrix} 2 & 0 & 1 \\ 0 & -3 & -5 \\ 4 & -1 & 3 \end{bmatrix}$, constant matrix $\begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}$

Note: When a term is missing from one of the equation of the system, the coefficient of that term is 0, and 0 is entered in the matrix.

Graph of a System of linear Equation

For any systems of linear equation as there are exactly three possibilities. If the system has.

(i) One solution: the lines are intersecting

(ii) Infinitely many solution: the lines are the same

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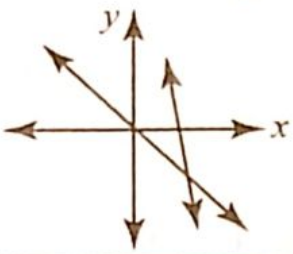
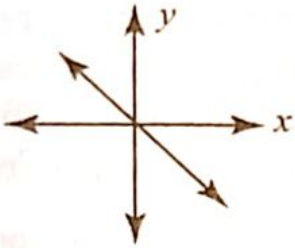
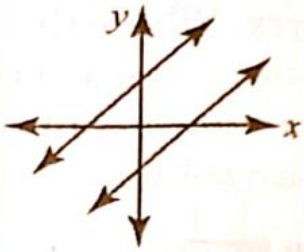
(iii) No solution: the lines are parallel.

	Two intersecting lines	Two same line	Two parallel lines
Graph			
Number of solution	Exactly one solution	Infinitely many solution	No solution
Term	consistent and independent	consistent and dependent	Inconsistent

A system of linear equation is called **consistent** if it has at least one solution and the system is called **inconsistent** if it has no solution. When we solve the system in augmented matrix as, in echelon form is as follow:

$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \end{bmatrix}$ | $\begin{bmatrix} 1 & a & b \\ 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 1 & a & b \\ 0 & 0 & c \end{bmatrix}$, $C \neq 0$

One solution | Infinitely many solution | No - solution

Solving Linear Systems Using Augmented Matrices

Here, we going to see the **Gauss Jordan Elimination** method for solving systems of linear equations. The method transforms an augmented matrix **AX = B** into a **row - echelon Form**.

- If the system transformed in to **Reduced - row echelon form** is called the **Gauss- Jordan Reduction** method.

Note: When we solve the system in Gauss - Jordan elimination, the nature of the solution set of the system of the augmented matrix in row - echelon form:

(i) $\begin{bmatrix} 1 & * & a \\ 0 & 1 & b \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & * & * & a \\ 0 & 1 & * & b \\ 0 & 0 & 1 & c \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & * & a \\ 0 & 0 & 0 \end{bmatrix}$

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(iv) $\left[\begin{array}{ccc|c} 1 & * & * & a \\ 0 & 1 & * & b \\ 0 & 0 & 0 & 0 \end{array}\right]$ (v) $\left[\begin{array}{cc|c} 1 & * & a \\ 0 & 0 & b \end{array}\right]_{b \neq 0}$ (vi) $\left[\begin{array}{ccc|c} 1 & * & * & a \\ 0 & 1 & * & b \\ 0 & 0 & 0 & c \end{array}\right]_{c \neq 0}$

Row Echelon Form: (i) and (ii) have unique solution (consistent and independent).

Row Echelon Form: (iii) and (iv) have infinitely many solutions, (consistent and Dependent).

Row Echelon Form: (V) and (Vi) No solution (inconsistent).

Illustrative Example

51. Solve, using augmented matrix methods and write the nature of solution (consistent, or inconsistent).

a. $\begin{cases} 2x - 5y = 16 \\ 4x - 3y = 11 \end{cases}$ b. $\begin{cases} 2x + 3y = -1 \\ 6x + 9y = -3 \end{cases}$ c. $\begin{cases} x + 3y = 2 \\ 2x + 6y = -3 \end{cases}$

Solution

a. Write the augmented matrix corresponded to the system:

$$\left[\begin{array}{cc|c} 2 & -5 & 16 \\ 4 & -3 & 11 \end{array}\right] \xrightarrow[\text{into the form}]{\text{transform}} \left[\begin{array}{cc|c} 1 & * & a \\ 0 & * & b \end{array}\right]$$

To get 1 in a_{11}, we multiply row 1 by $\frac{1}{2}$, and to get 0 in a_{21}, we multiply row 1 by (-4) and add to row 2.

$$\frac{1}{2}R_1 \longrightarrow R_1 \quad \left[\begin{array}{cc|c} 1 & -2.5 & 8 \\ 4 & -3 & 11 \end{array}\right] \longrightarrow \begin{cases} x - 2.5y = 8 \\ 4x - 3y = 11 \end{cases}$$

$$4R_1 + R_2 \longrightarrow R_2 \quad \left[\begin{array}{cc|c} 1 & -2.5 & 8 \\ 0 & 7 & -21 \end{array}\right] \longrightarrow \begin{cases} x - 2.5y = 8 \\ -7y = -21 \end{cases}$$

To get 1 in a_{22}, we multiply R_2 by $\frac{1}{7}$.

$$\frac{1}{7}R_2 \longrightarrow R_2 \quad \left[\begin{array}{cc|c} 1 & -2.5 & 8 \\ 0 & 1 & -3 \end{array}\right] \longrightarrow \begin{cases} x - 2.5y = 8 \\ y = -3 \end{cases}$$

To get a 0 in a_{12}, we multiply row 2 by 2.5 and add to row 1.

$$2.5R_2 + R_1 \longrightarrow R_1 \quad \left[\begin{array}{cc|c} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -3 \end{array}\right] \longrightarrow \begin{cases} x = \frac{1}{2} \\ y = -3 \end{cases}$$

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$\therefore$ The equation has unique solution $\left(\frac{1}{2}, -3\right)$

$\therefore$ The system is consistent and independent

b. The augmented matrix form $\left[\begin{array}{cc|c} 2 & 3 & -1 \\ 6 & 9 & -3 \end{array}\right]$

we need transform in to $\left[\begin{array}{cc|c} 1 & * & a \\ 0 & * & b \end{array}\right]$

To get a 1 in a_{11}, we multiply row 1 by $\frac{1}{2}$.

To get a 0 in a_{21}, we multiply row 1 by (-6) and add to row 2.

$$\frac{1}{2}R_1 \longrightarrow R_1 \left[\begin{array}{cc|c} 1 & 1.5 & -0.5 \\ 6 & 9 & -3 \end{array}\right] \longrightarrow \begin{cases} x + 1.5y = -0.5 \\ 6x + 9y = -3 \end{cases}$$

$$-6R_1 + R_2 \longrightarrow R_2 \left[\begin{array}{cc|c} 1 & 1.5 & -0.5 \\ 0 & 0 & 0 \end{array}\right] \longrightarrow \begin{cases} x + 1.5y = -0.5 \\ 0y + 0y = 0 \end{cases}$$

- If we end up with a row of 0 in an augmented matrix for a given system, the system is consistent, dependent there are infinitely many solution.

$\therefore$ The solution set is $\left\{(x, y) : y = \frac{2x + 1}{-3}\right\}$

c. $\begin{cases} x + 3y = 2 \\ 2x + 6y = -3 \end{cases} \longrightarrow \left[\begin{array}{cc|c} 1 & 3 & 2 \\ 2 & 6 & -3 \end{array}\right]$ Augmented matrix

To get a 0's in a_{21}, multiply row 1 by (-2) and add to row 2.

$$-2R_1 + R_2 \longrightarrow R_2 \left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & 0 & -7 \end{array}\right] \longrightarrow \begin{cases} x + 3y = 2 \\ 0 = -7 \end{cases}$$

Since $0 = -7$ is false

If we obtain all 0's to the left of the vertical bar and a non zero numbers to the right of the bar in a row of an augmented matrix, then system is inconsistent and there are no solution.

$\therefore$ The solution set is ø.

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52. Solve, using augmented matrix methods: $\begin{cases} x + y = 1 \\ 6x - 2z = -8 \\ 3y - z = -3 \end{cases}$

Solution

Augmented matrix form: $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 6 & 0 & -2 & -8 \\ 0 & 3 & -1 & -3 \end{array}\right]$

To get a 0's in a_{21}, we multiply row 1 by (-6) and to row 2.

$$-6R_1 + R_2 \longrightarrow R_2 \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & -6 & -2 & -14 \\ 0 & 3 & -1 & -3 \end{array}\right] \longrightarrow \begin{cases} x + y = 1 \\ -6y - 2z = -14 \\ 3y - z = -3 \end{cases}$$

To get $a_{22} = 1$, multiply row by $\left(\frac{-1}{6}\right)$.

$$\frac{1}{6}R_2 \longrightarrow R_2 \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & \frac{1}{3} & \frac{7}{3} \\ 0 & 3 & -1 & -3 \end{array}\right] \longrightarrow \begin{cases} x + y = 1 \\ y + \frac{1}{3}z = \frac{7}{3} \\ 3y - z = -3 \end{cases}$$

To get $a_{32} = 0$, we multiply row 2 by (-3) and add to row 3.

$$-3R_2 + R_3 \longrightarrow R_3 \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & \frac{1}{3} & \frac{7}{3} \\ 0 & 0 & -2 & -10 \end{array}\right] \longrightarrow \begin{cases} x + y = 1 \\ y + \frac{1}{3}z = \frac{7}{3} \\ -2z = -10 \end{cases}$$

By back substitytion: $-2z = -10$, $z = 5$

$$y = \frac{7}{3} - \frac{1}{3}z = \frac{7}{3} - \frac{5}{3} = \frac{2}{3}$$

$$x = 1 - y = 1 - \frac{2}{3} = \frac{1}{3} \longrightarrow \text{s.s is} \left\{\left(\frac{1}{3}, \frac{2}{3}, 5\right)\right\}$$

53. Solve the system:

$$\begin{cases} 2x + 6y - 4Z = -8 \\ 3x + 4y + Z = -3 \\ 5x + 3y + 8Z = 4 \end{cases}$$

Augmented matrix

$$\left[\begin{array}{ccc|c} 2 & 6 & -4 & -8 \\ 3 & 4 & 1 & -3 \\ 5 & 3 & 8 & 4 \end{array}\right]$$

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Solution

Our goal is transform the system into triangular form

- Multiply equation (1) by $\frac{1}{2}$

$$\begin{cases} x + 3y - 2z = -4 \\ 3x + 4y + z = -3 \\ 5x \ \ 3y + 8z = 4 \end{cases}$$

- Next, we want to eliminate x in 2nd and 3rd equation.
- Multiply 1st equation by - 3 and add the result to 2nd equation Again, multiply 1st equation by -5 and add the result to 3rd equation.

$$\begin{cases} x + 3y - 2z = -4 \\ -5y + 7z = 9 \\ -12y + 18z = 24 \end{cases}$$

Now, we want to eliminate y in 3rd equation Multiply third equation by $\frac{-1}{12}$.

Inter change 2nd and 3rd equation

$$\begin{cases} x + 3y - 2Z = -4 \\ y - \frac{3}{2}Z = -2 \\ -5y + 7Z = 9 \end{cases}$$

Our goal is to transform the matrix into the echelon form

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{bmatrix}$$

To find $a_{11} = 1$, multiply row 1 by $\frac{1}{2}$

$$\xrightarrow{\frac{1}{2}R_1 \to R_1} \begin{bmatrix} 1 & 3 & -2 & -4 \\ 3 & 4 & 1 & -3 \\ 5 & 3 & 8 & 4 \end{bmatrix}$$

- Next we want to produce $a_{21} = 0$ and $a_{31} = 0$
- Multiply row 1 by 3 and add the result to row 2. Again, multiply row 1 by - 5 and add the result to row 3

$$\xrightarrow[-5R_1 + R_3 \to R_3]{-3R_1 + R_2 \to R_2} \begin{bmatrix} 1 & 3 & -2 & -4 \\ 0 & -5 & 7 & 9 \\ 0 & -12 & 18 & 24 \end{bmatrix}$$

- Change a_{22} to 1 and change a_{31} to 0
- Multiply row 3 by $\frac{-1}{12}$
- Interchange row 2 and row 3.

$$\xrightarrow[R_2 \leftrightarrow R_3]{-\frac{1}{12}R_3 \to R_3} \begin{bmatrix} 1 & 3 & -2 & -4 \\ 0 & 1 & -\frac{3}{2} & -2 \\ 0 & -5 & 7 & 9 \end{bmatrix}$$

- Change a_{32} to 0 Multiply row 2 by 5 and add the result to row 3

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- Multiply 2nd equation by 5 add the result to 3rd equation

$$\begin{cases} x + 3y - 2Z = -4 \\ y - \frac{3}{2}Z = -2 \\ -\frac{1}{2}z = -1 \end{cases}$$

Multiply 3rd equation by -2

$$\begin{cases} x + 3y - 2Z = -4 \\ y - \frac{3}{2}Z = -2 \\ Z = 2 \end{cases}$$

$$\xrightarrow{5R_2 + R_3 \to R_3} \begin{bmatrix} 1 & 3 & -2 & -4 \\ 0 & 1 & -\frac{3}{2} & -2 \\ 0 & 0 & -\frac{1}{2} & -1 \end{bmatrix}$$

Change a_{33} to 1

Multiply row 3 by (-2)

$$\xrightarrow{-2R_3 \to R_3} \begin{bmatrix} 1 & 3 & -2 & -4 \\ 0 & 1 & -\frac{3}{2} & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

At this point, you can use back substitution to find x and y.
Solving for y from 2nd equation:

$$y - \frac{3}{2}(2) = -2, \text{ and } y = -2 + 3 = 1$$

$\therefore$ The solution set is $\{(-3, 1, 2)\}$

54. Solve the system in elementary row operation:

$$\begin{cases} 2x + 2y - 4z = 4 \\ 4x + 5y - 9Z = 8 \\ 2x + 3y - 5y = 4 \end{cases}$$

Solution

$$A = \begin{pmatrix} 2 & 2 & -4 \\ 4 & 5 & -9 \\ 2 & 3 & -5 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} 4 \\ 8 \\ 4 \end{pmatrix}$$

First, write an augmented matrix represent the system $\left[\begin{array}{ccc|c} 2 & 2 & -4 & 4 \\ 4 & 5 & -9 & 8 \\ 2 & 3 & -5 & 4 \end{array}\right]$

Our goal is transform the matrix in echelon form:

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Now we need to change $a_{11} = 1, a_{21} = 0, a_{31} = 0$

- Multiply row 1 by $\frac{1}{2}$ $\xrightarrow{\frac{1}{2}R_1 \to R_1}$ $\left[\begin{array}{ccc|c} 1 & 1 & -2 & 2 \\ 4 & 5 & -9 & 8 \\ 2 & 3 & -5 & 4 \end{array}\right]$

Also, we need to find 0 below the leading element 1.

That is we need to obtain $a_{21} = 0$ and $a_{31} = 0$.

- To find $a_{21} = 0$, we multiply row 1 by - 4 and add the result to row 2, then replace the original row2 by new row2: $-4R_1 + R_2 \to R_2$
- To find $a_{31} = 0$, $-2R_1 + R_3 \to R_3$.

$$\left(\begin{array}{ccc|c} 1 & 1 & -2 & 2 \\ 4 & 5 & -9 & 8 \\ 2 & 3 & -5 & 4 \end{array}\right) \xrightarrow[-2R_1 + R_3 \to R_3]{-4R_1 + R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & 1 & -2 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right)$$

To find $a_{32} = 0$, : $\xrightarrow{-R_2 + R_3 \to R_3}$ $\left(\begin{array}{ccc|c} 1 & 1 & -2 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$

This matrix is in row echelon form. The bottom row is equivalent to the equation: $0x + 0y + 0z = 0$ is true statement, it has **infinitely many Solution**

The original system of equation reduces to: $x + y - 2z = 2$ and $y - z = 0$ Now, we express both x and y interns of z.

Therefore, $y - z = 0$ implies $y = z$

Also, $x + z - 2z = 2$ implies $x = 2 + z$

$\{(2 + z, z, z) \mid Z \in \mathbb{R}\}$

Let $z = t$, then the solution set is $\{(2 + t, t, t) \mid t \in \mathbb{R}\}$.

5. The solution set of the system $\begin{cases} 2x - y + 3z = 1 \\ x + y + 3z = 5 \\ x - y + z = -1 \end{cases}$ is

A. $\{(-4 - 2k, k, 2 - k): k \in \mathbb{R}\}$ C. $\{(2 - 2k, 3 - k, k): k \in \mathbb{R}\}$

B. $\{(k + 1, 2k, k - 1): k \in \mathbb{R}\}$ D. $\{(k, k + 1, k - 1): k \in \mathbb{R}\}$

Solution: We proceed as follows, beginning with the suggested matrix.

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$$\begin{pmatrix}2 & -1 & 3 & 1\\ 1 & 1 & 3 & 5\\ 1 & -1 & 1 & -1\end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix}1 & 1 & 3 & 5\\ 2 & -1 & 3 & 1\\ 1 & -1 & 1 & -1\end{pmatrix} \xrightarrow[-R_1 + R_3 \to R_3]{-2R_1 + R_2 \to R_2} \begin{pmatrix}1 & 1 & 3 & 5\\ 0 & -3 & -3 & -9\\ 0 & -2 & -2 & -6\end{pmatrix}$$

$$\begin{pmatrix}1 & 1 & 3 & 5\\ 0 & -3 & -3 & -9\\ 0 & -2 & -2 & -6\end{pmatrix} \xrightarrow{-\frac{1}{3}R_2 \to R_2} \begin{pmatrix}1 & 1 & 3 & 5\\ 0 & 1 & 1 & 3\\ 0 & -2 & -2 & -6\end{pmatrix} \xrightarrow{2R_2 + R_3 \to R_3} \begin{pmatrix}1 & 1 & 3 & 5\\ 0 & 1 & 1 & 3\\ 0 & 0 & 0 & 0\end{pmatrix}$$

This matrix is in row echelon form. The bottom row is equivalent to $0x + 0y + 0z = 0$ is true for all x, y, and z, so that the system has infinite solution. The original equation $x + y + 3z = 5$ and $y + z = 3$

$\Rightarrow$ $y = 3 - z$, and $x + 3 - z + 3z = 5$

$\therefore$ $x = 5 - 3 - 2z = 2 - 2z$,

The solution set is $\{(2 - 2z, 3 - z, z)\}$,

Let $z = k \in R$, $\{(2 - 2k, 3 - k, k)k \in R\}$

Answer: C

56. Determine whether the system has no solution, or infinitely many solution. If the system has solution, determine the solution.

a. $\left[\begin{array}{cc|c} 3 & -2 & 5\\ 0 & 0 & 4\end{array}\right]$ b. $\left[\begin{array}{cc|c} 1 & 4 & -3\\ 0 & 1 & 2\end{array}\right]$ c. $\left[\begin{array}{cc|c} 1 & 3 & -2\\ 0 & 0 & 0\end{array}\right]$

Solution:- All the matrix a - *f* are in row - echelon form

a. $\left[\begin{array}{cc|c} 3 & -2 & 5\\ 0 & 0 & 4\end{array}\right]$ The bottom row is equivalent to the equation: $0x + 0y = 4$ is false statement

$\therefore$ The system has **no solution.**

b. $\left[\begin{array}{cc|c} 1 & 4 & -3\\ 0 & 1 & 2\end{array}\right]$ The bottom row is equivalent to the equation $y = 2$ and $x + 4y = -3$, to find for x,

$\Rightarrow$ $x + 4(2) = -3$ implies $x = -3 - 8 = -11$

$\therefore$ The solution set is $\{(-11, 2)\}$.

c. It has infinitely many solution. The solution set is $\{(-2 - 3y, y)\}$,
Let $y = t \in R$

$\therefore$ The solution set is $\{(-2 - 3t, t) \backslash t \in R\}$.

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57. What is the solution set of the system $\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 1 \\ 3x + 4y + 5z = 3 \end{cases}$

A. $\{(1, 0, 0)\}$ C. $\{(-3k + 1, k, k) \setminus k \in R\}$

B. $\{(k + 1, -k, k) / k \in R\}$ D. $\emptyset$

Solution

Step1: In this system, it seems easiest to eliminate x in equation 2. Multiply each side of the equation 1 by (-1) and add the result to equation 2.

Step 2: To eliminate x in equation 3 Multiply equation 1 by -3 and add the result to equation 3

Step 1 : $-x - y - 2z = -1$

Add : $x + 2y + z = 1$

$0 \quad y - z = 0$

Step 2 : $-3x - 3y - 6z = -3$

Add : $3x + 4y + 5z = 3$

$y - z = 0$

$\longrightarrow \begin{cases} x + y + 2z = 1 & (*) \\ y - z = 0 & (*) \\ y - z = 0 & (*) \end{cases}$

To eliminate y in (*)

$y - z = 0$

$y - z = 0$ multiply by (-1) $\longrightarrow$ add $\begin{cases} -y + z = 0 \\ y - z = 0 \\ \hline 0 = 0 \end{cases}$

This is true statement, so that the system has infinite solution.

To solve for x and y interms of z

$y - z = 0 \Rightarrow y = z$ and $x + y + 2z = 1$

Let $z = k \in R$, then the solution set is $\{(1 - 3k, k) / k \in R\}$.

Answer: C

58. What should be the value of k does the system $\begin{cases} x - y + z = 1 \\ -x + 5y - 4z = 1 \\ 2x + 2y - z = k \end{cases}$ has a solution?

A. 0 B. 1 C. -4 D. 4

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Solution: Using row operation

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ -1 & 5 & -4 & 1 \\ 2 & 2 & -1 & k \end{array}\right]$$

- Change a_{21} to 0 : $(R_2 + R_1 \to R_2$ (New))
- Change a_{31} to 0 : $(-2R_1 + R_3 \to R_3$ (New))

$$\xrightarrow[-2R_1 + R_3 \to R_3]{R_2 + R_1 \to R_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -3 & 2 \\ 0 & 4 & -3 & -2 + k \end{array}\right]$$

Change a_{32} to 0 : $(-R_2 + R_3 \to R_3)$

$$\xrightarrow{-R_2 + R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -3 & 2 \\ 0 & 0 & 0 & -4 + k \end{array}\right]$$

The system has an infinite solution if $-4 + k = 0$. then $k = 4$

Answer: D

59. For what values λ and μ so that the system of equation. $\begin{cases} x + y + z = 6 \\ x + 2y + 3z = 10 \\ x + 2y + \lambda z = \mu \end{cases}$ have infinite solution

A. $\lambda \neq 3$ and $\mu = \mathbb{R}$

B. $\lambda = 3$ and $\mu \neq 10$

C. $\lambda = 3$ and $\mu = 10$

D. $\lambda = \mathbb{R}$ and $\mu = 10$

Solution: In matrix notation $AX = B$

$$\text{Where } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

Augmented matrix $[A \vdots B]$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & b \end{array}\right] \xrightarrow[R_3 - R_1 \to R_3]{R_2 - R_1 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{array}\right]$$

$$\xrightarrow[R_3 - R_2 \to R_3]{R_1 - R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{array}\right]$$

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(i) The system has unique solution if $\lambda \neq 3$

(ii) The system has infinite solution if $\lambda = 3$ and $\mu = 10$.

(iii) The system has no solution if $\lambda = 3$ and $\mu \neq 10$.

Answer: (

3.7 Homogeneous System of Linear Equation

A system of a linear equations is called homogeneous if all the constant terms are zero. Some of a typical **homogeneous** linear equation has the form:

i. $\begin{cases} x + 2y = 0 \\ x - 3y = 0 \end{cases}$ ii. $\begin{cases} x - 3y = 0 \\ 2x + 4y = 0 \end{cases}$ iii. $\begin{cases} 3x + 2y + z = 0 \\ 2x - 3y + 5z = 0 \\ x - 4y + 5z = 0 \end{cases}$

- Homogeneous linear equation with n - variable $x_1, x_2, x_3, \ldots x_n$ has the form: $a_1 x_1 + a_2 x_2 + a_3 x_3 + \ldots + a_n x_n = 0$
- Because the constants are all zero, every homogeneous system always has the **trivial solution:** $x_1 = 0, x_2 = 0, \ldots x_n = 0$
- All non - zero solution to the system are called **non - trivial solution**, for example, (1, -1, -1) is non - trivial solution for equatio(iii).

Every homogeneous system of linear equations has either:

(i) Only the trivial solution: $x_1 = 0, x_2 = 0 \ldots x_n = 0$

(ii) Or trival solution and an infinite number of non - trivial solutions. $x_1 \neq 0, x_2 \neq 0 \ldots x_n \neq 0$

Illustrative Example

60. Determine whether each of the following homogeneous system of equation has:

(i) trivial solution only (ii) trivial and non - trivial solution.

a. $\begin{cases} x - 3y = 0 \\ 4x + 2y = 0 \end{cases}$

b. $\begin{cases} x - 4y + 5y = 0 \\ 2x - 3y + 5z = 0 \\ 3x + 2y + z = 0 \end{cases}$

c. $\begin{cases} x + 3y - 4z = 0 \\ 2x + 7 + z = 0 \\ 3x - 5y - 2z = 0 \end{cases}$

d. $\begin{cases} x + 3y + z = 0 \\ x + y - z = 0 \\ x - 2y - 4z = 0 \end{cases}$

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Solution

a. The given system in matrix form is :

$$\begin{pmatrix} 1 & -3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The augmented matrix form is $\left(\begin{array}{cc|c} 1 & -3 & 0 \\ 4 & 2 & 0 \end{array}\right)$. reduced as follow:

$$\xrightarrow{-4R_1 + R_2 \to R_2} \left(\begin{array}{cc|c} 1 & -3 & 0 \\ 0 & 14 & 0 \end{array}\right) \xrightarrow[3R_2 + R_1 \to R_1]{\frac{1}{14}R_2 \to R_2} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right)$$

$x = 0,\ y = 0$

$\therefore$ This homogeneous system of linear equation has only trivial solution (0,0), but no - non trivial solution.

b. The augmented matrix for the system is $\left[\begin{array}{ccc|c} 1 & -4 & 5 & 0 \\ 2 & -3 & 5 & 0 \\ 3 & 2 & 1 & 0 \end{array}\right]$

Now, we need to change a_{21} to 0 and a_{31} to 0:

$$\xrightarrow[-3R_1 + R_3 \to R_3]{-2R_1 + R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & -4 & 5 & 0 \\ 0 & 5 & -5 & 0 \\ 0 & 14 & -14 & 0 \end{array}\right)$$

We need to change a_{22} to 1 and a_{32} to 0

$$\xrightarrow{\frac{1}{5}R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & -4 & 5 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 14 & -14 & 0 \end{array}\right) \xrightarrow[4R_2 + R_1 \to R_1]{-14R_2 + R_3 \to R_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The last matrix corresponds to the system $\begin{cases} x + z = 0 \\ y - z = 0 \\ 0 = 0 \end{cases} \Rightarrow \begin{cases} x = -z \\ y = z \end{cases}$

The system has infinite number of solution and let $y = z = t \in \mathbb{R}$

$\therefore$ The system has a general solution of $\{(-t, t, t) \mid t \in \mathbb{R}\}$

$\therefore$ This system of homogeneous equation has a trivial solution (0, 0, 0) and a non - trivial solution of $\{(-t, t, t) \mid t \in \mathbb{R}\}$.

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c. The augmented matrix for the system $\left(\begin{array}{ccc|c} 1 & 3 & -4 & 0 \\ 2 & 7 & 1 & 0 \\ 3 & -5 & -2 & 0 \end{array}\right)$

The augmented matrix is reduced as follow: We need to change a_{21} to 0 and a_{31} to 0.

$$\xrightarrow[-3R_1 + R_3 \to R_3]{-2R_1 + R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & 3 & -4 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & -14 & -10 & 0 \end{array}\right)$$

$$\xrightarrow[\frac{1}{116} R_3 \to R_3]{14R_2 + R_3 \to R_3} \left(\begin{array}{ccc|c} 1 & 3 & -4 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow[\substack{4R_3 + R_1 \to R_1 \\ -3R_2 + R_1 \to R_1}]{-9R_3 + R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$x = 0, y =$ and $z = 0$.

$\therefore$ The system has only trivial solution (0, 0, 0)

d. The augmented matrix form is $\left(\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -2 & -4 & 0 \end{array}\right)$

We need to change a_{21} to 0 and a_{31} to 0.

$$\xrightarrow[-R_1 + R_3 \to R_3]{-R_1 + R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & -5 & -5 & 0 \end{array}\right) \xrightarrow[-\frac{1}{5} R_3 \to R_3]{-\frac{1}{2} R_2 \to R_2} \left(\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right)$$

We need to change a_{32} to 0 and a_{12} to 0

$$\xrightarrow[-3R_2 + R_1 \to R_1]{-R_2 + R_3 \to R_3} \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \Rightarrow \begin{cases} x - 2z = 0 \\ y + z = 0 \\ 0 = 0 \end{cases}$$

The system has an infinite number of solution $\{(2z, -z, z)\}$, let $z = t \in \mathbb{R}$

$$\{(2t, -t, t) | \, t \in \mathbb{R}\}$$

$\therefore$ The system has a trivial solution (0, 0, 0) and non trivial solution

$$\{(2t, -t, t) | \, t \in \mathbb{R}\}$$

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3.8 The Inverse of Matrix

If the product of the real numbers a and b is 1, then a and b are multiplicative inverse. If A is non - singular matrix, then it has an inverse denoted by A^{-1}

- The product of matrix and its inverse gives the identity matrix.

DEFINITION: Multiplicative Inverse of Matrix.

If A and B are n × n matrices, and AB = BA = I, then B is multiplicative inverse of A, written A^{-1}.

$$AA^{-1} = A^{-1}A = I_n$$

- An invertible matrix has one and only one inverse

Let's use definition of inverse matrices to **find A^{-1}**, if it exists for

$$A = \begin{bmatrix} 5 & 9 \\ 1 & 2 \end{bmatrix} \text{ and let } A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

By the definition of matrix inverse: $AA^{-1} = A^{-1}A = I$.

$$\underset{A}{\begin{bmatrix} 5 & 9 \\ 1 & 2 \end{bmatrix}} \underset{A^{-1}}{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5a + 9c & 5b + 9d \\ a + 2c & b + 2d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Which is true only if

$5a + 9c = 1$ and $5b + 9d = 0$

$a + 2c = 0$ $\quad b + 2d = 1$

Solving these two systems, we find a = 2, b = -9, c = - 1 and d = 5.

$$\therefore \quad A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2 & -9 \\ -1 & 5 \end{bmatrix}$$

Illustrative Example

61. Determine whether two matrices are inverses of each other

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a. $A = \begin{bmatrix} 1 & 3 \\ 4 & 13 \end{bmatrix}$ and $B = \begin{bmatrix} 13 & -3 \\ -4 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ -7 & 6 \end{bmatrix}$

Solution: We must show that $AB = I_2$ and $BA = I_2$

a. $A = \begin{bmatrix} 1 & 3 \\ 4 & 13 \end{bmatrix} \begin{bmatrix} 13 & -3 \\ -4 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 \times 13 + 3 \times (-4) & 1 \times (-3) + 3 \times 1 \\ 4 \times 13 + 13 \times (-4) & 4 \times (-3) + 13 \times 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\therefore$ A and B are multiplicatives inverse of each other.

b. $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ -7 & 6 \end{bmatrix} \begin{bmatrix} 2 \times 4 + 1 \times (-7) & 2 \times (-3) + 1 \times 6 \\ 3 \times 4 + 4 \times (-7) & 3 \times (-3) + 4 \times 6 \end{bmatrix} = \begin{bmatrix} 1 & \\ -16 & \end{bmatrix}$

$AB \neq I$, therefore, A and B are not inverse of each other.

Finding the Inverse of a Matrix Using Elementary Row Operations

Let A be a square matrix of size n and let I_n be identity matrix

To find A^{-1}:

STEP 1 : First Form the augmented matrix $[A|I_n]$

STEP 2 : Perform row operation $[A|I_n]$ to get row operation the form $[I$

STEP 3 : Then matrix $B = A^{-1}$

Symbol : $[A|I_n] \longrightarrow [I_n|B] = [I_n|A^{-1}]$

Illustrative Example

62. Determine whether each of the following are invertible (non - singular) or non - invertible (singular) matrices. If it is invertible, fi their inverses.

a. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ b. $A = \begin{bmatrix} -2 & 4 \\ 1 & -2 \end{bmatrix}$ c. $A = \begin{bmatrix} 5 & 3 \\ 4 & 2 \end{bmatrix}$

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Solution

a. We first form the matrix $[A|I] = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 1 \end{bmatrix}$

$$\xrightarrow{-3R_1 + R_2 \to R_2} \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{bmatrix} \xrightarrow{-\frac{1}{2}R_2 \to R_2} \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

change a_{12} to 0.

$$\xrightarrow{-2R_2 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

A is invertible and the inverse:

$B = A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$ is non-singular matrix.

b. We first form the matrix:

$$[A|I] = \begin{bmatrix} -2 & 4 & 0 & 0 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

change a_{11} to 1, and a_{21} to 0

$$R_1 \leftrightarrow R_2 \begin{bmatrix} 1 & -2 & 0 & 1 \\ -2 & 4 & 1 & 0 \end{bmatrix} \xrightarrow{2R_1 + R_2 \to R_2} \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

The matrix $B = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$ can not reduced into identity matrix.

∴ Matrix A is not invertible (singular)

c. We first form the matrix: $[A|I] = \begin{bmatrix} 5 & 3 & 1 & 0 \\ 4 & 2 & 0 & 1 \end{bmatrix}$

$$R_1 \leftrightarrow R_2 \begin{bmatrix} 4 & 2 & 0 & 1 \\ 5 & 3 & 1 & 0 \end{bmatrix} \xrightarrow{\frac{1}{4}R_1 \to R_1} \begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{4} \\ 5 & 3 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{-5R_1 + R_2 \to R_2} \begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{4} \\ 0 & \frac{1}{2} & 1 & \frac{5}{4} \end{bmatrix} \xrightarrow[2R_2 \to R_2]{-R_2 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & \frac{5}{2} \end{bmatrix}$$

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$\therefore \quad B = A^{-1} = \begin{bmatrix} -1 & -1 \\ 2 & 5/2 \end{bmatrix}$ is invertible (non - singular)

Illustrative Example

63. Find A^{-1} if $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$

Solution

Merge the given matrix with the identity matrix I_2.

$$\left[\begin{array}{cc|cc} 3 & 5 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 3 & 5 & 1 & 0 \end{array}\right] \xrightarrow{-3R_1 + R_2 \to R_2} \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & -3 \end{array}\right]$$

Change a_{12} to 0 and a_{22} to 1

$$\left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & -3 \end{array}\right] \xrightarrow[-R_2 \to R_2]{2R_2 + R_1 \to R_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -5 \\ 0 & 1 & -1 & 3 \end{array}\right]$$

$$\therefore \quad A^{-1} = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$$

64. Find A^{-1} if $A = \begin{bmatrix} -2 & 2 & 3 \\ 1 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix}$.

Solution

Form the matrix $[A|I] \longrightarrow [I|A^{-1}]$ is as follows:

$$\left[\begin{array}{ccc|ccc} -2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ -2 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow{2R_1 + R_2 \to R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 3 & 1 & 2 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_2 \leftrightarrow R_3]{R_3 + R_1 \to R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \\ 0 & 0 & 3 & 1 & 2 & 0 \end{array}\right] \xrightarrow{\frac{1}{3}R_3 \to R_3}$$

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$$\left[\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & 0 \end{array}\right] \xrightarrow[-4R_3 + R_1 \to R_1]{-4R_3 + R_2 \to R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{4}{3} & -\frac{5}{3} & 0 \\ 0 & 1 & 0 & -\frac{4}{3} & -\frac{8}{3} & 1 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & 0 \end{array}\right]$$

$$\therefore \quad A^{-1} = \frac{1}{3}\begin{bmatrix} -4 & -5 & 0 \\ -4 & -8 & 3 \\ 1 & 2 & 0 \end{bmatrix}$$

65. Let $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ find A^{-1} for $a, \neq 0, b \neq 0$.

Solution

$$[A|I_2] = \left[\begin{array}{cc|cc} a & 0 & 1 & 0 \\ 0 & b & 0 & 1 \end{array}\right] \xrightarrow[\frac{1}{b}R_2 \to R_2]{\frac{1}{a}R_1 \to R_1} \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{a} & 0 \\ 0 & 1 & 0 & \frac{1}{b} \end{array}\right]$$

$$\therefore \quad A^{-1} = \begin{bmatrix} \frac{1}{a} & 0 \\ 0 & \frac{1}{b} \end{bmatrix}$$

66. Let $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ $a, \neq 0, b \neq 0, c \neq 0$ find A^{-1}

Solution

$$\left[\begin{array}{ccc|ccc} a & 0 & 0 & 1 & 0 & 0 \\ 0 & b & 0 & 0 & 1 & 0 \\ 0 & 0 & c & 0 & 0 & 1 \end{array}\right] \xrightarrow[\frac{1}{c}R_3 \to R_3]{\frac{1}{a}R_1 \to R_1 \\ \frac{1}{b}R_2 \to R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{a} & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{1}{b} & 1 \\ 0 & 0 & 1 & 0 & \frac{1}{c} & 0 \end{array}\right]$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{1}{a} & 0 & 0 \\ 0 & \frac{1}{b} & 0 \\ 0 & 0 & \frac{1}{a} \end{bmatrix}$$

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67. If $A^{-1} = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$. Find matrix B such that $AB = \begin{bmatrix} 2 & -4 \\ 0 & -3 \end{bmatrix}$

Solution

We know that $A^{-1}A = AA^{-1} = I$

$$(A^{-1}A)B = A^{-1}(AB) = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & -4 \\ 0 & -3 \end{bmatrix}$$

$$IB = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{bmatrix} 2\times 2 + (-5\times 0) & 2\times(-4)+(-5)\times(-3) \\ -1\times 2 + 3\times 0 & -1\times(-4)+3\times(-3) \end{bmatrix}$$

$$\therefore \quad B = \begin{bmatrix} 4 & 7 \\ -2 & -5 \end{bmatrix}$$

68. If $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}\begin{bmatrix} -7 & 8 & 8 \\ 8 & -10 & -4 \\ 4 & k & -2 \end{bmatrix}$ then what are the values of m and k ?

A. $m = 3, k = 2$

B. $m = 5, k = -2$

C. $m = 1, k = -3$

D. $m = 2, k = -3$

Solution: We use $AA^{-1} = I$, (identity)

$$A^{-1}A = \frac{1}{6}\begin{bmatrix} -7 & 8 & 8 \\ 8 & -10 & -4 \\ 4 & k & -2 \end{bmatrix}\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_{11} = \frac{1}{6}(-7\times 2 + 8\times 0 + m\times 4) = 1$$

$$\frac{1}{6}(-14 + 4m) = 1 \text{ implies } 4m - 14 = 6$$

$$4m = 20, \text{ and } m = 5$$

$$C_{32} = \frac{1}{6}[4\times 1 + k\times(-1) + (-2)\times 3]$$

$$4 - k - 6 = 0, \; k = -2$$

Answer: B

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69. If $A=\begin{pmatrix}0 & x & 0\\ 1 & -1 & 1\\ 0 & y & -1\end{pmatrix}$ and $A^{-1}=\begin{pmatrix}1 & 1 & 1\\ 3 & 0 & 0\\ 2 & 0 & -1\end{pmatrix}$ then what are the values of x and y?

A. $x=3,\ y=-2$

B. $x=\frac{2}{3},\ y=\frac{1}{3}$

C. $x=-3,\ y=2$

D. $x=\frac{1}{3},\ y=\frac{2}{3}$

Solution: The product of A and A^{-1} I

$$C=AA^{-1}=\begin{pmatrix}0 & x & 0\\ 1 & -1 & 1\\ 0 & y & -1\end{pmatrix}\begin{pmatrix}1 & 1 & 1\\ 3 & 0 & 0\\ 2 & 0 & -1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

$$C_{111}=((0\times\ 0))\begin{pmatrix}1\\ 3\\ 2\end{pmatrix}=((0))((1))+3((x))+0((2))=1,\ 3x=1,\ x=\frac{1}{3}$$

$$C_{311}=((0\ y\ -1))\begin{pmatrix}1\\ 3\\ 2\end{pmatrix}=0+3y-2=0,\ 3y=2,\ y=\frac{2}{3}$$

Answer: D

70. Show that the matrix $A=\begin{bmatrix}1 & 3 & -2\\ 1 & 2 & 0\\ 2 & 8 & -8\end{bmatrix}$ is not invertible (singular)

Solution

The reduction of $[A\|I]\longrightarrow[I\|B]$

$$[A\|I]=\left[\begin{array}{ccc|ccc}1 & 3 & -2 & 1 & 0 & 0\\ 1 & 2 & 0 & 0 & 1 & 0\\ 2 & 8 & -8 & 0 & 0 & 1\end{array}\right],\ \begin{matrix}\text{change } a_{21}\to 0\\ \text{change } a_{31}\to 0\end{matrix}$$

$$\begin{matrix}-R_1+R_2\to R_2\\ -2R_1+R_3\to R_3\end{matrix}\left[\begin{array}{ccc|ccc}1 & 3 & -2 & 1 & 0 & 0\\ 0 & -1 & 2 & -1 & 1 & 0\\ 0 & 2 & -4 & -2 & 0 & 1\end{array}\right]\therefore \text{change } a_{32}\to 0$$

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$$\xrightarrow[-R_2 \to R_2]{2R_2 + R_3 \to R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 0 & -4 & 2 & 0 \end{array}\right]$$

Since A will never be transformed to the identity matrix by elementary row operation, A is not invertible, (singular matrix).

PROPERTIES OF INVERSE MATRIX

Let A and B are invertible square matrices. Then:

(i) $A^{-1} = A$

(ii) $(A^t)^{-1} = (A^{-1})^t$

(iii) $(CA)^{-1} = \frac{1}{C} A^{-1}, C \neq 0$

(iv) $(AB)^{-1} = B^{-1} A^{-1}$

(v) $(A^{-n}) = (A^{-1})^n$

IMPORTANT: LET A DENOTE A SQUARE TRIANGULAR MATRIX:

1. A is invertible if and only if no entry on the main diagonal is zero.
2. If A is upper triangular, then A^{-1} is also upper triangular.

Example Let $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, $a \neq 0$, $c \neq 0$

Then $A^{-1} = \begin{bmatrix} \frac{1}{a} & \frac{-b}{ac} \\ 0 & \frac{1}{c} \end{bmatrix}$

Illustrative Example

71. Let $A^{-1} = \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$ then find

a. $(4\ AB)^{-1}$ b. $\left(\frac{1}{2} A^t\ B\right)^{-1}$

Solution

$$\text{a.}(4\ AB)^{-1} = \frac{1}{4}\ B^{-1}\ A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -4 & -3 \end{pmatrix}$$

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$$= \frac{1}{4}\begin{bmatrix} 9-20 & 6-15 \\ 3+8 & 2+6 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -11 & -11 \\ 11 & 8 \end{bmatrix}$$

b. $\left(\frac{1}{2}A^t B\right)^{-1} = 2B^{-1}\left(A^{-1}\right)^t = 2\begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$

$$= 2\begin{bmatrix} 9+10 & -12-15 \\ 3-4 & -4+6 \end{bmatrix} = \begin{bmatrix} 38 & -54 \\ -2 & 4 \end{bmatrix}$$

Illustrative Example

72. Let $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, then find A^{-2}

Solution

The fact that $A^{-2} = (A^{-1})^2 = A^{-1} \cdot A^{-1}$

Then $A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$

$\therefore \quad A^{-2} = A^{-1} \cdot A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix}$

73. If $A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$, then find the value of matrix $(4AB)^{-1}$

Solution : $(4AB)^{-1} = \frac{1}{4} B^{-1} \cdot A^{-1}$

$$= \frac{1}{4}\begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 16 & -9 \\ -7 & 4 \end{pmatrix}$$

74. **Conceptual Skill:**

a. Is the sum of two invertible matrices necessarily invertible? Give example.

Answer : NO. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}$

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$\therefore \quad A^{-1} + B^{-1} = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, this is not invertible.

b. Is the product of two invertible matrices always invertible.

Answer: Yes

Examples: Involving Application of matrices.

75. Conceptual Analysing: Two bottles of water and 1 bottle of juices sell for Rs. 25.50. Two bottles of juice and 1 bottle of milk sell for 24.50. One bottle water, 1-bottles of juice, and 1-bottle of milk sells for Rs. 23.00 Find the price of the water, juice, and milk.

Solution

Let x, y and z to represent the various number of bottles, of water, juice and milk respectively.

$$\begin{cases} x + y + z = 23 \\ 2y + z = 24.50 \\ 2x + y = 25.50 \end{cases} \longrightarrow$$

To eliminate x in 3rd equation.

Multiply 1st equation by –2 and the result to 3rd equation.

$$\begin{cases} x + y + z = 23 \\ y + z = 24.50 \\ -y + 2z = 20.5 \end{cases} \longrightarrow$$

Interchange 2nd and 3rd equation

$$\begin{cases} x + y + z = 23 \\ y + 2z = 20.50 \\ 2y + z = 24.50 \end{cases} \longrightarrow$$

Eliminate y in 3rd equation, multiply 2nd equation by (–2) and add the result equation

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 23 \\ 0 & 2 & 1 & 24.50 \\ 2 & 1 & 0 & 25.50 \end{array}\right]$$

To change a_{31} to 0 Multiply row 1 by –2 and add the result to row – 3

$$\xrightarrow{-2R_1 + R_3 \longleftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 23 \\ 0 & 2 & 1 & 24.50 \\ 0 & 1 & 2 & 20.50 \end{array}\right]$$

To change a_{23} to 1, interchange row 2 and row 3.

$$\xrightarrow{R_2 \longleftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 23 \\ 0 & 1 & 2 & 20.50 \\ 0 & 2 & 1 & 24.50 \end{array}\right]$$

To change a_{32} to 0, multiply row 2 by (–2) and add the result to 3rd equation

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$$\begin{cases} x + y + z = 23 \\ y + 2z = 20 - 50 \\ -3z = -16.50 \end{cases} \longrightarrow \xrightarrow[\frac{1}{3}R_3 \to R_3]{-2R_2 + R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 23 \\ 0 & 1 & 2 & 20.50 \\ 0 & 0 & 1 & 5.50 \end{array}\right]$$

$-3z = -16.50$

$\therefore\ Z = \dfrac{16.5}{3} = 5.50$

At this point, you can use back substitution to find x and y.

$\therefore\quad y + 2(5.50) = 20.50,\ y = 20.50 - 11 = 9.50$

To solve for x, $x = 23 - y - z = 23 - 9.50 - 5.50 = 8$

$\therefore$ The price of water is Birr 8, the price of juice is Birr 9.50, and the price of milk is Birr 5.50.

76. Conceptual Analyzing: A fruit market packages fruit in three different ways for gift baskets.

The economy basket E has 8 oranges, 6 apples, and 4 grape fruits.

The standard basket S has 6 oranges, 7 apples and 5 grape fruits.

The luxury basket L has 10 oranges 8 apples and 5 grape fruits.

The cost is Birr 40, Birr 70 and Birr 50 for each orange, apple and grape fruits.

What is the cost of preparing each basket of fruit?

Solution: We arrange the given data in matrix form as follows:

Cost per fruit			Number of items in each basket type		
Orange	Apple	grape fruit	E	S	L

$$A = \text{Birr}\begin{bmatrix} 40 & 70 & 50 \end{bmatrix} \text{ and } B = \begin{matrix} \text{Apple} \\ \\ \\ \end{matrix}\begin{bmatrix} 8 & 6 & 10 \\ 6 & 7 & 8 \\ 4 & 5 & 5 \end{bmatrix}$$

To find the cost of each basket multiply matrix B by matrix A to get:

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$$AB = \begin{bmatrix} 40 & 70 & 50 \end{bmatrix} \cdot \begin{matrix} E \\ S \\ L \end{matrix} \begin{bmatrix} 8 & 6 & 10 \\ 6 & 7 & 8 \\ 4 & 5 & 5 \end{bmatrix} = \begin{bmatrix} 940 & 980 & 1210 \end{bmatrix}$$

$C_{11} = 40 \times 8 + 70 \times 60 + 50 \times 4 = 940$ Birr (Economy)

$C_{12} = 40 \times 6 + 70 \times 7 + 50 \times 5 = 980$ Birr (Standard)

$C_{13} = 40 \times 10 + 70 \times 8 + 50 \times 5 = 1210$ Birr (Luxury)

$\therefore$ The costs of each economy, standard and luxury basket are Birr 940, Birr 980, and Birr 1210, respectively.

77. **Mixture Problem :** How many gallons of 20% alcohol solution and 50% alcohol solution must be mixed to get 9 gallons of 30% alcohol solution.

Solution

Let x = the number of gallons of 20% alcohol solution needed.

y = the number of gallons of 50% alcohol solution needed.

Total number of gallons: $x + y = 9$

Alcohol solution on gallons: $0.2x + 0.5y = (0.30)(9) = 2.7$

Solving together: $\begin{cases} x + y = 9 \\ 2x + 5y = 27 \cdot \text{ multiplying by } 10 \end{cases}$

Multiply 1st equation by (-2) and add the result to 2nd equation

$$\begin{cases} x + y = 2 \\ \quad y = 3 \end{cases} \qquad \text{and } x = 9 - y = 9 - 3 = 6$$

$\therefore$ It takes 6 gallons of 20% alcohol solution and 3 gallons of 50% alcohol solution to produce 9 gallons of 30% alcohol solution.

78. **Mixture problem**: How many liters each of 15% acid and 33% acid should be mixed to get 40 liters of 21% acid.

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Solution

(i) $\begin{pmatrix}\text{Voume of}\\ \text{Solution } x\end{pmatrix} + \begin{pmatrix}\text{Volume of}\\ \text{solution } y\end{pmatrix} = \text{Volume of mixture}$

$x + y = 40$

(ii) $\begin{pmatrix}\text{Acid in}\\ \text{Solution } x\end{pmatrix} + \begin{pmatrix}\text{Acid in}\\ \text{Solution} y\end{pmatrix} = \text{Acid in mixture}$

$0.15x + 0.33y = 40(0.21) = 8.4$

Solving together: $\begin{cases} x + y = 40 \\ 0.15x + 0.33y = 8.4 \end{cases}$

Using substitution to solve for x,

$x = 40 - y$

$0.15\,(40 - y) + 0.33y = 8.4$

$6 - 0.15y + 0.33y = 8.4$

$0.18y = 8.4 - 6 = 2.4$

$y = \frac{2.4}{0.18} = 13\frac{1}{3}$ and $x = 26\frac{2}{3}$

$\therefore$ $26\frac{2}{3}$ liters each of 15% acid and $13\frac{1}{3}$ liters each of the 33% acid were used.

79. Motion problem: A car travels 250km in the same time that a truck travels 225km. If the speed of the car is 8km/hr faster than the speed of the truck. Find the speed of car and truck

Solution

Let x = the speed of the car and y = the speed of the truck

	distance	Speed (r)	time (t)
Car	250km	x	
Truck	225km	y	

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$$\text{time} = \frac{\text{distance}}{\text{Speed}}$$

since both vehicles travels for the same time:

$$\frac{250}{x} = \frac{225}{y} \text{ implies } 250y = 225x$$

And the car travels 8km/hr faster than the truck: $x = y + 8$

Solving together: $\begin{cases} x - y = 8 \\ 225x - 250y = 0 \end{cases}$

Multiply 1st equation by - 225 and the result to 2nd equation.

$\begin{cases} x - y = 8 \\ y = 72, \text{ then } x = 8 + 72 = 80 \end{cases}$

$\therefore$ The speed of the car is 80km/hr and speed of the truck is 72km/hr

Review Exercise 3

1. a. What is matrix ?

   b. What is the order of a matrix?

   c. What is a squire matrix?

   d. What is a row matrix?

   e. What is a column matrix?

   f. What are the requirements for two matrices to be equal?

   g. What are the requirements to multiply two matrices A and B?

2. Given two matrices A and B, how can you determine the order of AB, if AB is defined?

3. If matrix A is an m × n matrix and matrix B is an n × P matrix, then What is the size (order) of the product matrix AB?

4. If A is 4 × 8 matrix and B is 8 × 6, then what will be the order of AB.

5. Solve for matrix A

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a. $4A + 2\begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 4 & 2 \end{bmatrix}$

b. $4A - 2\begin{bmatrix} -1 & 3 \\ 2 & -5 \end{bmatrix} = \begin{bmatrix} 13 & -9 \\ 4 & 16 \end{bmatrix}$

6. Solve for x and y

a. $\begin{bmatrix} -2 & 1 \\ 3 & 2 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 4 & x \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -6 \\ 19 & 9 \\ 8 & y \end{bmatrix}$

b. $\begin{bmatrix} 4 & 1 \\ -2 & x \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix}\begin{bmatrix} 9 & -2 \\ 2 & 1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} y & 5 \\ -13 & 11 \end{bmatrix}$

7. let $A = \begin{pmatrix} -2 & 1 & 3 \\ 4 & -1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 4 \\ 1 & -1 \\ 3 & 2 \end{pmatrix}$

Find a. AB b. BA

8. If $\begin{pmatrix} 0 & 4 & 3 \\ 2x & 0 & 7 \\ 3 & 2-k & 0 \end{pmatrix}$ is a skew symmetric matrix then find the value of x and k.

9. Determine whether each of the following statement is true or false.

a. If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, then $A^2 = 2A$

b. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, then $A^2 = \begin{bmatrix} 4 & 9 \\ 1 & 16 \end{bmatrix}$

c. Every matrix has an additive inverse

d. Every square matrix has a multiplicative inverse

e. Let the matrices A, B, and C be square matrices of order n. If AB = AC,

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then B = C

f. If A and B are matrices of order n, then AB = BA

g. $(A - B)^2 = A^2 - 2AB + B^2$

h. $(A - B)(A + B) = A^2 - B^2$

i. If AB = 0 (the zero matrix), then A = 0 or B = 0.

10. If $A = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix}$, find a matrix E such that E + B = A

11. Find the matrix A such that $\left\{3A^t - 2\begin{bmatrix} 0 & 2 \\ -5 & 2 \\ 1 & 3 \end{bmatrix}\right\}' = \begin{bmatrix} 7 & 0 & -1 \\ 2 & 1 & 5 \end{bmatrix}$

12. Explain the difference between a matrix in row - echelon form and reduced row - each form

13. In each of the following determine if the matrix is in row - echelon form.

a. $\left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0 & 2 & 5 \end{array}\right]$ b. $\left[\begin{array}{ccc|c} 1 & 3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 2 & 1 & 3 \end{array}\right]$ c. $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 4 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right]$ d. $\left[\begin{array}{ccc|c} 1 & 3 & 4 & -5 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 1 & 0 \end{array}\right]$

14. In each case, determine if the matrix is reduced row- echelon form. If not explain why.

a. $\left[\begin{array}{ccc|c} 1 & 0 & 2 & -3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right]$ b. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ c. $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ e. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

15. In each case, determine the system as consistent or inconsistent

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a. $\begin{cases} -3x + y = 7 \\ 3x + 2y = 4 \end{cases}$ b. $\begin{cases} y - 3x = 4 \\ y - 3x = -2 \end{cases}$ c. $\begin{cases} y - 3x = 6 \\ 6x - 2y = -12 \end{cases}$

16. Determine whether the matrices are multiplicative inverses.

a. $A = \begin{bmatrix} 2 & 0.5 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 1 \\ 10 & -4 \end{bmatrix}$

b. $A = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$

c. $A = \begin{bmatrix} -3 & 2 \\ 3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 5 & 2 \end{bmatrix}$

d. $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -3 \\ -1 & 4 \end{bmatrix}$

17. Write the augmented matrix for each of the following equations:

a. $\begin{cases} 2x - 3y = 9 \\ -3x + y = -1 \end{cases}$ c. $\begin{cases} x - y = 4 \\ 2x + y = 3 \end{cases}$

b. $\begin{cases} x - y + z = 1 \\ x + y - 2z = 3 \\ y - 3z = 4 \end{cases}$ d. $\begin{cases} x + y = 2 \\ y - 3z = 5 \\ -3x + 2z = 8 \end{cases}$

18. Find the solution set of each of the following augmented matrix.

a. $\left[\begin{array}{cc|c} 5 & 1 & -8 \\ 0 & 1 & -3 \end{array}\right]$ b. $\left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -1 \end{array}\right]$ c. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ -1 & 0 & 1 & -3 \\ 1 & 1 & 0 & 1 \end{array}\right]$ d. $\left[\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & 5 & 15 \end{array}\right]$

Answer To Review Executrices 3

1. a. A matrix is a rectangular array of numbers.

b. The order of a matrix is the number of rows and columns

c. A square matrix is a matrix with the same number of rows and columns.

d. A row matrix is a matrix with only one row.

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e. A column matrix is a matrix with only one column.

f. The order of the matrices must be the same, and the corresponding elements must be equal.

g. The number of columns of A must equal to the number of rows of B, for the matrix product AB.

2. The order of product AB will be the number of rows of A by the number of columns of B.

3. The size (order) of the product matrix AB is an $m \times p$;

4. The size (order) of AB is the number of rows of A by the number of column of B.

5. a. $A = \frac{1}{4}\begin{bmatrix}10 & 0\\ 4 & 2\end{bmatrix} - 2\begin{bmatrix}3 & 4\\ -2 & 1\end{bmatrix} = \begin{bmatrix}1 & -2\\ 2 & 0\end{bmatrix}$

b. $A = \frac{1}{4}\begin{bmatrix}13 & -9\\ 4 & 16\end{bmatrix} + 2\begin{bmatrix}-1 & 3\\ 2 & -5\end{bmatrix} = \frac{-1}{4}\begin{bmatrix}11 & -3\\ 8 & 6\end{bmatrix}$

6. a. $C_{11} = -2 + x + 6 = 6, \quad x = 2$

$C_{32} = 0 - 4 - 4 = y, \quad y = -8$

b. $C_{11} = (4 \quad 1)\begin{pmatrix}9\\ 2\\ -1\end{pmatrix} = 36 + 2 - 3 = y = 35$

$C_{21} = (-2 \quad x \quad 1)\begin{pmatrix}9\\ 2\\ -1\end{pmatrix} = -18 + 2x - 1 = -13, \; x = 3$

7. a. $AB = \begin{pmatrix}14 & -3\\ -3 & 24\end{pmatrix}$, b. $BA = \begin{pmatrix}20 & -6 & 2\\ -6 & 2 & 1\\ 2 & 1 & 13\end{pmatrix}$

8. If $a_{12} = 4$, then $a_{21} = -4, \Rightarrow 2x = -4, x = -2$

If $a_{23} = 7$, then $a_{32} = -7, \Rightarrow 2 - k^2 = -7, k = \pm 3$.

9. a. True b. False, $A^2 = AA = \begin{bmatrix}2 & 3\\ 1 & 4\end{bmatrix}\begin{bmatrix}2 & 3\\ 1 & 4\end{bmatrix} = \begin{bmatrix}7 & 18\\ 6 & 19\end{bmatrix}$

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c. True d. False, $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ has no inverse

e. False, as an example, $A = \begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 1 & 5 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 7 \\ 3 & 11 \end{bmatrix}$

$AB = AC$, but $B \neq C$

f. False, as an example, $A = \begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 5 \end{bmatrix}$ $AB \neq BA$

g. False, $AB \neq BA$

h. False, $A^2 - B^2 = A^2 + AB - BA - B^2$ since $AB \neq BA$

i. False, as an example, $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ $A \neq 0$, $B \neq 0$,

but $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

10. $E = A - B = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 6 & 0 \end{bmatrix}$

11. $\left\{3A^t - 2\begin{bmatrix} 0 & 2 \\ -5 & 2 \\ 1 & 3 \end{bmatrix}\right\}^t = (3A^t)^t - 2\begin{bmatrix} 0 & -5 & 1 \\ 2 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 7 & 0 & -1 \\ 2 & 1 & 5 \end{bmatrix}$

$$A = \frac{1}{3}\left\{\begin{bmatrix} 7 & 0 & -1 \\ 2 & 1 & 5 \end{bmatrix} + 2\begin{bmatrix} 0 & -5 & 1 \\ 2 & 2 & 3 \end{bmatrix}\right\} = \frac{1}{3}\begin{bmatrix} 7 & -10 & 1 \\ 6 & 5 & 11 \end{bmatrix}$$

12. In reduced echelon form all the elements above the leading 1's must be zero.

13.a. Not in row - echelon form, because the element on the main diagonal in the second row is not 1

b. Not in row - echelon form, because the element below the leading 1 in the second row should be zero.

c. row - echelon form. d. row - echelon form.

14.a. No, the element in the first row 3rd column should be zero.

b. Yes c. Yes d. Yes e. No

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15. a. Consistent, (it has one solution)

b. In consistent (It has no - solution)

c. Consistent (It has infinite solution)

16. a. $AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$ inverse of each other

b. $AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$, multiplicative inverse

c. $AB = \begin{bmatrix} 13 & 4 \\ 17 & -8 \end{bmatrix}$, not multiplicative inverse

d. not multiplicative inverse, $AB = \begin{bmatrix} 7 & -8 \\ 4 & -1 \end{bmatrix} \neq I_2$

17. a. $\left[\begin{array}{cc|c} 2 & -3 & 9 \\ -3 & 1 & -1 \end{array}\right]$ b. $\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 2 & 1 & 3 \end{array}\right]$ c. $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 1 & 1 & -2 & 3 \\ 0 & 1 & -3 & 4 \end{array}\right]$ d. $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & -3 & 5 \\ -3 & 0 & 2 & 8 \end{array}\right]$

18. a. $\begin{cases} 5x + y = -8 \\ y = -3 \end{cases}$ and $5x - 3 = -8$, $x = -1$

The solution set is $\{(-1, -3)\}$

b. $x = 4$ and $y = -1$, the solution set is $\{(4, -1)\}$

c. $\begin{cases} x = 6 \\ -x + z = -3 \\ x + y = 1 \end{cases} \Rightarrow \begin{cases} -6 + z = -3 \Rightarrow z = 3 \\ 6 + y = 1 \Rightarrow y = -5 \end{cases}$

The solution set is $\{(6, -5\ 3)\}$

d. $\begin{cases} x - y = -1 \\ 5y = 15, \text{ and } y = 3 \text{ and } x = -1 + 3 = 2 \end{cases}$

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UNIT 4

4. Determinants

In this units, we going to present another method to solve a linear system with one unique solution. This method, called **cramer's rule**. Before studying cramer's rule, we need to become familar with the determinant of a square matrix. Determinants have many applications in mathematics including:

- Finding Inverse of matrix.
- Solving system of linear equations, (cramer's rule)
- Finding the area of a triangle
- Determining whether three points are collinear.

What is Determinant?

Associated with every square matrix is a real number called the **determinants** of the matrix. A determinant of a square matrix A, denoted by $|A|$, is written by the elements of the matrix with in two vertical bars. The determinant of 2×2 matrix is defined as follows.

4.1 Determinant of a 2 × 2 Matrix

The determinant of the matrix is

$$\det |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Note: If A is a square matrix of order 1×1, then A has only one element. Thus $A = [a_{11}]$, then $|A| = |a_{11}| = a_{11}$.

Illustrative Example

1. Find the determinate of the matrix.

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a. $A=\begin{bmatrix}-5 & 4\\ -3 & 2\end{bmatrix}$ b. $B=\begin{bmatrix}6 & 4\\ -3 & 2\end{bmatrix}$ c. $E=\begin{bmatrix}a\\ b\end{bmatrix}$

Solution

a. $\det(A) = |A| = \begin{vmatrix}-5 & 4\\ -3 & 2\end{vmatrix} = (-5)(2) - (4)(-3) = -10 + 12 = 2$

b. $\det(B) = |B| = \begin{vmatrix}6 & 4\\ -3 & 2\end{vmatrix} = 6 \times 2 - 4(-3) = 12 + 12 = 24$

c. $\det(E) = |E| = \begin{vmatrix}a & -a\\ b & -b\end{vmatrix} = (a)(-b) - (-a)(b) = -ab + ab = 0$

2. Let $A=\begin{pmatrix}2x & 1\\ x & 3\end{pmatrix}$. For What values of x, $\det(A) = 10$?

Solution

$\det(A) = |A| = \begin{vmatrix}2x & 1\\ x & 3\end{vmatrix} = 6x - x = 10,\ x = 2$

Minors and Cofactors

Each elements of a determinant are related two quantities the minor and the cofactor of that element. The minor of an elemen is the determinant obtained by deleting the row and the colum in which that element lies.

To define the determinant of a matrix of order 3 × 3, we first need to defi the **minor** and the **cofactor** of an element of the matrix.

DEFINITION:

Let $A = (a_{ij})$ of matrix of order: $n \times n$

(i) The minor M_{ij} of the element a_{ij} is the determinant of the matrix obtained by deleting the i^{th} row and j^{th} column of A.

Hence $M_{ij} = |A_{ij}|$

(ii) The cofactor C_{ij} of the element a_{ij} is

$C_{ij} = (-1)^{i+j} M_{ij} = (-1)^{i+j} |A_{ij}|$, where M_{ij} is the minor of a_{ij}.

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Finding the minor and cofactor of a Matrix.

Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

To find the minor of an element a_{11}, a_{12}, and a_{23} is M_{11}, M_{12} and M_{23} respectively

- The minor M_{11} of an element a_{11} is the determinant of the matrix obtained by deleting the 1st row and 1st column from A
- The minor of an element a_{11} is $M_{11} = |A_{11}|$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}\,a_{33} - a_{32}\,a_{32}$$

- The cofactor of the element a_{ij} is the minor of a_{ij} multiplied be $(-1)^{i+j}$
- The cofactor of a_{11} is $C_{11} = (-1)^{1+1} M_{11} = (-1)^2 |A_{11}| = a_{22}\,a_{33} - a_{23}\,a_{32}$

The minor of an element a_{12} is the determinant of the matrix obtained by deleting the 1st row and 2nd column and $M_{12} = |A_{12}|$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = a_{21}\,a_{33} - a_{23}\,a_{31}$$

- The cofactor of a_{12} is $C_{12} = (-1)^{1+2} M_{12} = (-1)^3 |A_{12}| = -(a_{21}\,a_{33} - a_{23}\,a_{31})$

The minor of an element a_{23} is the determinant of the matrix obtained by deleting the 2nd row and 3rd column of matrix A and $M_{23} = |A_{23}|$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} = a_{11}\,a_{32} - a_{12}\,a_{31}$$

- The cofactor of a_{23} is $C_{23} = (-1)^{2+3} M_{23} = (-1)\,|A_{23}| = -(a_{11}\,a_{32} - a_{12}\,a_{31})$

To find the cofactors of an element a_{11}, a_{12}, and a_{23} is given by

$$C_{11} = (-1)^{1+1} M_{11} = (-1)^{1+1} |A_{11}|$$

$$C_{12} = (-1)^{1+2} M_{12} = (-1)^{1+2} |A_{12}|$$

$$C_{23} = (-1)^{2+3} M_{23} = (-1)^{2+3} |A_{23}|$$

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Illustrative Example

3. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ find the minor and the cofactor of a,b,c, and d

Solution

(i) The minor of a is d, and minor of b is c

The minor c is b and minor of d is a

(ii) The cofactor a is $C_{11} = (-1)^{1+1}(d) = d$

The cofactor of b is $C_{12} = (-1)^{1+2}(C) = -c$

The cofactor of c is $C_{21} = (-1)^{2+1}(b) = -b$

The cofactor of d is $C_{22} = (-1)^{2+2}(a) = a$

$\therefore$ The cofactor of each element is

$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$, and minor $= \begin{pmatrix} d & c \\ b & a \end{pmatrix}$

4. Let $A = \begin{pmatrix} 3 & 4 \\ 1 & 5 \end{pmatrix}$ find the cofactor and minor of each element.

The cofactor $C_{ij} = \begin{pmatrix} 5 & -1 \\ -4 & 3 \end{pmatrix}$ and the minor $M_{ij} = \begin{pmatrix} 5 & 1 \\ 4 & 3 \end{pmatrix}$

5. Let $A = \begin{pmatrix} -1 & 2 & -3 \\ 3 & 8 & 1 \\ 1 & 0 & -2 \end{pmatrix}$

Find the minors and the cofactors of the element 3, 8, 0 -2 and 1

Solution

(i) An element 3 is in 2nd row and 1st column of A. To find the minor of 3, delete the second row and first column of A then evaluate the determinant of 2×2.

- Minor of 3 is $M_{21} = \begin{vmatrix} 2 & -3 \\ 0 & -2 \end{vmatrix} = 2(-2) - (-3)\cdot 0 = -4$
- The cofactor of 3 is $C_{21} = (-1)^{2+1} M_{21} = (-1)\,-4 = 4$

(ii) To find the minor of 8, delete the 2nd row and 2nd column of A, the

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evaluate the determinant of 2×2, denoted by M_{22}.

$$M_{22} = \begin{vmatrix} -1 & -3 \\ 1 & -2 \end{vmatrix} = (-1)(-2) - (-3)(1) = 2 + 3 = 5$$

$\therefore$ The minor an element 8 is 5

The cofactor of 8 is $C_{22} = (-1)^{2+2} M_{22} = 5$

(iii) To find the minor of 0, delete the 3rd row and 2nd column of A, then evaluate the determinant of the remaining 2×2, matrix denoted by M_{32}

$$M_{32} = \begin{vmatrix} -1 & -3 \\ 3 & 1 \end{vmatrix} = (-1)(1) - (-3)(3) = -1 + 9 = 8$$

$\therefore$ The minor of an element 0 is 8

The cofactor 0 is $C_{32} = (-1)^{3+2} (M_{32}) = (-1)(8) = -8$

(iv) To find the minor of -2, delete the 3rd row and 3rd column of A, then evaluate the determine of 2×2:

$$M_{32} = \begin{vmatrix} -1 & 2 \\ 1 & 0 \end{vmatrix} = (-1)(0) - (2)(1) = 0 - 2 = -2$$

$\therefore$ The minor of an element -2 is -2

The cofactor of -2 is $C_{33} = (-1)^{3+3}(-2) = -2$

6. Let $A = \begin{pmatrix} -2 & 0 & 5 \\ x & -1 & 1 \\ 2x+1 & 2 & -x \end{pmatrix}$.

If the minor of an element 0 is -16 then the possible value of x is

A. 7, -3 B. 1, 3 C. 3, -5 D. -3, 5

Solution

0 is in row 1, column 2 the minor of this element is found by deleting the 1st row and 2nd column and then evaluating the determinant of the remaining 2×2 matrix:

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$$M_{12} = \begin{vmatrix} x & 1 \\ 2x+1 & -x \end{vmatrix} = x(-x) - (1)(2x+1) = 16$$

$$-x^2 - 2x - 1 = -6 \text{ implies } x^2 + 2x - 15 = 0$$

$$\Rightarrow (x-3)(x+5) = 0 \text{ implies } x = 3 \text{ or } x = -5$$

Answer:(

7. From example #6 if the cofactor of an element 5 is 25, then the value of x is

A. -1 B. 0 C. 6 D. -6

Solution

The element 5 corresponds to a_{13}. The minor of this element is found by deleting the 1st row and 3rd column.

$$C_{12} = (-1)^4 \begin{vmatrix} x & -1 \\ 2x+1 & 2 \end{vmatrix} = 25$$

$$2x + 2x + 1 = 25$$

$$4x = 24, \quad x = 6$$

Answer:(

8. Let $A = \begin{pmatrix} x^2 & 3 \\ x & 1 \end{pmatrix}$. For what values of x, $\det(A) = 4$?

Solution

$$\det(A) = |A| = (x^2)(1) - 3x = 4$$

$$\Rightarrow x^2 - 3x - 4 = 0$$

$$\Rightarrow (x-4)(x+1) = 0$$

$$\therefore \quad x = 4 \text{ or } x = -1$$

Let $A = \begin{pmatrix} 2 & 3x \\ x & 2 \end{pmatrix}$. For what values of x, $|A| = 11x$?

Solution

$$\det(A) = (2)(2) - (3x)(x) = 11x$$

$$\Rightarrow 4 - 3x^2 - 11x = 0, \text{ the same as } 3x^2 + 11x - 4 = 0$$

$$\Rightarrow 3x^2 + 12x - x - 4 = 0, \text{ the same as } (3x-1)(x+4) = 0$$

$$\therefore \quad x = \frac{1}{3} \text{ or } x = -4$$

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10. What is the difference between the minor of an element a_{ij} and the cofactor of the element?

Explanation:- The cofactor is the product of the minor by $(-1)^{i+j}$. That is cofactor $C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the minor of a_{ij}.

4.2 Determinant of a 3 × 3 Matrix

Cofactors are used to evaluate the determinant of a matrix of order 3 × 3 matrix. The method used to evaluate a determinant by using cofactors is called **expanding by cofactors.**

The value of a determinant can be obtained by expansion by cofactors.

Determinants by Expanding by cofactors

Given the square matrix A of order 3, the value of the determinant of A is the sum of the products of the elements of any row or column and their cofactors.

For the 1st row of A, the value of the determinant of A is

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

A long column I: $|A| = a_{11}c_{11} + a_{21}c_{21} + a_{31}c_{31}$

The value of the determinant is the same in each case.

Finding the determinant of a 3 × 3 matrix:

Step - 1: Choose any row or column.

Step - 2: Multiply each element in selected or column by its cofactor.

Step - 3: The value of the determinant is the sum of the product from step 2.

For example: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then find the determinant of A by expanding cofactors, in first row

- The cofactor of a is d and The cofactor of b is -c

$\therefore |A| = a_{11}C_{11} + a_{12}C_{12} = a \cdot d + b(-c) = ad - bc$

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Illustrative Example

11. Let $A = \begin{pmatrix} 2 & 6 & 4 \\ 1 & 2 & 1 \\ 3 & 8 & 6 \end{pmatrix}$ Find the determinant of A

a. by expanding first row b. by expanding third column

Solution

a. The determinant of A by expanding first row.

$$|A| = 2\,C_{11} + 6\,C_{12} + 4\,C_{13}$$

$$= 2(-1)^{1+1} M_{11} + 6(-1)^{1+2} M_{12} + 4\,(-4)^{1+3} M_{13}$$

$$= 2\begin{vmatrix} 2 & 1 \\ 8 & 6 \end{vmatrix} - 6\begin{vmatrix} 1 & 1 \\ 3 & 6 \end{vmatrix} + 4\begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix}$$

$$\therefore \quad |A| = (2)(4) - 6(3) + 4(2) = -2$$

b. The determinant of A by expanding 3rd column.

$$|A| = 4\,C_{13} + 1C_{23} + 6\,C_{33}$$

$$= 4(-1)^{1+4} M_{13} + 1(-1)^{2+3} M_{23} + 6\,(-1)^{3+3} M_{33}$$

$$= 4\begin{vmatrix} 1 & 2 \\ 3 & 8 \end{vmatrix} - 1\begin{vmatrix} 2 & 6 \\ 3 & 8 \end{vmatrix} + 6\begin{vmatrix} 2 & 6 \\ 1 & 2 \end{vmatrix}$$

$$\therefore \quad |A| = 4(2) - 1(-2) + 6(-2) = -2$$

In both cases, the value of the determinant of A is the same whether we expanded by cofactors the elements of a row or by cofactors of the elements of a column.

Note: When evaluating a determinant choose the most convenient row or column, which usually is the row or column containing the most zeros.

12. Let $A = \begin{pmatrix} 1 & 2 & 5 \\ 0 & 8 & 4 \\ 3 & 7 & 6 \end{pmatrix}$, by expanding cofactors.

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Solution

Because $a_{21} = 0$, we select row 2 or column 1, which is easy to evaluate.

$$|A| = 0.C_{21} + 8\,C_{22} + 4C_{23} \longleftarrow \text{Along row 2}$$

$$= 0 + 8(-1)^{2+2} M_{22} + 4\,(-1)^{2+3} M_{23}$$

$$= 0 + 8\begin{vmatrix} 1 & 5 \\ 3 & 6 \end{vmatrix} + 4(-1)\begin{vmatrix} 1 & 2 \\ 3 & 7 \end{vmatrix}$$

$$\therefore \quad |A| = 0 + 8(6 - 15) + -4\,(7 - 6) = -76$$

Along column: 1: $|A| = 1 \cdot C_{11} + 0 \cdot C_{21} + 3C_{31}$

$$|A| = 1(20) + 0 + 3(-32) = 20 - 96 = -76$$

In both cases, the value of $|A|$ is the same.

Sign Array of Determinant

When we evaluate a determinant using expansion by minors about any row or column, the sign of the coefficients of the minors alternate according to the **sign array** that follows.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$= a_{11}\,(\text{minor of } a_{11}) - a_{12} \cdot (\text{minor of } a_{12}) + a_{13} \cdot (\text{minor of } a_{13})$$

Illustrative Example

13. Find the determinant of a matrix by using expansion by minors about second row and 3rd column.

a. $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 1 \\ 3 & 1 & 5 \end{bmatrix}$, b. $A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & -1 & 1 \\ 3 & 1 & 5 \end{bmatrix}$

Solution

a. Determinant by expansion of minors along row 2. The prefix sign (- + -)

$$|A| = -2 \cdot (\text{minor of } 2) + 3 \cdot (\text{minor of } 3) - 1 \cdot (\text{minor of } 1)$$

$$= -2\begin{vmatrix} 1 & 2 \\ 1 & 5 \end{vmatrix} + 3\begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix}$$

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$|A| = -2(5-2)+3(5-6)-1(1-3) = -6-3+2 = -7$

Determinant by expansion of 3rd column, try by yourself.

b. Determinant by expansion by minors about 3rd column with prefix sign (+, -, +)

$|A| = 4\cdot(\text{minor of } 4) - 1\cdot(\text{minor of } 1) + 5\cdot(\text{minor of } 5)$

$$= 4\begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} - 1\begin{vmatrix} 3 & -1 \\ 3 & 1 \end{vmatrix} + 5\begin{vmatrix} 3 & -1 \\ 2 & -1 \end{vmatrix}$$

$= 4(2+3) - 1(3+3) + 5(-3+2) = 20 - 6 - 5 = 9$

Remark: the minor of 4 is 5

the minor of 1 is 6

the minor of 5 is -1

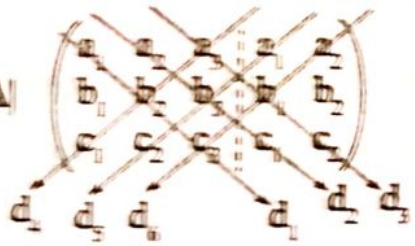
Tip: The determinant of a 3 × 3 matrix A can also be evaluated by using the "**method of diagonals**".

Step 1: Repeated columns 1 and 2 to the right of the matrix.

Step 2: Multiply the elements on the diagonals through $d_1, d_2, d_3, d_4, d_5, d_6$

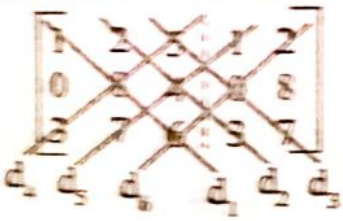
Step 3: Value of determinant is $|A| = (d_1 + d_2 + d_3) - (d_4 + d_5 + d_6)$

Let $A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$ then $|A|$



$|A| = (a_1b_2c_3 + a_2c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_1b_3c_2 + a_2b_1c_3)$

The determinant from example 12 is evaluated as follows:



$= [(d_1 + d_2 + d_3) - (d_4 + d_5 + d_6)]$

$= [(1)(8)(6) + (2)(4)(3) + (5)(0)(7)] - [5\cdot8\cdot3 + 1\cdot4\cdot7 + 2\cdot0\cdot7 + 2\cdot0\cdot8]$

$|A| = (48 + 24 + 0) - (120 + 28 + 0) = -76$

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14. Let $A=\begin{bmatrix}1 & 1 & 1\\ 1 & b & b\\ 1 & b & c\end{bmatrix}$, then the value of $|A|$:

A. $(1-b)(c-b)$ C. $(b-1)(c-b)$

B. $(1-c)(b-c)$ D. $(c-1)(b-c)$

Solution

By expanding 1st row: $|A| = 1C_{11} + 1C_{12} + 1C_{13}$

$$|A| = 1\begin{vmatrix} b & b\\ b & c\end{vmatrix} + 1(-1)^{1+2}\begin{vmatrix} 1 & b\\ 1 & c\end{vmatrix} + 1\begin{vmatrix} 1 & b\\ 1 & b\end{vmatrix}$$

$$= bc - b^2 - (c-b) + (b-b)$$

$$= b(c-b) - 1(c-b) = (b-1)(c-b)$$

Answer: C

15. Let $A=\begin{bmatrix}1 & x & x^2\\ x^2 & 1 & x\\ x & x^2 & 1\end{bmatrix}$ and the det$(A) = 49$ then find the values of x.

Solution

Expanding by cofactor along row 1:

$$|A| = 1C_{11} + xC_{12} + x^2C_{13} = 1(-1)^{1+1}M_{11} + x(-1)^{1+2}M_{12} + x^2M_{13}$$

$$= 1\begin{vmatrix} 1 & x\\ x^2 & 1\end{vmatrix} - x\begin{vmatrix} x^2 & x\\ x & 1\end{vmatrix} + x^2\begin{vmatrix} x^2 & 1\\ x & x^2\end{vmatrix}$$

$$= 1 - x^3 - 0 + x^2(x^4 - x)$$

$$= 1 - x^3 + x^5 - x^3 = x^3(x^3 - 1) - 1(x^3 - 1)$$

$$= (x^3 - 1)(x^3 - 1) = (x^3 - 1)^2 \text{ and } |A| = 49$$

$$|A| = (x^3 - 1)^2 = 49 \text{ implies } x^3 - 1 = \pm\sqrt{49} = \pm 7$$

$$\therefore \quad x = \sqrt[3]{8} = 2 \text{ or } x = \sqrt[3]{-6}$$

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4.3 Properties of Determinants

Elementary rows (column) operation on the value of determinants affects in the following ways: Let A be a square matrix of order n.

P-(1) : If a matrix B is obtained from A by interchanging two rows (columns) then $|B| = -|A|$.

Symbol : If $A \xrightarrow{R_i \leftrightarrow R_j} B$, then $|B| = -|A|$

P - 2 : If a matrix B is obtained from A by multiplying every element of one row (or columns of A by a real number K, then $|B| = K|A|$

Symbol : If $A \xrightarrow{KR_j \to R_j} B$, then $|B| = K|A|$

P - 3 : If B is obtained from A by a constant K multiple of one row (column) of A is added to another row the $|B| = |A|$

Symbol : If $A \xrightarrow{KR_i + R_j \to R_j} B$, then $|B| = |A|$

Illustrative Example

16. Let $A = \begin{bmatrix} 4 & -8 \\ 2 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -7 \\ 4 & -8 \end{bmatrix}$

a. Evaluate $|A|$ and $|B|$

b. How are A and B related?

c. How are $|A|$ and $|B|$ related?

d. What do you conclude?

Solution

a. $|A| = 4(-7) - (-8)(2) = -28 + 16 = -12$

$|B| = 2(-8) - (-7)(4) = -16 + 28 = 12$

b. $R_1 \leftrightarrow R_2$ (Inter changing rows)

c. $|B| = -|A|$.

d. We conclude that interchanging any two row will change the sign of the determinant change.

17. Let $A = \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & 6 \\ 5 & 1 \end{bmatrix}$

a. Evaluate $|A|$ and $|B|$

b. How are matrix A and matrix B related?

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c. How are |A| and |B| related?

d. What do you conclude?

Solution

a. $|A| = 3 - 10 = -7$ and $|B| = 9 - 30 = -21$

b. Row 1 of matrix B is 3 times row 1 of A. $(3R_1 \rightarrow R_1)$

c. $|B| = 3|A|$

d. If B is obtained from A by multiplying row by a number K, then det (B) = K det (A). This property is also true for column.

Symbols : If $A \xrightarrow{3R_1 \rightarrow R_1} B$, then $|B| = 3|A|$

If $A \xrightarrow{KR_1 \rightarrow R_1} B$, then $|B| = K|A|$

18. Let $A = \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 14 & 7 \end{bmatrix}$

a. Evaluate |A| and |B|

b. How are matrix A and matrix A related?

c. How are |A| and |B| related?

d. What do you conclude?

Solution

a. $|A| = 3 - 10 = -7$ and $|B| = 21 + 28 = -7$

b. $3R_1 + R_2 \rightarrow R_2$. (Row - 2 of matrix B is same as the sum of 3 times row 1 of A and row 2 of A)

c. $|A| = |B|$

d. If B is obtained from A by adding K times any row (column) of A to another row (column) of A to another row (column) of A does not change the value of |A|.

Symbols. If $A \xrightarrow{3R_1 + R_2 \rightarrow R_2} B$, then $|B| = |A|$

19. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and det A = 5 then find the determinant of each of the following matrix.

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a. $B = \begin{pmatrix} c & d \\ a & b \end{pmatrix}$

b. $B = \begin{pmatrix} c & d \\ 2a & 2d \end{pmatrix}$

c. $B = \begin{pmatrix} a & b \\ 2a+c & 2b+d \end{pmatrix}$

d. $B = \begin{pmatrix} 3a & 3b \\ 2c & 2d \end{pmatrix}$

e. $B = \begin{pmatrix} a+3c & b+3d \\ 4c & 4d \end{pmatrix}$

f. $B = \begin{pmatrix} -3a & -3b \\ -3c & -3d \end{pmatrix}$

Solution

a. Row 1 and 2 are interchanged between matrix A and matrix B

b. $A \xrightarrow[R_1 \leftrightarrow R_2]{2R_1 \to R_2} B$, then $|B| = -2|A| = -2(5) = -10$. By property (1) and 2

c. $A \xrightarrow{2R_1 + R_2 \to R_2} B$, then $|B| = |A| = 5$. Refer property 3

d. $A \xrightarrow{3R_1 \to R_1} B$, then $|B| = 3 \cdot 2|A| = 6 \times 5 = 3$

By factoring common factor (By property 2)

e. $|B| = 4\begin{vmatrix} a+3c & b+3d \\ c & d \end{vmatrix}$ ⟵ By propert 2

$= 4\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ ⟵ By propert 3 $3R_2 + R_2 \to R_1$

$= 4(5) = 20$

f. $|B| = -3(-3)\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 9(5) = 45$

20. Let A be a square matrix of order 3. If a matrix B is obtained from A by multiplying row 1 by 6 and det (B) = 18, what is det (A)?

A. 18 B. 6 C. 3 D. 108

Solution

Use property 2 to help understand.

If $A \xrightarrow{6R_1 \to R_1} B$, then $|B| = 6|A|$

$\therefore \quad |A| = \frac{|B|}{6} = \frac{18}{6} = 3$

Answer: C

21. Let A be a square matrix of order 3. If a matrix B is obtained from A by

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adding -3times row 1 to row 2 and det A = 4, what is determinant of B ?

A. -12 B. 4 C. 12 D. -4

Solution

Using property 3:

If $A \xrightarrow{-3R_1 + R_2 \to R_2} B$, then $|B| = |A|$

Answer: B

22. If a matrix B is obtained from a square matrix A by adding row 1 to row 3 and det (A) = 3, what is det (B)?

A. 6 B. -6 C. 3 D. -3

Solution:- Using property 3:

Answer: C

P - 4: Scalar or Multiplication Property of a determinant.

(i) If the elements of any row (column) of a determinant are multiplied by K, then the value of the determinant is multiplied be K.

(ii) If A is a square matrix of order n and K is scalar product of matrix then: $|KA| = K^n |A|$

Note: The matrix KA has K as a common factor in each of its n - rows (property 2).

As an example:

a. Let $A = \begin{bmatrix} Ka & Kd \\ c & d \end{bmatrix}$, then $|A| = \begin{vmatrix} Ka & Kb \\ c & d \end{vmatrix} = K \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

b. Let $A = \begin{bmatrix} Ka & Kb \\ Kc & Kd \end{bmatrix}$, then $|A| = K \cdot K \begin{vmatrix} a & b \\ c & d \end{vmatrix} = K^2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

c. Let $A = \begin{bmatrix} 6 & -15 \\ 1 & 4 \end{bmatrix}$, then $|A| = 3 \begin{vmatrix} 2 & -5 \\ 1 & 4 \end{vmatrix}$

d. Let $A = \begin{bmatrix} 6 & -15 \\ 3 & 12 \end{bmatrix}$, then $|A| = 3 \cdot 3 \begin{vmatrix} 2 & -5 \\ 1 & 4 \end{vmatrix} = 3^2 \begin{vmatrix} 2 & -5 \\ 1 & 4 \end{vmatrix}$

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P - 5: Product and transpose property of determinant.

(i) If A and B are square matrices, then $|AB| = |A|\,|B|$

(ii) $|A^k| = |A|^k$, K is any real number.

(iii) $|A^t| = |A|$

P - 6: Identical rows (columns) determinant

(i) If two rows (columns) of a square matrix A are identical, then $|A| = 0$

(ii) If one row (column) is a constant multiple of another row (column) of A, then $|A| = 0$

As an example:

a. Let $A = \begin{pmatrix} 2 & 5 \\ 4 & 10 \end{pmatrix}$, then $|A| = 20 - 20 = 0$

b. Let $A = \begin{pmatrix} 3 & 4 \\ 3 & 4 \end{pmatrix}$, then $|A| = 12 - 12 = 0$

23. Let $A = \begin{pmatrix} a & a & ac \\ b & -b & bc \\ c & c & -c^2 \end{pmatrix}$ then $|A|$ is equal to:

A. $2abc^2$ B. $4abc^2$ C. $-4abc^2$ D. 0

Solution

Removing common factor from row 1, row 2, row 3 and column 3

$$|A| = \begin{vmatrix} a & a & ac \\ b & -b & bc \\ c & c & -c^2 \end{vmatrix} = a \cdot b \cdot c \cdot c \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$|A| = abc^2 \left[1\begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} \right]$$

$$= abc^2 \left[(1 - 1) - (-1 - 1) + (1 + 1) = 4abc^2 \right]$$

Answer: B

24. Let $A = \begin{bmatrix} 14 & -6 & 4 \\ 4 & -5 & 12 \\ -21 & 9 & -6 \end{bmatrix}$, then find $|A|$

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Solution

- 2 is a common factor of row 1 (property -4)
- -3 is a common factor of row 3

$$|A| = 2(-3)\begin{vmatrix} 7 & -3 & 2 \\ 4 & -5 & 12 \\ 7 & -3 & 2 \end{vmatrix}.$$

$\therefore$ $|A| = 0$, two rows are identical, (Property - 6).

P - 7: A triangular matrix determinant

- If a square matrix A is triangular, then the determinant of matrix A is the product of the element on the main diagonal. That is $|A| = a_{11}\, a_{22}\, a_{33} \ldots a_{nn}$
- **As an example:**
- Let $A = \begin{pmatrix} a & x & y \\ 0 & b & c \\ 0 & 0 & c \end{pmatrix}$, then $|A| = abc$
- Let $A = \begin{pmatrix} 3 & 0 & 0 \\ 2 & -5 & 0 \\ 4 & 6 & -2 \end{pmatrix}$, then $|A| = (3)(-5)(-2) = 30$
- Let $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$, then $|A| = 3 \cdot 2 \cdot 4 = 24$

Illustrative Example

25. Let $A = \begin{pmatrix} a & b \\ x & y \end{pmatrix}$, and det (A) = 4

then find the determinantof matrix B obtained from matrix A

a. $B = \begin{pmatrix} x & y \\ a & b \end{pmatrix}$,

b. $B = \begin{pmatrix} a+3x & b+3y \\ 2x & 2y \end{pmatrix}$

c. $B = \begin{pmatrix} 3a+x & 3b+y \\ -2x & -2y \end{pmatrix}$

d. $B = \begin{pmatrix} 5a & 5b \\ 5x & 5y \end{pmatrix}$

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Solution

a. Matrix B is obtained from A by interchanging row1 and row 2

$\therefore$ det(B) = -det (A) = -4

b. Matrix B is obtained from A, 2 times row 2 of A. $(2R_2 \rightarrow R_2)$, $|B| = 2|A|$

$R_1 + 3R_2 \rightarrow R_1$ (No change in value).

$\therefore$ $|B| = 2|A| = 2(4) = 8.$

c. Matrix B is obtained from A: $3R_1 \rightarrow R_1$ and $-2R_2 \rightarrow R_2$.
(Adding row 2 to row 1, determinant value does not change)

$\therefore$ $|B| = 3(-2)\,|A| = -6\,(4) = -24.$

d. $|B| = 5^2\,|A| = 25\,(4) = 100$

26. Which of the following statements about the determinants of the matrices A and B is true?

A. If B is obtained from A by interchanging two rows, then det (B) = det (A).

B. If B is obtained from A by multiplying one row of A by some scalar K, det (B) = K det (A).

C. If any two rows in A are identical, det (A) = 1

D. If B is obtained from A by multiplying one row and adding the result to another row, det (B) = - det (A)

Answer: B

27. Let $A = \begin{pmatrix} a & b & c \\ x & y & z \\ 3 & 1 & 2 \end{pmatrix}$, and det A = 5 then find the determinant of matrix B and state the property used.

a. $B = \begin{pmatrix} a & b & c \\ 3 & 1 & 2 \\ 2x & 2y & 2z \end{pmatrix}$

b. $B = \begin{pmatrix} a & b & c \\ -9 & -3 & -6 \\ x & y & z \end{pmatrix}$

c. $B = \begin{pmatrix} 3 & 1 & 2 \\ x-3 & y-1 & z-2 \\ a & b & c \end{pmatrix}$

d. $B = \begin{pmatrix} a+2x & b+2y & c+2z \\ 3x & 3y & 3z \\ 3 & 1 & z \end{pmatrix}$

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Solution

a. Row 2 and row 3 are interchanged between A and B, so that it changes the sign of $|B| = -|A|$

- Multiplying row of A by -2, then $|B| = 2|A|$

$\therefore |B| = -2|A| = -2(5) = -10$ We used property 1 and 2.

b. By property 2: then $|B| = -3|A| = -3(5) = -15$. And by property 1, then $|B| = 15$

c. By property 1 and 3: $|B| = -5$

d. By property 2 and 3: $|B| = 3|A| = 3 \times 5 = 15$

28. Let $A = \begin{pmatrix} 1 & -a & b-c \\ 1 & b & a+b \\ 0 & c & c+c \end{pmatrix}$, then $|A|$ is equal to

A. 0 B. -2ac C. 2bc D. 2c(b-a)

Solution

$$|B| = |A| = \begin{vmatrix} 1 & -a & b \\ 1 & b & a \\ 0 & c & c \end{vmatrix} = 1C_{11} + 1C_{21} + 0.C_{31}$$

$$= 1\begin{vmatrix} b & a \\ c & c \end{vmatrix} + (-1)^3\begin{vmatrix} -a & b \\ c & c \end{vmatrix} = (bc - ca) - (-ac - bc) = 2bc$$

Answer: C

29. Let A be a square matrix of order 3. If $|A| = 4$ then value of $|2A|$ will be equal to

A. 8 B. 24 C. 32 D. 16

Solution

Using property 4: $|KA| = K^n |A|$, $n = 3$

$|2A| = 2^3 |4| = 8 \times 4 = 32$

Answer: C

30. Find det (A) if A if A is 3×3 and det (2A) = 12. What is det (A), is 4×4

Solution

Since A is 3×3 det (2A) $2^3 |A| = 12$

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$\therefore \quad |A| = \frac{12}{8} = \frac{3}{2}$

If A is 4×4, then $\det(2A) = 2^4 |A| = 16|A|$

31. If $|A| = \begin{pmatrix} x & -1 \\ 3x & 1 \end{pmatrix}$, then for what value of x is $\det(2A) = x^2$

A. 16 B. 8 C. 4 D. -8

Solution:- $|A| = x - (-1)(3x) = 4x$

$\det(2A) = 2^2 |A| = 4(4x) = x^2$

$\Rightarrow x^2 - 16x = 0$ and $x(x - 16) = 0$

$\therefore \quad x = 0$ or $x = 16$

Answer: A

32. If $M = \begin{pmatrix} 0 & 1 & 2 \\ 3 & -1 & 0 \\ 5 & 2 & 4 \end{pmatrix}$ and $A^T M = 2I$, when A is 3×3 matrix and I is the identity matrix of order 3, then what is det (A)?

A. 0.2 B. 0.8 C. $\frac{4}{17}$ D. $\frac{1}{17}$

Solution : Since $A^T M = 2I$, then $|A^T M| = |2I| = 2^3 |I|$.

$|M| = 0.C_{11} + 1.C_{12} + 2.C_{12} = 1(-1)(12) + 2(11) = 10$

$|A^T M| = |A^T||M| = 2^3 |I|$ implies, $|A^T| = \frac{8|I|}{|M|}$

$\therefore \quad |A| = \frac{8(1)}{10} = 0.8$

Answer: B

33. Let A and B are a square matrices of order 3.
If the det (A) = 4 and det (B) = 2

Find a. $|2A|$ b. $|3B|$ c. $|2AB|$

d. $|3A^tB|$ e. $|B^5|$

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Solution

a. $|2A| = 2^3 |A| = 8(4) = 32$

b. $|3A| = 3^3 |B| = 27\ (2) = 54$

c. $|2AB| = 2^3 |A|\ |B| = 8(4)\ (2) = 64$

d. $|2A^tB| = 3^3 |A^t|\ |B| = 3^3\ (4)\ (2) = 216$

e. $|B^5| = |B|^5 = 2^5 = 32$

34. Let A be a square matrix of order n. What can be said about the value of det (A) if :

a. $A^2 = I$ b. $A^3 = A$ c. $A^2 = 3A$ d. $A^3 = I$

Solution

a. $|A^2| = |I|$ implies $|A|^2 = 1$

$\therefore \quad |A| = 1$ or $|A| = -1$

b. $|A^3| = |A|$, the same as $|A|^3 = |A|$

$\Rightarrow |A|^2 = 1$, so that $|A| = \pm 1$

c. $|A^2| = |3A|$, implies $|A|^2 = 3^n |A|$

$\Rightarrow |A| = 3^n$, or $|A| = 0$

d. $|A^3| = |I| \Rightarrow |A|^3 = 1$, so that $|A| = 1$

What is the determinant of inverse matrix?

The determinant of inverse matrix A^{-1} is the reciprocal of determinant of matrix A. In symbol: $\left|A^{-1}\right| = \frac{1}{|A|}, \ |A| \neq 0.$

The fact that $A^{-1} A = I$, then $|A^{-1} A| = |I| = 1$

$|A^{-1}|\ |A| = 1$, this implies $\left|A^{-1}\right| = \frac{1}{|A|}$

35. Let A and B are square matrices of order n.

If det (A) = -2, and det (B) = 3 Find det $(A^3 B^{-1} A^t B^2)$

Solution

$$\left|A^3 B^{-1} A^t B^2\right| = \left|A^3\right|\left|B^{-1}\right|\left|A^t\right|\left|B^2\right| = |A|^3 \cdot \left(\frac{1}{|B|}\right) \cdot |A||B|^2 = (-2)^3 \cdot \frac{1}{3} \cdot (-2)(3)^2 = -48$$

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36. Let A be 3×3 matrix and $|A| = -2$. Then what is the value of

a. $|2A'A|$ b. $|2A'A^{-1}|$ c. $|4A^2|$ d. $|(2A)^{-1}|$

Solution

a. $|2A'A| = 2^3 |A'| |A| = 2^3 . (-2)(-2) = 32$

b. $\left|2A' A^{-1}\right| = 2^3 \left|A'\right| \left|A^{-1}\right| = \dfrac{8|A|}{|A|} = 8$

c. $|4A^2| = 4^3 |A|^2 = 64 (-2)^2 = 256$

d. $\left|(2A)^{-1}\right| = \dfrac{1}{|2A|} = \dfrac{1}{2^3 \times (-2)} = -\dfrac{1}{64}$

37. Let A and B are square matrices of order 4. If $|A| = 3$ and $|B| = 2$, then what is the value of $|2(AB)^{-1}|$?

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{4}{3}$ D. $\frac{8}{3}$

Solution

$$\left|2(AB)^{-1}\right| = 2^4 \left|(AB)^{-1}\right| = 16 \left|AB\right|^{-1} = \frac{16}{|AB|}$$

$$= \frac{16}{|A||B|} = \frac{16}{(3)(2)} = \frac{8}{3}$$

Answer: D

38. Let A and B are square matrices of order 3 and I is the identity matrix of order 3 such that $AB = 2(I)$. If $\det(B) = 12$, then What is the $\det A$?

A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{4}{3}$ D. $\frac{1}{6}$

Solution

$|AB| = |2I|$, Using the product property of determinant

$|A||B| = 2^3 (1)$, gives $|A| = \dfrac{8}{|B|} = \dfrac{8}{12} = \dfrac{3}{4}$

Answer: B

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39. Suppose A is 3×3 matrix and $B = \begin{pmatrix} 0 & 4 & 1 \\ 5 & 0 & 6 \\ 0 & 4 & 2 \end{pmatrix}$.

If $\det(A) = 4$ then $\det(2A^{-1}B^t)$ is equal to

A. -40 B. -10 C. 40 D. 10

Solution

To find the determinant of B expanding in column 1, is easy to evaluate, because $a_{11} = 0$ and $a_{31} = 0$.

$$\therefore \quad |B| = 0 \cdot C_{11} + 5C_{21} + 0 \cdot C_{31} = 0 + (-1)(5)\begin{vmatrix} 4 & 1 \\ 4 & 2 \end{vmatrix} = -20$$

$$\text{Again, } \left|2A^{-1}B^t\right| = \frac{2^3|B|}{|A|} = \frac{8(-20)}{4} = -40$$ **Answer: A**

40. Conceptual skill:

a. Under what condition $\det(-A) = \det(A)$

b. Under what condition $\det(-A) = -\det(A)$

Answer:

a. $\det(-A) = \det(A)$ if the order of the matrix is even, such as $2 \times 2, 4 \times 4$.

b. $\det(-A) = -(\det A)$ if the order of the matrix is odd.

4.4 Adjoint of a Square Matrix

The adjoint matrix of A is obtained by replacing the elements of A by its cofactors and then transposing. If A be a square matrix of order n. Then **adjoint of A** is defined as the transpose of the matrix of cofactors of A:

Adj.A transpose of cofactors of matrix A

$$\text{adj} \cdot A = (C_{ij})^t = \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}^t = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$$

Note

$C_{11} = (-1)^{1+1}|A_{11}|$, and $C_{12} = (-1)^{1+2}|A_{12}| = -|A_{12}|$

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Illustrative Example

41. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then find the adjoint of A.

Solution

Adjoint of A is the transpose of the cofactor of matrix A.

The cofactor of A: C_{ij} (A)

$C_{11} = (-1)^{1+1} |A_{11}| = d$ $\quad$ $C_{21} = (-1)^{2+1} |A_{21}| = -b$

$C_{12} = (-1)^{1+2} |A_{12}| = -c$ $\quad$ $C_{22} = (-1)^{2+2} |A_{22}| = a$

$$\therefore \quad \text{Adj.A} = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}^t = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}^t = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

42. Let $A = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix}$, then find the adj·A

Solution

The cofactor of matrix A is

$C_{11} = (-1)^{1+1} |A_{11}| = 1$ and $C_{12} = (-1)^{1+2} |A_{12}| = -1(4) = -4$

$C_{21} = (-1)^{2+1} |A_{21}| = (-1)(5) = -5$ and $C_{22} = (-1)^{2+2} |A_{22}| = 2$

$$\therefore \quad \text{Adj} \cdot A = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}^t = \begin{pmatrix} 1 & -4 \\ 5 & 2 \end{pmatrix}^t = \begin{pmatrix} 1 & 5 \\ -4 & 2 \end{pmatrix}$$

43. Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & -1 \\ 5 & -2 & -3 \end{bmatrix}$, then find adj · A.

Solution

First we determine the cofactor of matrix A.

$$C_{11} = (-1)^{1+1} |A_{11}| = \begin{vmatrix} 2 & -1 \\ -2 & -3 \end{vmatrix} = 2(-3) = ((-2)(-1)) = -8$$

$$C_{12} = (-1)^{1+2} |A_{12}| = (-1) \begin{vmatrix} 1 & -1 \\ 5 & -3 \end{vmatrix} = -(-3 + 5) = -2$$

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$$C_{13} = (-1)^{1+3}|A_{13}| = (1)\begin{vmatrix} 1 & 2 \\ 5 & -2 \end{vmatrix} = -2 - 10 = -12$$

$$C_{21} = (-1)^{3}|A_{21}| = -1\begin{vmatrix} 1 & 3 \\ -2 & -3 \end{vmatrix} = -(-3 + 6) = -3$$

$$C_{22} = (-1)^{4}|A_{22}| = 1\begin{vmatrix} 2 & 3 \\ 5 & -3 \end{vmatrix} = -6 - 15 = -21$$

$$C_{23} = (-1)^{5}|A_{23}| = -1\begin{vmatrix} 2 & 1 \\ 5 & -2 \end{vmatrix} = -(-4 - 5) = 9$$

Similarly, the cofactors of the element in 3rd row.

$C_{31} = -7$, $C_{32} = 5$ and $C_{33} = 3$

$$\therefore \quad \text{Adj}\cdot A = \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}^{t} = \begin{pmatrix} -8 & -2 & -12 \\ -3 & -21 & 9 \\ -7 & 5 & 3 \end{pmatrix}^{t} = \begin{pmatrix} -8 & -3 & -7 \\ -2 & -21 & 5 \\ -12 & 9 & 3 \end{pmatrix}$$

44. Let $A = \begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}$, then detemine:

a. $|A|$ b. adj·A c. A (adj·A) d. $|A(\text{adj}\cdot A)|$

e. How are A (adj·A) and $|A|$ related.

Solution

a. $|A| = 12 - 10 = 2$

b. $\text{adj}\cdot A = \begin{pmatrix} 4 & -5 \\ -2 & 3 \end{pmatrix}$

c. $A(\text{adj}\cdot A) = \begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}\cdot\begin{pmatrix} 4 & -5 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

d. $|A(\text{adj}\cdot A)| = 2 \times 2 = 4$

e. $A\ (\text{adj}\cdot A) = |A|I$

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4.5 Finding the Inverse of A Matrix

A procedure for finding the inverse if invertible matrix, we use determinant, cofactor and adjoint of a matrix. The procedure will be illustrated as follow:

Let A be any square matrix of order n:

STEP 1: Find the determinant of A.

STEP 2: Find the cofactor of matrix A.

STEP 3: Find the transpose of cofactor in step 2 of A (It is called adjoint matrix).

Last step: If $|A| \neq 0$, then the inverse of matrix A will be:

$$A^{-1} = \frac{\text{adj} \cdot A}{|A|}$$

To proof: we have the fact that

$$A^{-1} = I$$

$$A(\text{adj} \cdot A) = |A| I$$

$$A(\text{adj} \cdot A) = |A| (A^{-1} A) \leftrightarrow I = A^{-1} A$$

$$\boxed{A^{-1} = \frac{\text{adj} \cdot A}{|A|}}$$

Singular matrix and Non - singular matrix.

(i) A **singular matrix** is a matrix that does not have a multiplicative inverse. The determinant of a singular matrix is zero. (That is $|A| = 0$)

(ii) A **non - singular matrix** is a matrix that has a multiplicative inverse. The determinant of a non - singular matrix is not zero. (That is $|A| \neq 0$).

Note: A Square matrix A has an inverse if and only if $|A| \neq 0$.

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Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then cofactor of $A = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$

and Adjoint of $A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Definition : Inverse of a 2×2 matrix.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

If $\det(A) = 0$, then A has an inverse. If $\det(A) = 0$, then

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$ provided that $(ad - bc) \neq 0$.

Note: To find the adjoint: change the sign of b and c.

Switch (interchange) position of a and d in matrix A.

Illustrative Example

45. Determine whether each matrix has an inverse. If an inverse matrix exists, find it classify as singular or non - singular

a. $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$ c. $A = \begin{bmatrix} 8 & 2 \\ 4 & 1 \end{bmatrix}$ e. $A = \begin{bmatrix} 4 & -8 \\ -3 & 6 \end{bmatrix}$

b. $A = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix}$ d. $A = \begin{bmatrix} 5 & 2 \\ 4 & 1 \end{bmatrix}$ f. $A = \begin{bmatrix} 4 & 8 \\ -3 & 6 \end{bmatrix}$

Solution

a. $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$, $|A| = 3 \cdot 0 - (-2)(1) = 0 + 2 = 2$

since $|A| \neq 0$, the inverse of A exists, it is a non - singular matrix.

$$\therefore \quad A^{-1} = \frac{\text{adj} \cdot A}{|A|} = \frac{1}{2}\begin{bmatrix} 0 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -0.5 & 1.5 \end{bmatrix}$$

b. $|A| = (4)(3) - (3)(2) = 12 - 6 = 6$

$\therefore$ The matrix A is non - singular. Since $|A| \neq 0$, the inverse of A exists.

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$$\therefore \quad A^{-1} = \frac{\text{adj} \cdot A}{|A|} = \frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & \frac{-1}{3} \\ \frac{-1}{2} & \frac{1}{2} \end{bmatrix}$$

c. $|A| = 8 - 4(2) = 8 - 8 = 0 \leftrightarrow$ singular matrix.

$\therefore \quad A^{-1} = \begin{bmatrix} 8 & 2 \\ 4 & 1 \end{bmatrix}$ has not inverse, because $|A| = 0$

d. $|A| = (5)(1) - 2(4) = 5 - 8 = -3$

$\therefore$ The matrix is a non - singular, because $|A| = 6$. Since $|A| \neq 0$, the inverse of A exists.

$$\therefore \quad A^{-1} = \frac{1}{-3}\begin{bmatrix} 1 & -2 \\ -4 & 5 \end{bmatrix}$$

- Change the sign of 2 and 4, interchange the position of 5 and 1.

e. $|A| = (4)(6) - (-8)(-3) = 24 - 24 = 0$

$\therefore$ matrix A is singular, b/c, $|A| = 0$

Since $|A| = 0$, the inverse of A does not exist.

f. $|A| = 4(6) - (-3)(8) = 24 + 24 = 48$

The determinant different from zero, hence the matrix is non - singular.

$$\therefore \quad A^{-1} = \frac{\text{adj} \cdot A}{|A|} = \frac{1}{48}\begin{bmatrix} 6 & -8 \\ 3 & 4 \end{bmatrix}$$

46. Let $A = \begin{bmatrix} -1 & 3 & 1 \\ 2 & 5 & 0 \\ 3 & 1 & -2 \end{bmatrix}$

a. Evaluate $|A|$

b. Find the cofactor and adj·A.

c. Find A^{-1}

d. Evaluate $|\text{adj} \cdot A|$

e. How are $|A|$ and $|\text{adj} \cdot A|$ related ?

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Solution

a. $|A| = 2C_{21} + 5C_{22} + 0C_{23}$

$$= 2(-1)^3 \begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} + 5(-1)^4 \begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix} + 0$$

$$\therefore \quad |A| = -2(-6 - 1) + 5(2 - 3) = 14 - 5 = 9$$

b. The cofactor each element of matrix A is

$$C_{11} = (-1)^{1+1}|A_{11}| = \begin{vmatrix} 5 & 0 \\ 1 & -2 \end{vmatrix} = -10$$

$$C_{12} = (-1)^{1+2}|A_{12}| = (-1)\begin{vmatrix} 2 & 0 \\ 3 & -2 \end{vmatrix} = 4$$

$$C_{13} = (-1)^4|A_{13}| = \begin{vmatrix} 2 & 5 \\ 3 & 1 \end{vmatrix} = 2 - 15 = -13$$

Similarly, the cofactor of the elemnt in 2nd row:

$$C_{21} = (-1)^3\begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} = 7, \; C_{22} = \begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix} = -1, \text{ and } C_{21} = 10$$

The cofactor of the element 3rd row:

$$C_{31} = (-1)^4\begin{vmatrix} 3 & 1 \\ 5 & 0 \end{vmatrix} = -5, \; C_{32} = (-1)^5\begin{vmatrix} -1 & 1 \\ 2 & 0 \end{vmatrix} = 2, \text{ and } C_{33} = -11$$

The cofactor of matrix A is given by:

$$= \begin{pmatrix} -10 & 4 & -13 \\ 7 & -1 & 10 \\ -5 & 2 & -11 \end{pmatrix}$$

Adjoint of matrix is transpose of cofactor:

$$\text{Adj} \cdot A = \begin{pmatrix} -10 & 7 & -5 \\ 4 & -1 & 2 \\ -13 & 10 & -11 \end{pmatrix}$$

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c. $A^{-1} = \frac{\text{adj}\cdot A}{|A|} = \frac{1}{9}\begin{pmatrix} -10 & 7 & -5 \\ 4 & -1 & 2 \\ -13 & 10 & -11 \end{pmatrix}$

d. $\text{adj}\cdot A = |A|A^{-1} = 9A^{-1}$

$$\therefore \quad |\text{adj}\cdot A| = |9A^{-1}| = \frac{9^3}{|A|} = \frac{9^3}{9} = 81$$

e. $|\text{adj}\cdot A| = |A|^{n-1}$, where n is order of matrix.

47. Let A be 3×3 matrix and $|A| = -4$. Then what is the value of $|\text{adj}\,A|$?

A. -64 B. 16 C. $\frac{-1}{4}$ D. -4

Solution

Use $A^{-1} = \frac{1}{|A|}\text{adj}\cdot A$ implies $\text{adj}\cdot A = |A|A^{-1}$

$$|\text{adj}\cdot A| = |-4A^{-1}| = \frac{(-4)^3}{-4} = 16$$

Answer: B

48. Let A be square matrix of order 4 and $|A| = -2$, then what is the value of

a. $|2A^{-1}((A^{-1})^2)^{-1}|$ b. $|\text{adj}\cdot A|$ c. $|A(\text{adj}\cdot A)|$

Solution

a. $|2A^{-1}((A^{-1})^2)^{-1}| = \frac{2^4|A|}{|A|^2} = \frac{16}{|A|} = \frac{16}{-2} = -8$

b. We have $A^{-1} = \frac{\text{adj}\cdot A}{|A|}$, gives $A^{-1}|A| = \text{adj}\cdot A$

Taking determinant on both sides

$$|A^{-1}||A| = |\text{adj}\cdot A| \text{ this implies } \frac{|A|^n}{|A|} = |\text{adj}\cdot A|$$

$$\therefore \quad |\text{adj}\cdot A| = |A|^{n-1} = (-2)^{4-1} = (-2)^3 = -8$$

c. $|A(\text{adj}\cdot A)| = |A||\text{adj}\cdot A| = (-2)(-8) = 16$

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49. Suppose A and B are order 3 matrices, I is the identity matrix with order 3 such that $AB = -2I$. If $|A| = 4$, then what is the det (B^t)?

Solution:- Use facts: $|B^t| = |B|$

$|AB| = |-2I|$, this implies $|A|\,|B| = (-2)^3\,(1)$

$$|B^t| = |B| = \frac{-8}{|A|} = \frac{-8}{4} = -2$$

50. For what values of x, the given matrix A has an inverse.

a. $A = \begin{pmatrix} x & 3 \\ 4 & x-4 \end{pmatrix}$ b. $A = \begin{pmatrix} 2x & 3 \\ 3 & 2x \end{pmatrix}$

Solution

a. $|A| = (x)(x-4) - (3)(4) = x^2 - 4x - 12$

$\therefore \quad |A| = (x-6)(x+2)$

A square matrix A has an inverse if and only if $|A| \neq 0$

$\therefore$ Matrix A has an inverse if $x \neq 6$ and $x \neq -2$

That is $R\backslash\{6, -2\}$.

b. $|A| = (2x)(2x) - (3)(3) = 4x^2 - 9$

$|A| = (2x+3)(2x-3)$

A square matrix A has an inverse if and only if $|A| \neq 0$

$\therefore$ Matrix A has an inverse if $x \neq \frac{-3}{2}$ and $x \neq \frac{3}{2}$

That is $\mathbb{R} \setminus \left\{\frac{-3}{2}, \frac{3}{2}\right\}$

51. Find the value of 'a' for which matrix $A = \begin{bmatrix} a & -1 & 2 \\ a & a & 1 \\ 1 & a & 1 \end{bmatrix}$ has an inverse.

Solution

$|A| = aC_{11} + (-1)\,C_{12} + 2C_{13}$

$$= a\begin{vmatrix} a & 1 \\ a & 1 \end{vmatrix} + (-1)^{1+2}(-1)\begin{vmatrix} a & 1 \\ 1 & 1 \end{vmatrix} + (-1)^4\,2\begin{vmatrix} a & a \\ 1 & a \end{vmatrix}$$

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$= a(0) + (a - 1) + 2(a^2 - a) = (a - 1) + 2a(a - 1)$

$\therefore \quad |A| = (a - 1)(2a + 1)$ and $|A| = 0$, if $a = 1$ or $a = \frac{-1}{2}$

Hence A^{-1} exists for any value of 'a' except $\frac{-1}{2}$ and 1.

52. Let A and B are square matrices of order 3.

If $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix}$ and det (B) = 10. What is $\det\left((2A)^{-1}B^t\right)$

A. 1 B. 4 C. $\frac{1}{4}$ D. 50

Solution

$$|A| = (2)(5)\left(\frac{1}{2}\right) = 5$$

$$\left|(2A)^{-1}B^t\right| = \frac{1}{|2A|}\left|B^t\right| = \frac{|B|}{2^3|A|} = \frac{10}{(8)(5)} = \frac{1}{4}$$

Answer: C

53. Find the matrix A, if $(A^{-1} - 3I)^t = 2\begin{bmatrix} -1 & 2 \\ 5 & 4 \end{bmatrix}$

Solution

We use property of transpose: $(B^t) = B$. First transpose both side:

$$\left\{(A^{-1} - 3I)^t\right\}^t = \left\{2\begin{bmatrix} -1 & 2 \\ 5 & 4 \end{bmatrix}\right\}^t = 2\begin{bmatrix} -1 & 5 \\ 5 & 4 \end{bmatrix}$$

$$A^{-1} - 3I = \begin{bmatrix} -2 & 10 \\ 4 & 8 \end{bmatrix}$$

$$A^{-1} = 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -2 & 10 \\ 4 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 4 & 11 \end{bmatrix},$$

and so the fact that $(A^{-1})^{-1} = A$

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$$A = (A^{-1})^{-1} = \frac{1}{11 - 40}\begin{bmatrix} 11 & -10 \\ -1 & 1 \end{bmatrix} = \frac{1}{29}\begin{bmatrix} 11 & -10 \\ -4 & 1 \end{bmatrix}$$

54. If A is 3×3 and det (A) = 2 then the value of det $[-A^2(\text{adj}\cdot A)]$

A. 8 B. -8 C. 16 D. -16

Solution

$$\left|-A^2(\text{adj}\cdot A)\right| = \left|-A^2\right|\left|A^{-1}\right|\left|A\right| = (-1)^3\left|A\right|^2 \cdot \left|2A^{-1}\right|$$

$$= (-1)(2)^2(2^3)\cdot\frac{1}{|A|} = \frac{-32}{2} = -16$$

Answer: D

Critical thinking example

55. If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, then $(AB)^{-1}$ is equal to

A. $\begin{pmatrix} 4 & -3 \\ 4 & -5 \end{pmatrix}$ B. $\begin{pmatrix} -2 & 5 \\ 2 & -4 \end{pmatrix}$ C. $\begin{pmatrix} -3 & 11 \\ 1 & -3 \end{pmatrix}$ D. $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$

Solution

$(AB)^{-1} = B^{-1}A^{-1}$, and A^{-1} and $A^{-1} = \frac{1}{6 - 7}\begin{bmatrix} 3 & -7 \\ -1 & 2 \end{bmatrix}$

$(AB)^{-1} = B^{-1}A^{-1}$, and $A^{-1} = \frac{1}{6 - 7}\begin{bmatrix} 3 & -7 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & -7 \\ 1 & -2 \end{bmatrix}$

$$(AB)^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} -3 & -7 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} -2 & 5 \\ 2 & -4 \end{bmatrix}$$

Answer: B

56. If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, then What is the value of

a. $(4AB)^{-1}$ b. $\left(\frac{1}{4}AB\right)^{-1}$ c. $(2AB)^{-1}$

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Solution

We have the value factor that $(4AB)^{-1} = \frac{1}{4} B^{-1} A^{-1}$

a. $(4AB)^{-1} = \frac{1}{4} B^{-1} A^{-1} = \frac{1}{4}\begin{pmatrix}1 & 1\\ 0 & 2\end{pmatrix}\begin{pmatrix}-3 & 7\\ 1 & -2\end{pmatrix} = \begin{pmatrix}\frac{-1}{2} & \frac{5}{4}\\ \frac{1}{2} & -1\end{pmatrix}$

b. $\left(\frac{1}{4}AB\right)^{-1} = 4B^{-1} A^{-1} = 4\begin{pmatrix}-2 & 5\\ 2 & -4\end{pmatrix} = \begin{pmatrix}-8 & 20\\ 8 & -16\end{pmatrix}$

c. $(2AB)^{-1} = \frac{1}{2} B^{-1} A^{-1} = \frac{1}{2}\begin{pmatrix}-2 & 5\\ 2 & -4\end{pmatrix}$

57. If $2\begin{pmatrix}2x & x\\ -5 & -3\end{pmatrix}^{-1} = \begin{pmatrix}3 & 2\\ -5 & 4\end{pmatrix}$, then what is the value of x

A. -2 B. 2 C. $\frac{1}{2}$ D. $\frac{-1}{2}$

Solution

$$\begin{bmatrix}2x & x\\ -5 & -3\end{bmatrix}^{-1} = \frac{1}{-6x + 5x}\begin{bmatrix}-3 & -x\\ 5 & 2x\end{bmatrix}$$

$$2\begin{pmatrix}2x & x\\ -5 & -3\end{pmatrix}^{-1} = 2\left(\frac{1}{-x}\right)\begin{bmatrix}-3 & -x\\ 5 & 2x\end{bmatrix} = \begin{bmatrix}3 & 2\\ -5 & 4\end{bmatrix}$$

$\frac{2}{-x}\begin{pmatrix}-3 & -x\\ 5 & 2x\end{pmatrix} = \begin{pmatrix}3 & 2\\ -5 & 4\end{pmatrix}$, and so the fact that the matrix is equal if $x = 2$

Answer: B

58. **Fast learner skill.** If $A = \begin{bmatrix}3 & 0 & -1\\ 4 & 7 & 3\\ -2 & 8 & 5\end{bmatrix}$ then what will be the (2,3) entry of A^{-1}? (Extreme 2013 edition)

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A. $\frac{-3}{2}$ B. 1 C. -1 D. -8

Solution

$$|A| = 3C_{11} + 0.C_{12} + (-1)\,C_{13} = 3\begin{vmatrix} 7 & 3 \\ 8 & 5 \end{vmatrix} + 0 + (-1)\begin{vmatrix} 4 & 7 \\ -2 & 8 \end{vmatrix}$$

$$3\,(35 - 24) - (32 + 14) = -13$$

The $(2,3)$ entery of $A^{-1} = \frac{\text{adj} \cdot A}{|A|} = \left[\!\left[C_{23}\,|A_{23}| \right]\!\right]^T$ is the $(3,2)$ entery of the transpose matrix equals

$$A^{-1} = \frac{\left[\!\left[C_{32}\,|A_{32}| \right]\!\right]}{|A|} = \frac{1}{-13}(-1)^5 \begin{vmatrix} 3 & -1 \\ 4 & 3 \end{vmatrix} = \frac{-13}{-13} = 1$$

Answer: B

What will be the $(3, 1)$ entry of A^{-1} of example #57 ?

Ans. $\left(\frac{-46}{13}\right)$

59. In each case determine whether the statement is true or false. If it is false given example to show that the statement is false. Assume, A and B denote square matrices.

a. If adj-A exists then A is invertible

b. If det $(A) = 0$, then A has two identical rows.

c. Det $(-A) = -\det(A)$

d. If det $(A) = \det(B)$, then $A = B$

e. If the main diagonal of matrix A consists of zero then det $(A) = 0$

f. det $(A + B) = \det(A) + \det(B)$

g. If A and B are invertible, then $(AB)^{-1} = A^{-1}B^{-1}$

Solution

a. False,as an example $A = \begin{pmatrix} 8 & 4 \\ 2 & 1 \end{pmatrix}$ adj. $A = \begin{pmatrix} 1 & -4 \\ -2 & 8 \end{pmatrix}$, but A^{-1} does not exist

b. False, as an example, $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$

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c. False,as an example, $\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$ and $\begin{pmatrix} -3 & -4 \\ -1 & -2 \end{pmatrix}$ $\det(-A) = (-1)^n \cdot \det(A)$

(det A) , where n is order of matrix.

d. False, as an example, $A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 9 & 7 \\ 5 & 4 \end{pmatrix}$

e. False, as an example, $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$

f. False

g. False, $(AB)^{-1} = B^{-1} A^{-1}$

60. Determine whether each of the following statement is True or false. If the statement is false given an example to show that the statement is false.

a. Every square matrix has a multiplicative inverse.

b. Every non singular matrix has an inverse.

c. A 3 × 2 matrix has a multiplicative inverse.

d. Every singular matrix has an inverse.

e. The matrix $\begin{bmatrix} a & b \\ -a & -b \end{bmatrix}$ is invertible.

f. The inverse of the identity matrix is itself.

Solution

a. False, as an example: $\begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix}$

b. True

c. False, A 3 × 2 is not square matrix.

d. False, $\begin{pmatrix} 4 & 8 \\ 1 & 2 \end{pmatrix}$

e. False, determinant is zero

f. False

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Using Inverse Matrices to Solve System of Linear Equation

A system of linear equation: $\begin{cases} a_1x + b_1y = C_1 \\ a_2x + b_2y = C_2 \end{cases}$

A matrix equation: $\underbrace{\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} C_1 \\ C_2 \end{bmatrix}}_{B}$

Where the coefficient matrix $A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$ and the variable, $X = \begin{pmatrix} x \\ y \end{pmatrix}$

and the constant matrix $B = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$

If the inverse of matrix A exists, you can use it to solve matrix equations of the form:

$AX = B$, has a unique solution: $X = A^{-1}B$.

$(A^{-1}A)X = A^{-1}B$ • Multiply each side by A^{-1}

$IX = A^{-1}B$ • Definition of multiplicative inverse

$X = A^{-1}B$ ← Definition of multiplicative identity

Illustrative Example

61. Solve each equation:

a. $\begin{cases} 3x + 2y = -4 \\ 7x + 5y = -9 \end{cases}$

d. $\begin{cases} 3x - 8y = -1 \\ -4x + 11y = 2 \end{cases}$

Solution

$$\begin{matrix} A & X & B \\ \updownarrow & \downarrow & \downarrow \end{matrix}$$

a. Matrix equation: $\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} \quad \begin{bmatrix} -4 \\ -9 \end{bmatrix}$

The matrix equation has the form: $AX = B$

First find A^{-1}:

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$$A^{-1} = \frac{1}{(3)5 - 2(7)}\begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$$

Use the equation: $X = A^{-1} B$

$$X = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}\begin{bmatrix} -4 \\ -9 \end{bmatrix} = \begin{bmatrix} 5(-4) + (-2)(-9) \\ -7(-4) + 3(-9) \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\therefore$ The solution set is $\{(-2, 1)\}$

b. Matrix equation: $\begin{bmatrix} 3 & -8 \\ -4 & 11 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$\uparrow$ $\uparrow$ $\uparrow$

A X = B

The matrix equation has the form: $AX = B$ First find A^{-1}

$$A^{-1} = \frac{1}{33 - 32}\begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$$

To find X, use the equation: $X = A^{-1} B$

$$X = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -11 + 16 \\ -4 + 6 \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$$

$\therefore$ The solution set is $\{(-5, 2)\}$.

62. Let $A = \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$. If $AX = B$, for the 2×2 matrix X, then what is the value of matrix X ?

A. $\begin{bmatrix} 13 & 7 \\ -11 & 6 \end{bmatrix}$ B. $\begin{bmatrix} 3 & -13 \\ -2 & 7 \end{bmatrix}$ C. $\begin{bmatrix} -3 & 13 \\ 2 & -7 \end{bmatrix}$

Solution

$AX = B$ implies that $X = A^{-1} B$

First find A^{-1}:

$$A^{-1} = \frac{1}{20 - 21}\begin{bmatrix} 5 & -7 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix}$$

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$$X = A^{-1}B = \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix} \cdot \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 13 \\ 2 & -7 \end{bmatrix}$$ **Answer: C**

63. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. If X is a 2×2 matrix such that $AX - A^t = 2A$, then what is the value of X .

A. $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ B. $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 6 \\ 6 & 9 \end{pmatrix}$ D. $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

Solution :- Given that $A^t = A$:

$AX = 2A + A^t$, mutliply both side by A^{-1} :

$(A^{-1} A) X = 2A^{-1}A + A^{-1}A^t$

$IX = 2I + I = 3I$

$$X = 3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$ **Answer: D**

4.6 Cramer's Rule

We have learned several methods to solve a system of linear equations: the substitution method, the addition method, Gausian elimination method. We now present another method called **cramer's rule.** Here, we see determinants can be used to solve system of linear equations in two and three variables.

Cramer's Rule to solve a system of two Equations containing two variables.

The solution to the system : $\begin{cases} a_1x + b_1y = C_1 \\ a_2x + b_2y = C_2 \end{cases}$

is given by $x = \dfrac{D_x}{D}$ and $y = \dfrac{D_y}{D}$, where

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \quad D_x = \begin{vmatrix} C_1 & b_1 \\ C_2 & b_2 \end{vmatrix} \text{ and } D_y = \begin{vmatrix} a_1 & C_1 \\ a_2 & C_2 \end{vmatrix}$$

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Note:

(i) The determinant D is the determinate of the coefficients of x and y.

(ii) The determinant D_x has the column of x term coefficients replaced by C_1 and C_2

(iii) The determinate D_y has the column of y - term coefficients replaced by C_1 and C_2.

Illustrative Example

64. Use cramer's rule to solve each system

a. $\begin{cases} x + 3y = -10 \\ 3x + 7y = -2 \end{cases}$ b. $\begin{cases} 2x + 8y = 2 \\ 3x + 10y = 5 \end{cases}$ c. $\begin{cases} 3x - 6y = 9 \\ 2x - 4y = 6 \end{cases}$

Solution

a. For this system, $a_1 = 1, b_1 = 3, c_1 = -10$

$a_2 = 3, b_2 = 7$, and $c_2 = -22$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 3 & 7 \end{vmatrix} = 7 - 9 = -2$$

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = \begin{vmatrix} -10 & 3 \\ -22 & 7 \end{vmatrix} = -70 + 66 = -4$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = \begin{vmatrix} 1 & -10 \\ 3 & -22 \end{vmatrix} = -22 + 30 = 8$$

Therefore, $x = \frac{D_x}{D} = \frac{-4}{-2} = 2$ and $y = \frac{D_y}{D} = \frac{8}{-2} = -4$

The solution set is $\{(2, -4)\}$

b. $$x = \frac{D_x}{D} = \frac{\begin{vmatrix} 2 & 8 \\ 5 & 10 \end{vmatrix}}{\begin{vmatrix} 2 & 8 \\ 3 & 10 \end{vmatrix}} = \frac{20 - 40}{20 - 24} = \frac{-20}{-4} = 5$$

$$y = \frac{D_y}{D} = \frac{\begin{vmatrix} 2 & 2 \\ 3 & 5 \end{vmatrix}}{-4} = \frac{10 - 6}{-4} = \frac{4}{-4} = -1$$

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Therefore, the solution set is {(5, -1)}

c. $D = \begin{vmatrix} 3 & -6 \\ 2 & -4 \end{vmatrix} = -12 + 12 = 0$, cramer's rule does not apply

Also, $D_x = \begin{vmatrix} 9 & -6 \\ 6 & -4 \end{vmatrix} = -36 + 36 = 0$, $D_y = 0$. So that the system hasinfinitely many solution.

To find the solution, we use row operation:

$$\left(\begin{array}{cc|c} 3 & -6 & 9 \\ 2 & -4 & 6 \end{array}\right) \xrightarrow{\frac{1}{3}R_1 \to R_1} \left(\begin{array}{cc|c} 1 & -2 & 3 \\ 2 & -4 & 6 \end{array}\right)$$

To change $a_{21} = 0$, multiply row 1 by -2 add the result to row 2.

$$\xrightarrow{-2R_1 + R_2 \to R_2} \left(\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 0 & 0 \end{array}\right) \quad x - 2y = 3, \text{ and } x = 3 + 2y$$

So if we let K represent any real number, we can write as $y = k$

∴ The solution set is {(3 + 2k, k)}

Cramer's rule can also be a plied to systems of linear equations in three variable with three equation

The solution set the system $\begin{cases} a_1x + b_1y_1 + c_1z = d_1 \\ a_2x + b_1y + c_2z = d_2 \\ a_2x + b_3y + c_3z = d_3 \end{cases}$

is given by $x = \frac{D_x}{D}$, $y = \frac{D_y}{D}$ and $z = \frac{D_z}{D}$

Where

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} \text{ and } D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

Provided that $D \neq 0$

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Note:

(i) If $D \neq 0$, the system has unique solution and the system is consistent

(ii) If $D = 0$, and $D_x \neq 0$, $D_y \neq 0$ and $D_z \neq 0$ the system is inconsistent and it has no solution

(iii) If $D = 0$, and $D_x = 0$, and $D_y = 0$ and $D_z = 0$ the system is dependent and has infinitely many solution.

Illustrative Example

65. Solve the system of linear equation:

a. $\begin{cases} 2x - y + z = 6 \\ x + 2y + 3z = 3 \\ 3x + y - z = 4 \end{cases}$ b. $\begin{cases} x - y = 2 \\ 3x + z = 11 \\ y - 2z = -3 \end{cases}$

Solution

a. The determinant of the coefficient matrix D is given by:

$$D = \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & -1 \end{vmatrix} = 2C_{11} - 1C_{12} + 1C_{13}$$

$$D = 2\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} - 1(-1)^{1+2}\begin{vmatrix} 1 & 3 \\ 3 & -1 \end{vmatrix} + 1(-1)^4\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$$

$$= 2(-2 - 3) + (-1 - 9) + 1(1 - 6) = -10 - 10 - 5 = -25$$

$$D_x = \begin{vmatrix} 6 & -1 & 1 \\ 3 & 2 & 3 \\ 4 & 1 & -1 \end{vmatrix} = 6C_{11} - 1C_{12} + 1C_{13}$$

$$= 6\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} - 1(-1)^{1+2}\begin{vmatrix} 3 & 3 \\ 4 & -1 \end{vmatrix} + 1\begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$$

$$= 6(-2 - 3) + 1(-3 - 12) + (3 - 8) = -30 - 15 - 5 = -50$$

$$D_y = \begin{vmatrix} 2 & 6 & 1 \\ 1 & 3 & 3 \\ 3 & 4 & -1 \end{vmatrix} = 2C_{11} + 6C_{12} + C_{13}$$

$$= 2(-3 - 12) + 6(-1)(-1 - 9) + (4 - 9) = 25$$

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$$D_z = \begin{vmatrix} 2 & -1 & 6 \\ 1 & 2 & 3 \\ 3 & 1 & 4 \end{vmatrix} = 2C_{11} - 1C_{12} + 6C_{13}$$

$= 2(8 - 3) - 1(-1)(4 - 9) + 6(1 - 6) = 10 - 5 - 30 = -25$

Now we use cramer's rule to get the solution

$$x = \frac{D_x}{D} = \frac{-50}{-25} = 2,\ y = \frac{D_y}{D} = \frac{25}{-25} = -1,\ Z = \frac{D_z}{D} = \frac{-25}{-25} = 1$$

$\therefore$ The solution set is $\{(2, -1, 1)\}$

b. $$D = \begin{vmatrix} 1 & -1 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & -2 \end{vmatrix} = 1C_{11} - C_{12} + 0C_{13}$$

$= 1(0 - 1) - (1)(-1)(-6 - 0) + 0 = -1 - 6 = -7$

$$D_x = \begin{vmatrix} 2 & -1 & 0 \\ 11 & 0 & 1 \\ -3 & 1 & -2 \end{vmatrix} = 2C_{11} - 1C_{12} + 0C_{13}$$

$= 2(0 - 1) - 1(-1)(-22 + 3) + 0 = -2 + -19 = -21$

$$D_y = \begin{vmatrix} 1 & 2 & 0 \\ 3 & 11 & 1 \\ 0 & -3 & -2 \end{vmatrix} = 1C_{11} + 2C_{12} + 0C_{13}$$

$= 1(-22 + 3) + 2(-1)^3(-6 - 0) = -19 + 12 = -7$

$$D_z = \begin{vmatrix} 1 & -1 & 2 \\ 3 & 0 & 11 \\ 0 & 1 & -3 \end{vmatrix} = 1C_{11} + -1C_{12} + 2C_{13}$$

$= 1(0 - 11) - 1(-1)(-9 - 0) + 2(3 - 0) = -11 - 9 + 6 = 14$

Using the cramer's rule

$$x = \frac{D_x}{D} = \frac{-21}{-7} = 3,\ y = \frac{D_y}{D} = \frac{-7}{-7} = 1 \text{ and } Z = \frac{D_z}{D} = \frac{-14}{-7} = 2$$

$\therefore$ The solution set is $\{(3, 1, 2)\}$.

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66. If $\begin{vmatrix} a & b & c \\ a & -a & a \\ a & a & -a \end{vmatrix} = a^3$, and $a \neq 0$, then What is the solution of

$$\begin{pmatrix} a & b & c \\ a & -a & a \\ a & a & -a \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} ?$$

A. $x = 0,\ y = 2a,\ z = 2a$

C. $x = \frac{1}{a},\ y = \frac{-2}{a},\ z = \frac{2}{a}$

B. $x = \frac{1}{a},\ y = 2a,\ z = 2a$

D. $x = 0,\ y = \frac{2}{a},\ z = \frac{2}{a}$

Solution:- We have given that $D = a^3$

$$D_x = \begin{vmatrix} 1 & b & c \\ 0 & -a & a \\ 0 & a & a \end{vmatrix} = 1C_{11} + 0.C_{21} + 0.C_{31} = \begin{vmatrix} -a & a \\ a & -a \end{vmatrix} = 0$$

$$D_y = \begin{vmatrix} a & 1 & c \\ a & 0 & a \\ a & 0 & -a \end{vmatrix} = 1C_{12} + 0.C_{22} + 0.C_{32} = (-1)^3 (1) \begin{vmatrix} a & a \\ a & -a \end{vmatrix}$$

$$D_y = -1(-a^2 - a^2) = 2a^2$$

$$D_z = \begin{vmatrix} a & b & 1 \\ a & -a & 0 \\ a & a & 0 \end{vmatrix} = 1C_{13} + 0.C_{23} + 0.C_{33} = 1 \begin{vmatrix} a & -a \\ a & a \end{vmatrix}$$

$$D_z = a^2 + a^2 = 2a^2$$

$$\therefore \quad x = \frac{D_x}{D} = \frac{0}{a^3} = 0,\ y = \frac{D_y}{D} = \frac{2a^2}{a^3} = \frac{2}{a},\ z = \frac{D_z}{D} = \frac{2a^2}{a^3} = \frac{2}{a}$$

$$\therefore \quad \text{The solution set is } \left\{\left(0, \frac{2}{a}, \frac{2}{a}\right)\right\}$$

Answer: D

67. Determine whether the system of linear equation inconsistent, dependent or consistent.

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a. $\begin{cases} 3x - 2y = 8 \\ -6x + 4y = 1 \end{cases}$

b. $\begin{cases} 5x + 2y = -7 \\ 3x + 5y = 11 \end{cases}$

c. $\begin{cases} 4x - 3y = 1 \\ \frac{1}{3}x - \frac{1}{4}y = \frac{1}{12} \end{cases}$

d. $\begin{cases} x + y = -1 \\ 2x - z = 3 \\ y + 2z = -1 \end{cases}$

Solution

As we use the three possibility of solutions of a linear system.

a. $D = \begin{vmatrix} 3 & -2 \\ -6 & 4 \end{vmatrix} = 3(4) - (-6)(-2) = 12 - 12 = 0$

$D_x = \begin{vmatrix} 8 & -2 \\ 1 & 4 \end{vmatrix} = 8(4) - (1)(-2) = 32 + 2 = 34$

$D_y = \begin{vmatrix} 3 & 8 \\ -6 & 1 \end{vmatrix} = 3(1) - 8(-6) = 3 + 48 = 51$

since $D = 0$, $D_x \neq 0$, and $D_y \neq 0$, and that the system has no solution and it is inconsistent.

b. $D = \begin{vmatrix} 5 & 2 \\ 3 & 5 \end{vmatrix} = 25 - 6 = 19$, here $D \neq 0$

In this case, the system is called consistent and it has unique solution. To find the solution of the system:

$D_x = \begin{vmatrix} -7 & 2 \\ 11 & 5 \end{vmatrix} = -35 - 22 = -57$

$D_y = \begin{vmatrix} 5 & -7 \\ 3 & 11 \end{vmatrix} = 55 + 21 = 76$

$\therefore \quad x = \frac{D_x}{D} = \frac{-57}{19} = -3, \; y = \frac{D_y}{D} = \frac{76}{19} = 4$

c. $D = \begin{vmatrix} 4 & -3 \\ \frac{1}{3} & -\frac{1}{4} \end{vmatrix} = -4\left(\frac{1}{4}\right) - (-3)\left(\frac{1}{3}\right) = -1 + 1 = 0$

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$$D_x = \begin{vmatrix} 1 & -3 \\ \frac{1}{12} & -\frac{1}{4} \end{vmatrix} = 1\left(-\frac{1}{4}\right) - (-3)\left(\frac{1}{12}\right) = \frac{-1}{4} + \frac{1}{4} = 0$$

$$D_y = 4\left(\frac{1}{12}\right) - 1\left(\frac{1}{3}\right) = \frac{1}{3} - \frac{1}{3} = 0$$

Here, $D = 0$, $D_x = 0$ and $D_y = 0$, in this case, the system is called dependent and it has infinitely many solution.

To find the solution of the system we solve for y in terms of x : $4x - 3y = 1$, the same as $y = \frac{1 - 4x}{-3}$

Now, we let t represent any real number ($x = t$)

$\therefore$ The Solution set is $\left\{\left(t, \frac{1 - 4t}{-3}\right)\right\}$

d. The system has 3equation with 3 - variables

$$D = \begin{vmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 2 \end{vmatrix} = 1C_{11} + 1C_{12} + 0C_{13} = \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} + (-1)^3 \begin{vmatrix} 2 & -1 \\ 0 & 2 \end{vmatrix}$$

Here, $D = 1 - 4 = -3$, in this case, the system has unique solution and it is consistent.

$$D_x = \begin{vmatrix} -1 & 1 & 0 \\ 3 & 0 & -1 \\ -1 & 1 & 2 \end{vmatrix} = -1C_{11} + 1C_{12} + 0C_{13} = -6$$

$$D_y = \begin{vmatrix} 1 & -1 & 0 \\ 2 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 9 \text{ and } D_z = \begin{vmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 0 & 0 & -1 \end{vmatrix} = -3$$

$$\therefore \quad x = \frac{D_x}{D} = \frac{-6}{-3} = 2,\ y = \frac{D_y}{D} = \frac{9}{-3} = -3,\ Z = \frac{D_z}{D} = \frac{-3}{-3} = 1$$

68. For the system of equation: $\begin{cases} 3x - 4y = 1 \\ 6x + ay = b \end{cases}$

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Find all values of a and b for which the system has:

a. Unique solution
b. infinitely many solution
c. no solution

Solution

$$D = \begin{vmatrix} 3 & -4 \\ 6 & a \end{vmatrix} = 3a + 24, \quad D_x = \begin{vmatrix} 1 & -4 \\ b & a \end{vmatrix} = a + 4b$$

$$D_y = \begin{vmatrix} 3 & 1 \\ 6 & b \end{vmatrix} = 3b - 6$$

a. We solve for $D \neq 0$, and D : and $3a + 24 \neq 0$, it gives $a \neq -8$

∴ The system has a unique solution for all values of a and b and $a \neq -8$.

b. We solve for $D = 0$, $D_x = 0$ and $D_y = 0$ Thus $3a + 24 = 0$, $a = -8$ $a + 4b = 0$, $a = -4b$, $-8 = -4b \Rightarrow b = 2$

∴ The system has infinitely many solution if $a = -8$ and $b = 2$

c. We solve for $D = 0$, $D_x \neq 0$, $D_y \neq 0$

∴ The system has no solution if $a = -8$ and $b \neq 2$.

69. For what values of k does the system of equations:

$$\begin{cases} 7x + 2y = -5 \\ kx - 4y = -7 \end{cases}$$ have a unique solution?

A. -14 B. 14 C. R D. R\{-14}

Solution

$$D = \begin{vmatrix} 7 & 2 \\ k & -4 \end{vmatrix} = -28 - 2k$$

we solve for $D \neq 0$, so that $-2K - 28 \neq 0$

$\Rightarrow -2k \neq 28$ and $K \neq -14$

Answer: D

70. For what values of a and b does the system of equations

$$\begin{cases} ax - 4y = -4 \\ x - ay = b \end{cases}$$ have a uniqation solution

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A. $a = 1$ and $b = \mathbb{R}\setminus\{-1\}$

B. $a = 2$ and $b = \mathbb{R}\setminus\{-2\}$

C. $a \neq 2$ and $a \neq -2$, $b = \mathbb{R}$

D. $a = 2$, $a = -2$, and $b = \mathbb{R}$

Solution

$$D = \begin{vmatrix} a & -4 \\ 1 & -a \end{vmatrix} = -a^2 + 4, \text{ and set } D \neq 0$$

$4 - a^2 \neq 0$, this implies $a \pm 2$

$\therefore$ The system have unique solution if $a \neq 2$ and $a \neq -2$, and $b = \mathbb{R}$

Answer: C

71. What are the values of λ and μ so that the system

$$\begin{cases} x + y + z = 6 \\ x + 2y + 3z = 10 \\ x + 2y + \lambda z = \mu \end{cases}$$ has infinitely many solution

A. $\lambda \neq 3$ and $\mu \in \mathbb{R}$

B. $\lambda = 3$ and $\mu \neq 10$

C. $\lambda = 3$ and $\mu = 10$

D. $\lambda \in \mathbb{R}$ and $\mu = 10$

Solution

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 1C_{11} + 1C_{12} + 1C_{12}$$

$$= 1\begin{vmatrix} 2 & 3 \\ 2 & \lambda \end{vmatrix} + (-1)^3\begin{vmatrix} 1 & 3 \\ 1 & \lambda \end{vmatrix} + 1\begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix}$$

$$= (2\lambda - 6) - 1(\lambda - 3) + 0$$

$$= \lambda - 6 + 3 = \lambda - 3$$

$$D_z = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 10 \\ 1 & 2 & \mu \end{vmatrix} = 1C_{11} + 1C_{12} + 6C_{13}$$

$$= \begin{vmatrix} 2 & 10 \\ 2 & \mu \end{vmatrix} + (-1)^3\begin{vmatrix} 1 & 10 \\ 1 & \mu \end{vmatrix} + 6\begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix}$$

$$= (2\mu - 20) - (\mu - 10) + 0 = \mu - 10$$

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The system has an infinitely many solution if $D = 0$, $D_x = 0$ and $D_z = 0$

That is $D = 0$ means $\lambda - 3 = 0$, $\lambda = 3$

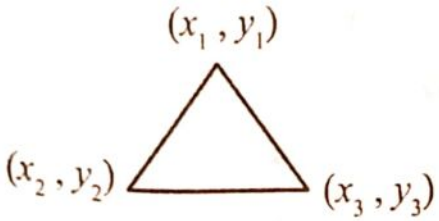
$D_z = 0$ means $\mu - 10 = 0$, $\mu = 10$ **Answer: C**

Areas of Triangles Using Determinants

Determinants provide a simple way to calculate the area of a triangle in the coordinate plane.

Area of A Triangle

If a triangle in the coordinate plane has vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) can be given as the absolute value of the determinant, then its area is



$$\text{Area} = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Illustrative Example

72. Using determinant, find the area of each triangle (ΔABC)

a. A (1, -2), B (3,-4), C (4,2) b. A (-3, 4), B (4, 2), C(1, -2)

Solution

a. $$\text{Area } (\Delta ABC) = \frac{1}{2}\begin{vmatrix} 1 & -2 & 1 \\ 3 & -4 & 1 \\ 4 & 2 & 1 \end{vmatrix} = \frac{1}{2}(1C_{11} - 2C_{12} + 1C_{13})$$

$$= \frac{1}{2}\left[1\begin{vmatrix} -4 & 1 \\ 2 & 1 \end{vmatrix} - 2(-1)^3\begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} + 1\begin{vmatrix} 3 & -4 \\ 4 & 2 \end{vmatrix}\right]$$

$$= \frac{1}{2}[(-4 - 2) + 2(3 - 4) + 6 + 16] = \frac{1}{2}(14) = 7$$

$\therefore$ Area of ΔABC is 7 square unit

b. $$\text{Area } (\Delta ABC) = \frac{1}{2}\begin{vmatrix} -3 & 4 & 1 \\ 4 & 2 & 1 \\ 1 & -2 & 1 \end{vmatrix} = \frac{1}{2}(-3C_{11} + 4C_{12} + 1C_{13})$$

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$$= \frac{1}{2}\left[-3\begin{vmatrix}2 & 1\\ -2 & 1\end{vmatrix} + 4(-1)^3\begin{vmatrix}4 & 1\\ 1 & 1\end{vmatrix} + 1\begin{vmatrix}4 & 2\\ 1 & -2\end{vmatrix}\right]$$

$$= \frac{1}{2}(-3(2+2) - 4(4-1)) + (-8-2) = \frac{1}{2}(-34) = -17$$

$\therefore$ Area of ΔABC is 17 square units.

73. Find a,b, and c, so that $\begin{vmatrix}6 & 3 & x\\ 1 & 2 & y\\ 4 & 1 & z\end{vmatrix} = ax + by + cz$

Solution

Expanding along column 3 : $xC_{13} + yC_{23} + ZC_{33}$

$$x\begin{vmatrix}1 & 2\\ 4 & 1\end{vmatrix} + (-1)y\begin{vmatrix}6 & 3\\ 4 & 1\end{vmatrix} + Z\begin{vmatrix}6 & 3\\ 1 & 2\end{vmatrix}$$

$$= x(1-8) - y(6-12) + Z(12-3) = -7x + 6y + 9z$$

$\therefore$ a = -7, b = 6 and c = 9

Collinear Points and Determinants

If three distinct (x_1, y_1), (x_2, y_2) and (x_3, y_3) are collinear (lie on the same line) if and only if :

$$\begin{vmatrix}x_1 & y_1 & 1\\ x_2 & y_2 & 1\\ x_3 & y_3 & 1\end{vmatrix} = 0$$

Illustrative Example

74. Use a determinant to check whether each set of points is collinear.

a. (-2, 1), (3, -9), and (-4, 5) b. (-3, -10), (1, 2), and (3, 1)

Solution

To test whether each point is collinear or not

a. $\begin{vmatrix}-2 & 1 & 1\\ 3 & -9 & 1\\ -4 & 5 & 1\end{vmatrix} = -2C_{11} + 1C_{12} + 1C_{13}$

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$$= -2\begin{vmatrix}-9 & 1\\ 5 & 1\end{vmatrix} + 1(-1)^3\begin{vmatrix}3 & 1\\ -4 & 1\end{vmatrix} + \begin{vmatrix}3 & -9\\ -4 & 5\end{vmatrix}$$

$$-2(-9-5) - 1(3+4) + (15-36) = 28 - 7 - 21 = 0$$

$\therefore$ (-2, 1), (3, - 9), and (-4, 5) are collinear points

b. $$\begin{vmatrix}-3 & -10 & 1\\ 1 & 2 & 1\\ 3 & 1 & 1\end{vmatrix} = -3\,C_{11} - 10\,C_{12} + 1C_{13}$$

$$= -3\begin{vmatrix}2 & 1\\ 1 & 1\end{vmatrix} - 10\,(-1)^3\begin{vmatrix}1 & 1\\ 3 & 1\end{vmatrix} + \begin{vmatrix}1 & 2\\ 1 & 1\end{vmatrix}$$

$$= -3(2-1) + 10(1-3) + (1-6)$$

$$= -3(1) + 10(-2) + (-5) = -28$$

Since the determinant value is not zero, so that the given points are not collinear.

Interpolating Polynomials:

We can use a system of equations to find a polynomial of degree 'n' whose graph passes through (n + 1) points.

- If n = 1, polynomials of degree 1, then it needs two coordinate points (x_1, y_2), so that $P(x) = ax + b$
- If n = 2, polynomials of degree 2, then it needs 3 points (x_1, y_1), (x_2, y_2), and (x_3, y_3), so that $P(x) = ax^2 + bx + c$

Illustrative Example

75. Find a polynomial that passes through the points whose coordinates are:

a. (-2 - 7) and (1, -1) and find P(6)

b. (0, 3), (2, 1) and (-1, 10) and find P(3)

Solution

a. The polynomial passes through two points, in this case, it is polynomial of degree 1 (linear equation). Thus $P(x) = ax + b$

$$P(-2) = -2a + b = -7 \text{ and } P(1) = a + b = -1$$

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$$\begin{cases} -2a + b = -7 \\ a + b = -1 \xrightarrow{\times 2} \end{cases} + \begin{cases} -2a + b = -7 \\ 2a + 2b = -2 \end{cases}$$

$0 + 3b = -9,\ b = -3$

and $a = -1 + 3 = 2$

$\therefore p(x) = 2x - 3$ and $P(6) = 2(6) - 3 = 9$

b. Since the polynomial passes through three given points, in this case, it is polynomial of degree 2.

$P(x) = ax^2 + bx + c$

$P(0) = 0 + 0 + c = 3$, this implies $C = 3$

$P(2) = 4a + 2b + c = 1$, this implies $4a + 2b + c = 1$

$P(-1) = a - b + c = 10$, this implies $a - b + c = 10$

$$\begin{cases} c = 3 \\ 4a + 2b + c = 1 \Rightarrow 4a + 2b + 3 = 1 \\ a - b + c = 10 \Rightarrow a - b + 3 = 10 \end{cases}$$: Solving together

$$\begin{cases} 4a + 2b = -2 \\ a - b = 7 \xrightarrow{\times 2} \end{cases} \begin{cases} 4a + 2b = -2 \\ 2a - 2b = 14 \end{cases}$$

$6a = 12$

$a = 2,\ b = -5$

$\therefore \quad P(x) = 2x^2 - 5x + 3$, and $P(3) = 2(9) - 5(3) + 3 = 7$

Equation of a Line and Determinant

An equation of the line containing the two points (x_1, y_1) and (x_2, y_2) may be expressed as the determinant

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Illustrative Example

76. Find the equation of the line containing the points (-1, -7) and (3, 5).

Solution

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$$\begin{vmatrix} x & y & 1 \\ -1 & -7 & 1 \\ 3 & 5 & 1 \end{vmatrix} = 0, \text{ implies that } xC_{11} + yC_{12} + 1C_{13}$$

$$x\begin{vmatrix} -7 & 1 \\ 5 & 1 \end{vmatrix} - y\begin{vmatrix} -1 & 1 \\ 3 & 1 \end{vmatrix} + 1\begin{vmatrix} -1 & -7 \\ 3 & 5 \end{vmatrix} = 0$$

$$x(-7 - 5) - y(-1 - 3) + 1(-5 + 21) = 0$$

$$-12x + 4y + 16 = 0 \Leftrightarrow y = 3x - 4$$

Review Exercise 4

1. What is a determinate?
2. What is a minor of 3 × 3 matrix ?
   c. How do you find this minor?
   d. What is the difference between the minor of an element a_{ij} and the cofactor of the element
3. What is an adjoint of a matrix.
4. Find the minor, the cofactor and adjoint of 2 × 2 matrix.

   a. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ b. $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ c. $\begin{bmatrix} 5 & -3 \\ 2 & -1 \end{bmatrix}$

5. If $A = a_{ij} = \begin{bmatrix} 4 & -3 & 7 \\ 0 & 1 & -6 \\ 3 & -2 & 5 \end{bmatrix}$,

   then find the minor, cofactor and adjoint of matrix A.
6. Find the determinant of the following matrices

   a. $A = \begin{bmatrix} -3 & 4 \\ 2 & 5 \end{bmatrix}$ c. $A = \begin{bmatrix} \cos x & \sin x \\ x - 3 & x + 2 \end{bmatrix}$

   b. $A = \begin{bmatrix} x - y & y \\ x & x + y \end{bmatrix}$ d. $A = \begin{bmatrix} x + 2 & x - 3 \\ x - 3 & x + 2 \end{bmatrix}$

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7. Let $A = \begin{pmatrix} 3 & 8 & 6 \\ 1 & 2 & 1 \\ 2 & 6 & 4 \end{pmatrix}$, then find det (A)

8. If a matrix B is obtained from matrix A by multiplying row 1 by 4 and det (B) = 10, What is det (A)?

9. If a matrix B is obtained from A by interchanging row 1 and row 3 and det (A) = 6, what is det (B)?

10. If a matrix B is obtained from A by adding -4 times row 1 to row 2, and det (A) = 6, What is det (B)?

11. Let $A = \begin{pmatrix} 1 & -2 \\ x & x-2 \end{pmatrix}$ and $|A| = x + 1$, then what is the value of x

12. Find the inverse of each of the following matrix, if it exists.

a. $A = \begin{pmatrix} 3 & 6 \\ -1 & 2 \end{pmatrix}$

b. $A = \begin{pmatrix} 3 & 6 \\ 2 & 4 \end{pmatrix}$

c. $A = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$

d. $A = \begin{pmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{pmatrix}$

13. If A is a 3 × 3 matrix and det (A) = 4, then find

a. $|-2A|$ b. $|adj \cdot A|$ c. $|A \cdot adj \cdot A|$ d. $|(2A)^{-1} A^t|$

14. Suppose A and B are order 3 matrices, I is the identity matrix with order 3, such that AB = 4I. If |A| = 12, then what is the det (B^t)?

15. Let $A = \begin{pmatrix} 2 & 5 & 4 \\ 1 & 4 & 3 \\ 1 & -3 & -2 \end{pmatrix}$ find A^{-1}

16. Solve the equation

a. $\begin{bmatrix} 3 & 6 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ -5 \end{bmatrix}$

b. $\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5 \\ -4 \end{bmatrix}$

17. If $|A| \neq 0$, what is the solution to AX = B?

18. Use cramer's rule to solve the linear system

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a. $\begin{cases} 3x + 6y = -3 \\ -x + 2y = -5 \end{cases}$ b. $\begin{cases} 4x + 3y = -5 \\ 3x + 2y = -4 \end{cases}$

c. $\begin{cases} x - 2y = 11 \\ 2x + 5y = -14 \end{cases}$ d. $\begin{cases} 2x - 4y = 7 \\ -x + y = 1 \end{cases}$

19. For what values of K does the system $\begin{cases} kx + 2y = 6 \\ 8x + ky = 1 \end{cases}$ has a unique solution

20. For what values of a and b does system $\begin{cases} x + y - z = 6 \\ x - 2y = -4 \\ x + 4y - az = b \end{cases}$

(i) has a unique solution (ii) has no solution

21. If det (A) = 2 and det (B) = -3 Find det ($A^2 B^{-1} A B^2$)

22. Let A $\begin{pmatrix} a & b \\ x & y \end{pmatrix}$ and det (A) = 3. then find the determinant of matrix B such that

a. B = $\begin{pmatrix} a + 2x & b - 2y \\ x & y \end{pmatrix}$ b. B = $\begin{pmatrix} 2a & 2b \\ 3x + a & 3y + b \end{pmatrix}$

c. B = $\begin{pmatrix} x & y \\ a & b \end{pmatrix}$ d. B = $\begin{pmatrix} a - x & b - y \\ x & y \end{pmatrix}$

23. Find the area of a triangle with vertices:

a. (-2, 1), (-1, -6) and (-8, -5)

b. (1, 0), (4, -5), and (7, -2)

Answer to Review Exercise 4

1. A determinant is a real number associated with a square matrix.

2. a. A minor for an element in a 3 × 3 matrix is the determinant of a 2 × 2 matrix.

b. a minor for an element is obtained by deleting the row and column of

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the element and finding the determinant of 2 × 2 matrix that remains.

c. Cofactor of the element a_{ij} is $C_{ij} = (-10)^{i+j} M_{ij} = (-1)^{i+j} |A_{ij}|$ where M_{ij} is the minor of a_{ij}

3. Adjoint of matrix is the transpose of cofactor of a matrix.

4. a. $C_{11} = |A_{11}| = d$, $C_{12} = (-1)^3 |A_{12}| = -C$

$C_{21} = (-1)^3 |A_{21}| = -b$ $C_{22} = |A_{22}| = a$

$$\therefore \quad \text{Adj} \cdot A = (\text{Cofactor})^t = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}^t = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

b. $\text{Adj} \cdot A = \begin{pmatrix} 2 & -4 \\ -1 & 3 \end{pmatrix}$, c. $\text{Adj} \cdot A = \begin{pmatrix} -1 & 3 \\ -2 & 5 \end{pmatrix}$

5. Minor of $A = a_{ij} = \begin{pmatrix} 7 & 18 & -3 \\ -1 & -1 & -1 \\ 11 & -24 & 4 \end{pmatrix}$

cofactor of A we use sign array in minor

$$\therefore \quad \text{cofactor} = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} = \begin{pmatrix} 7 & -18 & -3 \\ 1 & -1 & -1 \\ 11 & 24 & 4 \end{pmatrix}$$

$$\text{Adj } A = (\text{cofactor})^t = \begin{pmatrix} 7 & 1 & 11 \\ -18 & -1 & 24 \\ -3 & -1 & 4 \end{pmatrix}$$

6. a. $|A| = -15 - 8 = -23$

b. $|A| = x^2 - y^2 - xy$

c. $|A| = \cos^2 x + \sin^2 x = 1$

d. $|A| = 10x - 5$

7. $|A| = 3C_{11} + 8C_{12} + 6C_{13} = 2$

8. $\det(B) = 4\det(A)$, this implies $\det(A) = \dfrac{\det B}{4} = \dfrac{10}{4} = \dfrac{5}{2}$

9. $\det(B) = -\det(A) = -6$

10. $\det(B) = \det(A) = 6$

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11. $|A| = x - 2 + 2x = x + 1$, this implies $x = \frac{3}{2}$

12. a. $A^{-1} = \frac{1}{2}\begin{pmatrix} 2 & -6 \\ -1 & 3 \end{pmatrix}$ b. since $|A| = 0$, A^{-1} doesn't exist

c. $A^{-1} = \frac{1}{-2}\begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}$ d. $A^{-1} = \frac{1}{1}\begin{pmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{pmatrix}$

13.a. $|-2A| = (-2)^4 (4) = 64$

b. $|adj \cdot A| = |detA \cdot I| = |4I| = 4^3 \cdot 1 = 64$

c. $|Aadj \cdot A| = |A| \cdot |adj \cdot A| = (4)(64) = 256$

d. $\left|(2A)^{-1}A^t\right| = \frac{|A^t|}{|2A|} \quad \frac{4}{(2^3)|A|} = \frac{4}{8(4)} = \frac{1}{8}$

14. $|AB| = |A|\,|B|$ and $|B| = \frac{|4I|}{|A|} = \frac{4^3}{12} = \frac{64}{12} = \frac{16}{3}$

15. $A^{-1} = \begin{pmatrix} -1 & 2 & 1 \\ -5 & 8 & 2 \\ -7 & -11 & -3 \end{pmatrix}$

16. a. $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} 3 & 6 \\ -1 & 2 \end{bmatrix}^{-1} \begin{pmatrix} -3 \\ -5 \end{pmatrix} = \frac{1}{12}\begin{pmatrix} 2 & -6 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} -3 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \cdot 5 \end{pmatrix}$

b. $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}^{-1} \begin{pmatrix} -5 \\ -4 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 3 & -4 \end{pmatrix}\begin{pmatrix} -5 \\ -4 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

17. $X = A^{-1} B$

18. a. $x = \frac{D_x}{D} = \frac{\begin{vmatrix} -3 & 6 \\ -5 & 2 \end{vmatrix}}{12} = \frac{-6 + 30}{12} = 2$

$y = \frac{D_y}{D} = \frac{18}{12} = \frac{3}{2}$

b. $x\ \frac{D_x}{D} = \frac{2}{-1} = -2$ and $y = \frac{D_y}{D} = \frac{-1}{-1} = 1$

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c. $x = \frac{D_x}{D} = \frac{27}{9} = 3$ and $y = \frac{D_y}{D} = \frac{-36}{9} = -4$

d. $x = \frac{D_x}{D} = \frac{-11}{2}$ = and $y = \frac{D_y}{D} = \frac{-9}{2}$

19. $D = K^2 - 16 \neq 0$, $K \neq 4$ and $K \neq -4$.

20. i. Determinant of coef· matrix is $D = 3a - 6\ 0$, $a \neq 0$, $a \neq 2$, $b = R$

ii. $a = 2$ and $b \neq 16$

21. $\left|A^2\right| \left|B^{-1}\right| \left|A^1\right| \left|B^2\right| = \frac{(2)^2\ (2)\,(-3)^2}{(-3)} = \frac{72}{-3} = -24$

22. a. $|B| = |A| = 3$

b. $|B| = 2{\cdot}3\ |A| = 2{\cdot}3\ (3) = 18$

c. $|B| = -|A| = -3$

d. $|B| = |A| = 3$

23. a. Area = 24 sq unit

b. Area = 12 sq. unit

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UNIT 5

5. Vectors

Hello there! Wel come to this interesting unit where you will able to; define vectors, add and subtract vectors, multiply vectors by scalars and another vectors, describe unit vector and norm of a vector, find out equation of a line, equation of a circle and equation of a tangent and so much more. Enjoy!

5.1 What are Vectors ?

Physical quantities that are measured directly or indirectly can be described using numbers, units and directions.

However, not all physical quantities require direction for their complete description. For example, we may say, a velocity of 50km/hr due north, a force of 500N towards east. But, without attaching direction, we can say a 5kg of sugar, a 37^0c body temperature and the time is now 6:30 AM, ... etc. Therefore, based on the requirement of direction for their complete description, physical quantities are classified as scalars and vectors.

Scalar Quantity: Is a quantity that can be completely described using number and unit alone.

For example; mass, time, temperature, length area, speed, volume ... etc are **scalar quantities.**

Vector Quantity: Is a quantity that requires number, unit and direction for its complete description.

For example; displacement, velocity, acceleration, force, electric field ... etc are **vector quantities.**

Illustrative Example

1. Classify the following physical quantities as scalar and vector quantity.

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Work, energy, power, momentum, magnetic field, electric current density

Answer:

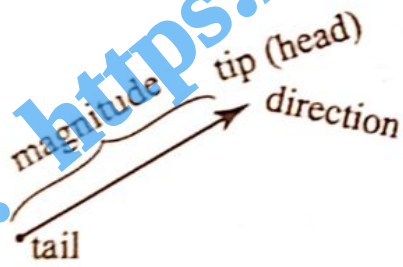
Scalar	Vector
Work	Momentum
Energy	Current density
Power	Magnetic field
Electric current	

2. Vector quantities are expressed using magnitude and direction. But scalar quantities need magnitude only. True or False?

Answer: True

5.2 Representation of Vectors

Vectors can be represented graphically or analytically. Graphically, a vector is represented using a directed line segment, where the length represents magnitude and the arrow indicating its direction as shown below.



Where: Tail = initial point of the vector

Tip (head) = final point of the vector

Illustrative Example

3. Graphically, represent a 500N force acting on a body at an angle of 37^0 from the horizontal.

Solution:- Now, using a ruler let the scale be 1cm ≡ 100N

Hence, 500N ≡ 5cm on paper, and using a protractor, measure 37^0 from the horizontal. Therefore, draw 5cm at an angle of 37^0 using a ruler starting from the origin

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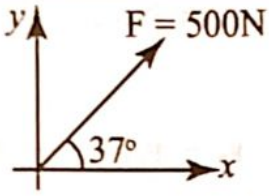
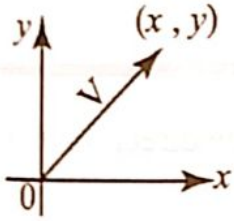


Algebraically, a vector can be represented by a bold face letter or a letter with an arrow over its head as $\overline{V}$ and can be expressed interms of its components in coordinate or in column forms as shown below:

If a vector $\overline{V}$ is in a plane whose initial point is the origin (0,0) and whose terminal point is (x,y) then we have:



$\overline{V} = (x, y)$ ⟵ coordnate form

$\overline{V} = \begin{pmatrix} x \\ y \end{pmatrix}$ ⟵ column form of the vector

The direction of the vector is obtained by:

$$\text{Slope} = \tan\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Where: The numbers x and y are called components of the vector.

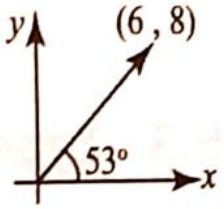
Illustrative Example

4. Find the coordinate form, column form and direction of a vector have initial point (0, 0) and terminal point (6, 8).

Solution :- Let the vector be $\overline{V}$

$\overline{V} = (x, y) = (6,8)$ ⟵ coordinate form

$\overline{V} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ ⟵ column form



And, its direction is Slope $= \tan\theta = \frac{y}{x} = \frac{8}{6}$

$$\theta = \tan^{-1}\left(\frac{8}{6}\right) = \tan^{-1}(1.33) = 53^0$$

In general, if $\overline{V}$ is a vector in the x - y plane whose initial point is (x_1, y_1)

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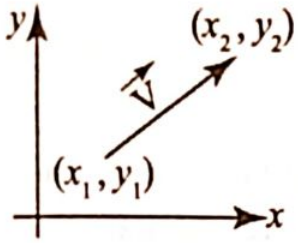
and whose terminal point is (x_2, y_2) then we have:



$\vec{V} = (x_2 - x_1, y_2 - y_1)$ ⟵ coordinate form

$\vec{V} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$ ⟵ column form of the vector

The direction of the vector is obtained by:

$$\text{slope} = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1}, \text{ or } \theta = \tan^{-1}\left(\frac{y_2 - y_1}{x_2 - x_1}\right)$$

Illustrative Example

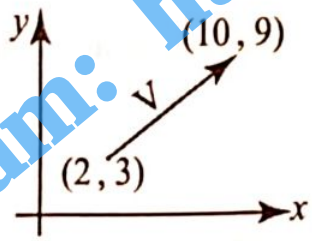
5. Find the coordinate form, the column form and the direction of a vector that initial point (2, 3) and terminal point (10, 9).

Solution:- Let the vector be $\vec{V}$

$\vec{V} = (x_2 - x_1, y_2 - y_1) = (10 - 2, 9 - 3) = (8, 6)$ ⟵ coordinate form

$\vec{V} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix} = \begin{pmatrix} 10 - 2 \\ 9 - 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$ ⟵ column form

And, the direction of the vector is detemined as:



$$\text{Slope} = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{9 - 3}{10 - 2} = \frac{6}{8}$$

$$\theta = \tan^{-1}(0.75) = 37^0$$

6. If the initial point and terminal points of a vector are (-2, 4) and (-6, -3) respectively. What is the coordinate form of the vector?

Answer: $\vec{V} = (-4, -7)$. How?

5.3 Types of Vectors

Position Vectors:- Is a vector that gives the position of an object with respect to the origin. For example $\overrightarrow{OP}$ is a position vector with initial point 0 at (0, 0) and terminal point P at (x,y)

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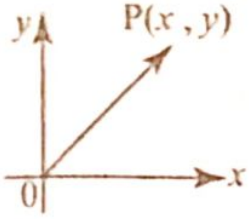
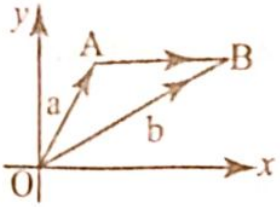


Moreover, we can have:



$\overline{AB}$ = Position vector of B relative to A

$\overline{OB}$ = Position vector of B relative to A

$\overline{OA} + \overline{AB} = \overline{OB}$

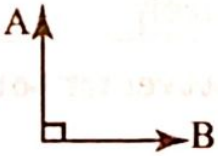
$\overline{AB} = \overline{OB} - \overline{OA}$

Note: The magnitude of $\overline{AB}$ is obtained by: $|\overline{AB}| = |b - a|$

- **Equal Vectors:** Are vectors of equal magnitude and the same direction

- **Negative Vectors:** Are vectors of equal magnitude but opposite directions.

- **Zero or Null Vector:** Is a vector of zero magnitude and undefined direction.
- **Orthogonal Vectors:** Are vectors that act on a body perpendicular to each other



- **Collinear Vectors:** Are vectors that act an a body parallel to each other.

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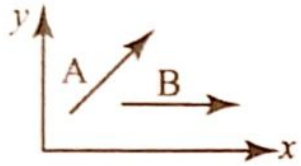
• Coplanar vectors: Are vectors that lie in the same plane.



Illustrative Example

Match the items in column "A" with those in column "B"

A	B
7. Position Vector	A. Negative of a vector
8. Unit Vector	B. Vectors on the same plane
9. Null Vector	C. Indicate position with respect to the origin
10. Orthogonal Vectors	D. Magnitude zero, arbitrary direction
11. Collinear Vectors	E. Magnitude one, particular direction
12. Coplanar Vectors	F. Perpendicular vectors
13. Concurrent Vectors	G. Parallel vectors
	H. Vectors acting at a point

Answer

7. C	9. D	11. G	13. H
8. E	10. F	12. B	

5.4 What are Unit Vectors?

A vector whose magnitude is one and pointing along the x, the y or the z axis is called **unit vector**.

The standard unit Vectors are i = (1, 0) and j = (0, 1) pointing along the x and the y axis perspectively.

Any Vector can be expressed in the unit vector notation as follows

Let $\overline{V}$ = (a, b) be any vector. Then, it can be expressed using the standard unit vector as;

$$\overline{V} = (a, b) = a(1, 0) + b(1, 0) = ai + bj$$

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Hence $\vec{V} = (2,3) = 2(1, 0) + 3(0,1) = 2i + 3j$

Illustrative Examples

14. Express the vector $\vec{V} = (3, 5)$ interims of the standard unit vectors i and j

Solution:- $\vec{V}$ $(3, 5) = 3(1, 0) + 5(0, 1) = 3i + 5j$

15. Express vector $\overrightarrow{PQ}$ interms of unit vectors i and j when the end points are P(3, 1) and Q (1, 4)

Solution: The coordinate form of vector $\overrightarrow{PQ}$ is

$\overrightarrow{PQ} = (1 - 3, 4 - 1) = (-2, 3)$

$\overrightarrow{PQ} = (-2, 3) = -2 (1, 0) + 3(0, 1) = -2i + 3j$

16. Let $\vec{V}$ be a vector with initial point (-2, 3) and terminal point (-5, 2). Then:

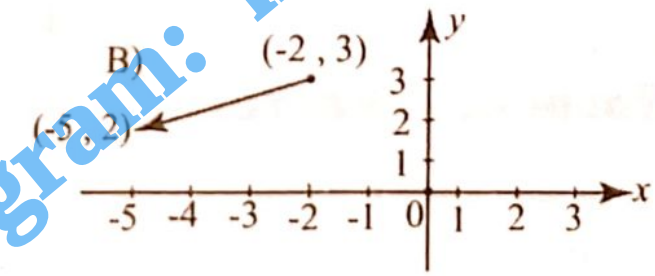
A. Write $\vec{V}$ interims of the standard unit vectors

B. Represent vector $\vec{V}$ graphically

Solution: A. $\vec{V} = (-5-(-2), 2 - 3) = (-3, -1)$

$\vec{V} = -3 (1, 0) + -1 (0, 1) = -3i - j$

B)



17. Write the following vectors in a coordinate form

A. $i + 3j$ B. $-3i - j$ C. $0.5i + 2j$ D. $\frac{-3}{4}i - 3j$

Solution

A. $i + 3j = 1(1, 0) + 3 (0, 1) = (1 + 0, 0 + 3) = (1, 3)$

B. $-3i - j = -3 (1, 0) - 1 (0, 1) = (-3, 0) + (0, -1) = (-3 + 0, 0 + (-1)) = (-3, -1)$

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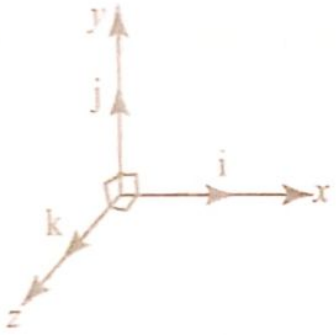
C. $0.5i + 2j = 0.5\,(1,0) + 2\,(0,1) = (0.5 + 0, 0 + 2) = (0.5, 2)$

D. $\frac{-3}{4}i - 3j = \frac{-3}{4}(1,0) - 3\,(0,1) = \left(\frac{-3}{4} + 0, 0 - 3\right) = \left(\frac{-3}{4}, -3\right)$

In space, unit vectors are represented by i,j and k pointing along the x,y and z axes respectively.



where : i (1,0,0)

j = (0, 1, 0)

k = (0, 0, 1)

- i, j and k are mutually perpendicular to each other.
- The unit vectors i,j and k are perpendicular to each other. Using the unit vector notation, any vector can be expressed as:

$$\bar{V} = V_1 i + V_2 j + V_3 K = (V_1, V_2, V_3)$$

Hence, if we have

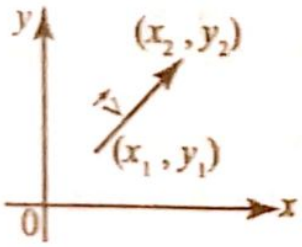
$\bar{V} = (2, -3, 1)$ in coordinate form, then

$\bar{V} = 2i - 3j + K$, in unit vector notation

5.5 What is Norm of a Vector?

The normal of a vector is also known as magnitude of the vector. And, if a vector $\bar{V}$ is represented by initial point (x_1, y_1) and terminal point (x_2, y_2) then, norm or magnitude of the vector is denoted by $|\bar{V}|$ and is obtained by:

$$\boxed{|\bar{V}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$



Where; coordinates of the vector are

$$\bar{V} = (x_2 - x_1, y_2 - y_1)$$

- And, if the vector is given on a plane in unit vector notation as;

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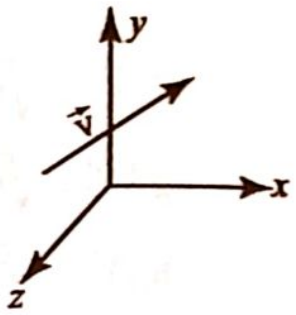
$\vec{V} = V_1 i + V_2 j$, then norm of the vector is $\boxed{|\vec{V}| = \sqrt{V_1^2 + V_2^2}}$

- And, if the vector is in space, $\vec{V} = V_1 i + V_2 j + V_3 k$

then, its magnitude is obtained by:

$$\boxed{|\vec{V}| = \sqrt{V_1^2 + V_2^2 + V_3^2}}$$



The unit vector in the direction of any non-zero vector $\vec{V} = V_1 i + V_2 j$ is obtained by:

$$\boxed{\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{V_1 i + V_2 j}{\sqrt{V_1^2 + V_2^2}}}$$

Where: $\hat{V}$ is read as "V - CaP"

Illustrative Example

18. Find the normal of the following vectors

a. $\vec{V} = (1, 2)$

b. $\vec{V} = (-2, 3)$

c. $\vec{V} = (5, 0)$

d. $\vec{V} = \left(\frac{1}{4}, \frac{3}{4}\right)$

e. $\vec{V} = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{5}\right)$

f. $\vec{V} = i + 3j$

g. $\vec{V} = -3i - 4j$

h. $\vec{V} = 2j - j + 3k$

i. $\vec{V} = -4i + k$

Solution

a. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{(1)^2 + (2)^2} = \sqrt{5}$ units

b. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{(-2)^2 + (3)^2} = \sqrt{13}$ units

c. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{(5)^2 + (0)^2} = 5$ units

d. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{3}{2}\right)^2} = \frac{\sqrt{37}}{4}$ units

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e. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{3}}{5}\right)} = \sqrt{\frac{31}{50}}$ units

f. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{(1)^2 + (3)^2} = \sqrt{10}$ units

g. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{(-3)^2 + (-4)^2} = 5$ units

h. $|\vec{V}| = \sqrt{V_1^2 + V_2^2 + V_3^2} = \sqrt{(2)^2 + (-1)^2 + (3)^2} = \sqrt{14}$ units

i. $|\vec{V}| = \sqrt{V_1^2 + V_2^2 + V_3^2} = \sqrt{(-4)^2 + (0)^2 + (1)^2} = \sqrt{14}$ units

19. Check whether each of the following are unit vectors or not

a. $\vec{V} = \left(\frac{2}{55}, \frac{1}{\sqrt{5}}\right)$ b. $\vec{V} = \left(\frac{3}{5}, \frac{-4}{5}\right)$ c. $\vec{V} = -i + j$

Solution

a. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{\left(\frac{2}{55}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)}$

$|\vec{V}| = 1$, $\vec{V}$ is a unit vector

b. $|\vec{V}| = \sqrt{V_1^2 + V_2^2} = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{-4}{5}\right)^2}$

$|\vec{V}| = 1$, $\vec{V}$ is a unit vector

c. $\vec{V} = \sqrt{V_1^2 + V_2^2} = \sqrt{(-1)^2 + (1)^2}$

$|\vec{V}| = \sqrt{2}$, $\vec{V}$ is not a unit vector

20. Given that; $\vec{V} = (-3, 4)$. Find a unit vector in the direction

a. of vector $\vec{V}$ b. Opposite to vector $\vec{V}$

Solution

a. $\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{V_{1i} + V_{2j}}{\sqrt{V_1^2 + V_2^2}} = \frac{-3i + 4j}{\sqrt{(-3)^2 + (4)^2}} = \left(\frac{-3}{5}, \frac{4}{5}\right)$

b $\hat{V} = \frac{-\vec{V}}{|\vec{V}|} = \frac{-V_{1i} + (-V_2)j}{\sqrt{V_1^2 + V_2^2}} = \frac{-(-3)i + (-4)j}{5} = \left(\frac{3}{5}, \frac{-4}{5}\right)$

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21. Find a unit vector in the direction of:

a. $\vec{V} = i + 3j$ b. $\vec{V} = 2i - j + 3k$ d. $\vec{V} = i + j - k$

c. $\vec{V} = 2i - j + 3k$ e. $\vec{V} = 2i + 3k$

Solution

a. $\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{(3, 6)}{\sqrt{(3)^2 + (6)^2}} = \frac{(3,6)}{3\sqrt{5}} = \left(\frac{2}{55}, \frac{3}{\sqrt{5}}\right)$

b. $\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{i + 3j}{\sqrt{(1)^2 + (-1)^2}} = \frac{i + 3j}{\sqrt{0}} = \frac{1}{\sqrt{14}}(i + 3j)$

c. $\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{i + 3j}{\sqrt{(1)^2 + (-1)^2}} = \frac{i + 3j}{\sqrt{0}} = \frac{1}{\sqrt{14}}(i + 3j)$

d. $\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{i + j - k}{\sqrt{(1)^2 + (1)^2 + (-1)^2}} = \frac{i + j - k}{\sqrt{3}} = \frac{1}{\sqrt{3}}(i + j - k)$

e. $\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{2i + 3k}{\sqrt{(2)^2 + (3)^2}} = \frac{2i + 3k}{\sqrt{13}} = \frac{1}{\sqrt{13}}(2i + 3k)$

22. Find a unit vector in the direction of vector $\vec{V} = 4i - 3k$

Solution

$$\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{V_1 i + V_2 j + V_3 K}{\sqrt{V_1^2 + V_2^2 + V_3^2}} = \frac{4i + 0j + (-3k)}{\sqrt{(4)^2 + (0)^2 + (-3)^2}}$$

$$\hat{V} = \frac{4i - 3k}{5} = \frac{4}{5}i - \frac{3}{5}K$$

5.6 Addition of Vectors

Vectors can be added graphically or algebraically Graphically, we can add vectors using triangle law, parallelogram law or polygon law.

Triangle Law of Vector Addition

To find the resultant or sum of two vectors using the triangle law, we follow

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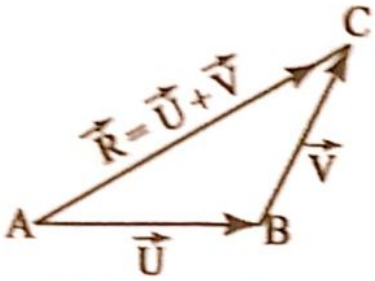
the following procedure.

Join the vectors 'head to tail' and the resultant is drawn from the tail of the first vector to the head of the second vector.

23. Find the resultant of the two vectors shown below using the triangle law

Solution:- Join the tail of vector $\overline{V}$ to the head of vector $\overline{V}$ to the head of vector $\overline{U}$ and then, draw the resultant from the tail of vector $\overline{U}$ to the head of vector $\overline{V}$ as shown below;



- $\overline{AB} + \overline{BC} = \overline{AC}$
- $\overline{AB} - \overline{CB} = \overline{AC}$

Parallelogram Law of Vector Addition

To find the resultant of two vectors, using the parallelogram law we follow the following procedure.

Join the vectors "tail to tail" and form a parallelogram. The diagonal of the parallelogram. The diagonal of the parallelogram then represents the resultant of the vectors

24. Use the parallelogram law to find the resultant of the resultant of the two vectors given below.

Solution: Join the vectors $\overline{U}$ and $\overline{V}$ tail to tail and form a parallelogram. Then, the diagonal of the parallelogram given the resultant of the vector as shown below.

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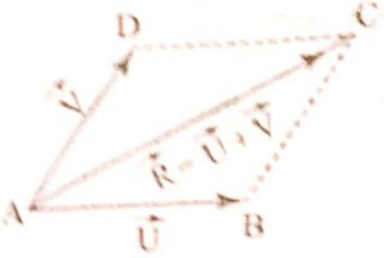


- $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$
- $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} = \overrightarrow{AC}$

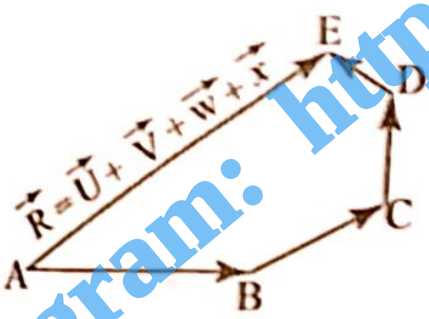
Polygon law of Vector Addition

To find the resultant of two or more vectors graphically, we use the polygon law as follows:

Join the vectors "head to tail" and then draw the resultant from tail of the vector to the head of the last vector.

25. Use the polygon law and find the resultant of the following vectors.

Solution: Join the vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ and $\vec{x}$ head to tail and finally draw the resultant from the tail of the first vector $\vec{U}$ to the head of the last vector $\vec{x}$ as shown below.



- $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{AE}$
- $\overrightarrow{AC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{AE}$
- $\overrightarrow{AD} + \overrightarrow{DE} = \overrightarrow{AE}$
- $\overrightarrow{AD} - \overrightarrow{AE} = \overrightarrow{ED} = -\overrightarrow{DE}$

Properties of Vector Addition

Addition of vectors is **commutative**

$$\vec{u} + \vec{v} = \vec{v} + \vec{u}$$

Addition of vectors is **associative**

$$(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$$

That is, the order of vector addition does not have an effect on the final result.

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5.7 Subtraction of Vectors

The subtraction of vectors is the addition of a vector to a negative vector. That is, if we have two vectors $\vec{U}$ and $\vec{V}$ then;

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$$

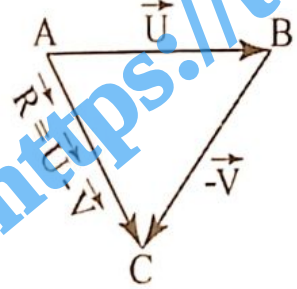
- The difference of the vectors $\vec{U}$ and $\vec{V}$ is the sum of $\vec{U}$ and $-\vec{V}$

26. Given that:

Find : $\vec{u} - \vec{v}$

Solution :- $\vec{u} - \vec{v} = \vec{u} + (-\vec{V})$

Using the triangle law; we have:



- $\overline{AB} + \overline{BC} = \overline{AC}$
- $\overline{AC} - \overline{AB} = \overline{BC}$
- $\overline{BC} - \overline{BA} = \overline{AC}$
- $\overline{CB} - \overline{CA} = \overline{AB}$

Properties of Vector Subtraction

Subtraction of vectors is not **commutative.**

$$\vec{u} - \vec{v} \neq \vec{v} + \vec{u}$$

Subtraction of vectors is **not associative.**

$$(\vec{u} - \vec{v}) - \overline{W} \neq (\vec{u} - \overline{W})$$

Analytically, Vectors can easily be added or subtracted in component form or in a unit vector notation.

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That is, if we have two vectors $\vec{U}$ and $\vec{V}$ such that;

$\vec{U} = (u_1, u_2)$, and $\vec{V} = (V_1, V_2)$, then

$\vec{U} + \vec{V} = (u_1 + V_1, u_2 + V_2)$

$\vec{U} - \vec{V} = (u_1 - V_1, u_2 - V_2)$

Illustrative Example

27. Given that $\vec{u} = (2, 4)$, $\vec{V} = (5, -1)$ and $\vec{W} = (-4, 3)$. Then, find

a. $\vec{u} + \vec{V}$ c. $\vec{u} + \vec{V} - \vec{W}$

b. $\vec{u} - \vec{V}$ d. $\vec{u} - (\vec{V} + \vec{W})$

Solution

a. $\vec{u} + \vec{V} = (u_1 + v_1, u_2 - v_2) = (2+5, 4+(-1)) = (7, 3)$

b. $\vec{u} - \vec{V} = (u_1 + v_1, u_2 - v_2) = (2 - 5, 4-(-1)) = (-3, 5)$

c. $\vec{u} + \vec{V} - \vec{W} = (7, 3) - (-4, 3) = (7-(-4), 3-(3)) = (11, 0)$

d. $\vec{u} - (\vec{V} + \vec{W}) = (2, 4) - (5+(-4)), (-1) + 3) = (2 - 1, 4 - 2) = (1, 2)$

28. Given that: $\vec{U} = 3i + 4j$ and $\vec{V}$ $7i - j$ Then, find

a. $\vec{u} + \vec{V}$ b. $\vec{u} - \vec{V}$ c. $|\vec{u} + \vec{V}|$

d. a unit vector in the directio of $\vec{u} - \vec{V}$

Solution

a. $\vec{u} + \vec{V} = (3 + 7)i + ((4) + (-1))j = 10i + 3j$

b. $\vec{u} - \vec{V} = (3 - 7)i + [(4) - (-10]j = -4i + 5i$

c. $|\vec{u} + \vec{V}| = \sqrt{(10)^2 + (3)^2} = \sqrt{109}$ units

d. $\left(\widehat{u - v}\right) = \dfrac{\vec{u} - \vec{v}}{\sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}}$

$= \dfrac{-4i + 5j}{\sqrt{(-4)^2 + (5)^2}} = \dfrac{1}{\sqrt{41}}(-4i + 5j)$

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29. Given that: $\vec{U} = 2i - 5j + 10k$ $\vec{V} = -i + 2j - 9k$

Then find:

a. $\vec{u} + \vec{V}$ b. $\vec{u} - \vec{V}$ c. $|\vec{u} + \vec{V}|$

d. unit vector in the directio of $\vec{u} + \vec{V}$

Solution

a. $\vec{u} + \vec{V} = [(2) + (-1)]i + [(-5) + (2)]j + [(10) + (-9)]k$

$\vec{u} + \vec{V} = i - 3j + k$

b. $\vec{u} - \vec{V} = [(2) - (-1)]i + [(-5) - (2)]j + [(10) + (-9)]k$

$\vec{u} - \vec{V} = 3i + 7j + 19k$

c. $|\vec{u} + \vec{v}| = \sqrt{(1)^2 + (-3)^2 + (1)^2} = \sqrt{11}$ units

d. $\left(u \hat{+} v\right) = \dfrac{\vec{u} + \vec{v}}{|\vec{u} + \vec{v}|} = \dfrac{i + 3j + k}{\sqrt{(1)^2 + (-3)^2 + (1)^2}} = \dfrac{1}{\sqrt{5}}(i - 3j + k)$

30. If the resultant of vector $\vec{U}$ and vector $\vec{V} = 2i + K$ is $i - 2j$, then vector $\vec{U}$ is ?

Solution : $\vec{u} + \vec{V} = i - 2j$

$(u_1 + v_1)i + (u_2 + V_2)j + (u_3 + V_3)k = i - 2j$

$\rightarrow u_1 + v_1 = 1$ and $u_2 + v_2 = -2$

$u_1 + 2 = 1$ and $u_2 + 0 = -2$

$u_1 = -1$ and $u_2 = -2$

$\rightarrow u_3 + v_3 = 0$ and $u_3 + 1 = 0$

$u_3 = -1$

$\therefore \quad \vec{u} = u_1 i + u_2 j + u_3 k$ and $\vec{u} = -i - 2j - k$

31. Given that: $\vec{U} = 3i - j$ and $\vec{U} + \vec{V} = 0$ Then, find $\vec{V}$

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Solution :- $\vec{u} + \vec{V} = 0$, $\vec{u} + \vec{V} = 0$

$\therefore \quad \vec{V} = -3i + j$

5.8 Multiplication of Vectors

A vector may be multiplied by a scalar or by another vector.

Multiplication of a Vector by a Scalar

When a vector is multiplied by a scalar, the result is a vector.

The product of vector $\vec{V}$ and scalar K $\vec{V}$ is vector K with magnitude $|K|$ times the magnitude of $\vec{V}$ and with direction same as $\vec{V}$ if $K > 0$ and opposite to $\vec{V}$ if $K < 0$.

In scalar multiplication, if $\vec{V} = 0$ or $k = 0$ then $K\vec{V} = 0$, null vector

- If two vectors are collinear or parallel, then any one of them can be expressed as scalar multiple of the other.

  That is, if $\vec{u} \,||\, \vec{V}$, then, $\vec{U} = K\vec{V}$

Properties of Scalar Multiplication

If $\vec{U}$ and $\vec{V}$ are vectors and K_1 and K_2 are scalars, then the following are true.

- The magnitude of the vector multiplied by a scalar is equal to the magnitude of the scalar times the magnitude of the vector.

  $|K\,u| = |K|\,|u|$

- Associative property : $K_1\left(K_2\vec{u}\right) = \left(K_1\ K_2\right)\vec{u}$

- Distributive property: $\left(K_1 + K_2\right)\vec{u} = K_1\,\vec{u} + K_2\,\vec{u}$

  $K_1\left(\vec{u} + \vec{v}\right) = K_1\,\vec{u} + K_1\,\vec{u}$

- Identity property : $1\left(\vec{u}\right) = \vec{u}$

- Multiplication by zero : $0\left(\vec{u}\right) = 0$, null vector

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Illustrative Example

32. Given that; $\left(\vec{u}\right) = (1, -3)$. Find: $\vec{u} = (1, -3)$

a. $2\vec{u}$ b. $-2\vec{u}$ c. $\left|2\vec{u}\right|$

Solution

a. $2\vec{u} = 2(1,-3) = (2(1), 2(-3)) = (2, -6)$

b. $-2\vec{u} = -2(1, -3) = (-2(1), -2(-3)) = (-2, 6)$

c. $\left|2\vec{u}\right| = \sqrt{(-2)^2 + (6)^2} = \sqrt{40} = 2\sqrt{10}$ units

33. Given that : $\vec{u} = 3\ 3i + j$ and $\vec{V} = j - 4j$ Then, find

a. $\frac{1}{2}\vec{u} - \vec{V}$ d. $-2\left(\vec{u} - \vec{V}\right)$ g. $-2\left(3\vec{u}\right)$

b. $\vec{u} + 3\vec{V}$ e. $(3 + 2)\vec{u}$

c. $3\left(2\vec{V}\right)$ f. $3\left(\vec{u} + \vec{V}\right)$

Solution

a. $\frac{1}{2}\vec{u} - \vec{V} = \frac{1}{2}(3i + j) - (i - 4j)$

$\frac{1}{2}\vec{u} - \vec{V} = \left(\frac{3}{2}i + \frac{1}{2}j\right) - (i - 4j)$

$\frac{1}{2}\vec{u} - \vec{V} = \left(\frac{3}{2} - 1\right)i + \left(\frac{1}{2} + 4\right)j = \frac{1}{2}i + \frac{9}{2}j$

b. $\vec{u} + 3\vec{V} = (3i + j) + 3(i - 4j) = (3 + 3)i + (1 - 12)j = -11j$

c. $3\left(2\vec{V}\right) = 6(i - 4j) = 6i - 24j$

d. $-2\left(\vec{u} - \vec{V}\right) = -2(3 - 1)i + -2(1 - (-4))j = -4i - 10j$

e. $(3 + 2)\vec{u} = 5\vec{u} = 5(3i + j) = 15i + 5j$

f. $3\left(\vec{u} + \vec{V}\right) = 3[(3 + 1)i + (1 + (-4))i] = 12i - 9j$

g. $-2\left(3\vec{u}\right) = -2[3(3i + j)] = -18i - 6j$

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34. Let $\vec{U} = (1, -3, 2)$ and $\vec{V} = (-4, -1, 0)$. Then find

a. $\vec{u} + 2\vec{V}$ b. $3\vec{u} - \vec{V}$ c. $\frac{-1}{2}\vec{V}$

Solution

a. $\vec{u} + 2\vec{V} = (1, -3, 2) + 2(-4, -1, 0)$

$\vec{u} + 2\vec{V} = (1 + (-8), -3 + (-2), 2 + 0) = (-7, -5, 2)$

b. $3\vec{u} - \vec{V} = 3(1, -3, 2) - (-4, -1, 0)$

$= (3, -9, 6) - (-4, -1, 0) = (7, -8, 6)$

c. $\frac{-1}{2}\vec{V} = \frac{-1}{2}(-4, -1, 0) = \left(2, \frac{1}{2}, 0\right)$

35. Let $\vec{U} = 6i - 2j + 4k$ and $\vec{V} = -3i + j - 2k$

Determine whether $\vec{U}$ and $\vec{V}$ are parallel or not

Solution:- Two vectors $\vec{U}$ and $\vec{V}$ are parallel if one is the scalar multiplication of the other.

$\vec{U} = 6i - 2j + 4k = -2(-3i + j + 2k)$

$\therefore \vec{U} = -2\vec{V}$, they are parallel.

Multiplication of Vector by another Vector

Two vectors may be multiplied to give a scalar quantity or vector quantity

5.9 What is Dot Product?

The dot product of two vectors is also called **inner product** or scalar **product** since the resulting quantity is a scalar.

The scalar product of two vectors is equal to the product of the magnitude of the vectors and the cosine of the angle between them.

That is, if we have the vectors $\vec{U}$ and $\vec{V}$

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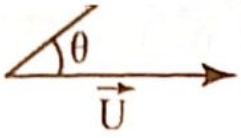

Then, their dot product is obtained by:

$$\vec{u} \cdot \vec{V} = |u|\,|V| \cos\theta$$

Where: $\vec{U} \cdot \vec{V}$ is read as "u dot V", and $0 \leq \theta \leq \pi$

And, if the vectors are given in unit vector notation :

$$\vec{U} = u_1 i + u_2 j + u_3 k \text{ and } \vec{V} = V_1 i + V_2 j + V_3 k$$

Then, their dot product is obtained by: $\vec{u} \cdot \vec{V} = u_1 V_1 + u_2 V_2 + u_3$

Properties of Dot Product

- The dot product a unit vector with itself is one: $i \cdot i = j \cdot j = k \cdot k = 1$
- The dot product of a unit vector with other unit vector is zero: $i \cdot j = j \cdot k = i \cdot k$
- Dot product is commutative: $\vec{U} \cdot \vec{V} = \vec{V} \cdot \vec{U}$
- Dot product of perpendicular vectors is zero if, $\vec{u} \cdot \vec{v} = 0$, then $\theta = 9$
- Dot product of parallel vectors is maximum if $\vec{u} \cdot \vec{v} = -|\vec{u}||\vec{v}|$, then $\theta = 0$
- Dot product of anti parallel vectors is negative if, $\vec{u} \cdot \vec{v} = -|\vec{u}||\vec{v}|$, then $\theta = 180^0$
- Distributive property: $\vec{u} \cdot (\vec{v} + \vec{W}) = \vec{u} \cdot \vec{v}$, then $\theta = 180^0$
- Distributive property: $\vec{u} \cdot (\vec{v} + \vec{W}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{W}$
- Scalar multiplication: $K(\vec{u} + \vec{v}) = k\vec{u} \cdot \vec{v} = \vec{u} \cdot K\vec{v}$
- If θ is acute, then: $\vec{U} \cdot \vec{V} > 0$
- If θ is obtuse, then: $\vec{U} \cdot \vec{V} < 0$
- $\vec{U} \cdot \vec{U} = |\vec{U}||\vec{U}| = |\vec{U}|^2$

  $\therefore \quad |U| = \sqrt{\vec{U} \cdot \vec{U}}$

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- $|\vec{U}+\vec{V}|^2 = \sqrt{(\vec{U}+\vec{v})\cdot(\vec{U}+\vec{v})} = \sqrt{|\vec{U}|^2 + 2\vec{U}\cdot\vec{v} + |\vec{v}|^2}$
- $|\vec{U}-\vec{V}|^2 = \sqrt{(\vec{U}-\vec{v})\cdot(\vec{U}-\vec{v})} = \sqrt{|\vec{U}|^2 - 2\vec{U}\cdot\vec{v} + |\vec{v}|^2}$

Illustrative Example

36. Given that;

a. $\vec{U} = 4i + j$, $\vec{V} = -3i + 21j$ b. $\vec{U} = 8i - 3j$, $\vec{V} = 2i + 7j$

Find $\vec{U} \cdot \vec{V}$ and state the nature of the angle

Solution

a. $\vec{U} \cdot \vec{V} = u_1 v_1 + u_2 v_2 = (4)(-3) + (1)(21) = 9$

θ is a obtuse angle, $\theta > 90^0$

b. $\vec{U} \cdot \vec{V} = (8)(2) + (-3)(7) = -5$

θ is obtuse angle, $\theta < 90^0$

37. Find the dot product of the following vectors:

a. $\vec{U} = (0, 3)$ and $\vec{V} = (4, -5)$

b. $|\vec{u}| = 4$ and $|\vec{V}| = 3$ and $\theta = 37^0$

c. $|\vec{u}| = 5$ and $|\vec{v}| = 4$ and $\theta = \frac{\pi}{3}$

d. $\vec{U} = 2i + 5j$ and $\vec{V} = 5i - 3j$

e. $\vec{U} = i - k$ and $\vec{V} = 2i + 4j$

f. $\vec{U} = i + j - 2k$ and $\vec{V} = 3i - 2j + k$

Solution

a. $\vec{U} \cdot \vec{V} = (0)(4) + (3)(-5) = -15$

b. $\vec{U} \cdot \vec{V} = |\vec{u}| \cdot |\vec{v}| \cos 37^0 = 9.6$

c. $\vec{u} \cdot \vec{v} = (5)(4)\cos\frac{\pi}{3} = 10$

d. $\vec{U} \cdot \vec{V} = (2)(5) + (5)(-3) = -5$

e. $\vec{U} \cdot \vec{V} = (i - k) \cdot (2i + 4j) = (1)(2) + (-1)(4) = -2$

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f. $\vec{U} \cdot \vec{V} = (i+j-2k)\cdot(3i-2j+k) = (1)(3)+(1)(-2)+(1)(-2)+(-2)(1)$

38. Give that;

a. $\vec{U} = (-2, 2)$ $\vec{V} = (-1, -1)$ and

b. $\vec{U} = (4, 3)$ and $\vec{V} = (1, 7)$

Find the angle between $\vec{U}$ and $\vec{V}$

Solution

a. $\vec{u} \cdot \vec{v} = |\vec{u}|\,|\vec{v}| \cos\theta$

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{|\vec{u}|\,|\vec{v}|} = \frac{u_1v_1 + u_2v_2}{\left(\sqrt{u_1^2 + u_2^2}\right)\left(\sqrt{v_1^2 + v_2^2}\right)}$$

$$\cos\theta = \frac{(-2)(-1) + (2)(-1)}{4} = \frac{0}{4} = 0$$

$$\theta = \cos^{-1}(0) = \frac{\pi}{2}$$

b. $\vec{u} \cdot \vec{v} = |\vec{u}|\,|\vec{v}| \cos\theta$

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{|\vec{u}|\,|\vec{v}|} = \frac{u_1v_1 + u_2v_2}{\left(\sqrt{u_1^2 + u_2^2}\right)\left(\sqrt{v_1^2 + v_2^2}\right)}$$

$$\cos\theta = \frac{(4)(1) + (3)(7)}{\left(\sqrt{25}\right)\left(\sqrt{50}\right)} = \frac{25}{25\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

39. Find the cosine of the angle between $\vec{U}$ and $\vec{V}$

a. $\vec{U} = i + j - k$ and $\vec{V} = 2i - 3j + 4k$

b. $\vec{U} = -4i + 5j + 7k$ and $\vec{V} = i - 2j + 2k$

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Solution

a. $\vec{u} \cdot \vec{v} = |\vec{u}|\,|\vec{v}| \cos\theta$

$$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}|\,|\vec{v}|} = \frac{(1)(2) + (1)(-3) + (-1)(4)}{\left(\sqrt{(1)^2 + (1)^2 + (-1)^2}\right)\left(\sqrt{(2)^2 + (-3)^2 + (4)^2}\right)}$$

$$\cos\theta = \frac{-5}{\sqrt{87}}$$

b. $\cos\theta = \dfrac{(-4)(1) + (5)(-2) + (7)(2)}{\sqrt{99}} = \dfrac{0}{\sqrt{99}} = 0$

40. Let $\vec{U} = (2, -2)$, $\vec{V} = (5,8)$ and $\vec{W} = (-4, 3)$ Then find $(\vec{U} \cdot \vec{V})\,\vec{W}$

Solution: $(\vec{U} \cdot \vec{V})\,\vec{W} = (2, -2)(5,8)(-4, 3)$

$= (10 - 16)(-4, 3) = -6(-4, 3) = (24, -18)$

41. Determine m such that the two vector are orthogonal

a. $\vec{U} = 5mi + 3j$ and $\vec{V} = 2i + 7j$

b. $\vec{U} = mi - 3j$ and $\vec{V} = mi + 2mj$

Solution

a. $\vec{U} \cdot \vec{V} = (5m)(2) + (3)(7) = 0$ implies, $10m + 21 = 0$

$m = -21$

b. $\vec{U} \cdot \vec{V} = (m)(m) + (-3)(2m) = 0$

$m^2 - 6m = 0$ implies that $m(m - 6) = 0$

$\therefore \quad m = 0$ or $m = 6$

42. Show that the vectors :

$\vec{U} = 2i - j + k$, $\vec{V} = i - 3j - 5k$, and $\vec{W} = 3i - 4j - 4k$

form a right angle triangle

Solution: $\vec{U} + \vec{V} = \vec{W}$, and $\vec{U} \cdot \vec{V} = 0$

Hence, they are sides of a right angled triangle

43. Determine the following vectors as perpendicular, parallel or neither

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a. $\vec{U} = (-5, 3)$ and $\vec{V} = (6, -8)$ c. $\vec{U} = (4, 6)$ and $\vec{V} = (-3, 2)$

b. $\vec{U} = (-1, 1/2)$ and $\vec{V} = (3, 6)$ d. $\vec{U} = (-2, -6)$ and $\vec{V} = (-3, -9)$

Solution

a. $\vec{U} \cdot \vec{V} = (-5)(6) + (3)(-8) = -54$, neither

b. $\vec{u} \cdot \vec{v} = (-1)(3) + \left(\frac{1}{2}\right)(6) = 0$, perpendicular

c. $\vec{U} \cdot \vec{V} = (4)(-3) + (6)(2) = 0$, perpendicular

d. $\vec{U} \cdot \vec{V} = (-2)(-3) + (-6)(-9) = 60$, parallel

44. Given that; $\theta = \frac{\pi}{3}$ and $|u| = 2, |v| = 4$. Find

a. $(\vec{u} - \vec{v}) \cdot (\vec{u} + \vec{v})$ c. $(\vec{u} + \vec{v}) \cdot (\vec{u} - 2\vec{v})$

b. $(\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v})$ d $(\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v})$

Solution

a. $(\vec{u} - \vec{v}) \cdot (\vec{u} + \vec{v}) = |u|^2 - |v|^2 = (2)^2 - (4)^2 = -12$

b. $(\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = |u|^2 + 2\vec{u} \cdot \vec{v} + |v|^2 = 2^2 + 2(2)(4)(0.5) + 4^2 = 28$

c. $(\vec{u} + \vec{v}) \cdot (\vec{u} - 2\vec{v}) = |u|^2 - \vec{u} \cdot \vec{v} - 2|v|^2 = (2)^2 - 2(4)(0.5) - 2(4)^2 = -32$

d $(\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = |u|^2 - 2u.v + |v|^2 = (2)^2 - 2(4)(2)(0.5) + (4)^2 = 12$

45. Let $|a| = 7$, $|b| = 3$ and $|a + b| = 6$, then determine

a. $\vec{a} \cdot \vec{b}$ b. $|\vec{a} - \vec{b}|$ c. $|2\vec{a} + 3\vec{b}|$

d. angle between $\vec{a}$ and $\vec{b}$

Solution

a. $|\vec{a} + \vec{b}|^2 = |a|^2 + 2\vec{a} \cdot \vec{b} + |b|^2$

$36 = 7^2 + 2\vec{a} \cdot \vec{b} + 3^2$

$\therefore \quad \vec{a} \cdot \vec{b} = \frac{36 - 49 - 9}{2} = \frac{-22}{2} = -11$

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b. $\left|\vec{a} - \vec{b}\right|^2 = |a|^2 - 2\,\vec{a} \cdot \vec{b} + |b|^2 = 7^2 + -2(-11) + 3^2 = 80$

$\therefore \quad \left|\vec{a} - \vec{b}\right| = \sqrt{80} = 4\sqrt{5}$

c. $\left|2\vec{a} + 3\vec{b}\right|^2 = 4|a|^2 + 12\,\vec{a} \cdot \vec{b} + 9\,|b|^2$

$= 4(49) + 12(-11) + 9(9) = 145$

$\therefore \quad \left|2\vec{a} + 3\vec{b}\right| = \sqrt{145}$

d. $\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{\left|\vec{a}\right|\left|\vec{b}\right|} = \dfrac{-11}{(7)(3)} = \dfrac{-11}{21}$

$\therefore \quad \theta = \cos^{-1}\left(\dfrac{-11}{21}\right) = 122^0$

46. Suppose a triangle has verities A (1,5), B(2, 2), and C(4, 6). what is the interior angle at C?

A. $\dfrac{\pi}{3}$ B. $\dfrac{\pi}{2}$ C. $\dfrac{\pi}{4}$ D. $\dfrac{2\pi}{3}$

Solution

$\overrightarrow{CA} = (1,5) - (4,6) = (-3, -1)$ and $\overrightarrow{CB} = (2,2) - (4,6) = (-2, -4)$

$\cos(C) = \dfrac{\overrightarrow{CA} \cdot \overrightarrow{CB}}{\left|\overrightarrow{CA}\right| \cdot \left|\overrightarrow{CB}\right|} = \dfrac{(-3, -1) \cdot (-2, -4)}{\sqrt{9+1} \cdot \sqrt{4+16}} = \dfrac{6+4}{\sqrt{200}} = \dfrac{10}{10\sqrt{2}} = \dfrac{1}{\sqrt{2}}$

$C = \cos^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{4}$

Answer: C

47. What is the work done by a force $\vec{F} = (2i + 3j + k)N$ while displacing a body to $\vec{S} = (5i + 6j)$ m

Solution

$W = \vec{F} \cdot \vec{S} = (2i + 3j - k) \cdot (5i + 6j)$

$= (2)(5) + (3)(6) + (1)(0) = 28$ J

48. What power is developed by the engine of a car that exerted a force of $\vec{F} = (5i + 8j - k)$ N to accelerate the car from rest to a speed $\vec{V}$

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$(10i + j)$ m/s

Solution

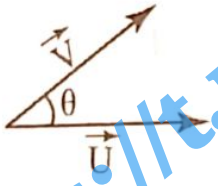
$$P = \vec{F} \cdot \vec{V} = (5i + 8j - k)\,(10i + j)$$
$$= (5)(10) + (8)(1) + (-1)(0) = 58 \text{ W}$$

5.10 What is Cross Product of Vectors?

The cross product of two vectors is also called vectors product since the resulting quantity is a vector, this vectors does not lie in the plane of given vectors, but **perpendicular** to each other.

The magnitude of the cross product of two vectors is equal to the product of their magnitudes and the sine of the angle between them.

That is, we have the vectors.



Then, the magnitude of their cross product is $|\vec{u} \times \vec{v}| = |u|\,|v| \sin\theta$

When the vectors are given in unit vector notation:

$$\vec{U} = u_1 i + u_2 j + u_3 k \text{ , and } \vec{V} = v_1 j + v_2 j + v_3 k$$

Their cross product is obtained by

$$\vec{u} \times \vec{v} = (u_2\ v_2 - u_3\ v_2)i + (u_3\ v_1 - u_1\ v_3)j + (u_1 v_2 - u_2 v_1)k$$

The above relation can easily be remembered using the determinant notation as follows:

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} i - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} j + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} K$$
$$= (u_2 v_3 - u_3 v_2)i + (u_3 v_1 - u_1 v_3)j + (u_1 v_2 - u_2 v_1)k$$

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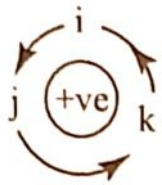
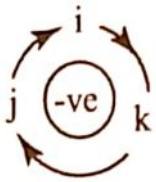
With the fact that:



and



Therefore , $i \times j = k$
$j \times k = i$
$k \times i = j$

$j \times i = -k$
$k \times j = -i$
$i \times k = -j$

Properties Cross Product

Cross product of unit vector with itself is zero

$$i \times j = j \times j = k \times k = 0$$

Cross product of unit vector with other unit vector is one

$$|i \times j| = |j \times k| = |i \times k| = 1$$

Cross product is **Not commutative**

$$\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$$

Cross product of parallel vectors is **zero**

If , $\vec{u} // \vec{v}$, then $\vec{u} \times \vec{v} = 0$

Distributive property

$$\vec{u} \times \left(\vec{v} + \vec{W}\right) = \vec{u} \times \vec{v} + \vec{u} \times \vec{W}$$

$$\vec{u} \cdot \left(\vec{v} \times \vec{W}\right) = \left(\vec{u} \times \vec{v}\right) \cdot \vec{W} \longleftarrow \text{scalar triple product}$$

49. Find the cross product of the following vectors

a. $\vec{u} = (1,3)$ and $\vec{v} = (-1, 2)$

b. $\vec{u} = 3i - k$ and $\vec{v} = 2i + 4j$

c. $\vec{u} = 3$ and $\vec{v} = 8$ and $\theta = \frac{\pi}{6}$

d. $\vec{u} = i - 2j + 2k$ and $\vec{v} = 3i + j - 2k$

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Solution

a. $\bar{u} \times \bar{v} = \begin{vmatrix} i & j & k \\ 1 & 3 & 0 \\ -1 & 2 & 0 \end{vmatrix} = \begin{vmatrix} 3 & 0 \\ 2 & 0 \end{vmatrix} j - \begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} j + \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} k$

$= [(3)(0) - (0)(2)]i + [(0)(-1) - (1)(0)]j + [(1)(2) - (3)(-1)]K$

$= 0i + oj + 5K = 5K$

b. $\bar{u} \times \bar{v} = \begin{vmatrix} i & j & k \\ 3 & 0 & -1 \\ 2 & 4 & 0 \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 4 & 0 \end{vmatrix} i - \begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix} j + \begin{vmatrix} 3 & 0 \\ 2 & 4 \end{vmatrix} k$

$= [(0)(0) - (-1)(4)]i + [(-1)(2) - (3)(0)]j + [(3)(4) - (0)(2)]K$

$= 4i - 2j + 12k$

c. $\bar{u} \times \bar{v} = |\bar{u}||\bar{v}| \sin\theta = (3)(8)(0.5) = 12$ units

d. $\bar{u} \times \bar{v} = \begin{vmatrix} i & j & k \\ 3 & -2 & 2 \\ 3 & 1 & -2 \end{vmatrix} = \begin{vmatrix} -2 & 2 \\ 1 & -2 \end{vmatrix} i - \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} j + \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} K$

$= [(2)(3) - (1)(-2)]i + [(2)(3) - (1)(-2)]j + [(1)(1) - (-2)(3)]K$

$= 2i + 8j + 7k$

50. Give that;

$\bar{U} = 3i + j\,2k$ and $\bar{V} = 2i - 2j + 4k$

Find a unit vector perpendicular to both $\bar{U}$ and $\bar{V}$

Solution

$\bar{u} \times \bar{v} = \begin{vmatrix} i & j & k \\ 3 & 1 & 2 \\ 2 & -2 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ -2 & 4 \end{vmatrix} i - \begin{vmatrix} 3 & 2 \\ 2 & 4 \end{vmatrix} j + \begin{vmatrix} 3 & 1 \\ 2 & -2 \end{vmatrix} K$

$= [(1)(4) - (2)(-2)]i + [(2)(2) - (3)(4)]j + [(3)(-2) - (1)(2)]$

Let this be $\bar{W} = 8i - 8j - 8k$

$\hat{W} = \dfrac{\bar{W}}{|W|} = \dfrac{8i - 8j - 8K}{8\sqrt{3}} = \dfrac{i - j - k}{\sqrt{3}}$

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51. Find the triple vector product of

$\vec{U} = i + 2j - k$

$\vec{V} = 3i + 2j - 4k$, and $\vec{W} = 2i - j + 3k$

Solution

$$\vec{u} \times \vec{v} \times \vec{W} = (\vec{u} \times \vec{v}) \times \vec{W}$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 1 & 2 & -1 \\ 3 & 2 & -4 \end{vmatrix} = -6i + j - 4K$$

$$(-6i + j - 4k) \times \vec{W} = \begin{vmatrix} i & j & k \\ -6 & 1 & -4 \\ 2 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & -4 \\ -1 & 3 \end{vmatrix} i - \begin{vmatrix} -6 & -4 \\ 2 & 3 \end{vmatrix} j + \begin{vmatrix} -6 & 1 \\ 2 & -1 \end{vmatrix} k$$

$$= -i + 10j + 4K$$

Geometric properties of the Cross Product

Let $\vec{U} \times \vec{V}$ be non zero vectors in space, and let θ be the angle between $\vec{U}$ and $\vec{V}$

1. $\vec{U} \times \vec{V}$ is orthogonal to both $\vec{U}$ and $\vec{V}$
2. $|\vec{U} \times \vec{V}| = |u||v| \sin\theta$
3. $\vec{U} \times \vec{V} = 0$ if and only if vector $\vec{U}$ and $\vec{V}$ are parallel
4. $|\vec{U} \times \vec{V}|$ = Area of parallelogram having $\vec{U}$ and $\vec{V}$ as adjacent sides.
5. $\dfrac{u \times v}{|u \times v|}$ = unit vector orthogonal to both $\vec{u}$ and $\vec{v}$
6. $\left|\vec{u} \cdot (\vec{v} \times \vec{W})\right|$ = Volume of a parallelepiped having $\vec{u}, \vec{v}$ and $\vec{W}$ as adjacent sides.

52. Given that

$\vec{U} = 4i - j + 3k$, and $\vec{V} = 2i + 3j - k$

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Then, find a vector perpendicular to both vectors.

Solution

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 4 & -1 & 3 \\ 2 & 3 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 3 \\ 3 & -1 \end{vmatrix} i - \begin{vmatrix} 4 & 3 \\ 2 & -1 \end{vmatrix} j + \begin{vmatrix} 4 & -1 \\ 2 & 3 \end{vmatrix} K$$

$$= [(-1)(-1) - (3)(3)]i + [(3)(2) - (4)(-1)]j + [(4)(3) - (-1)(2)]K$$

$$= -8i + 10j + 14K, \text{ this is it!}$$

53. Find the area of the triangle whose two sides are formed by the vectors:

$$\vec{U} = 2i + j \text{ and } \vec{V} = i + 3j + 2k$$

Solution

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 2 & 1 & 0 \\ 1 & 3 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} i - \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} j + \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} K$$

$$= [(1)(2) - (0)(3)]i + [(0)(1) - (2)(2)]j + [(2)(3) - (1)(1)]K$$

$$= 2i - 4j + 5K$$

$$\text{Area of triangle} = \frac{1}{2}|\vec{u} \times \vec{v}|$$

$$= \frac{1}{2}\left(\sqrt{(2)^2 + (-4)^2 + (5)^2}\right) = \frac{3\sqrt{5}}{2} \text{ units square}$$

54. Let P = (3, -2, 1), Q = (7, -3, 4), R = (5, 1, 0) be vertices of a triangle.

Find the area of the triangle

Solution

$$\overrightarrow{PQ} = 4i - j + 3k, \text{ and } \overrightarrow{PR} = 2i + 3j - k$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} i & j & k \\ 4 & -1 & 3 \\ 2 & 3 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 3 \\ 3 & -1 \end{vmatrix} i - \begin{vmatrix} 4 & 3 \\ 2 & -1 \end{vmatrix} j + \begin{vmatrix} 4 & -1 \\ 2 & 3 \end{vmatrix} K$$

$$= -8i + 10j + 14K$$

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Area of triangle $= \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{1}{2}\sqrt{(-8)^2 + (10)^2 + (14)^2}$

$= 3\sqrt{10}$ units square

55. Find the area of a parallelogram whose sides are:

$\vec{U} = i + 2j + 3k$ and $\vec{V} = 3i - 2j + k$

Solution

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 3 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix} i - \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} j + \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} K$$

$= 8i + 8j - 8k$

Area of parallelogram $= \left|\vec{U} \times \vec{V}\right| = \sqrt{(-8)^2 + (8)^2 + (-8)^2}$

$= 8\sqrt{3}$ units square

56. Let $\vec{U} = i - j$, $\vec{V} = 2i + 3j - k$ and $\vec{W} = -i + 2k$. Find the volume of the parallelopiped determined by the vectors $\vec{U}$, $\vec{V}$ and $\vec{W}$

Solution : $\vec{V} \times \vec{W} = \begin{vmatrix} i & j & k \\ 2 & 3 & -1 \\ -1 & 0 & 2 \end{vmatrix} = 6i - 3j + 3K$

Volume $= \vec{u} \cdot \left(\vec{V} \times \vec{W}\right) = (i - j) \cdot (6i - 3j + 3k) = 9$ units cube

57. Find the torque acting on a body by a force $\vec{F} = (i + 10j - k)$ N exerted at a distance of $\vec{r} = (10i - j + 5k)$ m from the pivot point

Solution

$$\vec{J} = \vec{F} \times \vec{r} = \begin{vmatrix} i & j & k \\ 1 & 10 & -1 \\ 10 & -1 & 5 \end{vmatrix}$$

$= [(50)(5) - (-1)(-1)]i + [-1(10) - 1(5)]\, j + [1(-1) - (10)(10)]K$

$= (249i - 15j - 101K)$ NM

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58. Determine whether the vectors

$\vec{U} = 6i - 2j + 4k$ and $\vec{V} = 3i - j + 2k$ are parallel or not

Solution

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 6 & -2 & 4 \\ 3 & -1 & 2 \end{vmatrix} = \begin{vmatrix} -2 & 4 \\ -1 & 2 \end{vmatrix} i - \begin{vmatrix} 6 & 4 \\ 3 & 2 \end{vmatrix} j + \begin{vmatrix} 6 & -2 \\ 3 & -1 \end{vmatrix} k$$

$$= [(-2)(2) - (-1)(4)]i + [(4)(3) - 6(2)]\, j + [1(6) - (-2)3]k$$

$$= 0, \text{ they are parallel indeed!}$$

59. Given the vectors : $\vec{u} = 4i - j + 3k$ and $\vec{V} = 2i + 3j - k$

Find, $\vec{u} \times \vec{V}$

Solution

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 2 & 3 & -1 \\ 4 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} i - \begin{vmatrix} 2 & -1 \\ 4 & 3 \end{vmatrix} j + \begin{vmatrix} 2 & 3 \\ 4 & -1 \end{vmatrix} k$$

$$= 8i - 10j - 14K$$

60. What can you say about the angle between non - zero vectors $\vec{U}$ and $\vec{V}$ if $\vec{U} \cdot \vec{V} = |\vec{U} \times \vec{V}|$

Solution

$$\vec{u} \cdot \vec{v} = |\vec{u} \times \vec{v}|$$

$$= |\vec{u}|\,|\vec{v}| \cos\theta = |\vec{u}|\,|\vec{v}| \sin\theta$$

$$\cos\theta = \sin\theta$$

$$\therefore \quad \theta = \frac{\pi}{4}$$

5.11 Vectors and Lines

A straight line in a plane is determined by its slope and a point through which it passes. Consider two points in line L such as $R_o(x_o, y_o)$ and $R_1(x_1, y_1)$ and let r_0 and r_1 are position vectors

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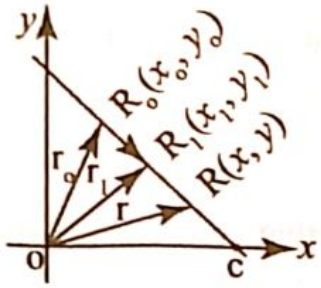
corresponding to the points R_o and R_1 respectively, then the vector from R_o to R_1 is;

$$V = r_1 - r_o = (x_1 - x_o, y_1 - y_o)$$



For $t \in R$, we have : $\overrightarrow{OR_0} + \overrightarrow{R_o R} = \overrightarrow{OR}$

$$r_o + tv = r$$

Where, t is a scalar and V is a direction vector of the line and $V \neq 0$

$$\boxed{r = r_0 + tv \longleftarrow \text{Vector Equation of the line}}$$

If $V = (P, q)$, $R(x,y)$ and $R_o(x_o, y_o)$, then we have:

$$r = r_o + tv$$

$$(x, y) = (x_o, y_o) + t(P, q)$$

$$(x, y) = (x_o, y_o) + (tp, tq)$$

$$(x, y) = (x_o + tP, y_o + tq)$$

Then, from equality of vectors, we have

$$\left.\begin{aligned} x &= x_o + tP \\ y &= y_o + tq \end{aligned}\right\} \text{ parametric Equation of the line}$$

$t \in R$ and $(p, q) \neq (0, 0)$

If P and q are both different from zero, solving for t we get;

$$x = x_0 + tp, \quad t = \frac{x - x_0}{P}$$

$$y = y_0 + tq, \quad t = \frac{y - y_0}{q}$$

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$\frac{x - x_0}{P} = \frac{y - y_0}{q}$ ⟵ standard Equation of the line

Illustrative Examples

61. Find vector equation of a line that contains (-1, 3) and is parallel to $2i - 3j$

Solution: $r = r_o + tV \Rightarrow (x, y) = (-1 - 3) + t(2, -3)$

62. Find the parametric equation of the line that contains (2, -4) and is parallel $3i + \frac{1}{2}j$

Solution :- $r = r_o + tV = (2, -4) + \left(3, \frac{1}{2}\right)t$

$$(x, y) = (2, -4) + \left(3t, \frac{1}{2}t\right)$$

$$\therefore \quad x = 2 + 3t \text{ and } y = -4 + \frac{1}{2}t$$

63. Find the standard equation of the line containing the points P(4, -6) and q (2, -3)

Solution

$$V = \overline{Pq} = (-2, 3)$$

$$r = r_0 + tV \Rightarrow (x, y) = (4, -6) + t(-2, 3)$$

$$x = 4 - 2t \text{ and } y = -6 + 3t$$

$$\therefore \quad t = \frac{x - 4}{-2} = \frac{y + 6}{3}$$

64. Consider the line passing through the point $R_o(1, 5)$ and $R(-2, -1)$ then find

a. The vector equation

b. The parametric equation

c. The standard equation of the line

Solution

a. The direction vector is obtained by;

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$V = \overrightarrow{R_oR} = R - R_o = (-2, -1) - (1, 5) = (-3, -6)$

$r = r_o + tv$ implies $(x, y) = (1, 5) + t(-3, -6)$

$\therefore \quad (x, y) = (1, 5) + t(-3, -6)$ ← Vector Equation

b. $x = 1 - 3t$ and $y = 5 - 6t$ ← Parametric Equation

c. $\frac{x - 1}{-3} = \frac{y - 5}{-6}$ ← Standerd Equation

65. Consider a line passing through the point $R_o(1, -2)$ and is parallel to the vector $V = (2, 4)$, then find

a. The vector equation

b. The parametric equation

c. The standard equation of the line

Solution

a. The direction vector is $V = (2, 4)$,

$r = r_o + tV = (1, -2) + t(2, 4)$

$(x, y) = (1, -2) + t(2, 4)$

b. $x = 1 + 2t$ and $y = -2 + 4t$

c. $\frac{x - x_o}{P} = \frac{y - y_o}{q} \Rightarrow \frac{x - 1}{2} = \frac{y + 2}{4}$

66. Consider a line passing through the point $R_o(3, -1)$ and is perpendicular to the vector $V = (4, 2)$ then find

a. The vector equation

b. The parametric equation

c. The standard equation of the line

Solution

a. Let $R(x, y)$ and $R_o(3, -1)$ be points on the line

$\overrightarrow{R_oR} = (x, y) - (3, -1) = (x - 3, y + 1)$

Since $\overrightarrow{R_oR}$ is perpendicular to the vector V, $\overrightarrow{R_oR} \cdot \overrightarrow{V} = 0$

$(x - 3, y + 1) \cdot (4, 2) = 0$

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$4x - 12 + 2y + 2 = 0$

$4x + 2y = 10$

$2x + y = 5$

The direction vector is obtained by setting: $2x + y = 0, y = -2x$

The set of all direction vectors is then.

$V = (x, y) = (x, -2x) = x(1, -2) = t(1, -2)$

$r = r_o + tv\ (x, y) = (3, -1) + t(1, -2)$

b. $x = 3 + t$ and $y = -1 - 2t$

c. $\frac{x - x_o}{P} = \frac{y - y_o}{q}$ it gives $\frac{x - 3}{1} = \frac{y + 1}{-2}$

67. Consider the line passing through the point $R_o(0, 3)$ and is parallel to the line, $x = -5 + t$, $y = 1 - 2t$, then find

a. The vector equation

b. The parametric equation

c. The standard equation of the line

Solution

a. $r = r_o + tv$, $v = (1, -2)$

$(x, y) = (0, 3) + t(1, -2)$

b. $x = x_o + tp = 0 + t = t$ and $y = y_o + tq = 3 - 2t$

c. $\frac{x - x_o}{P} = \frac{y - y_o}{q}$ it gives $\frac{x + 5}{1} = \frac{y - 1}{-2}$

68. If the line passes through the point (1, -4) and is parallel to the vector $\bar{V} = (-2, 5)$, then determine the points that lie on the line and find t

A. (-1, 1) B. (-3, 9)

Solution

A. $(x, y) = (1, -4) + t(-2, 5)$ ⟵ vector Equation

$x = 1 - 2t$, $y = -4 + 5t$ ⟵ parametric Equation

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$$\frac{x-1}{-2} = \frac{y+4}{5}$$

$$5x + 2y = -3$$

check for (-1, 1): $5x + 2y = -3$

$$5(-1) + 2(1) = -3$$

Point, (-1, 1) lies on the line:

$x = 1 - 2t$ or $y = -4 + 5t$

$-1 = 1 - 2t$ $1 = -4 + 5t$

$t = 1$ $t = 1$

B. Check for (-3, 9)

$5x + 2y \neq -3$

$5(-3) + 2(9) = -3$

Point (-3, 9) does not lie on the line :

$x = 1 - 2t$ and $y = -4 + 5t$

$-3 = 1 - 2t$ $9 = -4 + 5t$

$t = 2$ $t = \frac{13}{5}$

different t values

69. Determine whether the following points are collinear or not

(i) A (1, -1), B (-2, 5), C (3, -5)

(ii) A (2, 5), B (-1, -4), C (-2, -5)

Solution: The set of all points on the line is determined by the vector equation

(i) $(x, y) = (x_0, y_0) + tv$, $V = (-2, 5) - (1, -1) = (-3, 6)$

$(x, y) = (1, -1) + t(-3, 6)$

or $(x, y) = (1 - 3t, -1 + 6t)$

A. $(1, -1) = (1 - 3t, -1 + 6t) \Rightarrow 1 = 1 - 3t$ and $-1 = -1 + 6t$ in both cases, $t = 0$, hence $(1, -1) = (1, -1)$ Ok

B. $(-2, 5) = (1 - 3t, -1 + 6t) \Rightarrow -2 = 1 - 3t$ and $5 = -1 + 6t$ in both cases,

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t = 1 hence (-2, 5) = (-2, 5), Ok

C. $(3, -5) = (1 - 3t, -1 + 6t) \Rightarrow 3 = 1 - 3t$ and $-5 = -1 + 6t$ in both cases,

$t = \frac{-2}{3}$, hence (3, -5) = (3, -5) Ok.

Therefore, the points (1, -1), (-2, 5) and (3, -5) are collinear.

(ii) $(x, y) = (x_o, y_o) + tv$, V = (-1, -4) - (2, 5) = (-3, -9)

$(x, y) = (2, 5) + t(-3, -9) = (2 - 3t, 5 - 9t)$

A. $(2, 5) = (2 - 3t, 5 - 9t) \Rightarrow 2 = 2 - 3t$ and $5 = 5 - 9t$
in both cases, t = 0, hence (2, 5), Ok

B. $(-1, -4) = (2 - 3t, 5 - 9t) \Rightarrow -1 = 2 - 3t$ and $-4 = 5 - 9t$
in both cases, t = 1, hence (-1, -4) = (-1, -4), Ok

C. $(-2, -5) = (2, 3t, 5 - 9t) \Rightarrow t = \frac{4}{3}$ and $-5 = 5 - 9t \Rightarrow t = \frac{10}{9}$, not o

Therefore, the points (2, 5), (-1, -4) and (-2, -5) are not collinear

70. Find the parametric equation of the line

a. L : $3x + 4y = 10$ b. L : $\frac{x - 1}{5} = \frac{y + 6}{7}$

Solution

a. The set of all direction vector parallel to the line is $3x + 4y = 0$, $y = \frac{-3}{4}x$

$V = (x, y) = t\left(x, \frac{-3}{4}x\right) = t\left(1, \frac{-3}{4}\right) = t(4, -3)$

$3x + 4y = 10$, When $x = 2$, $y = 1 \Rightarrow (x_o, y_o) = (2, 1)$

$(x, y) = (2, 1) + t(4, -3)$

$x = 2 + 4t$ and $y = 1 - 3t \leftarrow$ parametric Equation

b. L : $\frac{x - 1}{5} = \frac{y + 6}{7}$, V = (5, 7) and R_o (1, -6)

$(x, y) = (1, -6) + t(5, 7)$

$x = x_o + t_p$ and $y = y_o + t_q$

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$x = 1 + 5t$ and $y = -6 + 7t$ ← parametric Equation

71. Consider a line passing through the point having vector $5i + 4j$ with direction vector $(-3, 4)$, then find:

a. The vector equation

b. The parametric equation

c. The standard equation

Solution

a. $r = r_o + tv$ and $(x, y) = (5, 4) + t(-3, 4)$

or, $r = (5 - 3t)i + (4 + 4t)j$

b. $x = x_o + t_p = 5 - 3t$ and $y = y_o + t_q = 4 + 4t$

c. $\frac{x - x_o}{P} = \frac{y - y_o}{q}$ implies $\frac{x - 5}{-3} = \frac{y - 4}{4}$

72. Find an equation of the line in slope - intercept form whose parametric equations are:

a. $x = 3 - 4t$ and $y = -2 + 5t$ b. $x = 3t - 5$ and $y = 4t$

Solution

a. Solve each parametric equation for t.

$x = 3 - 4t$, implies that $-4t = x - 3$ and $t = \frac{x - 3}{-4}$

$y = -2 + 5t$, implies that $t = \frac{y + 2}{5}$

Equating for t: $\frac{x - 3}{-4} = \frac{y + 2}{5}$

$5(x - 3) = -4(y + 2)$

$5x - 15 = -4y - 8$, this implies $5x + 8 - 15 = -4y$

$\therefore\ y = \frac{-5}{4}x + \frac{7}{4}$

b. $x = 3t - 5$, equal to $t = \frac{x + 5}{3}$

$y = 4t$, equal to $t = \frac{y}{4}$

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